The Corporation of the Town of Ajax

GENERAL GOVERNMENT COMMITTEE

Tuesday, April 11, 2023
Open Meeting at 1:00 p.m.
Hybrid Meeting
Council Chambers, Town Hall
65 Harwood Avenue South

AGENDA

Alternative formats available upon request by contacting:
accessibility@ajax.ca or 905-619-2529 ext. 3347

Electronic Meeting: In-person attendance is permitted at this meeting of the General Government Committee. This meeting will be livestreamed for viewing at ajax.ca/live. Electronic participation is permitted pursuant to the Municipal Act and Council's Procedure By-law.

Online Agenda: Anything in **blue** denotes an attachment/link. By clicking the links on the agenda page, you can jump directly to that section of the agenda.

R. Tyler Morin, Chair
M. Crawford, Vice Chair

Open Meeting

1. Call To Order & Acknowledgement of Traditional Treaty Lands

2. Disclosure of Pecuniary Interest

3. Adoption of Minutes

3.1 **Regular Meeting – March 20, 2023** 3

3.2 **Closed Session Meeting – March 20, 2023** .....(circulated separately)
- *Any discussion will be held in the Closed Session*

4. Consent Agenda – *Considered to be routine, these items may be approved by one motion. Items may be separated and referred to the Discussion Agenda*

4.1 **OES-2023-06: Contract Award – Ajax Library Makerspace Renovation**
D. Meredith, Director of Operations & Environmental Services / L. Sinclair, Supervisor of Facility Capital Projects & Asset Management 9

4.2 **OES-2023-07: Contract Award – T23012 Road Resurfacing & Geotechnical Services**
D. Meredith, Director of Operations & Environmental Services / M. Boyle, Capital Projects Coordinator 29

4.3 **OES-2023-10: Preliminary Report on a Potential Over-Expenditure – MCC Pool Renovations**
D. Meredith, Director of Operations & Environmental Services / L. Sinclair, Supervisor of Facility Capital Projects & Asset Management 37

4.4 **OES-2023-08: Contract Award – St. Andrew's Site Improvements**
D. Meredith, Director of Operations & Environmental Services/ A. Jackson, Supervisor, Infrastructure Design and Construction 41

4.5 **PDS-2023-14: Update to the Residential Parking Permit Feasibility Study**
G. Romanowski, Director of Planning & Development Services / D. Pandya, Manager of Transportation Planning 48

4.6 **RCCD-2023-03: Bus Transportation Services**
C. Vida, Director of Recreation & Culture / A. Kushner, Community Recreation Supervisor.....55

4.7 **LIS-2023-05: Corporate Information Technology Strategic Plan – Year Three Update**
N. Cooper, Director of Legislative & Information Services/Clerk / C. Walker, Manager of Technology and Innovation 58

4.8 **LIS-2023-06: Integrity Commissioner Services – Recommendation to Appoint Principles Integrity**
N. Cooper, Director of Legislative & Information Services/Clerk.....67

5. Discussion

5.1 **PDS-2023-13: Traffic Calming Program Reporting Process**
G. Romanowski, Director of Planning & Development Services / D. Pandya, Manager of Transportation Planning 71

6. Presentations

6.1 **OES-2023-04: Audley Recreation Centre Fairgrounds Feasibility Study**
D. Meredith, Director of Operations & Environmental Services/ A. Jackson, Supervisor, Infrastructure Design and Construction 80

6.2 **PDS-2023-12: 2023 Transportation Demand Management Plan Update**
G. Romanowski, Director of Planning & Development Services / A. De Francia, Active Transportation Coordinator 145

6.3 **LIS-2023-07: Privacy Risk Management Policy**
N. Cooper, Director of Legislative & Information Services/Clerk / J. McWilliam, Manager of Legislative Services/Deputy Clerk 415

7. Closed Session

None

8. Adjournment

DRAFT MINUTES - Subject to Approval

Minutes of the General Government Committee Meeting Held Electronically At 1:00 p.m. on March 20, 2023

Alternative formats available upon request by contacting:
accessibility@ajax.ca or 905-619-2529 ext. 3347

Present:	Chair – Regional Councillor	J. Dies
	Regional Councillors	M. Crawford
		S. Lee
	Councillors	R. Tyler Morin
		L. Bower
		N. Henry
	Mayor	S. Collier

1. Call to Order & Acknowledgement of Traditional Treaty Lands

Chair Dies called the meeting to order at 1:00 p.m. and acknowledged traditional treaty lands.

2. Disclosure of Pecuniary Interest

There were no disclosures of pecuniary interests.

3. Adoption of Minutes

3.1 2023 Operating Budget and 2023 Capital Budget; 2024 – 2032 Long Range Capital Forecast Meeting

Moved by: R. Tyler-Morin

That the Minutes of the 2023 Operating Budget and 2023 Capital Budget; 2024 – 2032 Long Range Capital Forecast meeting of the General Government Committee held on February 16, 2023 be adopted.

CARRIED

3.2 Regular Meeting

Moved by: L. Bower

That the Minutes of the regular meeting of the General Government Committee held on February 21, 2023 be adopted.

CARRIED

3.3 **Closed Session**

Moved by: S. Lee

That the Minutes of the Closed Session of the General Government Committee held on February 21, 2023 be adopted.

CARRIED

4. Consent Agenda

Members separated items 4.1, 4.2, 4.3, 4.5, 4.6, 4.7, 4.10 and 4.11 for discussion.

4.1 **CAO-2023-02: Council Ward Constituent Newsletter Update**

Committee Members posed questions respecting the options identified by staff including the process undertaken to select the company identified in Option C, previous mechanisms employed by the Town to distribute newsletters, the total number of subscribers to the Town's electronic newsletters, availability of supplemental copies for distribution to senior's residences and opportunities to supply newsletters in languages other than English, as well as the ability to develop and distribute ward-specific content. D. Jarvis, Supervisor of Communications and Engagement, responded to questions from Committee Members.

Amendment

Moved by: S. Lee

1. That Council defer consideration of Report CAO-2023-02 to the Council meeting scheduled for March 27, 2023.

CARRIED

4.2 **CAO-2023-03: Vaccination Policy**

Committee Members posed questions respecting the number of staff impacted by the policy and opportunities to reinstate employees following suspension of the policy, the results of recent arbitration decisions involving other Ontario municipalities, the position of the Town's legal counsel on the proposed suspension and actions taken by other municipalities within the Region of Durham. Mary Lou Consentino, Manager, Human Resource Services and Shane Baker, Chief Administrative Officer, responded to questions from Committee Members.

Moved by: S. Lee

That Council approves the suspension of the Town's Vaccination Policy.

CARRIED

4.3 **CAO-2023-04: OMERS By-law**

Committee Members sought clarification concerning the timing of registered retirement

savings plan contributions, opt-out provisions, clarification on next steps, and the effective participation date following adoption of the by-law by Council. Dianne Valentim, Director of Finance/Treasurer, responded to questions from Committee Members.

Moved by: S. Collier

That the proposed OMERS By-law (ATT-1) which provides the Town of Ajax authorization to participate in the OMERS plan effective January 1, 2023, be brought forward to the March 27th, 2023 Council meeting for approval.

CARRIED

4.4 FIN-2023-06: 2022 Statement of Remuneration and Expenses for Members of Council and Council Appointees

Moved by: S. Collier

That the 2022 Statement of Remuneration and Expenses for Members of Council and Council Appointees be received for information.

CARRIED

4.5 OES-2023-09: Contract Award – AVL & Dashcams Award

Committee Members posed questions to staff respecting the changing value of the contract and anticipated benefits associated with the proposed system, one-time and ongoing costs required to support the system, the total number of vehicles to be equipped with this new technology, and opportunities to revise the system or seek alternatives in light of the projected costs. Dave Meredith, Director of Operations & Environmental Services, responded to questions from Members of the Committee.

Moved by: N. Henry

1. That Council award the contract for AVL Dashcam Solution, to DiCan in the amount of $710,523.15 (inclusive of all taxes).

2. That Council approve funding of $396,840.52 to be allocated to Capital Account No. 1031611 from the Capital Contingency Reserve.

3. That Council authorize Staff to renew the annual licensing, maintenance and support contract for an additional four, one-year periods, pending an analysis and satisfactory performance review at the anniversary date of the contract, in the estimated amount of $649,035.84 (inclusive of all taxes).

CARRIED

4.6 OES-2023-01: Contract Award – Ajax Cricket Site Improvements

Committee Members posed questions about the history and funding of the project. D. Meredith, Director of Operations & Environmental Services, responded to questions from Members of the Committee.

Moved by: S. Collier

1. That Council award the contract for Ajax Cricket Field Improvements, to Latitude 67 Limited in the amount of $4,367,463.56 (inclusive of all taxes).

2. That Council award the contract for Contract Administration ~ Ajax Cricket Field Improvements, to Landscape Planning Limited in the amount of $67,800.00 (inclusive of all taxes).

3. That Council approve funding of $1,362,793.78 to be allocated to Capital Account No. 1020111 from Canada Community Building Fund.

CARRIED

4.7 OES-2023-05: RFT No. 23003 – Municipal Vehicles – Brine Truck with Snowplow

Committee Members posed questions about the funding mechanism for the purchase, how many similar vehicles exist in the Town's fleet and if the new vehicle will require an increase in the Town's staffing levels to support its use in operational activities. D. Meredith, Director of Operations & Environmental Services, responded to questions from Committee Members.

Moved by: N. Henry

1. That Council award the contract for Municipal Vehicles ~ Brine Truck with Snowplow, to Premier Truck Group Inc. in the amount of $615,595.75 (inclusive of all taxes)

CARRIED

4.8 PDS-2023-06: Corporate Ban on Single-Use Plastics – Implementation Update

Moved by: S. Collier

1. That the report to General Government Committee, dated March 20, 2023, entitled "Corporate Ban on Single-Use Plastics – Implementation Update" be received for information; and

2. That Council endorse a shift from external enforcement of the corporate ban on single-use plastics, referenced in Phases III and IV of the Corporate Ban on Single-use Plastics Implementation Plan, to encouraging behavioural changes through public education and engagement initiatives.

CARRIED

4.9 PDS-2023-07: 2022 Pickering Village Community Improvement Plan: 2022 Façade Improvement Grant and Sign Grant Program Summary

Moved by: S. Collier

1. That the report to the General Government Committee, dated March 20, 2023 entitled "Pickering Village Community Improvement Plan: 2022 Façade Improvement Grant and Sign Grant Program Summary" be received for information.

CARRIED

DRAFT

4.10 PDS-2023-08: 2022 Year End Building Permit Activity and 2023 Projection Report

Committee Members sought information on whether these statistics will count towards the commitments made in the Town's Housing Pledge and the quantity of residential development that is classified as high density. Steven McReelis, Chief Building Officer/Manager of Building Approvals, responded to questions from Committee Members.

Moved by: M. Crawford

That the 2022 Year End Building Permit Activity and 2023 Projection Report be received for information.

CARRIED

4.11 PDS-2023-09: Development Charge Deferral Amending Agreement – 35-37 Kings Crescent

Moved by: S. Lee

That the Development Charge Deferral Amending Agreement for 35-37 Kings Crescent (ATT-1) be approved.

CARRIED

5. Discussion

None

6. Presentations

6.1 PDS-2023-05: Environmental Sustainability Projects and Programs

Cameron Richardson, Supervisor of Environmental Sustainability & Climate Change, delivered a presentation reviewing the contents of the written report and recommendations contained therein. C. Richardson discussed programs undertaken to advance the Town's environmental sustainability portfolio including a wetland rehabilitation project on Kingston Road, the Sustainable Neighbourhood Action Program (SNAP), and updates to the Town's Urban Forest Management Plan. C. Ricardson identified energy and greenhouse gas emissions related activities and the implementation of the Ajax Green Standards to guide future site plan and subdivision applications. Finally, C. Richardson discussed the 5Rs in the waste hierarchy, achievements witnessed through the textile diversion program, and work towards eliminating single use plastics within the Town.

Committee Members posed questions respecting changes associated with the province's Extended Producer Responsibility (EPR) program, the definition of single-use plastics adopted by the Town compared to definitions established by the Government of Canada, efforts underway within Town facilities to reduce single-use plastics and activities to combat phragmites within the Town's ecosystem, and feedback received as part of the Town's SNAP-related public engagement activities. C. Richardson responded to

DRAFT

questions from Committee Members.

Moved by: L. Bower

1. That the report entitled “Environmental Sustainability Projects and Programs – Status Update” be received for information.

CARRIED

Closed Session (3:13 p.m.)

7. Authority to Hold a Closed Meeting and Related In-Camera Session

Moved by: S. Collier

That Council convene to a Closed Session pursuant to Section 239 (2) (d), (i), (j) and (k) of the Municipal Act, 2001, as amended, to discuss labour relations or employee negotiations, a matter pertaining to a trade secret or scientific, technical, commercial, financial or labour relations information, supplied in confidence to the municipality or local board, which, if disclosed, could reasonably be expected to prejudice significantly the competitive position or interfere significantly with the contractual or other negotiations of a person, group of persons, or organization negotiations, a matter pertaining to a trade secret or scientific, technical, commercial or financial information that belongs to the municipality or local board and has monetary value or potential monetary value, and a matter pertaining to a position, plan, procedure, criteria or instruction to be applied to any negotiations carried on or to be carried on by or on behalf of the municipality or local board.

CARRIED

7.1 **Elexicon Corp Strategic Update**

7.2 **Fire Service Negotiations**

Open Meeting (4:55 p.m.)

Ratify Actions Taken In-Camera

Moved by: S. Collier

That all directions approved during the Closed Session be ratified.

CARRIED

8. Adjournment (4:57 p.m.)

Moved by: M. Crawford

That the March 20, 2023 meeting of the General Government Committee be adjourned.

CARRIED

DRAFT

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Lindsay Sinclair, Supervisor of Facility Capital Projects & Asset Management
Report #:	OES-2023-06
Subject:	**Contract Award – Ajax Library Makerspace Renovation**
Ward:	1
Date of Meeting:	April 11, 2023
Reference:	RFT No. T23009 - Ajax Library Makerspace Renovation RFS No. S23004 - Contract Administration - Ajax Library Makerspace Renovation Capital Account No: 1038511

Recommendation:

1. **That Council award the contract for Ajax Library Makerspace Renovation, to Area Construction Inc. in the amount of $794,687.19 (inclusive of all taxes).**

2. **That Council award the contract for Contract Administration ~ Ajax Library Makerspace Renovation, to Ward99 Architects in the amount of $24,143.50 (inclusive of all taxes).**

3. **That Council approve the additional funding of $222,724.91 to be allocated to capital account 1038511 from the Capital Contingency Reserve.**

4. **That the Library be directed to repay this additional funding through annual payments from in-year surpluses, grants, and/or sponsorship monies as funding is available.**

Background:

A makerspace is a dedicated place where people come together to collaborate, learn, problem solve, and create tangible or digital projects. Leading libraries incorporate makerspaces into their service delivery model to inspire residents' creativity and engage in non-traditional learning activities by allowing free, hands-on access to high-tech and low-tech tools.

In August 2021, The Ajax Public Library completed a Makerspace Study. The Study provides recommendations on the location, service offerings, and operations of a makerspace in Ajax. These recommendations were developed after completing an industry analysis and comprehensive community consultation.

Findings of the Makerspace Study included:
- Significant community interest and demand for a makerspace within Ajax;
- The makerspace should be a dedicated space with controlled access;

- Makerspace should offer free-of-charge access to hands-on tools, instruments, machines, and equipment – both low tech and high tech;
- Dedicated staff being paramount to customers use and adoption of a makerspace; and
- Recommended Makerspace location is the McLean Branch, MCC Community Room 1.

The Ajax Public Library intends to construct a 918 square foot Makerspace in Community Room #1 of the McLean Community Centre. Community Room #1 was built in 1993 and consists of a room approx. 36' x 27' with two entrances, one off the main entrance foyer, and one into the Library.

The scope of work for the renovation includes:

- Demolition of non-rated and fire-rated steel doors and screens and associated hardware and glazing, interior finishes, fitments and fixtures including millwork, counters, wall mounted display boards, non-load bearing masonry partitions, ceiling and floor finishes.
- Demolition of existing mechanical and electrical services to suit the new floor layout.
- Installation of new non-load bearing steel stud framed gypsum board finished interior partitions, rated and non-rated doors and associated hardware, glazed aluminum framed screens, glazed hollow metal doors, glazed screens and interior window.
- Installation of new finishes (floor, wall, ceiling) and built-in millwork.
- New electrical work including new electrical supply, devices, and light fixtures.
- New mechanical work including linear supply air diffusers, return air grilles, new wall mounted heating unit and direct rooftop exhaust.
- Minor relocation of sprinklers to suit new layout.
- New furniture including desks, tables, chairs, and stools to suit the new layout.

In 2022 the Town established a Library Capital account that is funded through annual Library surplus. The Library Board approved the transfer of $524,000 into the Library capital account to fund the Makerspace project. The Library and Town MOU only allows for the Library to allocate surpluses up to $250,000. The transfer of $524,000 was permitted on a one-time basis from the 2021 annual Library surplus as a funding solution for the Makerspace.

Discussion:

The Town undertook a Request for Pre-Qualification [RPQ No. PQ2206] for Pre-Qualification General Contractors ~ Library Makerspace in December 2022 to cover Library Makerspace Renovation requirements. This RPQ resulted in the appointment of 6 pre-qualified Bidders.

Request for Tender (RFT) documents were posted to the Towns' eBidding platform and the 6 pre-qualified bidders were invited to register for the opportunity. Of the invited bidders 6 registered for the opportunity with bids being received from all of these, prior to the closing on March 22, 2023. Listed below is a summary of the bids received:

NAME OF BIDDER	TOTAL TENDER AMOUNT
Area Construction Inc.	$ 794,687.19
Anacond Contracting Inc.	$ 873,160.04
Ritestart Limited	$ 883,886.00
Pre-Eng Contracting Ltd.	$ 888,745.00
West Metro Contracting Inc.	$ 1,288,426.00

Twin Contracting (855758 Ontario Limited)	$ 1,437,247.00

Financial Implications:

1038511 - Library - Makerspace Project

Approved Capital Budget		$524,000.00
2022 Library Surplus Contribution		$250,000.00
Previous Awards/Expenditures		
Detailed Design Consulting	$55,968.00	
I.T. Equipment	$6,382.53	
		$62,350.53
Available Budget		$711,649.47
Current Award		
Construction Contract Award (net of HST rebate)	$715,640.42	
Construction Contingency (10%)	$71,564.04	
CA Contract Award (net of HST rebate)	$22,641.60	
CA Contingency (20%)	$4,528.32	
Estimated** Equipment & Technology Costs	$120,000.00	$934,374.38
Over Budget		($222,724.91)

A 10% contingency has been identified to ensure sufficient funding is available to address unforeseen issues that can be encountered during construction.

There are additional expenditures for the equipment and technology for the makerspace such as computers, software, hardware and various large and small tools/equipment. The total estimated cost for the equipment, software and hardware is $120,000.

The Library is expected to have an annual surplus of approximately $324,000, which will be finalized through the annual financial statements. The Library Board intends to transfer an additional $250,000 from the 2022 annual surplus into the Library Makerspace Capital account with their approval of the audited financial statements. As per the established Memorandum of Understanding, the Library will transfer the remaining surplus to the Town to be used at Council's discretion.

Additional funding, not to exceed $222,724.91, will be allocated from the Capital Contingency Reserve through the Capital Account Closing Report as a loan to the Library that will be repaid through annual payments from in-year surpluses, grants, and/or sponsorship monies as funding is available.

The current Purchasing By-Law allows for purchases through negotiations, under certain conditions, more specifically, "where, at the discretion of the Department Head, in consultation with the Manager of Purchasing, it is deemed to be in the best interest of the municipality to negotiate with vendors."

It is in the best interest of the Town to single source the Contract Administration to Ward99 Architects as they have completed the detailed design and are familiar with the project requirements.

Contract commencement is May 1, 2023 and completion is expected by Aug 31, 2023.

Communication Issues:

The construction will commence in early Spring with completion by the end of summer. There will be project signs posted at the site and the Town will have construction information available and updated regularly on our Construction Projects webpage and social media platforms.

Relationship to the Strategic Plan:

The Ajax Public Library is the gateway to information and resources that improve the quality of life, foster life-long learning, encourage community involvement, and meet the varying educational and social needs of all members of its communities.

Ajax Public Library Board's Ends/Objectives:

1. Customers will have access to valuable Library services that support their financial, physical, social, and mental well-being.

2. The Library will attract new customers and satisfy existing customers leading to an increased adoption of services.

4. Customers will have safe, welcoming, and inclusive spaces to meet, interact with others, read, work quietly, or simply be.

6. The Library will increase customers' transliteracy skills with inspirational and creative opportunities.

This report aligns with the following section of the 2018-2022 Strategic Plan – Stronger Together:

Focus Area:	Connecting our Community
Goal 3:	Fostering a sense of place and celebrating our history
Goal 4:	Creating opportunities to foster a welcoming and inclusive community
Goal 5:	Promoting a range of programs, services, and supports for all ages and abilities
Focus Area:	Investing in our Community
Goal 3:	Building new municipal infrastructure
Goal 4:	Modernizing existing municipal infrastructure and other assets
Goal 6:	Enhancing the beauty and appeal of the Town

Conclusion:

It is the recommendation of staff that Area Construction Inc. be awarded the contract for the Ajax Library Makerspace Renovation project, being the lowest bidder meeting the minimum specifications.

It is also the recommendation of staff that Ward99 Architects be awarded the contract to complete contract administration services as a single source purchase.

Attachments:

ATT-1: Ajax Makerspace Design Concept

Prepared by:

Lindsay Sinclair - Supervisor of Facility Capital Projects & Asset Management

Submitted by:

Dave Meredith - Director of Operations & Environmental Services

Reviewed by:

Sarah Vaisler – Chief Librarian and Executive Officer

Approved by:

Shane Baker – Chief Administrative Officer

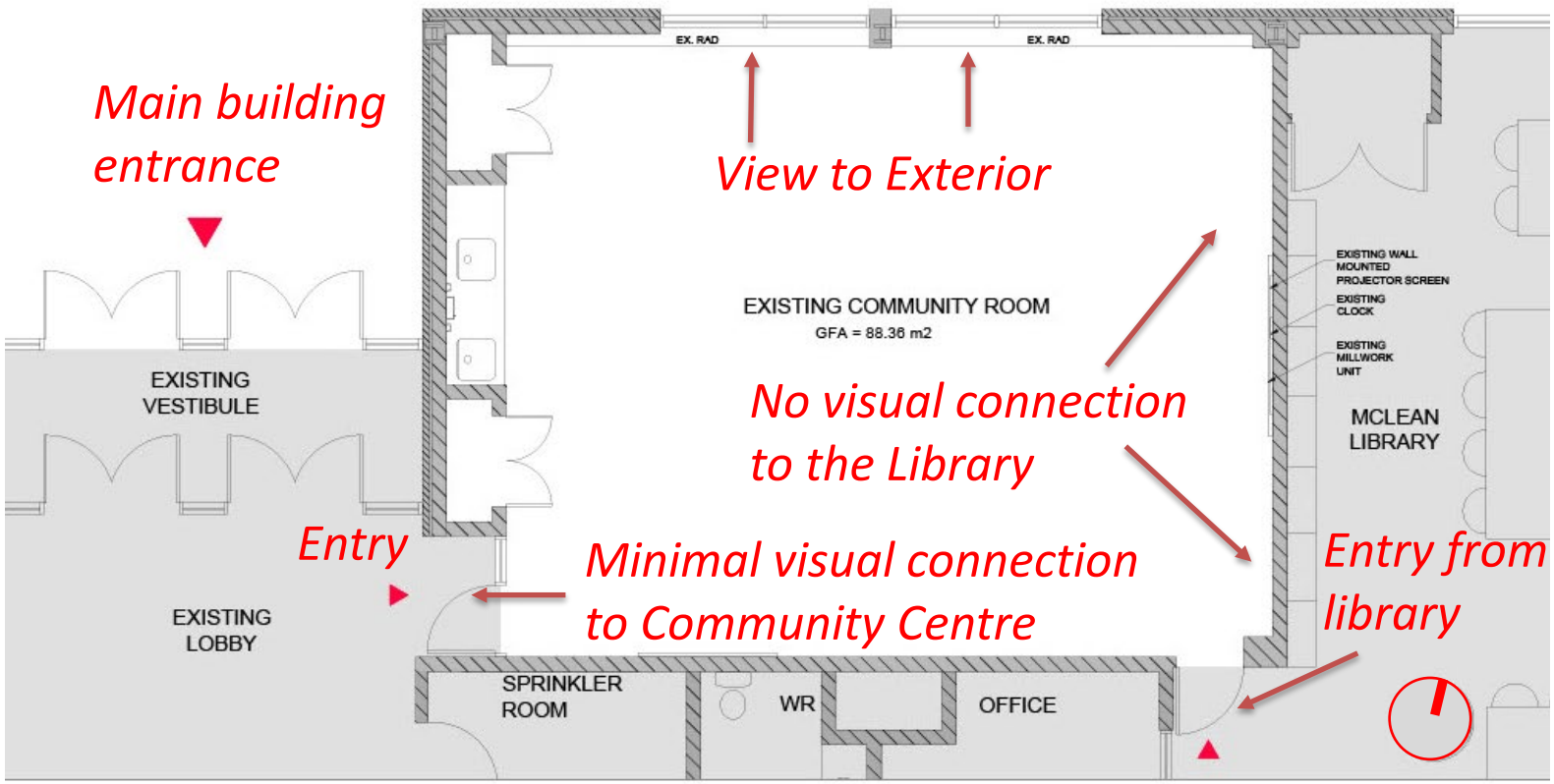
McLean Branch Makerspace

vision

The redesign must consider:

Identity – Establish visual connections and clearly identify the space as a makerspace.
Function – Functional, flexible, accessible spaces that support makerspace activities.
Engagement – Spaces that engage, inspire, support collaboration.

Existing Program Room



Current Floor Plan Layout

Existing Windows to Program Room

Existing Building Entrance

Existing Exterior Building Facade

Existing Entry Door

No views into the Library or Program Room

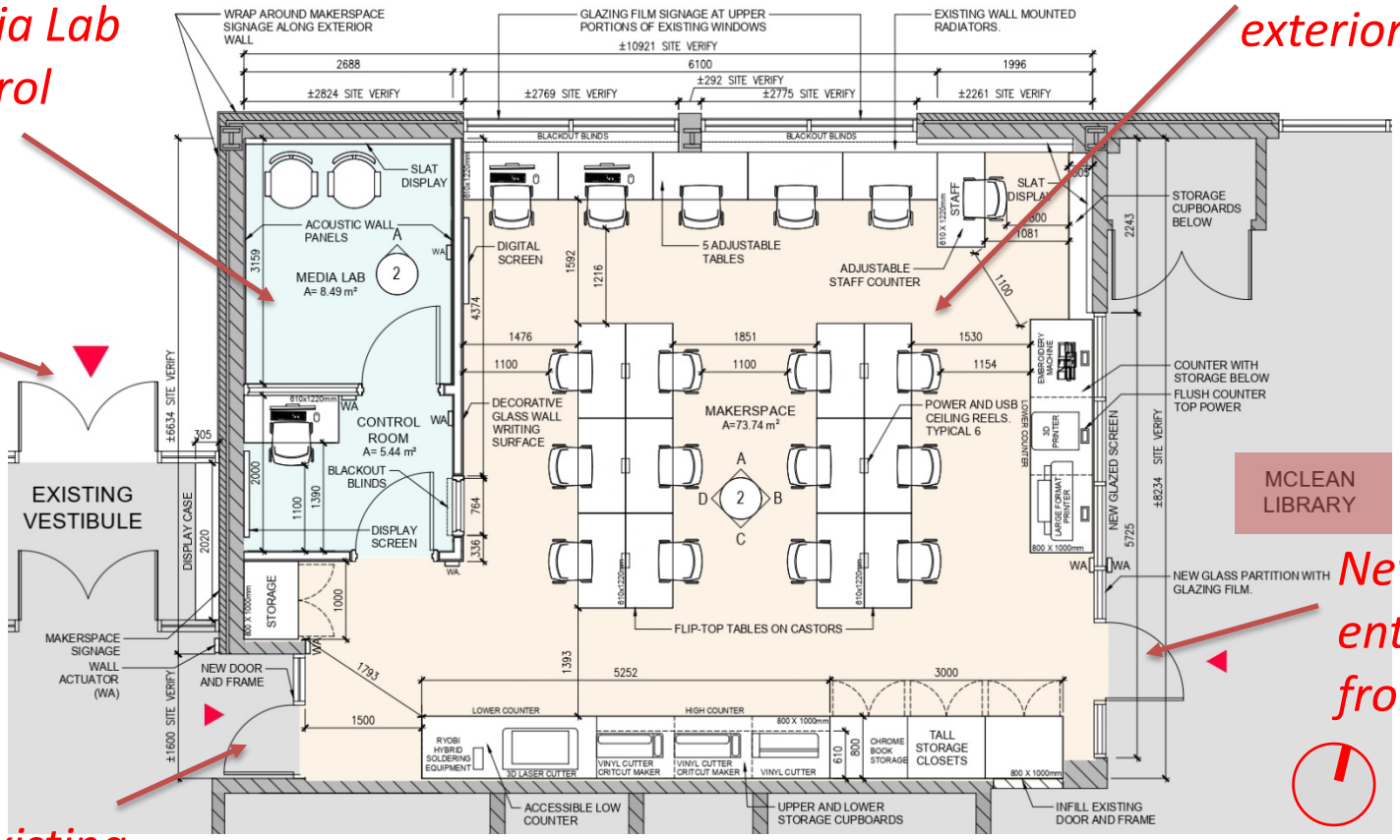
Existing Interior Entrance

Makerspace – Concept Plan

New Media Lab with Control Room

Existing Exterior Entry

Large open central space with views to exterior.



New entrance from library

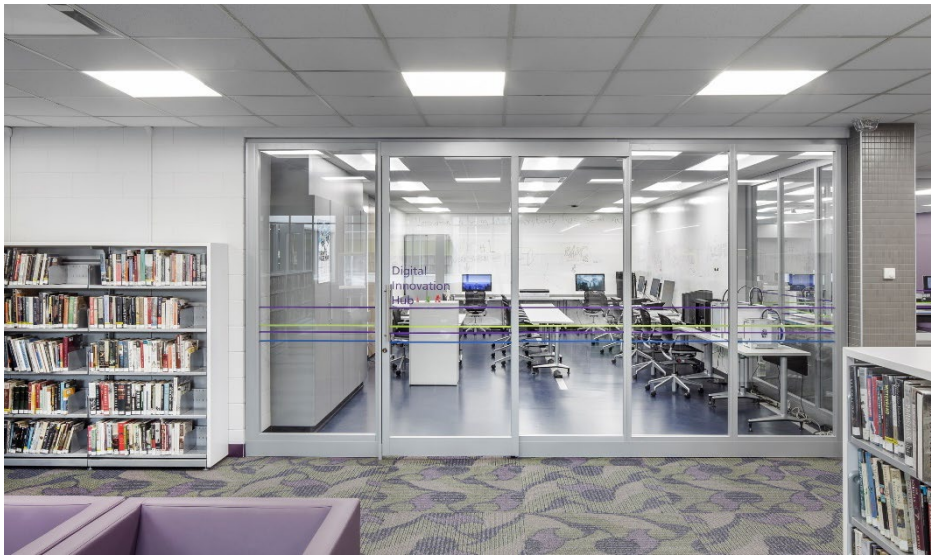
Maintain existing entrance from the community centre

Proposed Plan – Two Main Areas

Makerspace – Exterior Presence

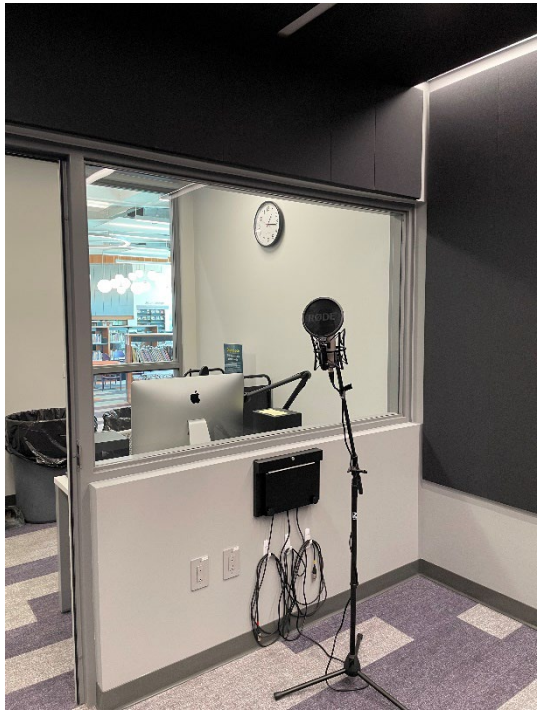
Inviting, Identifiable Exterior Signage

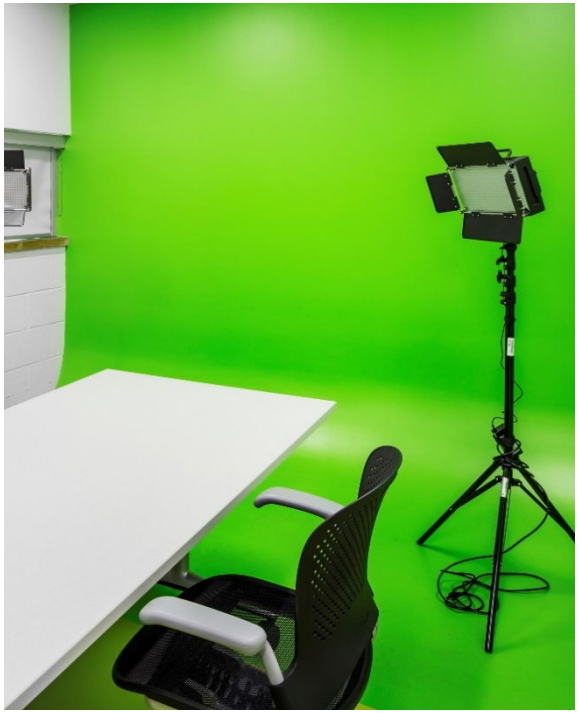
Makerspace – Identity and Visual Connections



Glazed walls between spaces provide transparency, that leads to marketability. Interiors on display.

Makerspace – Function and Services





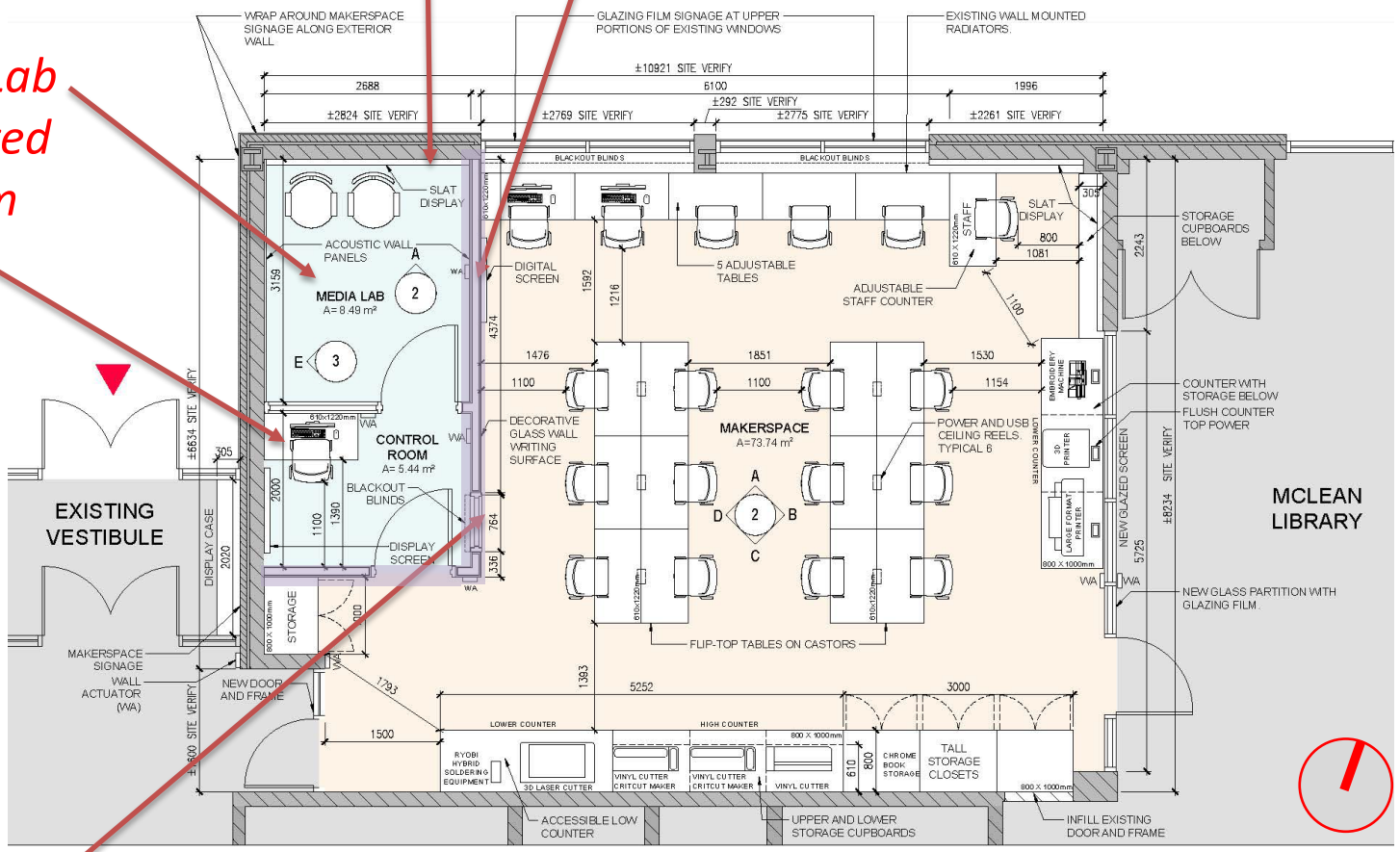
Dedicated space for a Media Lab; that supports a variety of activities.

Makerspace

Provision for high quality AV equipment

Soundproof walls and acoustic finishes

New Media Lab with connected Control Room



Views to open makerspace

Proposed Concept Plan

Makerspace

Acoustic suspended ceiling and wall panels

Counter with equipment

Accessible counter to access equipment

Secure storage closets and cupboards

Moveable flip-top tables and chairs on castors

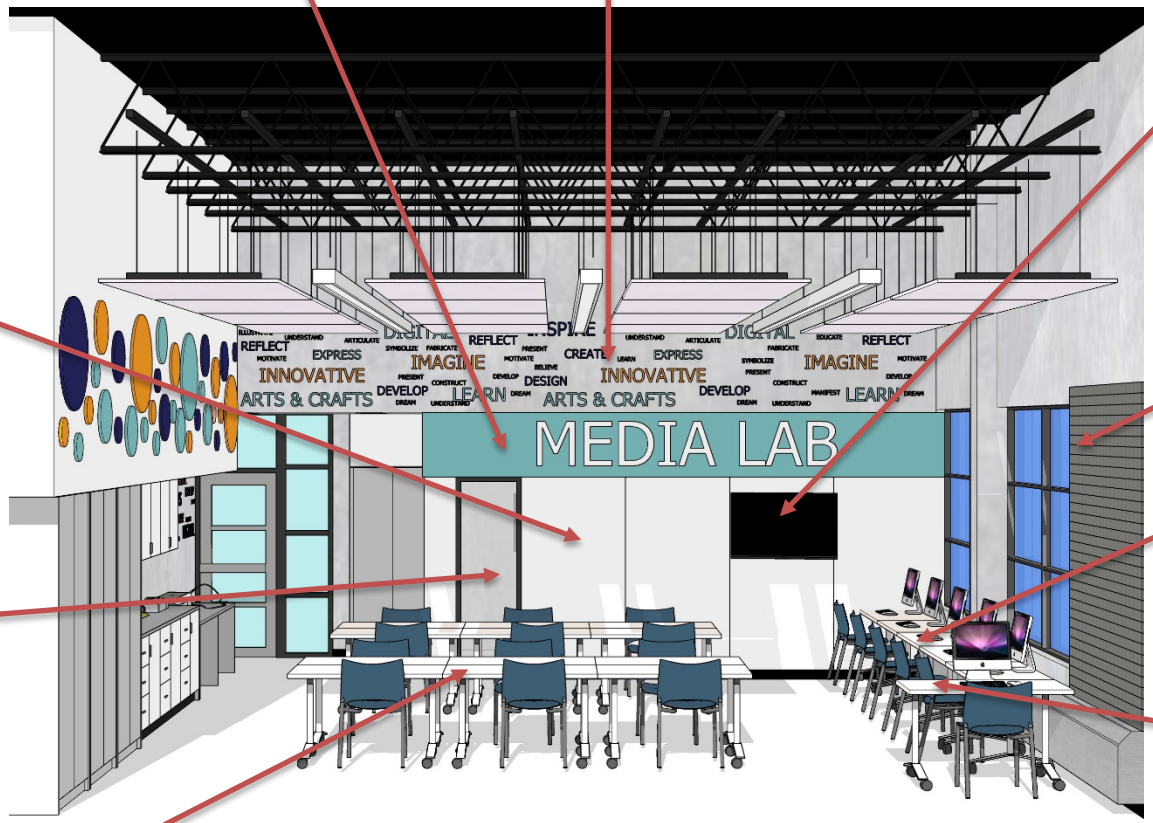
Adjustable counters at perimeter

Proposed Interior View 2C

Makerspace

Proposed Interior View

Makerspace



Interior sign identification

Decorative wall covering

Digital screen

Writable glass magnetic walls

Display wall

Adjustable counters

Window with view into media lab control room

Adjustable staff counter

Flexible reconfiguration of space possible with flip-top tables and chairs

Proposed Interior View 2D

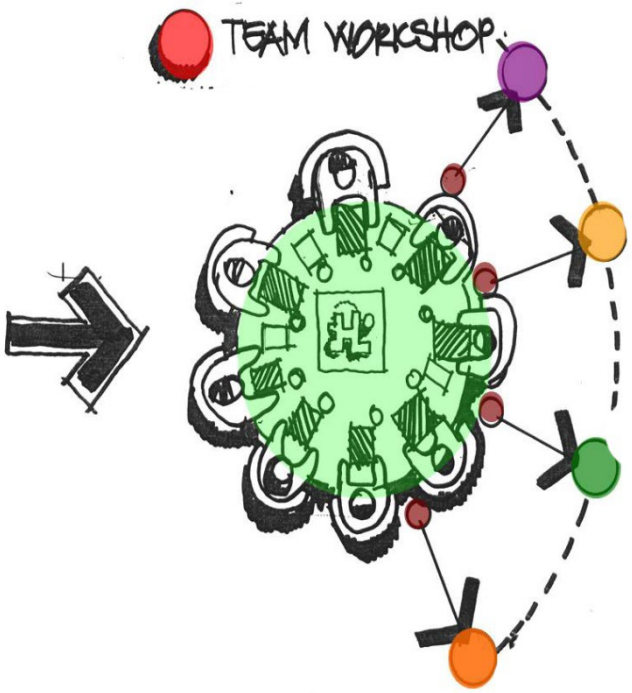
Makerspace

Glazed Wall between library and makerspace.

Display wall

Proposed Interior View 2B

Makerspace



Flexibility , Adaptability, Versatility of Space promotes engagement and collaboration.

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Mat Boyle, C.E.T., Capital Projects Coordinator
Report #:	OES-2023-07
Subject:	**Contract Award – T23012 Road Resurfacing & Geotechnical Services**
Ward(s):	2 & 3
Date of Meeting:	April 11, 2023
Reference:	Bid Document T23012 – Road ResurfacingPQ2005 – Geotechnical Consulting Services LC030000 – Road Resurfacing Program

Recommendation:

1. **That Council award the contract for Road Resurfacing, to Viola Management Inc. in the amount of $2,539,182.94 (inclusive of all taxes).**

2. **That Council award the contract for Geotechnical Services – Road Resurfacing, to Peto MacCallum Ltd. in the amount of $49,431.85 (inclusive of all taxes).**

Background:

The resurfacing of municipal roads and associated curb, sidewalk and culvert repairs are a critical component of the Town's infrastructure maintenance program. The roads to be resurfaced and/or reconstructed each year are prioritized based on the Town's Pavement Management System and coordinated with the Region of Durham's construction projects.

The following roads are part of the asphalt resurfacing, concrete repairs and restoration works within the Town of Ajax:

ROAD	FROM	TO
Delaney Drive	Church Street North	Ravenscroft Road
Ritchie Avenue	Westney Road North	Kingston Road West
Farrow Crescent	Parkes Drive	Parkes Drive
Fitzgerald Drive	Varley Drive	Parkes Drive
Varley Drive	Lake Driveway West	Parkes Drive
Rideout Street	Kings Crescent	Exeter Road

Discussion:

The Town undertook a Request for Pre-Qualification [RPQ No. PQ2304] for Pre-Qualification Road Contractors ~ Road Resurfacing in January 2023 to cover Road Resurfacing requirements. This RPQ resulted in the appointment of 7 pre-qualified Bidders.

Request for Tender (RFT) documents were posted to the Towns' eBidding platform and the 7 pre-qualified bidders were invited to register for the opportunity. Of the invited bidders 7 registered for the opportunity with bids being received from 6 of these, prior to the closing on March 22, 2023. Listed below is a summary of the bids received:

NAME OF BIDDER	TOTAL TENDER AMOUNT
Viola Management Inc.	$ 2,539,182.94
C. Valley Paving Ltd.	$ 2,589,915.93
Forest Contractors Ltd.	$ 2,895,795.29
D. Crupi & Sons Limited	$ 2,999,580.14
Brennan Paving & Construction Ltd.	$ 3,096,137.29
Dufferin Construction Company, A division of CRH Canada Group Inc.	$ 3,769,370.10

Contract commencement is May 15, 2023 and completion is expected by September 1, 2023.

Geotechnical Services

The current Purchasing By-Law allows for purchases through negotiations, under certain conditions, more specifically, "where, at the discretion of the Department Head, in consultation with the Manager of Purchasing, it is deemed to be in the best interest of the municipality to negotiate with vendors."

In 2020, the Town undertook a Request for Pre-Qualification (RPQ) for Geotechnical Engineering Consultants to cover material testing requirements. That RPQ resulted in a registry of 6 pre-qualified Bidders [Ref: PQ2005 Geotechnical Services].

In accordance with the Pre-Qualification No. PQ2005, Multiple Year Award for Material Testing, Geotechnical & Environmental Investigation Services Request for quotation documents for Geotechnical Services were issued to the 6 prequalified bidders with quotations being received back from 3. Listed below is a summary of the Quotations received:

VENDOR NAME	TOTAL QUOTE PRICE (INCL HST)
Peto MacCallum Ltd.	$49,431.85
Cambium Inc.	$64,916.24
Aecon Materials Engineering	$76,578.29

It is recommended that the Geotechnical Services be awarded to Peto MacCallum Ltd., being the lowest bidder.

Financial Implications:

LC030000 – Road Resurfacing Program – Canada Community Building Fund

Approved Capital Budget:		
23030001 Delaney Dr Resurf-Church to Ravenscroft	$ 870,200.00	
23030002 Ritchie Ave Resurf-Westney S to Kingston	$ 866,600.00	
23030003 Farrow Cr Resurf-Parkes to Parkes	$ 443,800.00	
23030004 Fitzgerald Dr Resurf-Varley to Parkes	$ 356,000.00	
23030005 Varley Dr Resurf-Parks to Lake Driveway	$ 288,300.00	
23040006 Rideout St Resurf-Exeter to Kings	$ 176,200.00	$ 3,001,100.00
Current Award:		
Viola Management Inc. (net of HST rebate)	$ 2,286,612.88	
Contingency (15%)	$ 342,991.93	
Peto MacCallum Ltd. (net of HST rebate)	$ 44,514.91	
Contingency (15%)	$ 6,677.24	$ 2,680,796.96
Under Budget		$ 320,303.04

LC030000 – Road Resurfacing Program – Roads Maintenance Reserve

Approved Capital Budget:	
23030011 Patching-Var Locations	$ 50,000.00
Current Award:	
Viola Management Inc. (net of HST rebate)	$ 50,000.00
Over/Under Budget	$ 0.00

A 15% contingency has been identified for Road Resurfacing due to the possibility of poor sub grade conditions and soft spots that could be identified once the asphalt surface has been removed. Over-runs may occur on the estimated quantities of asphalt and granular materials when repairing unforeseen deteriorated areas. Risks that can cause delays, necessitating the need for additional material and labor associated with the construction include poor weather conditions, unforeseen global supply chain shortages, contamination of soil, etc.

A 15% contingency has been identified for the Geotechnical Services to ensure sufficient funding is available to address site specific issues that can be encountered during the construction. Risk factors for this project include soft subgrade conditions that may require additional testing, additional chemical testing of bulk excavated material to be removed off site in accordance with the Provincial Excess Soils Regulation, and applicable recommendations by Peto MacCallum Ltd.

$50,000.00 has been allocated from the Roads Maintenance Reserve in the event staff are made aware of any unforeseen deteriorating segments of roadways that require large scale patching works, on any roadways within the Town not being resurfaced in 2023.

The Long-Range Capital Forecast Detail Sheet is attached for information.

Communication Issues:

Communications will include the delivery of a construction notice to all residents adjacent to the work, including the provision of a construction schedule on the Town's website. Construction notice signs will be posted on site prior to commencement of work. The Town's contractors will be instructed to ensure that adequate safety measures are in place to protect the public for this project.

Minor disruptions due to dust, noise, driveway access and localized lane closures, will be experienced by residents and motorists. Every effort, however, will be made to keep these inconveniences to a minimum throughout the process.

Signage and notices will be posted a minimum of 10 days in advance of construction and be arranged through the Town's Road Occupancy/Road Closure By-Law. As per the By-Law, notices will be sent to residents, emergency services and transit prior to the construction. All efforts will be made to complete the construction work as quickly and efficiently as possible to minimize the impact to public.

Relationship to the Strategic Plan:

The Town of Ajax the 2023-2026 Strategic Plan is currently being developed.

This report aligns with the following section of the 2018-2022 Strategic Plan – Stronger Together:

Focus Area: Investing in our Community
Goal 4: Modernizing existing municipal infrastructure and other assets

Conclusion:

It is the recommendation of staff that:

1. Viola Management Inc. be awarded the contract for T23012 - Road Resurfacing being the lowest bidder meeting minimum specifications.

2. Peto MacCallum Ltd. be awarded the contract for Geotechnical Services, being the lowest bidder meeting minimum specifications.

Attachments:

ATT-1: LC030000 – Road Resurfacing Program (see highlighted accounts)

Prepared by:

Mat Boyle – Capital Projects Coordinator

Submitted by:

Dave Meredith - Director of Operations & Environmental Services

Approved by:

Shane Baker – Chief Administrative Officer

TOWN OF AJAX

2023 CAPITAL BUDGET / 2024-2032 LONG RANGE CAPITAL FORECAST

DETAIL SHEET

Project Name **Road Resurfacing Program**
Project Number LC030000

Program Description

The Road Resurfacing program includes projects related to resurfacing of the Town's urban and rural roadways. Resurfacing projects include the removal of the existing surface layer and repaving with new asphalt to restore the road surface back to an excellent condition. Mill and Overlay (M&O) projects include grinding the existing surface and applying new asphalt to restore the road surface back to an excellent condition.

The scope of the projects may also include sidewalk replacement, curb and gutter replacement, storm sewer repairs, and minor road improvements. Minor road maintenance such as patching and crack sealing is also done through this program.

The ongoing monitoring of road conditions through the annual pavement management systems updates is also included within the scope of this program.

Timely road resurfacing and maintenance of the Town's road network, consisting of 675 lane kms, ensures public safety, reduces the lifecycle cost of operation and minimizes the risk of unplanned service disruptions.

EXPENDITURES / FUNDING

		2023	2024	2025	2026-2032	Total
Expenditures						
23030001	Delaney Dr Resurf-Church to Ravencroft	**870,200**	**-**	**-**	**-**	**870,200**
23030002	Ritchie Ave Resurf-Westney S to Kingston	**866,600**	**-**	**-**	**-**	**866,600**
23030003	Farrow Cr Resurf-Parkes DrW to Parkes	**443,800**	**-**	**-**	**-**	**443,800**
23030004	Fitzgerald Dr Resurf-Varley Dr to Parkes Dr	**356,000**	**-**	**-**	**-**	**356,000**
23030005	Varley Dr Resurf-Parkes to Lake Driveway	**288,300**	**-**	**-**	**-**	**288,300**
23030006	Rideout St Resurf-Exeter Rd to Kings Cres	**176,200**	**-**	**-**	**-**	**176,200**
23030007	Pavement Management System Update	**51,000**	**-**	**-**	**-**	**51,000**
23030008	Bowers Ct Recons-Brockman Cres to End	**48,000**	**396,900**	**-**	**-**	**444,900**
23030009	Brockman Cres Resurf-Delaney to Delaney	**-**	**489,900**	**-**	**-**	**489,900**
23030010	Crack Sealing-Var Locations	**100,000**	**-**	**-**	**-**	**100,000**
23030011	Patching-Var Locations	**50,000**	**-**	**-**	**-**	**50,000**
24030001	Harwood S Resurf-Clements W to Bayly E	**-**	**1,682,000**	**-**	**-**	**1,682,000**
24030002	Pickering Beach M&O-Beachview to Foord	**-**	**427,300**	**-**	**-**	**427,300**
24030003	Pickering Beach M&O-Foord to Emperor	**-**	**281,200**	**-**	**-**	**281,200**
24030004	Tipton Cres M&O - Brockman to Tipton	**-**	**251,500**	**-**	**-**	**251,500**
24030005	Pavement Management System Update	**-**	**48,000**	**-**	**-**	**48,000**
24030006	Hobson M&O-Brockman Cr. to Brockman	**-**	**43,300**	**-**	**-**	**43,300**
24030007	Crack Sealing-Var Locations 2024-2032	**-**	**100,000**	**100,000**	**700,000**	**900,000**
24030008	Patching-Var Locations 2024-2032	**-**	**50,000**	**50,000**	**350,000**	**450,000**

TOWN OF AJAX

2023 CAPITAL BUDGET / 2024-2032 LONG RANGE CAPITAL FORECAST

DETAIL SHEET

Project Name **Road Resurfacing Program**

Project Number LC030000

EXPENDITURES / FUNDING

		2023	2024	2025	2026-2032	Total
Expenditures						
25030001	Harwood Ave M&O-Kerrison to Rossland W	-	-	1,150,000	-	1,150,000
25030002	Large Cres Resurf - Pearce Dr to Pearce Dr	-	-	668,100	-	668,100
25030003	Mullen Dr Resurf - Latimer St to Latimer St	-	-	432,100	-	432,100
25030004	Dakin Dr Resurf - Large Cres to Large Cres	-	-	334,000	-	334,000
25030005	Burningham Cres Resurf - Lincoln to Lincoln	-	-	315,900	-	315,900
25030006	Linton Ave Resurf-Kearney to Old Kingston	-	-	307,500	-	307,500
25030007	Station St Resurf-Mills Rd to Harwood Ave	-	-	303,800	-	303,800
25030008	Windsor Dr Resurf-Old Kingston to Kearney	-	-	224,700	-	224,700
25030009	Sherwood Rd Resurf-Windsor to Church	-	-	212,200	-	212,200
25030010	Cameron St Resurf - Elizabeth to Windsor	-	-	172,700	-	172,700
25030011	Dingley Ct Resurf - Lincoln St to End	-	-	103,000	-	103,000
25030012	Pavement Management System Update	-	-	50,000	-	50,000
25030013	Latimer St Resurf - Mullen Dr to Large Cres	-	-	56,900	-	56,900
26030001	Church St N M&O-Meekings to Rossland W	-	-	-	848,000	848,000
26030002	Ashbury M&O - Shoal Point to Audley	-	-	-	501,000	501,000
26030003	Mandrake St Resurf - Doric to Salem	-	-	-	500,100	500,100
26030004	Humphrey Dr Resurf-Lake Dwy W to Hilling	-	-	-	418,700	418,700
26030005	Cedar St Resurf - Knapton to Tudor Ave	-	-	-	351,800	351,800
26030006	Dobson Dr Resurf - Angus to Mandrake	-	-	-	330,500	330,500
26030007	Kipling Resur-Lake Dwy W to Lake Dwy W	-	-	-	268,100	268,100
26030008	Fielding Court Resurf - Westney Rd to End	-	-	-	252,400	252,400
26030009	Bird Crescent Resurf - Dobson to Dobson	-	-	-	227,400	227,400
26030010	Walker Resurf-Lake Dwy W to Lake Dwy W	-	-	-	208,600	208,600
26030011	Hilling Resurf-Lake Dwy W to Humphrey	-	-	-	188,400	188,400
26030012	Warwick Resurf - Ashbury to Lilley	-	-	-	184,700	184,700
26030013	Vale Resurf-Lake Dwy W to Lake Dwy W	-	-	-	171,000	171,000
26030014	Simpson Rd Resurf-Clements to Emperor	-	-	-	163,500	163,500
26030015	Follett Court Resurf - Simpson Road to End	-	-	-	135,000	135,000
26030016	Hewison Court Resurf - Kingston Rd to End	-	-	-	81,800	81,800
26030017	George Jones St Resurf-Sherwood to	-	-	-	74,800	74,800
26030018	Pavement Management System Update	-	-	-	54,000	54,000
26030019	Oldfield Court Resurf - Delaney Dr to End	-	-	-	51,200	51,200
27030002	Pavement Management System Update	-	-	-	51,000	51,000
27030003	Harwood Ave S Resurf-Kings Cr to Bayly St	-	-	-	1,215,100	1,215,100
27030004	Shoal Point Rd Resurf-Ashbury to Duckfield	-	-	-	231,500	231,500
27030005	Mayor Cres Resurf-Range Line to End	-	-	-	374,900	374,900
27030006	Flazington Court Resurf-Mayor Cres to End	-	-	-	74,700	74,700
28030002	Pavement Management System Update	-	-	-	53,000	53,000
28030003	Road Resurfacing Program - Inc. Backlog	-	-	-	36,424,900	36,424,900
29030001	Road Resurfacing Program	-	-	-	15,814,500	15,814,500
29030002	Pavement Management System Update	-	-	-	57,000	57,000
30030001	Road Resurfacing Program	-	-	-	16,950,400	16,950,400
30030002	Pavement Management System Update	-	-	-	54,000	54,000
31030001	Pavement Management System Update	-	-	-	56,000	56,000
31030002	Road Resurfacing Program	-	-	-	16,099,600	16,099,600
32030001	Road Resurfacing Program	-	-	-	19,345,100	19,345,100
Total Expenditures		3,250,100	3,770,100	4,480,900	112,862,700	124,363,800

TOWN OF AJAX

2023 CAPITAL BUDGET / 2024-2032 LONG RANGE CAPITAL FORECAST

DETAIL SHEET

Project Name **Road Resurfacing Program**

Project Number LC030000

EXPENDITURES / FUNDING

	2023	2024	2025	2026-2032	Total
Financing					
Road Maintenance Reserve	**249,000**	**1,598,200**	**1,653,800**	**93,025,300**	**96,526,300**
Canada Community-Building Fund	**3,001,100**	**2,171,900**	**2,827,100**	**19,837,400**	**27,837,500**
Total Financing	**3,250,100**	**3,770,100**	**4,480,900**	**112,862,700**	**124,363,800**
Annual Operating Costs	**-**	**-**	**-**	**-**	**-**

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Lindsay Sinclair, Supervisor of Facility Capital Projects & Asset Management
Report #:	OES-2023-10
Subject:	**Preliminary Report on a Potential Over-Expenditure – MCC Pool Renovations**
Ward:	1
Date of Meeting:	April 11, 2023
Reference:	Bid Document No. S22007 Bid Document No. T22001 1019511.4108 1037611.4108

Recommendation:

1. **That this report, on the potential over-expenditure on Capital Project Number 1037611 – MCC Pool Filt & Pump Sys Repl, previously awarded to Canada Construction Limited, be received for information.**

2. **That this report, on the potential over-expenditure on Capital Project Number 1019511 – MCC - Repl Pool Slide & Refub Stairs, previously awarded to Canada Construction Limited, be received for information.**

Background:

Under the Capital Expenditure Control Policy, staff are required to advise Council on a potential over-expenditure exceeding the approved spending limits.

On April 11, 2022, Council awarded the contract for the replacement of the MCC pool slide and the filtration system associated with all pools, re-waterproofing and tiling of the hot tub.

Discussion:

The replacement of the pool filtration system and pool slide construction is underway. During the course of construction additional deficient items have been discovered that require replacement or repair, these items include:

- Replacement of the main electrical Motor Control Centre and all disconnect switches located in the basement.
- Corroded basement plumbing components not identified in scope.
- Corroded pipe support brackets, bands and ductwork.
- Replacement of the main pool and tot pool drains and drain piping.
- Hot tub structural modifications, replacement of all piping, jets and drains.
- Corroded watermain valve system.

These items have been found to be in very poor condition due to the open filtration system in the basement. These systems and elements have been in a high corrosive environment for over 30 years and have therefore reached the end of their usable life and require replacement. The replacement pool filtration system is a closed tank style which will result in an improved indoor air quality in the basement, which is expected to reduce the rate of corrosion and deterioration of the new equipment.

In addition to the deficient items discovered, a number of upgrades have been considered to meet current code, these items include:

- Installation of a slop sink and handwashing sink in the basement for proper chlorine disposal and handwashing.
- Replacement of the pool stair treads, handrails and guardrails to meet current code standards.

These upgrades will ensure the pool continues to operate in a safe and efficient manner. The opportunity to complete these upgrades while the pool is undergoing extensive renovation allowed the Town to better utilize pool time without further closure of recreation facilities.

Financial Implications:

1037611 - MCC Pool Filt & Pump Sys Repl

Approved Spending Limit		$1,002,336.00
Approved Spending Limit Contingency		$100,233.60
Additional Expenditures		
Replacement of Electrical Systems in Basement	$115,000.00	
Basement Plumbing Replacement/Repairs	$50,000.00	
Additional Hot Tub Repairs & Pool Drain Piping	$40,000.00	
Watermain Replacement	$50,000.00	
Estimated Additional Expenditures	$255,000.00	
Potential Expenditures Exceeding Approved Contingency		($154,766.40)

1019511 - MCC - Repl Pool Slide & Refub Stairs

Approved Spending Limit		$254,400.00
Approved Spending Limit Contingency		$25,440.00
Additional Expenditures		
Replacement of the Pool stairs (treads, handrails & guardrails)	$210,000.00	
Waterslide Structural Inspection/Engineering	$20,000.00	
Estimated Additional Expenditures	$230,000.00	
Potential Expenditures Exceeding Approved Contingency		($204,560.00)

A final report will be provided upon completion of the project. Any additional funding will be allocated at that time from the Canada Community Building Fund and the Building Maintenance Reserve.

Communication Issues:

The MCC Pools are currently closed for construction and unfortunately the additional work has resulted in additional time, the pool will not re-open April 1, 2023 as planned. The public notice is attached for reference on how this information has been communicated to the public and pool users.

Conclusion:

The original approved spending limit for the MCC Pool Renovation will be exceeded and a final report will be prepared in accordance with the Capital Expenditure Control Policy, upon project completion.

Attachments:

ATT-1: McLean Community Centre Pool Notice - Update

Prepared by:

Lindsay Sinclair - Supervisor of Facility Capital Projects & Asset Management

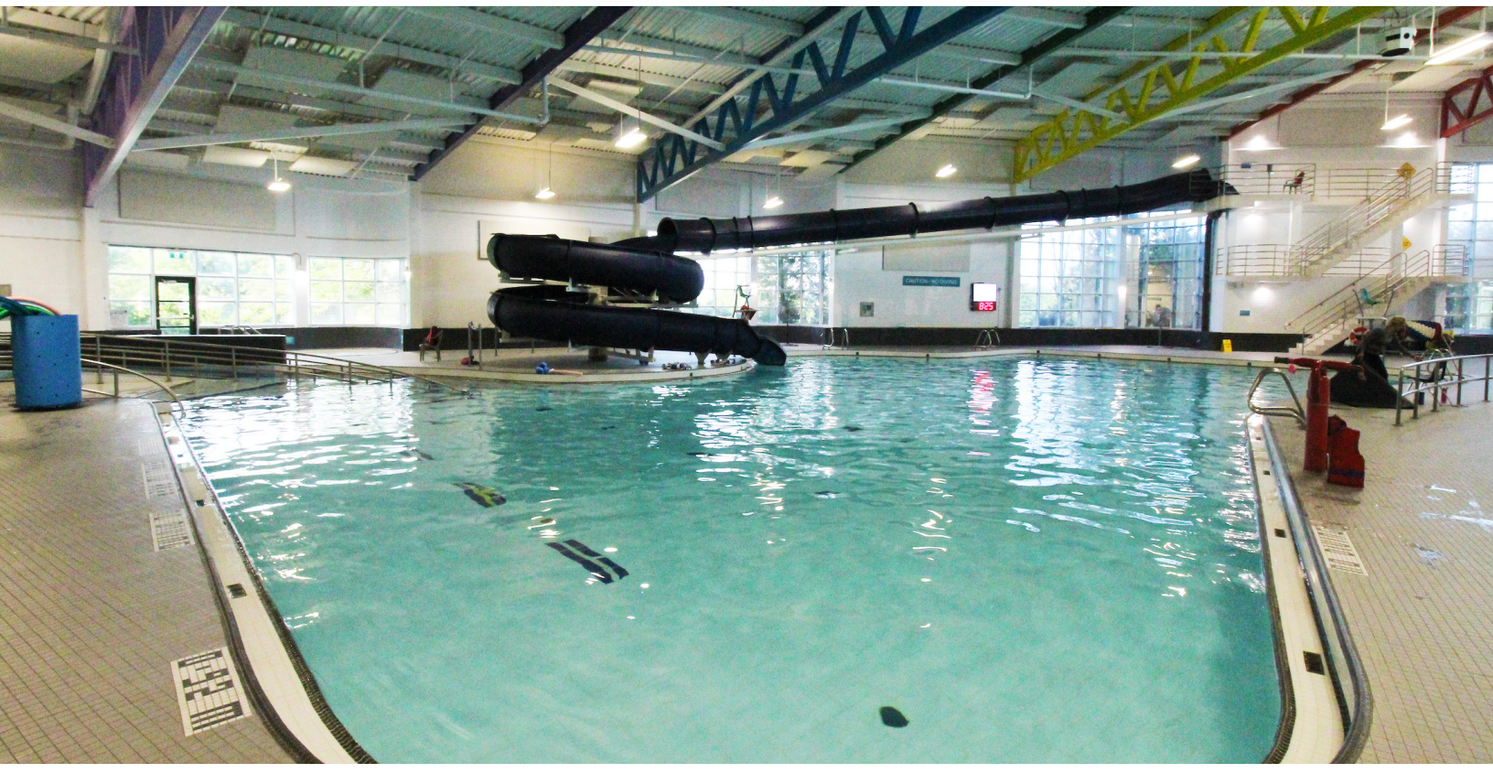
Submitted by:

Dave Meredith - Director of Operations & Environmental Services

Approved by:

Shane Baker – Chief Administrative Officer

Pool Closure Notice
McLean Community Centre (MCC)



The McLean Community Centre will remain closed until further notice. The pool filtration system including all pumps, controls and piping is being replaced due to age and condition. In addition, a new water slide will be installed on the pool deck, returning a very popular feature to McLean Community Centre.

For Aqua Fitness and Recreational Swimming participants:

All Aqua Fitness programs and recreational swimming opportunities will continue to take place at the Audley Recreation Centre (1955 Audley Rd. N.) and the Ajax Community Centre (75 Centennial Rd.) until further notice.

For registered Spring Swim Lesson participants:

- Most programs originally scheduled at the McLean Community Centre will be re-located to the Audley Recreation Centre (1955 Audley Rd. N.) for the same day of week and time as currently scheduled.
- The location change is for the entire 9-week Spring Session.
- A very small number of programs will be re-scheduled into alternate program spaces. These families have been contacted to discuss alternate program time options.

Aqua Fitness
Scan here or visit
ajax.ca/**Dropin**

Recreational Swimming
Scan here or visit
ajax.ca/**Swimming**

We will continue to provide updated re-opening information as it becomes available.

Thank you for your continued understanding and patience as we work to improve the facility.

ajax.ca/**Swimming**

Town of Ajax By the Lake

This content is available in alternative formats upon request by contacting 905-683-4550 or email contactus@ajax.ca

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Andrew Jackson, Supervisor, Infrastructure Design and Construction
Report #:	OES-2023-08
Subject:	Contract Award – St. Andrew's Site Improvements
Ward:	3
Date of Meeting:	April 11, 2023
Reference:	Capital Account 1035311.5850 – St. Andrew's Parking Lot (Resurfacing) Capital Account 1030011.5850 – St. Andrew's Outdoor Recreation Improvements RFT No. T22015 – St. Andrew's Site Improvements RFS No. S22005 – Contract Administration and Geotechnical Services RFS No. S23005 – St. Andrew's Site Improvements

Recommendation:

1. **That Council award the contract for St. Andrew's Site Improvements, to Wyndale Paving Co. Ltd. in the amount of $1,054,804.10 (inclusive of all taxes).**

2. **That Council approve the following funding to be allotted to Capital Account No. 1035311:**

Building Maintenance Reserve	**$ 261,447.59**

Background:

The approved 2022 Capital Budget included the provision for site improvements to the St. Andrew's Community Centre. The Centre is located on the east side of Exeter Road, south of King's Crescent. The project was awarded to the contractor in June 2022, with construction starting in August 2022. The contract was expected to be substantially completed by the end of 2022. During the course of construction, the contractor did not meet expectations and was contractually non-compliant. After several verbal warnings and two written warnings including two failing Contractor Performance Evaluations, the contract was terminated in December 2022.

During the winter of 2023, Town staff re-tendered the project as a single source to the second lowest bidder from the original tender call. The original scope of work will remain, with the expectation that the construction will be completed mid-summer, 2023.

Discussion:

The current Purchasing By-Law allows for purchases through negotiations, under certain conditions, more specifically, "where, at the discretion of the Department Head, in consultation with the Manager of Purchasing, it is deemed to be in the best interest of the municipality to negotiate with vendors."

Town staff prepared a single source tender to the second lowest bidder from the original tender. It is in the best interest of the Town to award the new contract to this contractor as they have the experience necessary to complete this project. The contractor has worked on several Capital projects for the Town spanning many years. They are fully aware of Town processes and have an excellent working relationship with Town staff.

Financial Implications:

Capital Account No. 1035311.5850 – St. Andrew's Parking Lot (Resurfacing)

Approved Capital Budget		$265,000.00
Approved Additional Budget (GGC June 13, 2022)		$130,584.00
		$395,584.00
Previous Awards/Expenditures		
Contract (net of HST rebate)	$72,534.52	
Consulting Fees – Contract Administration and Geotechnical Testing and Inspection	$9,194.16	$81,728.69
Available Budget		$313,855.31
Current Award		
Contract (net of HST rebate)	$523,002.64	
Contingency (10%)	$52,300.26	$575,302.90
Over Budget		($261,447.59)

Capital Account No. 1030011.5850 – St. Andrew's Outdoor Recreation Improvements

Approved Capital Budget		$535,000.00
Previous Awards/Expenditures		
Contract (net of HST rebate)	$13,812.14	
Consulting Fees – Golder Engineering	$9,056.63	
Signage	$91.58	
Site Security Fence	$5,308.56	
Consulting Fees – Contract Administration and Geotechnical Testing and Inspection	$18,565.97	$46,834.88
Available Budget		$488,165.12
Current Award		
Contract (net of HST rebate)	$426,881.16	
Contingency (10%)	$42,688.11	$469,569.27
Available Budget		$18,595.85

A 10% contingency has been identified to ensure sufficient funding is available to address site specific issues that can be encountered during construction. Risk factors for this project include:

- Unusual soil or base conditions, site conditions different than those anticipated,
- Unanticipated utilities encountered during the construction,
- Additional environmental management required such as sediment controls,
- De-watering requirements,
- Contamination of soil, and/or,
- Poor weather conditions.

These risks can cause delays, necessitating additional material and labour associated with construction.

The cost increase can be attributed to the current inflation rate, the increased cost of labour, construction materials and increased fuel prices. The increase can also be attributed to the additional quantities of soil removals and disposals, an increase in the need for additional base materials, geogrid fabric, increases to asphalt and concrete pricing, and drainage works.

Additional funding, not to exceed $261,447.59, will be allocated from Building Maintenance Reserve.

The Capital Detail Sheets are attached for information.

Communication Issues:

Information signage will be posted on the site during construction. A construction schedule will be posted on the Town's website upon contract award. Staff will continue to correspond with various user groups adjacent to the construction site, providing status updates as required.

Relationship to the Strategic Plan:

This report aligns with the following section of the 2018-2022 Strategic Plan – Stronger Together:

Focus Area:	Connecting our Community
Goal 3:	Fostering a sense of place and celebrating our history
Goal 4:	Creating opportunities to foster a welcoming and inclusive community
Goal 5:	Promoting a range of programs, services, and supports for all ages and abilities
Goal 7:	Enhancing connections to get people moving
Focus Area:	Investing in our Community
Goal 3:	Building new municipal infrastructure
Goal 4:	Modernizing existing municipal infrastructure and other assets
Goal 6:	Enhancing the beauty and appeal of the Town
Focus Area:	Leading in Our Community
Goal 4:	Becoming an age-friendly community

Conclusion:

It is the recommendation of staff that Wyndale Paving Co. Ltd. be awarded the contract for the St. Andrew's Improvements project, as a single source award.

Attachments:

Attch 1: Site Concept Plan
Attch 2: Long Range Capital Forecast Detail Sheet – 1035311
Attch 3: Long Range Capital Forecast Detail Sheet – 1030011

Prepared by:

Andrew Jackson – Supervisor, Infrastructure Design and Construction

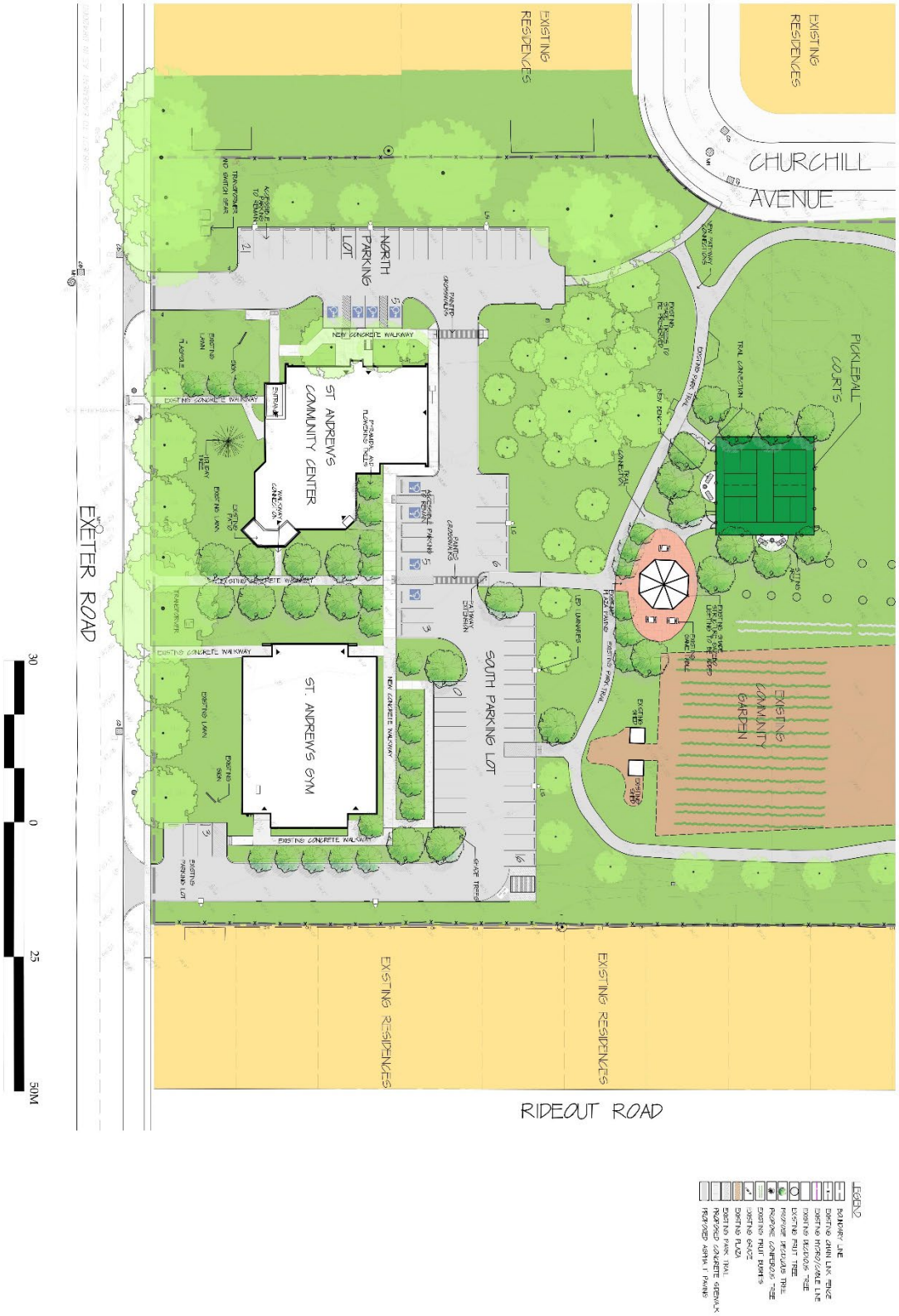
Submitted by:

Dave Meredith – Director Operations and Environmental Services

Approved by:

Shane Baker – Chief Administrative Officer

Attachment 1: Site Concept Plan



Attachment 2: Long Range Capital Forecast Detail Sheet

Municipal Parking Lifecycle Maintenance Program

TOWN OF AJAX
2022 CAPITAL BUDGET / 2023-2031 LONG RANGE CAPITAL FORECAST
DETAIL SHEET

Project Name **Municipal Parking Lifecycle Maintenance Program**
Project Number LC033000

Program Description

The Municipal Parking Lifecyle Maintenance program provides for maintenance and rehabilitation of municipal parking lots at various locations throughout the Town of Ajax in order to maintain these facilities in a state of good repair.

The Town currently maintains 4 6 parking lots associated to Town facilities and 29 parking lots within the park areas and on the waterfront.

EXPENDITURES / FUNDING

	2022	2023	2024	2025-2031	Total
Expenditures					
1035211 Town Hall Parking Lot (M&O)	**905,000**	-	-	-	**905,000**
1035311 St. Andrew's Pkg Lot (Resurf)	**265,000**	-	-	-	**265,000**
1035411 Mill St ComCtr PrkgLot (Recon)	**100,000**	-	-	-	**100,000**
ENG079 Lakeside CC - Parking Lot (Reconstruction)	-	-	**380,000**	-	**380,000**
ENG097 Hermitage Upper Lot at Griffiths (Resurf)	-	**300,000**	-	-	**300,000**
ENG121 Veterans Point Parking Lot (Mill & Overlay)	-	-	-	**70,000**	**70,000**
ENG122 MCC East & South Parking Lot (Recons)	-	**910,000**	-	-	**910,000**
ENG123 ACC North Parking Lot & Driveway (Resurf)	-	-	-	**705,000**	**705,000**
ENG125 ACC - West Arena Service Area (Resurf)	-	-	-	**100,000**	**100,000**
ENG233 Millers Creek Parking Lot (Resurfacing)	-	-	-	**140,000**	**140,000**
ENG234 Sportplex Handicap Parking Lot (Resurf)	-	-	-	**20,000**	**20,000**
ENG235 Rotary Park Mn Pkg Lot & N/S Driveway	-	-	-	**390,000**	**390,000**
Total Expenditures	**1,270,000**	**1,210,000**	**380,000**	**1,425,000**	**4,285,000**
Financing					
Building Maintenance Reserve	**1,170,000**	**910,000**	**380,000**	**805,000**	**3,265,000**
General Infrastructure Maintenance Reserve	**100,000**	**300,000**	-	**620,000**	**1,020,000**
Total Financing	**1,270,000**	**1,210,000**	**380,000**	**1,425,000**	**4,285,000**

Attachment 3: Long Range Capital Forecast Detail Sheet

St. Andrew's Outdoor Recreation Improvements

TOWN OF AJAX
2022 CAPITAL BUDGET / 2023-2031 LONG RANGE CAPITAL FORECAST
DETAIL SHEET

Department	Operations & Environmental Services	**Project Initiation Year**	2021
Section	Infrastructure Engineering		
Project Name	**St. Andrew's Outdoor Rec Impr**		
Submitted By	Ranil Fernando, Manager of Infrastructure & Asset Management		
Start Year	2022		
Project Number	1030011		

PROJECT DESCRIPTION / JUSTIFICATION

Scope: Construction of a new pickleball court, connecting the two independent parking areas, tree planting, and age appropriate AODA compliant site furniture. The proposed improvements will be located adjacent to the existing St. Andrew's Community Centre.

Rationale: In support of the Seniors Strategy, the introduction of a pickleball court will provide a desirable outdoor recreational opportunity for seniors. Parking lot connections will improve the internal circulation and flow for pedestrians and vehicles. Construction of a pickleball court and parking lot will align with the timing of the construction of St. Andrew's playground (ENG071) and with the resurfacing of the existing parking lot (ENG019).

References: This project is consistent with Section 2.2.6.1 of the Town of Ajax Official Plan which states that the Town of Ajax shall establish open space and parks with amenities that encourage physical activity, wellness, and informal use opportunities for a wide range of users. This project responds to the Recreation, Parks and Culture Master Plan's goal of assuming a leadership role in providing and facilitating recreation, parks, arts, and cultural services (Goal 2).

Schedule: Construction: 2022

EXPENDITURES / FUNDING

	2022	2023	2024	2025-2031	Total
Total Expenditures	**535,000**	**-**	**-**	**-**	**535,000**
Development Reserve	**535,000**	**-**	**-**	**-**	**535,000**
Total Funding	**535,000**	**-**	**-**	**-**	**535,000**
Annual Operating Costs	***2,500***	**-**	**-**	**-**	***2,500***

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Geoff Romanowski, MCIP, RPP, CPT Director, Planning & Development Services Dhaval Pandya, P. Eng., PTOE Manager of Transportation Planning
Report #:	PDS-2023-14
Subject:	**Update to the Residential Parking Permit Feasibility Study**
Ward(s):	All
Date of Meeting:	April 11, 2023
References:	Town of Ajax Residential Parking Review (Nov. 24/05 - GGC) Town of Ajax Residential Parking Review (Jun. 5/06 - CAP) RPP Feasibility Study: Findings and Analysis (Mar. 23/09 - CAP) 2016 Waterfront Management and By-law Enforcement Strategy (Apr. 11/16) – GGC) Occasional Use On-Street Parking Permit System (Jun. 12/17 - GGC) Occasional Use On-Street Parking Permit System (Oct. 2/17 - CAP) Town Boulevard and Private Driveway Alterations (Jun. 10/19 - GGC) Residential Boulevard Encroachments (Oct. 13/20 - GGC) Occasional Use Parking Permits (Mar. 9/20 - GGC) 2020 Waterfront Management and By-law Services Enforcement Strategy (Nov. 9/20 – GGC) Background Report for Town Initiated Zoning By-law Amendment: Residential Driveway Standards (Jan.4/21 - CAP) Town Initiated Zoning By-law Amendment Application Z1/21: Residential Driveway Standards (Apr. 6/21 - CAP) Waterfront Parking Management Program – Review of First Season and Proposed Program Changes (Jan. 10/22- GGC) Summary of Council Decisions – December (Dec. 12/22 - Council)

Recommendations:

1. **That the report titled "Update to the Residential Parking Permit Feasibility Study" be received for information; and**

2. **That staff prepare and post a Request for Information (RFI) for a "Town Parking and Parking Permit Feasibility Study" to create the necessary documentation for further direction.**

1. Background:

There have been several times over the past 20 years that staff (from all departments) have reported to Council on challenges with parking, parking permits, parking on certain streets or in certain neighbourhoods, parking on private property, and parking on boulevards and streets (of all classification). Internally at all levels, the discussion about parking has taken different views and asked many questions about how the Town will grow this aspect of the municipality, as we have many residents and visitors that drive and need space to park when they are out in the community, for various reasons, including shopping, visiting family and friends, enjoying events, or conducting business within the community.

1.1 Residential Parking Review & Residential Parking Permit Feasibility Study (2004 – 2009)

Prior to 2004, the Town of Ajax had not conducted any kind of parking or parking permit study. Parking issues were predominately a waterfront issue. On September 27th, 2004, Council passed a resolution to review the Town's minimum lot frontage and parking requirements for freehold dwelling units in response to concerns and complaints about the lack of residential parking within particular communities.

On November 24th, 2005, staff brought forward a report with recommendations regarding a Town of Ajax Residential Parking Review and asked that the review, which essentially provided suggestions on how to reconcile some of the competing demands that have been placed upon residential front yards and private garages, while other broader issues, which extend beyond zoning standards, such as, on-street visitor parking, width of road allowances, and the ability to accommodate parking and storage needs of homeowners.

Council directed staff to circulate this review to the Urban Development Institute (Durham Chapter), the Town of Ajax Transportation Advisory Committee, ratepayer groups, and to post it on the Town's website. Following this consultation, staff were directed to report back to the Community Affairs and Planning Committee by March 6th, 2006 with recommended changes to the Town's Zoning By-law, conditions of draft plan approval, and other related matters, to improve the supply of residential parking within Ajax.

On June 5th, 2006 (originally to be reported back on March 6th, 2006), staff reported back to the Community Affairs and Planning Committee seeking endorsement of proposed zoning by-law amendments related to driveway widths, garage widths, and curb cut widths, as well as adjustments to zoning requirements and policy related to new subdivisions. Another outcome from this Residential Parking Review, was that Council further direct staff to examine the feasibility of an overnight on-street parking by permit (or Residential Parking Permit) program.

Beginning in November 2007, and over the course of 2008, staff (with the assistance of LEA Consulting Ltd.) carried out a feasibility analysis of a Residential Parking Permit (RPP) program, which also included an overnight on-street parking analysis.

This analysis reviewed several factors:

- demand, supply, road/parking characteristics.
- how a RPP program would operate: support/permitting/processing/enforcement/ legal/staffing/costs.
- feedback and input form an appointed advisory committee.
- identified three (3) residential areas that were representative of the parking-related concerns within the Town (evaluation of participation rate, on-street parking supply, and cost to implement and operate program).

The results of this study/review brought forward recommendations that a Residential Parking Permit (RPP) program not be pursed at this time due to the high start-up costs and the projected low residential participation rate. This recommendation was approved by Council in March 2009.

1.2 Waterfront Parking Management & By-law Enforcement Program/Strategy (2016 – 2022)

The Town carried out two Waterfront Management and By-law Enforcement (WMBE) strategies, one in 2016 and another in 2020. Although these strategies discussed various issues, such as damaged infrastructure, conflicts with permit holders, waste management and litter, the steady increase in special events and permits, and non-permitted gatherings of large groups of people, these strategies also discussed on-street parking in and around the waterfront.

Preceding these strategies, specifically in 2004, Council directed staff to present a report to improve efficiencies in managing waterfront events with respect to parking. The results of this report led to the application of certain restrictions on specific streets surrounding Rotary Park, for a 2-week window from the middle of June to the beginning of July (Homeweek and Canada Day events). These restrictions took the form of “No Parking” signs being installed on one side of certain streets (Parkes Drive, Bartlett Drive, Varley Drive, Farrow Crescent, Fitzgerald Drive, and Love Crescent) and prohibited parking 24 hours a day, 7 days a week, however enforcement only took place during events at Rotary Park.

Through the 2016 WMBE strategy, additional streets were added (Simonds Drive, Milner Crescent, Anstead Crescent, and Cadby Road) and the window of time that parking restrictions would be in place also grew from a 2-week window to a 3-month window. The restrictions that were put in place at this time were “No Parking” signs being installed on one side of each street, and restricting on-street parking between May 15th and September 30th, on Saturdays, Sundays, and Holidays, between 8 a.m. and 7 p.m.

These parking restrictions continued until 2020, which is when a revised WMBE strategy was presented to Council. The 2020 WMBE strategy introduced (endorsed and approved by Council) a paid permit parking program along the entire waterfront. This program applies to all Town parking lots along the waterfront (Rotary Park, Bartlett Drive, Love Crescent, Lions Point, and Paradise Park), as well as the 10 seasonal on-street parking restrictions around Rotary Park with a slight adjustment to the parking restrictions, being May 15th to September 15th, including Saturdays, Sundays, and Holidays, from 8:30 a.m. to 9:00 p.m. The cost to park in any of the designated waterfront areas for non-residents was $20.00 in the initial phase of this program.

However in June 2021, at the direction of Council, and prior to staff reporting back on the progress of this aspect of the Waterfront Management and By-law Enforcement Strategy in January 2022, the fees for parking for non-residents was amended to $10.00 for 3 hours or less, and $20.00 for a full day, as well as a few other adjustments for those that own or lease property in Town and businesses, along with the removal of two streets from enforcement (Milner Crescent and Farrow Crescent). To-date this program is still operational.

1.3 Occasional Use On-Street Parking Permit System (2017)

Between 2009 and 2017 no studies or reports were brought forward to Council for their consideration.

On June 12th, 2017, a report was presented to Council from the Office of the CAO, Planning & Development Services, and Legislative and Information Services, and outlined many of the elements that were presented through the 2009 report, with the exception of some more

detailed analysis related to overnight on-street parking, which was the prevailing issue as identified by staff at the time as seen through this report.

The recommendations of this report were that staff survey residents on the potential implementation of an Occasional Use On-Street Parking Permit System and that staff report back in the Fall (2017) on the results to make recommendations on how to proceed.

Staff reported back on October 2nd, 2017, with the recommendations that the Town implement a one-year pilot for an Occasional Use On-Street Parking Permit System and that staff report back after the one-year pilot period.

On March 9th, 2020, staff reported back to Council and recommended that the Occasional Use Parking Permit System be implemented permanently, along with a few parameters (i.e., number of permits per license plate per year), and that the Fees and Charges By-law be amended accordingly.

These recommendations were endorsed, and the Occasional Parking Permit program is currently operational.

1.4 Residential Driveway Standard & Driveway Apron Amendments (2019 to 2021)

From 2019 to 2021, Planning staff collaborated with By-Law Services, the Office of the CAO and other Town Departments/Sections to identify opportunities for increasing the supply of off-street parking spaces. Two separate projects were undertaken concurrently in pursuit of this goal: 1) a Residential Driveway Standards Zoning By-law Amendment, and 2) the creation of a Residential Boulevard Encroachment Policy.

The Residential Driveway Standards Zoning By-law Amendment was designed to eliminate inefficiencies in the existing Zoning By-law and to maximize the amount of hard surface permitted in residential zones to be used for parking purposes. Much of the work was administrative in nature and focused on items such as reducing the size of a legal parking space and widening the maximum permitted driveway width standard to align with parking space width dimensions. These improvements helped to maximize the number of legal parking spaces that could fit onto existing properties but did little to provide opportunities to property owners to physically expand the areas for parking on their property.

To accommodate this issue, the Town introduced a new tool in the Zoning By-law called a Driveway-Adjacent Walkway (DAW). DAWs were introduced into all single-use residential zones and permitted property owners to install additional hard surfaces immediately adjacent to their driveways up to 1.8 meters-wide on the side of their driveway nearest the principal entrance to the dwelling and up to 0.6 meters wide on the opposite side. The result of this was an immediate expansion of the area that could be used for parking.

As the work on the Zoning By-law was being completed, Town staff were also working to implement a similar solution for driveway aprons – the portion of a driveway located on the Town's boulevard (right-of-way) between the front property line (private property limits) and the street. Up to this time, the Town had required anyone who wanted to alter their driveway apron to apply to the Town for an Apron Widening Permit. The creation of a Residential Boulevard Encroachment Policy (and subsequent amendments to the Town's Road Occupancy and Closure By-law) enabled property owners to alter their aprons to the degree outlined in Town by-laws without having to make an application to the Town. Specifically, it allowed property owners to install DAWs to the extent described above along the side of their driveway aprons. This additional width was especially valuable for enabling the parking of a vehicle on a driveway apron in a position parallel to the street, as By-law Services allows for vehicles to be parked in

this manner provided, they do not overhang onto the street or sidewalk and all tires are positioned on a hard surface.

Both of these amendments are formally in place as Council had approved these amendments to Zoning By-law 95-2003, as amended. These amendments are being consolidated into the Zoning By-law during its next consolidation.

Parking on private properties and public rights-of-way within the Town has been an area of focus for Council for almost 20 years, but at no time over the past 20 years has a Town-wide parking (public and/or private) study been carried out. Studies to-date have only looked at a few areas where problems have existed or persisted or where the parking characteristics have portrayed the parking related concerns over the years within Town. This limited scope has yielded limited results.

2. Discussion:

Council directed staff, through resolution, at the December 12th, 2022 Council meeting, to present a report no later than April 2023 on the cost, timeline, scope, and potential funding sources for an updated Town-wide Residential Parking Permit (RPP) Study.

The cost, timeline, scope, and potential funding source(s) should always be explored prior to investing time, resources, and finances into any kind of study, implementable program, or future deliverable, especially if the desire is to achieve a mutual benefit (Town and Community), efficiency in service (and enforcement), and buy-in from the community.

The following information outlines (at a high-level) what would be required to complete the study, from preparing the Request for Information (RFI) to reporting to Council the final recommendations. This information does not include any costs of implementation of any sort, as these costs, timelines, future project scopes, and potential funding sources would be determined at later dates through different capital projects.

2.1 Costs to complete an updated Residential Parking Permit Study

Based on a scan of municipalities that have completed or are about to take on some form of parking or parking permit study, studies of this nature range between $150,000 (Vancouver/Calgary/Whitby) and $500,000 (Mississauga/Hamilton/Markham). From this information and other information solicited from GTHA municipalities, it has been estimated that a Residential Parking Permit Study for the Town of Ajax would cost anywhere between $150,000 and $400,000, depending on the scope of the study.

Since the original study from 2009 was not a Town-wide study, the scope from then (2004 - 2009) to now (2023) would be significantly different in all aspects of the study. The study that Council direct staff to look into should be reviewing the entire Town and every street or opportunity within the Town.

A RFI should be prepared and issued to fully understand the true costs of carrying out a Town-wide Residential Parking Permit (RPP) Study or a “Town Parking and Parking Permit Feasibility Study”.

Staff will need the balance of 2023 to prepare the RFI, issue the RFI, and report back to Council for further direction.

2.2 Timeline to complete the Study

It is estimated that the RPP study would take up to two years to complete. This does not include the procurement process through an Request for Proposal (RFP).

2.3 Scope of the Study

The scope of a study of this nature would include, at a minimum:

- an examination of the existing on-street parking supply and parking opportunities (public or private) within the Town of Ajax;
- an investigation into the potential participation rates of residents/businesses within and visitors to the Town of Ajax;
- how an RPP could be integrated with the Town's existing on-street parking programs, services, and by-laws; and
- the cost recovery mechanisms that the Town could potentially utilize.

This study should also look at the entire Town, not just certain areas of the Town like previous studies had. Studying the entire Town will provide the biggest and clearest picture of the residential parking opportunities that are available and the parking permit system that could be implemented.

The study should also seek to bring forward recommendations to ensure that any system or opportunity, if supported and implemented, is functionally simple to use by all users, can adapt over time as the Town grows, and can be a benefit to the Town and its residents.

An RFI is an essential step to not only ensure that the project is properly costed out but also to establish that the scope of the study is manageable and achievable. An RFI will give the Town an opportunity to seek an understanding of the estimated costs and components of a RPP study for Ajax without spending any money. It can also identify other components that should be considered as part of the study, which may not be limited to just a parking permit system, but it could also look at other parking opportunities, such as public parking lots, public/private parking facilities within new developments, or other types of parking facilities within specific areas of the Town (i.e. Downtown, Uptown, Midtown, and Pickering Village).

2.4 Potential Funding Source for the Study

The funding sources available to support a Town Parking and Parking Permit Feasibility Study would be through the Strategic Initiatives Reserve.

3. Financial Implications:

There are no financial implications related to the recommendations in this report.

4. Communication Issues:

There are no communication issues to report because of this report.

5. Relationship to the Strategic Plan:

Pillar: Growing our Community
Priority: Embrace dynamic and sustainable growth
Action: Advance our vision for community growth

6. Conclusion:

Parking in general, whether it be on private property, boulevards, or streets, has been a long-standing topic of discussion amongst residents, municipal administrators, and elected representatives within the Town of Ajax. The cost, timeline, scope, and funding source to carry out a study of this nature is significant (both from an overall project delivery perspective, but also from an individual staff time perspective) and should be considered through a proper process. Staff have provided within this report a high-level estimate of the cost of the study, the length of time to complete the study, the general scope of the study, and the funding source for this study.

To fully understand the impacts of on-street parking, a parking permit system, or other types of parking facilities or opportunities, staff are recommending that a robust feasibility study is the necessary step in ensuring that any parking and the system to enforce parking, if implemented, is a system that is functionally simple to use by all users, can adapt over time as the Town grows, and can be a benefit to the Town and its residents.

Staff are also recommending that should the study proceed, it be conducted in a purposeful and team-like manner with participation by those departments that intersect with parking (private or public), parking by-laws, parking users, and operations around parking within our municipal rights-of-way and municipally owned lands.

Attachments: N/A

Prepared and Submitted by:

Dhaval Pandya, P. Eng., PTOE – Manager of Transportation Planning

Prepared and Submitted by:

Geoff Romanowski, MCIP, RPP, CPT – Director of Planning & Development Services

Approved by:

Shane Baker – Chief Administrative Officer

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Allison Kushner, Community Recreation Supervisor
Report #:	RCCD-2023-03
Subject:	**Bus Transportation Services**
Date of Meeting:	April 11, 2023
Reference:	RFT No. T23004 Bus Transportation Services Various business units

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Recommendation:

1. That Council award the contract for Bus Transportation Services, to Stock Transportation Ltd. in the estimated amount of $87,910.61 (inclusive of all taxes), for a period of approximately one year.

2. That Council authorize Staff to renew the contract for an additional five, one year periods, pending an analysis and satisfactory performance review at the anniversary date of the contract, in the estimated amount of $487,922.25 (inclusive of all taxes).

Background:

Bus Transportation Services support the operational needs of various programs and events within Recreation & Culture. These services include the following:

Older Adult/Seniors Programming & Events

- No cost weekly shuttle service is provided to and from several Ajax seniors residence buildings to the Pickering Village and St. Andrew's Community Centre Seniors' Clubs.

- Free shuttle service is provided for several Older Adult/Seniors events including the 55+ BBQ and Information & Wellness Fair. Venue parking becomes congested due to high attendance.

- No cost shuttle service is provided once a month to shuttle seniors from designated community pick-up locations to the St Francis Centre for Arts & Educational Programming.

Festivals and Events

- Free shuttle service is provided to and from Town Hall, Ajax GO Station and the Audley Recreation Centre to the annual Canada Day and Pumpkinville Special Events. Venue parking becomes congested due to high attendance.

School Break Camps

- Busing services are available from the McLean Community Centre to and from designated camp locations for those requiring Before and After Care or transportation to and from camp. Fees for these services are included annually on the Fees and Charges By-Law.

- Participants are transported to and from various camp locations to the Ajax Memorial Outdoor Pool several times a week for swimming. Participants attending Day Trippers and Youth on the Move camp travel to various trip locations located in Durham Region and the GTA. Bus costs are incorporated into the fee of the respective camp (as per the Council approved Fees and Charges By-Law).

Summer Playgrounds

- Busing services are available for the Summer Playgrounds program, to and from playground locations to the Audley Recreation Centre for swim trips and the Village Arena for special events. This service is free for participants.

Wheelchair accessible buses are available if required for any of the above services.

Discussion:

Request for Tender (RFT) documents were posted to the Towns' eBidding platform and 6 prospective bidders registered for the opportunity with bids being received from 2 of these, prior to the closing on March 10, 2023. Listed below is a summary of the bids received:

NAME OF BIDDER	TOTAL TENDER AMOUNT
Stock Transportation Ltd.	$ 575,832.86
FirstCanada ULC	$ 1,099,443.45

Financial Implications:

Funds for Bus Transportation Services are included in the Recreation & Culture operating budget.

Relationship to the Strategic Plan:

Strategic Focus Area: Leading in Our Community

Goal: Becoming an Age Friendly Community

Action:

- Explore the feasibility of providing transportation to seniors attending senior clubs at Town facilities

Conclusion:

It is the recommendation of staff that Stock Transportation Ltd. be awarded the contract for Bus Transportation Services, being the lowest bidder meeting minimum specifications.

Attachments: N/A

Prepared by:

Allison Kushner – Community Recreation Supervisor

Submitted by:

Chris Vida – Director, Recreation & Culture

Approved by:

Shane Baker – Chief Administrative Officer

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Chris Walker – Manager of Technology and Innovation
Report #:	LIS-2023-05
Subject:	**Corporate Information Technology Strategic Plan – Year Three Update**
Ward(s): All	N/A
Date of Meeting:	April 11, 2023
Reference:	Corporate Information Technology Strategic Plan (ITSP)

Recommendation:

That the report entitled "Corporate Information Technology Strategic Plan – Year Three Update" be received for information.

Background:

The Town approved a Corporate Information Technology Strategic Plan (ITSP) in early 2020. The plan provides a strategic roadmap for the Town's technology initiatives over a 5-year timeframe. The ITSP identified the need for significant work to be done at the various levels of the Town's Technology environment. The plan broke down the Town's Technology needs into four areas: Infrastructure, Business Systems, Integrations and Customer facing, identifying priorities in each of those areas.

The purpose of this report is to present the initiatives and projects that the Technology and Innovation Division undertook in year three (2022) as a result of the directions outlined within the ITSP, and to showcase the many accomplishments and advancements that have been made possible by Ajax Council's investment in this critical area of our business.

Discussion:

ITSP Recommendations**:**

The ITSP outlined a staged approach to implementation to address the needs found above by defining three distinct phases. These are:

- Phase 1 - New IT: First, before anything else can be achieved, the Town must reset the technology management model and establish a more collaborative environment.

- Phase 2 - Process Digitization and Mobilization: Next, the Town must focus on digitization of its core business processes (HR, Planning, Permitting, Asset and

Work Management, Finance) taking the opportunity to streamline and simplify aging paper-based processes.

- Phase 3 - Optimization through Data and Digital Services: Then, building upon the digitization of processes, the Town can focus on offering new digital services and using the data from process digitization to become a data-driven organization.

Table 1 – High Level ITSP Implementation Stages

	2020	2021	2022	2023	2024	2025
Phase 1: New IT						
Phase 2: Process Modernization						
Phase 3: Optimization & Digital						

Accomplishments - Phase 1: New IT

There were a number of key objectives identified in order to build the conditions for success which were a focus for the Technology and Innovation Division in 2022. Although these items were identified to take place from 2020-2021, the Technology and Innovation Division continues to focus on and implement better governance and Technology processes to realize success. Below is a summary of these ongoing objectives:

Objective #1: Improving corporate coordination and planning around technology and improving outcomes of business-technology projects:

Actionable Item	Status	Comments
Implement policies and processes to improve the rigor by which technology ideas are conceptualized, planned, funded and executed	Ongoing	The following policies have been established and/or updated in 2022: - Issuing a replacement MFA Token - Early Adopters Terms of Reference - Technology Use Policy (M365 Additions) - Access to Technology Resources while Travelling Internationally
Implement project management and portfolio management techniques to improve project outcomes	Ongoing	- Project and Portfolio policy and work instructions have been established. - Currently this team is leading multiple large corporate projects for example: o Microsoft 365 Ph.2 o Digital Plans Submissions Portal o Electronic Signatures o HRMS Implementation o AVL and Dashcam - Project Portfolio Management program: o 42 Active Projects o 66 Projects approved for 2023
Implement key performance indicators to monitor the quality and performance of IT service delivery	Ongoing	- Process improvements have been made to capture KPIs enabling improved reporting going forward. - A service review will be undertaken in 2023

Objective #2: Recognize the need to evolve the IT function and establish suitable resources to support the Town's vision

Actionable Item	Status	Comments

Augment internal IT and business resources by including contract and backfill staffing costs in capital project request.	Ongoing	- Contract Business Analyst hired in 2022 and 2023 - Contract Network Architect was hired in 2022 - Use of contracted staffing as an alternative to the traditional staffing model will be a common practice moving forward, where appropriate.
Increasingly use 3rd party experts to assist the Town in planning and implementing technologies as part of a hybrid IT service delivery model	Ongoing	- Hired a managed service to monitor our network traffic (SIEM) - Hired consultant to manage the CityWorks project - Hired a vendor to support implementation of M365 - Hired a vendor support Amanda Implementation - Hired a Managed Service to support the Town's financial system.

Objective #3: The Town must implement a range of foundational technologies to ensure robust and reliable technology services for staff and customers, and to mitigate known risks.

Actionable Item	Status	Comments
IT Security Enhancements	Ongoing	Many initiatives took place in 2022 and will continue; some highlights include: - Implemented MFA for all staff accessing e-mail and MS Office products. - Managed Security Service Provider onboarded. - IT Security Audit and Penetration Test completed. - Begun upgrade of firewall and redundant firewall. - Core Switch replacement and replacement of switches throughout the town - MS Exchange Upgrade - Procured and implemented new cybersecurity education program for staff which enhances the town's program for phishing tests and corporate benchmarking or our security posture. - Arranged cybersecurity training for Council members
IT System Management Projects	Ongoing	Some examples of system management: - Procured redundant network (Bell) - Increased network bandwidth - Core switch upgrade (increased network capacity) - Upgrade of Wireless Access Points (APs) at all town facilities - Cable Management - ACC Boardroom Upgrade
Business Solutions upgrades to current versions	Ongoing	Examples of business solutions that have been or are currently being upgraded: - Amanda - CityWorks - Meeting Room Upgrades – TV and Hybrid Cameras - Soft Phone for queues - HR and Finance Solutions - AVL and Dashcams
Implement Microsoft 365 to encourage collaboration and a modern workspace	Ongoing	- Email, and Office products were rolled out corporately in by June 2022 - MFA enabled for all staff - Roll out of Teams, OneDrive and SharePoint is in progress

Business Continuity and Disaster Recovery Planning	Ongoing	- Completed the Business Impact Assessment with all departments to identify the recovery time needed for each service during system outages. - Playbooks, incident response plans are currently being developed. - Tabletop exercise completed with Finance - Tabletop exercise with the rest of the departments in 2023

The following initiatives have not been started by 2022 which were identified by the ITSP, they are:

- Creation of a Web and Digital Steering Committee
- Corporate Wi-Fi review and strategy

These items have been delayed in favor of higher priority items but will be re-visited in the coming years.

Accomplishments - Phase 2: Process Modernization

There were a number of key objectives identified in the ITSP which were to commence in 2022 as part of Phase 2 of the ITSP implementation. In this phase the Town must focus on digitization of its core business processes (HR, Planning, Permitting, Asset and Work Management, Finance) taking the opportunity to streamline and simplify aging paper-based processes.

Actionable Item	Status	Comments

Fully Implement CityWorks (Work and Asset Management System)	Complete	- System has been fully implemented across the Operations and Environmental Services Department.
		- Supervisors assign work through the system and field staff complete work on mobile devices making the process more efficient and eliminating the need for paper and manual processes.
		- In 2022 there were:
		  ○ 15,724 Work Orders Created
		  ○ 11,375 Work Orders Completed

Implement an HRMS system (Human Resources Management System)	Ongoing	- The IT Projects Team began the assessment of this project in 2021 working with all town departments to understand requirements. - Project has been approved in the 2023 capital budget and scheduled to be implemented by 2025 - This will eliminate 15 disparate systems and improve workflows across the organization.
Work Towards a CRM (Customer Relationship Management)	Not Started	- This project has been identified in the capital budget for 2024. - Assessment will begin, pending completion of Customer Service Strategy in 2023
Enhancements to the Amanda System	Ongoing	- Amanda Upgrade scheduled for Mid-April 2023 - Digital Plans Submission project – Vendor Awarded, project scheduled to be completed in Q3 2023.
MOU with the Library	Complete	- Worked with the library to develop an MOU focusing on services and technology provided to clarify roles

		and responsibilities, and rationalize and enhance the technology relationship between the two groups
Corporate GIS Strategy	Complete	- Consultant worked with all departments to identify over 200 Action Items - Will be implemented over the next 5 years

The ITSP identified the need for the Town to modernize its processes and find efficiencies where possible. In 2022 the town undertook several modernization projects and have realized widespread efficiencies as identified in the table below:

Actionable Item	Status	Comments and Statistics

Electronic Signatures	Ongoing	- Adobe Sign is the awarded product:
		  ○ Over **1200** Digital Agreements have been created by town staff eliminating the need for paper and saving staff and end user time by signing digitally.
		  ○ Industry reports suggest on average users experience 37% improvement in productivity using e-signatures.
		- Staff hiring process was the initial focus, HR have been champions of this rollout.
		- Planned rollout throughout the organization in 2023, currently working with Finance and LIS.

Payment Gateway	Ongoing	- Awarded to 'Paymentus' - IT, Finance and Vendor working on onboarding various workloads in 2023. Focus will be on current inefficient processes to increase staff efficiency. Significant time will be saved by finance staff and improvements to how residents pay for services.
TCOL (Tax Certificates Online)	Complete	- Enable users to receive tax certificates electronically and save time for Finance staff (10 minutes per certificate) - Since October an estimated **28 Hours of staff time** has been saved (169 certificates) - In peak months, the Finance department could see 100 certificates per month.
Fire Apparatus Hardware Upgrade	Complete	- Modernization of hardware in the fire trucks, enabling AFES staff to respond to and manage incidents more effectively and efficiently.
Soft Phone – Queue setup	Complete	- All staff can now answer calls for their respective queue from home. Previously staff that were taking calls from a queue had to come into the office. This upgrade enables a more flexible staffing model and efficiencies in queue and workforce management.
Fire System GIS Mapping	Complete	- Development of a GIS application to enable AFES staff to visualize fire related incidents and incident patterns. This application assists staff with planning, preparedness, response, and risk reduction, allowing AFES to target its efforts where they will have the biggest impact.

Code Blue System Implementation	Compete	- Panic buttons in the Parking Lot at Town Hall and Rotary Park; improves staff and customer safety, modernizing the response to personal safety incidents
Open Developments Interactive Map	In progress/ Testing	- Online Map viewer and dashboard to communicate active development applications and planning information to the public and council.
InterviewStream – Talent Acquisition Video System Pilot	In Progress/ Testing	- Automated hiring process for an estimated 150 positions, saving HR and hiring managers significant time and effort during the recruiting process: o 58% reduction in time-to-offer o 80% decrease in time spent on each interview. o Reduce paper-based processes such as application status, interview evaluation and consolidation of results. o Automated scheduling of interviews o Real time reporting o Improved candidate experience o Overall reduction in hiring costs
Senior Snow GIS mapping application	Complete	- Field crews can view routes and homes that are part of the senior snow program. Once a house is completed, it will be marked as complete by the field crew and a picture is taken. Supervisors can track the completion of the program on a web map on their computer. - The previous process was completed on paper. - Estimate from Supervisor of the Senior Snow Program: o Saving 4 hours/day during a snow event from having to attend site visits (5-10 visits per day) o Saving 4 hours/day during a snow event from an administrative perspective such as looking up individual information, and improved customer services (30-80 calls during an event) o Improved liability control, which can help defend against claims against the town. ▪ Documented the crew who performed the work ▪ Date stamped with time of completion captured ▪ Record of photos showing completed work for every snow event

2022 Municipal Election:

The Technology and Innovation Division is a key contributor to the delivery of the Municipal Election. This was a top priority of the team in 2022 being responsible for several items, including but not limited to:

- Procurement of Computers used for voting
- Testing and configuring the "image" on the voting computers
- Setup and testing at voting centers
- Election Auditing
- Technical support at voting centres throughout the voting period
- Project Management

It is important to note that the above items take up significant time and effort from the Technology and Innovation Division and was an important deliverable for the team in 2022. Ajax continues to be a sector leader in the delivery of modern, technology-enabled, paperless municipal elections.

Looking Forward:

2023 will be year 4 of the ITSP and will focus primarily on Phase 2 and 3 of the plan as identified on page 2 of this report. Work will continue to enhance and maintain the Town's infrastructure and security to enable the improved services the Technology and Innovation is eager to provide, as well as establishing the necessary governance to ensure that technology is properly managed at the Town.

Based on the improvements that the Technology and Innovation Division has made to its infrastructure and building off the business solutions enhancements its already made, the Town is able to continue enabling technology for improved staff and customer facing solutions that will modernize processes. As processes are modernized and business solutions are improved, the Technology and Innovation division will lead the data focused initiatives that will enable departments, senior management and council to make more informed decisions with a goal of being a data driven organization.

Some of the noteworthy projects scheduled for 2023 include:

Project (s) Category	Examples
Infrastructure	- Complete Firewall and redundant firewall upgrade - Complete bandwidth Increase at all town facilities - Server Upgrades - Switch Upgrades for improved network traffic - Improvements and upgrades for Fire Dispatch - Support/Infrastructure for Library Makerspace - Complete Council Chambers Technology Upgrades - Finance System Infrastructure Upgrade - Windows 11 Deployment Planning - Improved boardroom technology, TVs and Hybrid meeting cameras

Business Systems and Integration	- Begin to focus on corporate data projects and improving corporate use of data, for example:
	  o Corporate Analytics and Dashboards
	  o GIS ETL Solution
	  o Improve reports from current systems
	  o CityWorks Reporting
	- Amanda Upgrade
	- Microsoft 365 Implementation for all staff (Teams, OneDrive, SharePoint)
	- Digital Plans Submissions Portal

	- E-Signatures - Payment Gateway - GIS Strategy Implementation - AVL/Dashcam Solution - Accommodation Strategy Technology Enablement - HRMS Implementation - Open Data Portal Improvements - Talent Acquisition Video Solution - Ajax.ca Website Project Support/Assessment - Online Assessment Lookup Tool - Online Resident Tax Portal
Organizational Changes	- Establishment of a GIS and Data Services Team to lead data focused projects. - Establishment of a Cybersecurity and Infrastructure team to lead to have a more focused team making technology infrastructure improvements and to have a focused internal cybersecurity group.

There are numerous projects slated for completion in future years of the ITSP that are sure to drive considerable efficiencies through better automation, and the replacement of manual, paper-based processes with technology enabled solutions. These solutions free up time, allowing staff to re-focus time and energy on more high-value activities. Staff will endeavor to track these improvements as carefully as possible to demonstrate the value of continued investment in technology.

Financial Implications:

There are no financial implications related to the recommendations in this report.

Communication Issues:

The ITSP identified that the Technology and Innovation Division's communication and profile within the broader organization was an area requiring attention and improvement. Throughout 2022, IT has made notable improvements in this area. For example:

- More Frequent OLA posts
- GIS Week Activities
- Development of an onboarding/new hire website
- Ongoing Cybersecurity correspondence and participation in Cybersecurity Month
- Internal project websites
- Technology and Innovation Division updates on the intranet (OLA)

Looking forward, IT will continue to increase their communications within the organization to ensure staff understand the changes that are taking place. Some items that are planned include:

- Technology Project Dashboard
- Increased communications regarding policy updates and work instructions
- Increased solution and security "on demand" training for staff to access at any time.

The Technology and Innovation Division will also ensure that Town staff, Senior Management, and Council are kept apprised of the progress being made in the ITSP, as well as exciting upcoming technology initiatives.

Relationship to the Strategic Plan:

The 2019 – 2022 Strategic plan states the promise “We will strive to be leaders in all that we do and foster a vibrant, innovative and connected community.” It further identifies several action tasks within each focus area that will be fully supported by the IT Strategic Plan, including:

- *Connecting Our Community* – creating a more engaged community, providing excellence in customer service
- *Investing in Our Community* – modernizing existing municipal infrastructure and other assets, ensuring the Town’s workforce effectively and efficiently meets the community’s needs
- *Leading in Our Community* – strengthening governance, building external partnerships to help achieve strategic goals.

Conclusion:

The Technology and Innovation Division have been able to successfully complete or start nearly all of the recommendations made from the ITSP. Multiple other projects have been delivered or are in progress which align with the theme of the ITSP and that is to make best use of our business solutions and to become a tech-savvy and data driven organization.

Attachments:

N/A

Prepared by:

Chris Walker – Manager of Technology and Innovation

Submitted by:

Nicole Cooper – Director of Legislative and Information Services/Clerk

Approved by:

Shane Baker – Chief Administrative Officer

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Nicole Cooper
Report #:	LIS-2023-06
Subject:	**Integrity Commissioner Services – Recommendation to appoint Principles Integrity**
Ward(s):	All
Date of Meeting:	April 11, 2023
Reference:	N/A

Recommendation:

That a by-law to appoint Principles Integrity as Integrity Commissioner for the Town of Ajax and to repeal By-law 12-2019 (ATT-1) be brought forward to the Council Meeting of April 17, 2023 for approval.

Background:

Under the Municipal Act, 2001 ("the Act"), Council is required to appoint an Integrity Commissioner. The Integrity Commissioner is an independent officer who reports directly to Council and whose powers and duties are set out in the Act.

The Integrity Commissioner has the following responsibilities as outlined under section 223.3 of the Act:

a. The application of the code of conduct for members of council and the code of conduct for members of local boards.

b. The application of any procedures, rules and policies of the municipality and local boards governing the ethical behaviour of members of council and of local boards.

c. The application of sections 5, 5.1 and 5.2 of the Municipal Conflict of Interest Act to members of council and of local boards.

d. Requests from members of council and of local boards for advice respecting their obligations under the code of conduct applicable to the member.

e. Requests from members of council and of local boards for advice respecting their obligations under a procedure, rule, or policy of the municipality or of the local board, as the case may be, governing the ethical behaviour of members.

f. Requests from members of council and of local boards for advice respecting their obligations under the Municipal Conflict of Interest Act.

g. The provision of educational information to members of council, members of local boards, the municipality, and the public about the municipality's codes of conduct for members of council and members of local boards and about the Municipal Conflict of Interest Act.

In addition, Ajax Council has delegated authority to the Integrity Commissioner to impose penalties in accordance with the Municipal Act, 2001 if in the opinion of the Integrity Commissioner, a Member of Council has contravened the Code of Conduct.

Aird and Berlis was appointed as Integrity Commissioner for the Town of Ajax in 2019, following a competitive bidding process, for a period of one year from Feb 27, 2019, to Feb 28, 2020 with the possibility of 7 one year extensions. The agreement may be terminated by the Town at any time and it is always within Council's authority to terminate the appointment of an Integrity Commissioner and appoint a new Integrity Commissioner. The contract with Aird and Berlis is currently up for renewal, and staff are recommending that the Town opt not to renew the contract.

In 2022, the Region of Durham selected a new Integrity Commissioner, Principles Integrity, through a competitive bidding process and subsequent appointment by Council. Principles Integrity is a well-known and respected provider of integrity commissioner services.

A "piggyback" clause was included in the Region's contract with its new Integrity Commissioner so that Area Municipalities could avail themselves of the contract should they wish to do so. As such, Clerks from various area municipalities were involved in the interview process for the Region's Integrity Commissioner. All lower-tier municipalities in Durham Region, with the exception of Ajax, have opted to piggyback on the contract with Principles Integrity. This means that all area municipalities have the same Integrity Commissioner except for Ajax.

Since late 2022, efforts have been underway to achieve better alignment between the lower-tier Codes of Conduct and the Region's Code of Conduct. The Clerks of all municipalities have held several meetings with an aim to bring better synchronization to Codes of Conduct within the Region, and many Durham municipalities have already, or will be, bringing forward changes to their Code of Conduct toward this end. The updated Codes of Conduct being proposed are based on a template supplied by Principles Integrity which reflects current norms and best practices within the municipal sector.

Discussion:

While LIS staff have been attending the above noted meetings regarding Code of Conduct harmonization, Ajax continues to be the outlier as the only municipality not utilizing the services of Principles Integrity, and with a Code of Conduct that follows a different form and structure than the example recommended by Principles Integrity.

Four members of Ajax's Council also sit on Regional Council, plus a fifth member who serves as a temporary alternate from time to time in the absence of Regional members. When fulfilling their duties as elected officials, it is not always clear whether members are acting in their capacity as a member of Regional Council or a member of Ajax Council. This can create understandable confusion for the member in trying to understand and comply with the rules under two different Codes of Conduct, and determine the standard of behavior that they are held to, where they may differ. This problem is best demonstrated as it relates to rules around the acceptance of gifts and benefits. Imagine a scenario where the Region's Code of Conduct

permitted the acceptance of hospitality (i.e. a ticket to a sporting event) up to a certain threshold, but the Town's Code of Conduct sets a lower acceptable threshold. If a Regional Councillor accepted hospitality that was allowable under the Region's Code, but not allowable under the Town's Code, it is not especially clear to which standard they would be held.

Further, one of the most important roles of the Integrity Commissioner is their advice-giving function to members as it relates to Code of Conduct and Conflict of Interest matters. At present, where an Ajax member of Regional Council requires advice, it may not be evident whether they should consult the Region's Integrity Commissioner or the Ajax Integrity Commissioner. For the convenience of Regional Council members, staff are of the view that it is beneficial to utilize the same service provider as the Region for these services so that seeking advice is a 'one-stop-shop' for Ajax's Regional Councillors.

Financial Implications:

Pending engagement of Principles Integrity, the Town will be required to pay a one-time retainer fee of $1,250. The go-forward costs for Integrity Commissioner services have the potential to be significantly reduced given the lower hourly rates of Principles Integrity as compared to Aird and Berlis; of course, overall costs will depend on usage of the service which can be unpredictable. Funds for Integrity Commissioner services are included in the LIS operating budget.

Communication Issues:

Pending the passage of a by-law to appoint a new Integrity Commissioner, the Town will update all public-facing materials to reference the contact information for Principles Integrity.

Relationship to the Strategic Plan:

N/A

Conclusion:

Staff are of the view that it is in the Town's interests to work in harmony with the rest of the Durham Region municipalities to seek alignment, wherever possible, on Council Codes of Conduct. Unifying service provision for Integrity Commissioner Services with the Region of Durham is also beneficial for the reasons outlined within this report. Pending approval of the recommendation contained herein, a by-law will be forwarded to Council to appoint Principles Integrity as Integrity Commissioner of the Town of Ajax, effective immediately. Under the terms of the Town's existing contract with Aird and Berlis, they are required to *"provide the Client's next appointed Integrity Commissioner with any and all materials, documents, or records related to any matters that are then on-going and will cooperate with the transfer of same."* Staff will work with both parties to ensure a smooth transition.

Attachments:

ATT-1: Proposed Bylaw XX-2023 – Appointment of Integrity Commissioner

Prepared and Submitted by:

Nicole Cooper – Director, Legislative and Information Services

Approved by:

Shane Baker – Chief Administrative Officer

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Dhaval Pandya, P.Eng., PTOE Manager of Transportation Planning
Report#	PDS-2023-13
Subject:	**Traffic Calming Program Reporting Process**
Ward(s):	All
Date of Meeting:	April 11, 2023
Reference:	Council Resolution, dated June 20, 2022

Recommendations:

1. **That the report entitled "Traffic Calming Program Reporting Process" be received for information; and**

2. **That staff be directed to report annually (in April) to Council on the Traffic Calming Program.**

1. Background:

Traffic calming is used to improve traffic safety for all road users primarily through physical features to encourage driver behaviour changes. The objectives of these traffic calming measures are to reduce the negative impacts of vehicle traffic on a specific road segment such as high vehicle speeds, high traffic infiltration / cut-through traffic and increased conflicts between road users. Traffic calming measures are selected depending on the specific concerns, geometric considerations, existing and future active transportation infrastructure considerations, and needs. The implementation of traffic calming should be applied consistently Town-wide and with careful attention to the network impacts to ensure volume and speed concerns are not transferred to adjacent streets.

Types of Traffic Calming Devices

Both Speed Humps and Speed Cushions raise a portion of a road which induces a vertical motion for vehicles, and discomfort that influences the driver to slow down. A speed hump is not as wide as a speed cushion but is higher resulting in a more abrupt deflection. Speed humps cover the entire width of the road while speed cushions provide an opening for wheel paths of certain vehicles (e.g. fire services, transit).

Temporary Bollards are flexible plastic posts located every 100m to 150m along a road. Each location has a bollard in the centre of the road and one bollard on either side of the road. This set-up visually narrows the road and alters driver behaviour. These bollards are removed before the Winter to ensure winter control operations are not impacted.

Three types of traffic calming devices, as mentioned above, were chosen to assess these products on different criteria, such as speed reduction, impact on vehicle operation while navigating through these products, impacts to emergency response and transit vehicles, winter maintenance operation, etc. in the 2022 Traffic Calming Project.

Traffic Calming Warrant

The Town's current Traffic Calming Warrant is a policy tool that outlines when and how traffic calming should be used. Initially, it was approved in 2007 and has since been updated twice (2015 and 2020). The Traffic Calming Warrant provides a process to evaluate all traffic calming requests received by staff. This process involves the evaluation of each segment and assessment of seven distinct characteristics as mentioned in the Traffic Calming Guidelines. If through the evaluation process, a road segment achieves the minimum threshold points, it is added to the list of warranted projects. A key component of the 2020 Traffic Calming Warrant Update was to improve the process so road segments, which were warranted for traffic calming, would see measures implemented in a shorter timeframe.

Traffic Calming Project 2022

Thirteen (13) traffic calming projects, that were identified between 2017 and 2020, were implemented in 2022. **Table 1** provides a list of traffic calming locations.

Table 1 – List of Traffic Calming Locations

Ward	Road	Segment
1	Elizabeth Street	Delaney Drive to Old Kingston Road
1	Linton Avenue	Kearney Drive to Sherwood Road
1	Old Harwood Avenue	Magill Drive to Chapman Drive
1	Simms Drive	Wilce Drive to Griffiths Drive
2	Ravenscroft Road	Taunton Road to Beaverton Road
2	Seaward Crescent	Williamson Drive to Williamson Drive
2	Warner Drive	Taunton Road to Williamson Drive
2	Williamson Drive	Gillett Drive to Warner Drive
3	Clover Ridge Drive	Harwood Avenue South to Dreyer Drive
3	Elm Street	Beatty Road to Windsor Avenue
3	Emperor Street	Harwood Avenue South to Pickering Beach Road
3	Lake Driveway East	Harwood Avenue South to McClarnan Road
3	Pickering Beach Road	Emperor Street to Rollo Drive

Resident and Council's Communication Concerns

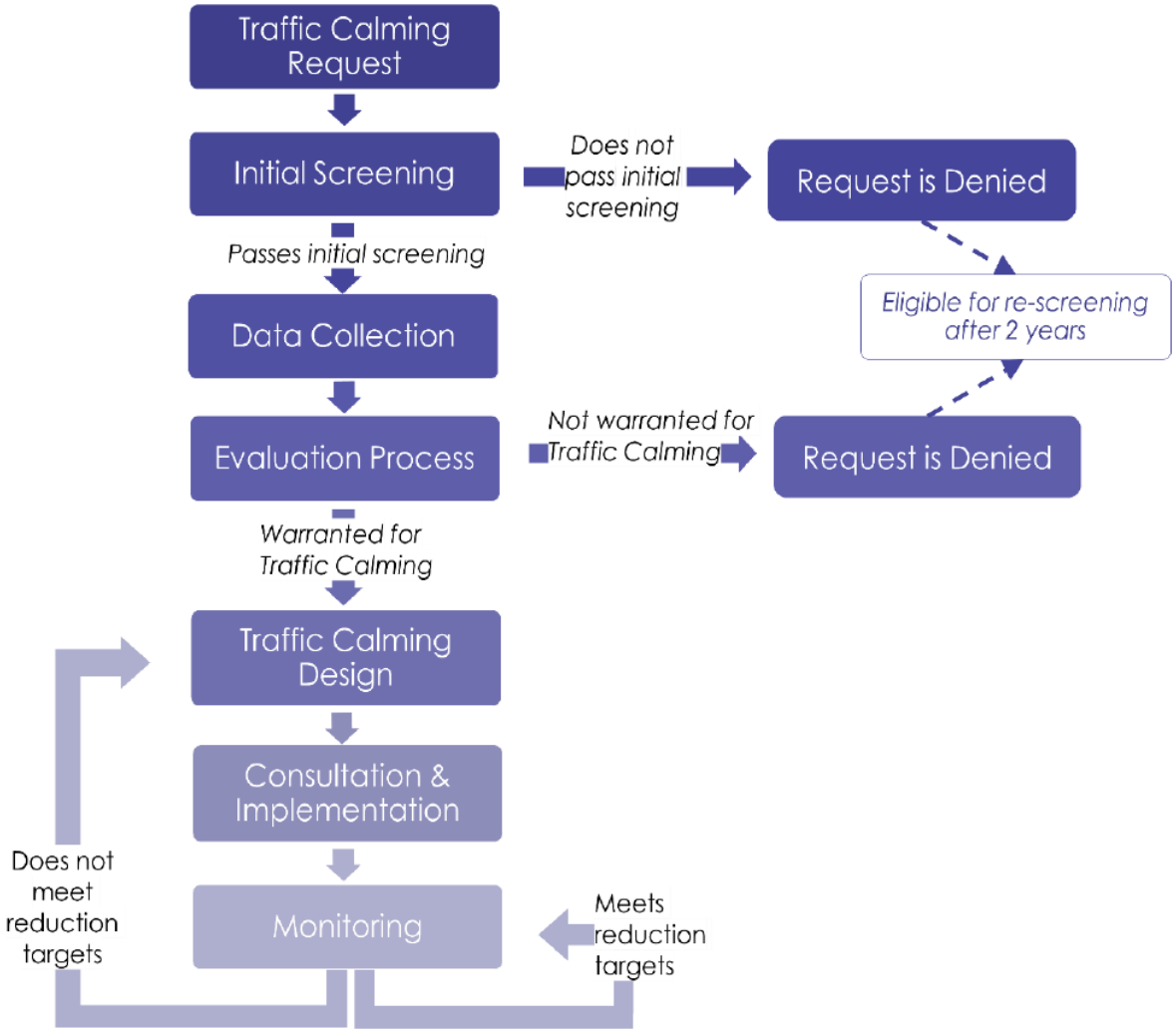
Since completion of the 2020 Traffic Calming Warrant Update, staff received input from residents and Council that overall, communication related to the Traffic Calming program was unclear. It was not always clear which road segments had been tested and which road segments were scheduled to be tested. Furthermore, once tested, it was not always clear which segments had met the criteria for traffic calming and which had not.

Subsequently on January 20, 2022, Council passed a resolution and directed staff to report back to Council in 2023 with a proposed traffic calming reporting process and begin regular traffic calming program reporting to Council up to twice a year, beginning in 2023.

2. Discussion:

Traffic Calming Process

The Traffic Calming process is a seven-step process, which is shown in the diagram below. A brief summary of each step follows the diagram:



Step 1: Traffic Calming Complaints/Request

Transportation Planning staff receive concerns/complaints related to speeding vehicles and/or requests for installing traffic calming devices. In 2022, staff received 224 traffic related complaints/concerns, out of which 96 complaints were related to speeding and/or requests for installation of traffic calming measures. Typically, these concerns/complaints are received via emails (traffic@ajax.ca or contactus@ajax.ca or through the council office), phone calls or in-person. With each request, staff collect basic information about the road segment and provide a brief overview of the next steps in the process. Many requests do not include full contact information (i.e., address and phone number) of the resident and as a result it becomes challenging for staff to communicate with the respective complainants/residents directly.

Step 2: Initial Screening

Once the request/complaint is received by Town staff, a road segment is screened using a list of initial criteria to confirm its eligibility for traffic calming. A road segment must pass this preliminary screening as per Town of Ajax Traffic Calming Policy to proceed to the full evaluation. Once the location is deemed eligible for traffic calming, it is added to the list of locations for traffic data collection. The speed on the road is also reviewed based on the historical traffic data reports.

Step 3: Data Collection

If data in the traffic database is more than two years old, consideration should be given to getting updated information to complete the Traffic Calming Warrant evaluation and the road sections are added to the list of traffic data collection. Typically, traffic data is collected during the Spring and Fall of each year. The timeline of traffic data collection also depends on the schedule/availability of the contractor. Generally, the Town receives detailed traffic data reports in 6-8 weeks after the traffic data collection period. The data collection process is focused on collecting two specific traffic characteristics for a given segment, the speed and volume of the traffic.

Step 4: Evaluation Process

Once the required data is collected, the next step in the process is to evaluate based on the warrant criteria as shown in **Table 2** (on the next page). Each warrant criterion has points assigned to it based on a road segment's characteristics. Elementary schools and major parks with playgrounds are considered major pedestrian generators for the traffic calming process analysis. In order for a segment to qualify for traffic calming, where applicable, it must meet the minimum threshold of a criteria and then overall it must score at least the minimum number of points based on its road classification as noted below.

- 35 points for a Local Road
- 40 points for a Collector Road
- 45 points for a Type 'C' Arterial Road

Step 5: Traffic Calming Design

The locations that score at least the minimum point threshold are eligible for traffic calming implementation. As per the Traffic Calming Guideline, Arterial A and Arterial B roads are in-eligible for traffic calming. Traffic calming measures can be sorted into two categories, minor-adjustment measures, and major-adjustment measures. The type of measure appropriate for a given location is identified based on the traffic and road characteristics and confirmed through consultation. Minor adjustment measures include a combination of horizontal deflection, pavement markings and the use of signage. **Table 3** (on the next page) shows the types of measures described in the Toolbox by category.

Based on the Town's available budget, each year the Town is implementing the appropriate traffic calming measures at as many locations as possible.

Table 2 - Warrant Criteria & Scoring

Criteria	Minimum Threshold	Points	Maximum Points
Operating Speed	85th percentile speed is higher than 10km/h above the posted speed limit	1 point for each km/h above posted speed PLUS 1 point for each 1% of vehicles over 15 km/h above posted speed	25
Traffic Volume	900 vehicles/day for Local roads 2,000 vehicles/day for Collector roads 5,000 vehicles/day for Type 'C' Arterial roads	1 point for each 50 vehicles above local threshold 1 point for each 100 vehicles above collector threshold 1 point for each 200 vehicles above arterial threshold	20
Collision History	Less than 6 (local) or 12 (collector / arterial) within the last 3 years	5 points for each qualifying collisions in excess of 3 *Qualifying collisions include collisions with vulnerable road users (pedestrians, cyclists) and collisions for which 'exceeding speed limit' or 'speed too fast for condition' is reported in the MVAR.	20
Pedestrian Generator		5 points for each school or park within the study area *Other Pedestrian Generators may be defined by Ajax and the study area includes the frontage of the road segment or within 300m	15
Pedestrian Facilities		10 points if there are no sidewalks in the study area 5 points if only on one side	10
Bicycle Facilities or Routes		5 points if bicycle lanes, sharrows, or routes are present on the road segment	5

Table 3: Types of Traffic Calming Measures

Vertical Deflection	**Horizontal Deflection**
• Raised Crosswalk • Raised Intersection • Speed Cushion • Speed Hump / Speed Table	• Chicane, 1-lane • Chicane, 2-lane • Curb Radius Reduction • Lateral Shift
Surface Treatment	**Road Narrowing**
• Sidewalk Extension • Textured Crosswalk	• Flexible Bollards • Curb Extension • Neckdown • Lane Narrowing • On-Street Parking • Raised Median Island • Road Diet
Obstruction	
• Directional Closure • Diverter • Full-Closure • Intersection Channelization	

Step 6: Consultation and Implementation

With all traffic calming projects, the public is invited to the public consultation process for reviewing the preliminary design. Residents and Council are notified of the consultation opportunity and are invited to provide feedback on the preliminary design. This gives residents and interested parties the opportunity to provide feedback and input on the proposed traffic calming design prior to implementation. At the consultation stage, staff also take the opportunity to speak with the public to see if they have any roads they want reviewed.

After a preliminary design is reviewed, staff revise the design based on the feedback received before preparing the tender documents. Upon completion of the tender process (tender invitation, tender evaluation, and tender awards), the awarded contractor completes the installation of the traffic calming measure on prioritized road segments.

Step 7: Monitoring

All traffic calming measures are reviewed and evaluated after one year of installation. The monitoring results determine whether additional traffic calming efforts are required. Generally, the evaluation is based on the change in traffic speed, volume, and impacts on the adjacent roadways. With new data collected after the traffic calming treatment has been installed and traffic has adjusted, the location is re-evaluated against the same criteria and point system. If the location scores enough points to still be eligible for traffic calming, additional measures are considered.

Traffic Calming Reporting/Communication Process

Current Issues with the Traffic Calming Reporting/Communication Process

As noted above, staff receive numerous traffic related complaints (224 complaints in 2022) from the residents and members of Council through various modes of communication, but predominantly via email. Most complaints or requests do not include full contact information (i.e., address and phone number) of the complainant/resident. Residents have also expressed that they do not feel comfortable sharing their full contact information with Town staff.

Transportation Planning staff are also responsible for development application reviews, transportation related capital projects, planning studies, etc. and receive numerous additional email communications regarding other work-related priority items on a daily basis. In the past, these complaints/requests have been saved in various locations in our system and not necessarily in the Town's work order tracking platform (i.e., AMANDA), which has led to some requests or responses being misplaced or not acted upon immediately. Also due to a significant amount of staff turnover in the transportation section of the Planning and Development Services Department, some information has been lost.

Based on the above noted information, it has been challenging for staff to track all complaints and provide a timely update/response to the members of Council and residents.

Traffic Calming Reporting/Communication Process Moving Forward

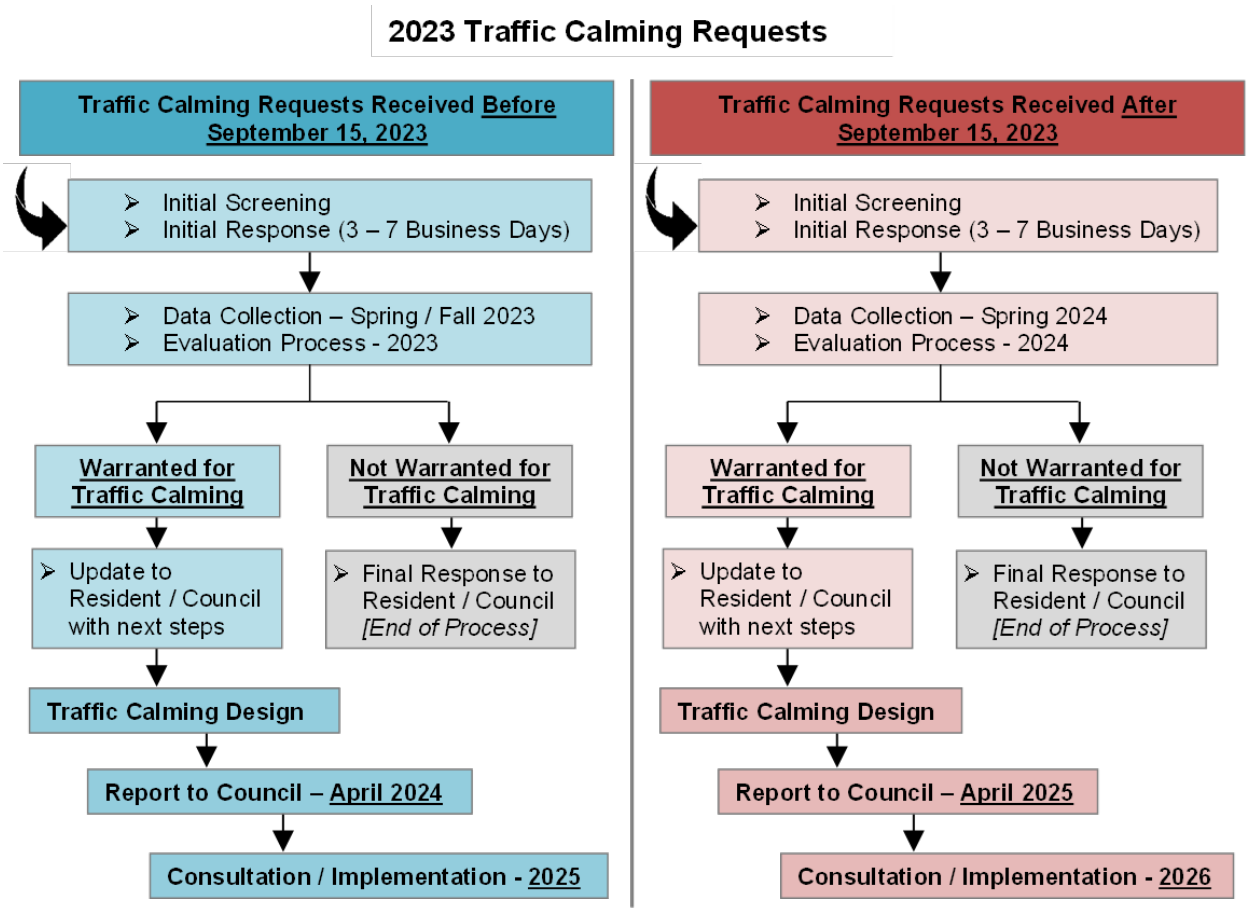
All locations that are deemed eligible for traffic calming will be added to a comprehensive list for traffic data collection. Per the Town's annual traffic data collection program, traffic data is collected during the Spring (May-June) and Fall (October-November) of each year. The actual timing of traffic data collection depends on the schedule or availability of the external contractor. Generally, the Town receives detailed traffic data reports about 6-8 weeks after traffic data

collection depending on contractor's schedule. Once the required data is collected, the next step in the process is to evaluate all road segments based on the warrant criteria.

Based on the times of the year (data collection, receipt of detailed traffic data reports, Summer and Christmas staff/Council vacation times), volume of warrant analysis (i.e., traffic calming, all-way stop control, PXO, traffic signal), time required for identification and preliminary design of traffic calming measures, and other ongoing work priorities, it is challenging for staff to provide traffic calming reporting twice a year. Based on these constraints, it is recommended that staff provide a one-time report on Traffic Calming in April of each year starting in 2024. This annual report will be presented to Council in advance of the annual capital budget process for the following year and be available publicly on the Town's website after being received by Council.

The requests/complaints received between January 1st and September 15th of each year will be included in the next report. Complaints received after the deadline of September 15th will be accessed and added to a Spring traffic data collection list of the following year.

A graphical representation of the TCP Reporting/Communication Process moving forward is provided below:



The annual traffic calming report will include the information on all road segments which were evaluated for traffic calming and the results of that evaluation and a summary of cost estimates for traffic calming implementation in the following year. The annual traffic calming report will also include details on locations with newly implemented traffic calming measures and the results of any monitoring completed. The traffic calming projects will be aligned with other Community Traffic Calming Initiatives (CSZs, ASE, soft Traffic Calming Measures such as implementation of radar message boards, signage and pavement markings, etc.).

Staff will prioritize the road sections for implementation of traffic calming for the following year. When prioritizing the projects, staff will consider multiple factors including, but not limited to, the following:

- what year was the segment first tested and identified as warranted;
- what ward is the segment in (with an effort, where possible, to balance projects between all wards annually);
- are there efficiencies if segments are implemented in the same year; and
- are there any other initiatives or road projects which would impact the implementation?

Cityworks

Based on the recent update, the Town's IT staff are in the process of initiating the Cityworks project for Transportation Planning (P&D) and Roadway Maintenance (O&ES) sections in Spring 2023. This is a user-friendly platform for entering, tracking, and monitoring work orders. Cityworks is also expected to improve the current communication and coordination gaps between the staff of the Planning and Development Services and Operations and Environmental Services Departments. Once full access of Cityworks is granted to both sections, it is expected that staff will be able to track and monitor all transportation/traffic related work orders, including traffic calming requests, and provide timely communication to the members of Council and general public.

Upon receiving full access to the Cityworks platform, all transportation/traffic related complaints or improvement requests received through various communication modes will be entered into Cityworks. Transportation Planning staff will provide an initial response to the resident(s) and/or members of Council within 3-7 working days. The initial response will include the information regarding the next steps in the process and the expected timeline for the next update and/or a final response. The subject Cityworks request/work order will be updated with the most recent communication with the residents, members of Council, and Roadway Maintenance Section of the Operations and Environmental Services Department.

3. Financial Implications:

There are no financial implications related to the recommendations in this report.

4. Communication Issues:

There are no communication issues as a result of this report.

5. Relationship to the Strategic Plan:

This report aligns with the following section of the Strategic Plan:

Focus Area: Connecting our Community
Priority 4: Foster a safe and welcoming community
Action 4.2: Continue to implement traffic calming initiatives

6. Conclusion:

Moving forward, staff will provide a one-time report on Traffic Calming in April of each year starting in 2024. This annual report will be presented to Council well in advance of the annual capital budget process for the following year and be available publicly on the Town's website after being received by Council.

The annual traffic calming report will include information on all road segments that were evaluated for traffic calming, the results of that evaluation, and a summary of cost estimates for traffic calming implementation in the following year. The annual traffic calming report will also include details on locations with newly implemented traffic calming measures and the results of any monitoring completed. The traffic calming projects will be aligned with other Community Traffic Calming Initiatives.

Additionally, access to Cityworks platform will enable Transportation Planning (P&D) and Road Operations (O&ES) staff to alleviate the current communication gap and improve overall coordination. It will also ensure improved communication with the residents and members of Council.

Attachments:

N/A

Prepared by:

Dhaval Pandya, P.Eng., PTOE – Manager of Transportation Planning

Submitted by:

Geoff Romanowski, CPT, MCIP, RPP – Director of Planning & Development Services

Approved by:

Shane Baker – Chief Administrative Officer

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Andrew Jackson
Report #:	OES-2023-04
Subject:	**Audley Recreation Centre Fairgrounds Feasibility Study**
Ward:	2
Date of Meeting:	April 11th, 2023
Reference:	Capital Account No.: 1040011.4108 S22032 Consulting Services – Feasibility Study ARC Fairgrounds Tourism Relief Fund Grant

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Recommendation:

1. **That Council receive the Audley Recreation Centre Fairgrounds Feasibility Study dated March 14, 2023, as prepared by MacLennan Jaunkalns Miller Architects Ltd. for information.**

2. **That staff be authorized to consider the detailed design for the proposed Audley Recreation Centre Fairground as part of the Town's 2024 Capital Budget and Long Range Capital Forecast process.**

Background:

As part of the master planning process for the phased Audley Recreation Centre ("ARC"), a portion of the open space area south of the facility was allocated for a future event space. As the Town continues to grow, there has become a demonstrated need for the Town to provide an outdoor space that can accommodate large-scale community events.

The Town was awarded a federal grant in 2022, through the Tourism Relief Fund Grant program, to develop a Fairground Feasibility Study for the Audley Recreation Centre, including a functional design concept. Upon approval of this grant to the Town, the consulting firm of MacLennan Jaunkalns Miller Architects Ltd. (MJMA) was hired as the project lead to prepare the study and functional plan. A condition of the grant stipulated that the study was to be completed by the end of March 2023.

The scope of the study was to confirm the feasibility of using the ARC grounds to hold future special events such as fairs, concerts, cultural events, seasonal events, rallies, and sporting events. It was anticipated that the existing infrastructure associated with ARC, such as washrooms, meeting rooms, parking, and banquet facilities, combined with the physical space and municipal services, would support the use of this area for large scale community events.

Discussion:

The Audley Recreation Centre Fairgrounds Feasibility Study, dated March 14, 2023 (Attachment 2), has been prepared by MJMA and their team of consultants to demonstrate the feasibility of adapting the existing Audley Recreation Centre lands into a permanent fairground and event venue. With the assistance of Town staff, the consultant determined the requirements and needs for various known Town events including: operational needs, existing condition assessments, a preliminary traffic and site circulation analysis, and initial engineering assessments.

Upon compiling and assessing these needs and requirements, a detailed event planning analysis was prepared to address site capacity, traffic management, circulation and access, staging, audio requirements, electrical, fireworks, washrooms, and emergency services. Upon completion of the event planning analysis, a functional design plan was created to address what additional infrastructure would be required to successfully transform the space into a fairground and event venue while maintaining the overall parkland aesthetic and function of the site.

The functional design plan was developed to address several features inclusive of the following:

- A large open green to serve as the major event space as well as a non-event multi-purpose parkland space for year-round use, including the potential use for outdoor sports, such as cricket;
- An infrastructure “loop” that includes access to power, lighting, water, and a maintenance track;
- A shelter belt surrounding the open green that acts as a planted berm creating a sense of enclosure, minimizing noise and fencing while incorporating the existing excess fill located on-site;
- A multi-functional plaza used as a space for stages, outdoor rinks, markets;
- A sensory friendly area used as a quiet zone separate from the active green area;
- A shuttle bus drop-off zone created along Audley Road to assist with traffic management and congestion;
- Public entry points to manage pedestrian access to the site, and
- A maintenance yard to assist with storage and staging, as well as emergency services.

The functional design plan also addresses such features as naturalized plantings, walkways, areas for active recreation, picnic and family gathering areas, stormwater management, water supply, and power supply.

The Audley Recreation Centre Fairgrounds Feasibility Study demonstrates the feasibility of transforming the site into a fairgrounds and event space venue. The study addresses various requirements, each of which supports the recommendation that the site is suitable as a fairground and event space. The various parts of the study include:

- Part 1: Summary and Background: An executive summary outlining the feasibility of developing the site as a fairgrounds and event space. This section also analyzes the existing site conditions, the existing conditions of the existing fairgrounds at Picov Downs, and future conditions at ARC inclusive of the future railway over-pass at Audley Road and the future next phase to the Audley Recreation Centre.

- Part 2: Needs Assessment: This section outlines the anticipated events and programs and assesses the site for opportunities in relation to future events.

- Part 3: Functional Design: This section uses the analysis of the various event needs and site conditions and demonstrates how the site could be developed to support the fairgrounds and event space. The functional plan incorporates various components into a design concept that supports the recommendation that the site is suitable to be used for a fairground and event space venue.

- Part 4: Phase One and Costs: A description of a base functional plan that could be considered for a first-phase of the fairgrounds project. The base plan is flexible enough to allow for future phases to be considered as needs are developed and funding becomes available. The first phase should incorporate re-grading of the existing berms to level the ARC field and to naturally define the perimeter of the event space. A cricket field could be considered as a viable use of the open field during non-event periods, further enhancing the use of the space. This section also includes an Order of Magnitude Cost Estimate that outlines a cost analysis based on limiting the extent of features within a phased approach. A first phase will include options for pricing so that decisions can be made to suit the budget and priorities of the Town. The estimate will be reviewed and refined through future detailed design phases.

- Part 5: Next Steps: A detailed analysis of the next steps required to proceed with developing the site into a fairground.

- Appendix A: Event Planning Considerations.
- Appendix B: Traffic Control and Vehicular Site Circulation.
- Appendix C: Grading and Stormwater Management and Considerations.
- Appendix D: Mechanical and Electrical Considerations.

The investment to develop the existing open field and infrastructure at the Audley Recreation Centre into a fairground, year-round event, and multi-purpose parkland space will add great value to the Town's green infrastructure. With the creation of this infrastructure, the Town's recreational and cultural needs will be further enhanced with the ability to continue to foster an engaged and active community.

Financial Implications:

In 2022, the Town received a federal Tourism Relief Grant to proceed with a feasibility study to assess the potenital of creating a fairground and event space at the Audley Recreation Centre. There are no further costs directly related to this project.

As per the recommendations of the study, additional costs will be required for detailed design and construction, which will be considered annually through the Town's annual budget process.

Relationship to the Strategic Plan (2018 – 2022):

Focus Area: Connecting our Community.
Goal 1: Creating a more engaged community.
Goal 3: Fostering a sense of place and celebrating our history.
Goal 6: Leveraging our arts and tourism opportunities.

Focus Area: Investing in our Community.
Goal 3: Building new municipal infrastructure.
Goal 4: Modernizing existing municipal infrastructure and other assets.
Goal 6: Enhancing the beauty and appeal of the Town.

Focus Area: Leading in our Community.
Goal 4: Becoming an Age-friendly Community.

Conclusion:

It is the recommendation of staff that the Audley Recreation Centre Fairgrounds Feasibility Study as prepared by MacLennan Jaunkalns Miller Architects Ltd. and dated March 14, 2023, be received for information.

Staff will consider the detailed design for the proposed Audley Recreation Centre Fairground as part of the Town's 2024 Capital Budget and Long-Range Capital Forecast process.

Attachments:

ATT-1: Audley Recreation Centre – South Field Air Photo

ATT-2: Audley Recreation Centre Fairgrounds Feasibility Study

Prepared by:

Andrew Jackson – Supervisor, Infrastructure Design and Construction

Submitted by:

Dave Meredith – Director Operations and Environmental Services

Approved by:

Shane Baker – Chief Administrative Officer

Attachment 1: Audley Recreation Centre, South Field Air Photo

DRAFT REPORT

Audley Recreation Centre Fairgrounds Feasibility Study

Town of Ajax

MJMA
ARCHITECTURE & DESIGN

Audley Recreation Centre Fairgrounds Feasibility Study

LEAD CONSULTANT

LANDSCAPE & ARCHITECTURE

MJMA
ARCHITECTURE & DESIGN

CONSULTANTS

EVENT PLANNING

DESIGN PARTNERS

TRAFFIC

HDR

CIVIL

EMC

ELECTRICAL & MECHANICAL

Smith + Andersen

REPORT PREPARED FOR

Town of Ajax By the Lake

LAND ACKNOWLEDGMENTS

We would like to acknowledge the ancestral lands and Indigenous peoples that have been present in the Town of Ajax for thousands of years. The Mississauga's of Scugog Island First Nation are the traditional stewards of this territory, and their cultural heritage is an integral part of the fabric of this community and this land. With gratitude, we strive to design public spaces that respect the teachings of this heritage and foster connections to the natural world.

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EMC GROUP LIMITED

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SMITH + ANDERSON

STANDARD LIMITATIONS

This Audley Recreation Centre Fairgrounds Feasibility Study for the Town of Ajax was prepared by MJMA, Architecture + Design ("MJMA"), along with the following Consultants: Design Partners of Canada, EMC Group Limited, HDR Inc and Smith + Andersen.

The recommendations of this study are presented as conceptual and have been based on general information provided by the Town and visual observations made while visiting the site. Professional judgement was exercised in assessing the information but no warranty is expressed or implied.

The disclosure of any information in this study is the sole responsibility of the Town. Any use that a Third Party makes of the information in this report will be the responsibility of such Third Parties.

MJMA and our Consultants accept no responsibility for damages, if any, suffered by a third party as a result of decisions made or actions made based on this study.

PART 1 - SUMMARY & BACKGROUND

- EXECUTIVE SUMMARY
- EXISTING SITE & BACKGROUND

EXECUTIVE SUMMARY

The work for this study is the next phase of the ongoing developments at the Audley Recreation Centre. Over the past 15 years, the Town has been consistently upgrading this site to respond according to the needs of the community, Eastern Greater Toronto Area's fastest growing population. In 2015, the Master Plan allocated a portion of the open field, south of the community centre, as a space appropriate to serve large scale Town events. As current spaces within the Town are potentially lost to future development, the priority to establish a permanent event space has been identified and the need to determine the potential of this site to serve that purpose is the aim of this study.

The MJMA Architecture + Landscape consultant team was engaged by the Town to study the feasibility of adapting the Audley site into a permanent Fairground, suitable for major townwide events such as fairs, concerts, cultural events, seasonal events, rallies and sporting events, while maintaining and enhancing the parkland that exists today. Town Staff assisted the Consultant Team in determining anticipated needs for the future Fairground, including a review of operations for events held at the Ajax Downs site, an assessment of existing conditions and operational challenges at the Audley site, a preliminary analysis of traffic management planning issues, and an initial engineering assessment to determine what accommodation would be required to host popular events such as the Canada Day Celebration.

A functional design plan was established with the compiled information and engineering considerations to determine what existing infrastructure could be used or upgraded, while identifying areas where additional infrastructure would be needed. This functional design plan structures the park concept design and formulates the design solutions that will make a new multi-use green space that can be both a Fairground and a community park for daily and all-season enjoyment.

The existing role of the park to the community is an important consideration for the future development of this site. A walking trail that meanders around the perimeter of the site leads pedestrians through the site's features: baseball fields, wetland, meadows, playgrounds, dog park, skate park, landscape mounds and at the south end, a naturalized meadow beside the site's irrigation and stormwater management ponds. Part of the vision for the park design concept is to enhance this experience by preserving the continuity of the perimeter walking trail and connecting future pedestrian paths to this loop.

The new spaces for the Fairground should be varied in experience. Different spatial scales, shading, lighting and planting will characterize the spaces from open and large to sheltered and quiet. The proposed design concept presented in this study is a new landscape design that is, first and foremost, a beautiful park, built to minimize its environmental footprint, detailed to ease operations and maintenance, and capable of transforming into a state-of-the-art event space for the Town to host premier events.

The concept design demonstrates the suitability of the Audley Recreation Centre site for development into the Town's new Fairground. Existing site infrastructure will allow the infrastructure of the new Fairground to "plug in", minimizing the scale of construction when compared to a greenfield project. Though future projects such as the Audley Road Overpass and the ARC Phase 3 expansion need careful consideration and traffic management will need further study, the investment to upgrade the existing open field into a year-round, multi-purpose parkland will add great value to the Town's green and social infrastructure.

As a site that serves both daily routines and remarkable events, the Audley Recreation Centre Fairgrounds will be a landmark site, a place for Ajax to gather outdoors and to celebrate together.

EXISTING SITE & BACKGROUND - COMMUNITY CENTRE & PARK

Occupying 100 acres of land, the Audley Recreation Centre was conceived as the anchor of a new neighbourhood with the intention of establishing and sustaining community. Combining overlapping indoor and outdoor programs, the Centre serves as a model for integrating diverse programs within a residential suburban context.

1 Open Field
2 Audley Recreation Centre: Aquatic Hall, Gymnasium, Community Hall, Active Living Studio, Dance Studio, Preschool Room, Youth Space, Community Spaces and Ajax Public Library ARC Branch
3 Existing Outdoor Program Spaces: Splashpad, Playground, Basketball Court, Tennis Court
4 Existing Facility Garbage room
5 Dog Park
6 Parking Lot
7 Loading Area
8 Service Driveway
9 Landscape Mounds
10 Walking Trail
11 Meadow Marsh Area
12 Sportsplex Ball Diamonds
13 Skateboard Park
14 Irrigation Pond
15 Stormwater Management Pond
16 Park Maintenance Building and Yard

EXISTING FAIRGROUND & BACKGROUND - AJAX DOWNS

Ajax Downs is currently being used to host Ajax's largest town event, Canada Day. Attended by 20,000 people, the event includes cultural performances, children's games and activities, interactive experiences, 2 stages for live entertainment, community displays and exhibitors, a sensory zone, food vendors, a beer garden and a special fireworks finale in the evening. The staff experience at Ajax Downs revealed valuable lessons for designing a new Fairgrounds at the Audley Recreation Centre site.

What works:

- Separate service entrance/exit:
  - Located behind site, away from attendee access
  - Vendors can load/unload and enter/exit site without getting stuck in traffic
- Multiple access points for security and emergency services
- Enclosed area makes access easy to control
- Clearly delineated Fairground area

What hurts:

- Congestion at the end of events
  - After a headline show, 18,000 peak occupancy exiting field through a single access to the parking lot
- Shuttlebuses
  - No clear area for waiting and lining up which leads to frustration for attendees
  - Mixing of shuttlebuses with cars adds to the traffic congestion
- Lack of Accessibility
  - Difficult for wheelchairs to access field because of grade difference between parking lot and Fairground
- Limited access to power
  - Access is limited to permanently stationed stage trailer with supplemental generators
  - Generators are loud and smelly

1 Fairground Lawn
2 Stage Area
3 Parking Lot / Shuttle Bus Access
4 Stair to Fairground
5 Operations and Emergency Vehicle Driveway

EXISTING SITE & BACKGROUND - FUTURE RAILWAY OVERPASS & ARC PHASE 3 DEVELOPMENT

The Town of Ajax plans to build a railway overpass at the south-west corner of the Fairground site in approximately 10 years. The extent of the proposed road embankment is highlighted with a white dashed line in the side figure. The proposed maintenance yard and building for the Fairground will sit outside the boundaries of the embankment and the railway corridor south of the site to avoid service interruption in the future. Once the overpass is completed, access to the maintenance yard from Audley Road could be replaced with a new access road terminating at the existing loading area of the recreation centre building. Alternatively, a new access drive could be located near the start of the embankment to reduce travel time and traffic congestion at the public parking lot entrance.

Locating the shuttle bus lay-by off Audley Road reduces the overlap of both vehicular and pedestrian traffic moving through the site. If the road is raised to pass over the railway, grade differences of up to 7m will need to be considered. Re-grading for barrier free pedestrian access between the Fairground and the drop-off / pick-up area will be challenging since the space available for re-grading is limited in this area.

A detailed study will be required to determine the best design strategies for accommodating the possible raising of Audley Road including location of the new maintenance building and infrastructure, planting and grading.

The Audley Recreation Centre Master Plan includes a third phase building expansion that includes a twin pad arena with an outdoor skating trail and an extension of the existing parking lot. Given the limited area available for development and the amount of space required for large public gatherings, the construction of ARC Phase 3 will greatly limit the size of events that can be held on this site. Furthermore, the footprint of the building will change the proportion of the Green to a more linear proportion, dividing the available open space into smaller areas. Designing the Fairground to accommodate a future arena building will require further detailed study to determine the most effective way to maintain flexibility for future options.

PART 2 - NEEDS ASSESSMENT

- ANTICIPATED EVENTS
- SITE OPPORTUNITIES AND FAIRGROUND NEEDS ASSESSMENT

ANTICIPATED EVENTS & PROGRAM

The possibility of moving many of the Town's major events to the new Fairgrounds was considered, though a definitive list was not finalized. It is anticipated that the new Fairgrounds will inspire new events or re-imagine existing events. As a starting point, current events that are run by the Town and could be moved to the Audley Site are:

- Pumpkinville
- Winterfest
- Ribfest
- Canada Day

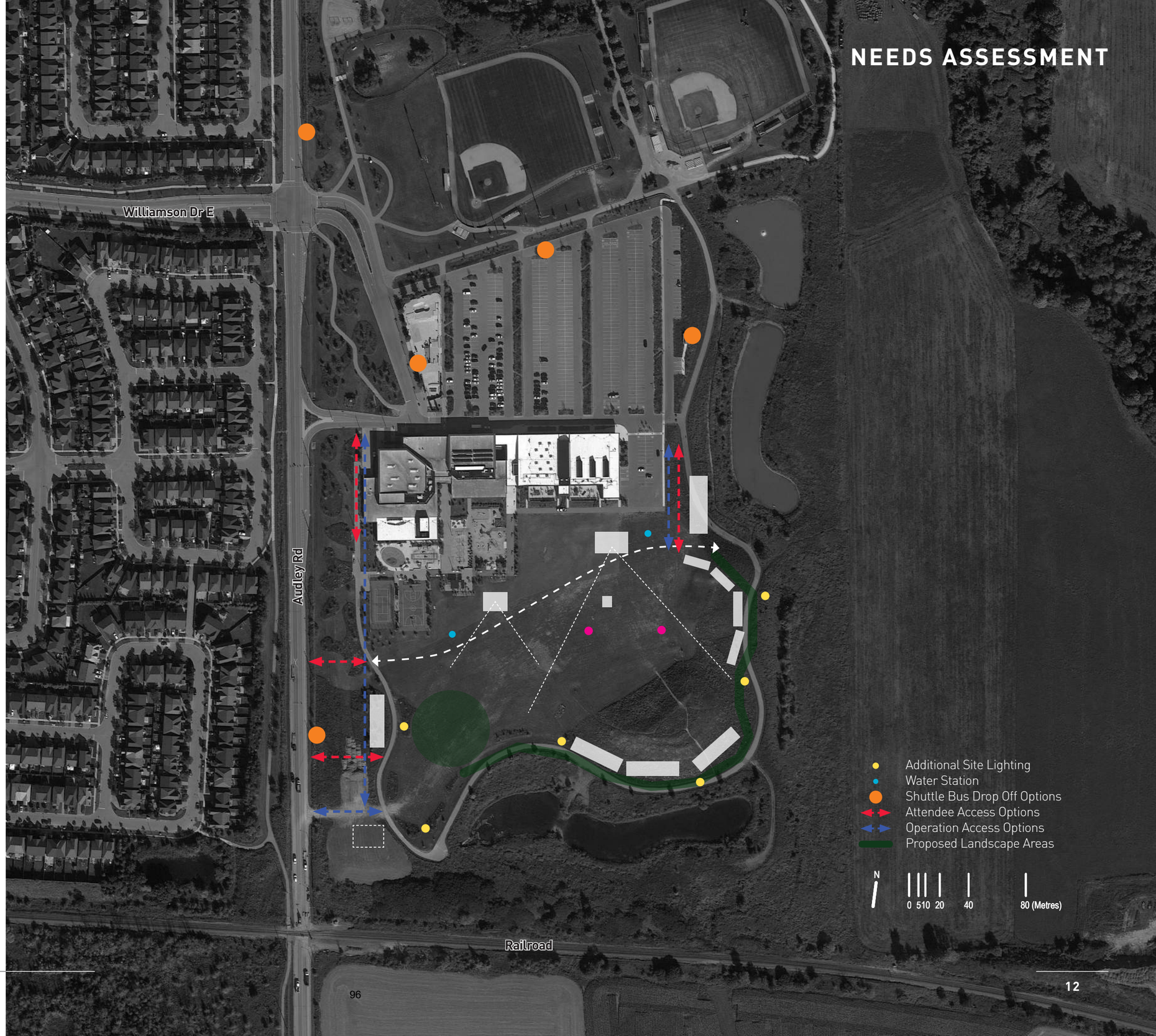
Since the Canada Day event attracts the largest crowd and would have the highest peak loads, it was used as a basis for developing the Fairground functional and concept design plans.

In addition to special events, temporary skating and amenities for day-to-day park use such as landscaping, benches, shading, picnic tables and open fields for recreation were also discussed for integration into the study.

SITE OPPORTUNITIES & FAIRGROUND NEEDS ASSESSMENT

The Consultant Team met with Town Staff and identified the following as important considerations for the future development of the ARC Fairground. The adjacent figure maps out a possible organization of the site

1. Existing conditions:
   - Existing landscape mounds are difficult to maintain and do not provide programmatic use.
   - The existing ARC loading area is configured to restrict vehicular access to authorized vehicles. This access could be improved to facilitate vehicular access for Operations while maintaining restricted access.
   - The naturalized experience of the existing trail should be maintained.
2. Fairgrounds:
   - The proposed Fairground green should have maximum openness for future growth.
   - A sensory zone away from the noise and crmain event space is necessary.
   - Fairgrounds should prioritize sustainable development and enhance or maintain existing protective measures on the site.
3. Visitor Access:
   - Shuttle bus stops should have adequate space for waiting and minimize overlap and disruption of traffic paths.
   - Staff should have the option to control access to Fairgrounds. Gates and fences could be used but a permanent naturalized landscape barrier would be preferred to reduce costs and operational challenges.
   - Fairgrounds should have more than one gate to avoid overcrowding and bottlenecks
4. Operations and Infrastructure
   - The existing walking trail could be upgraded to serve maintenance trucks and Fairground operations. Heavy duty pavement would be required for emergency vehicle access.
   - A new building for staging and storage will be needed. Parks Maintenance could also share the space. The building should fit into the architectural language of the ARC campus.
   - Staff parking and emergency vehicle access will be needed.
   - Power, water and lighting systems for the site are required. Temporty sound systems suit varied outdoor events better.



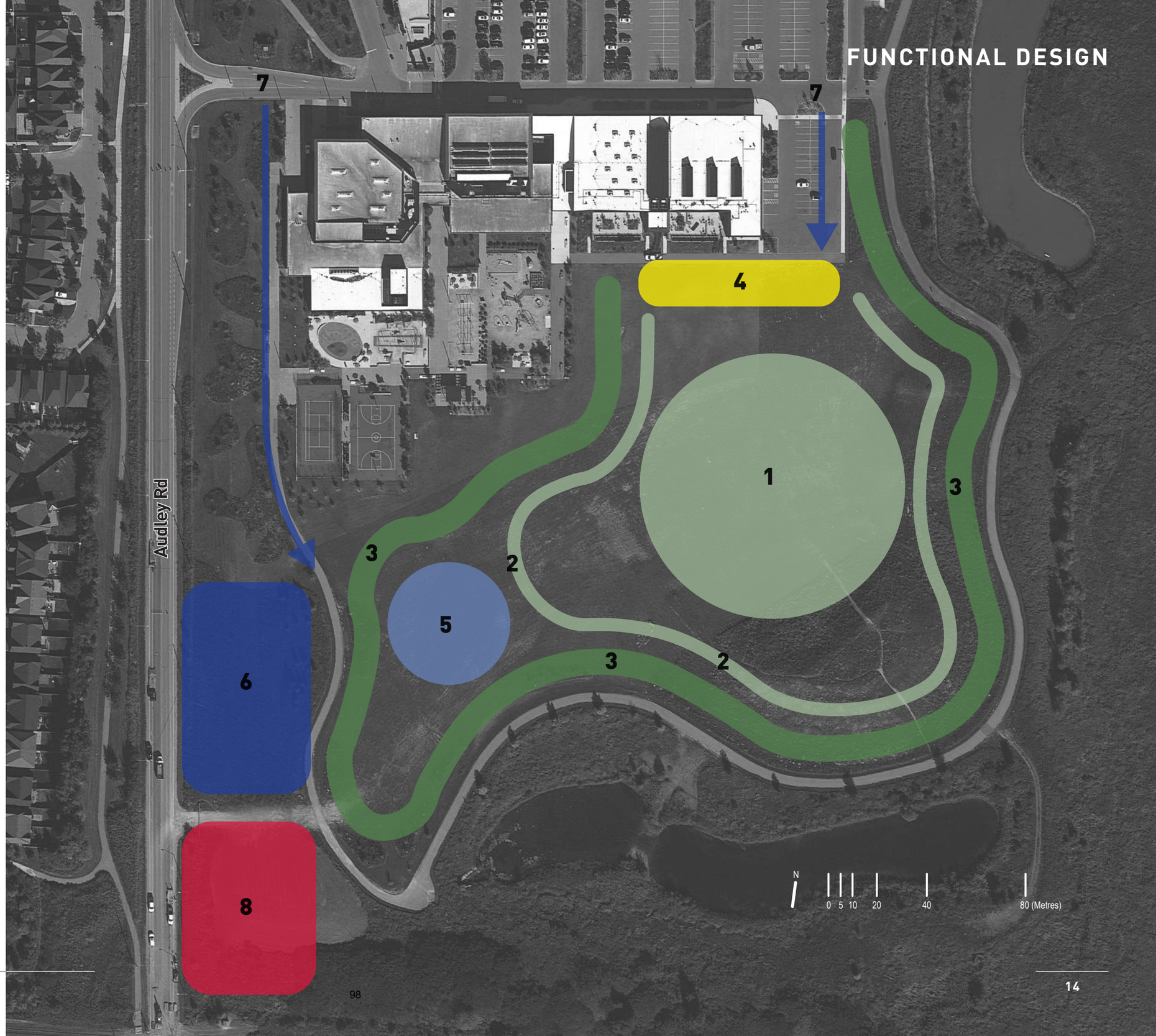
PART 3 - FUNCTIONAL DESIGN

- FUNCTIONAL DESIGN PLAN
- EVENT PLANNING
- PARK PLANNING

FUNCTIONAL DESIGN PLAN - CONCEPT

Based on the Site Review and Needs Assessment, a layout concept for a multi-use space that could function as a special event Fairground and a well loved neighbourhood park was developed.

1. The Green - A large open field at the centre of the space, serves as the major event space. The large field will also provide more recreation space for year-round use.
2. Infrastructure Loop - Events in the Green will be served by a loop of infrastructure at its perimeter, which includes access to power, lighting, water and a vehicular-grade track of reinforced sod. This loop provides flexibility for event layouts and accommodates events of different sizes and needs.
3. Shelter Belt - A planted berm rings the site, creating a sense of enclosure, minimizes fencing requirements, mitigates noise, provides casual seating, uses the site's excess fill and provides wind and sun mitigation.
4. Multi-Functional Plaza - A hardscaped area at the north of the site, provides a flexible space for events and installations requiring a paved surface, such as a stage, temporary skating rink or small farmers market.
5. Quiet Zone - A more intimate area separate from the Green, will function as the sensory-friendly zone during large events and throughout the year.
6. Shuttle Bus Drop-off - A shuttle bus lay-by with a reinforced sod terrace off Audley Road will allow for safe queuing for shuttle buses and will minimize traffic congestion in the parking lot.
7. Public Entry - These areas will serve as the controlled entry for people arriving by car or on foot.
8. Maintenance Yard - A maintenance building with parking will serve both park maintenance and event staff for storage and staging. This existing driveway will also serve emergency vehicles and VIP access.



FUNCTIONAL DESIGN PLAN - MASTER PLAN

The Master Plan envisions a beautiful public park with hidden infrastructure to allow transformation into a flexible space for thousands of people.

SUSTAINABILITY

In September of 2019, the Town of Ajax declared a Climate Emergency and followed up with further commitment in 2022 by adopting specific green house gas emission reduction targets and mandating sustainable development through its Ajax Green Standard.

The Audley Recreation Centre was certified as a LEED Gold building and its surrounding site developed with low impact to protect the meadow marsh area on the site. Development of the Audley Fairgrounds can continue with these high standards, demonstrating Town's commitment to lead by example.

Sustainable features such as low carbon building standards, renewable energy, permeable paving, resilient planting and water conservation can all be considered as part of the next stage of design development.

1 Public Access
2 Operations Access - Parks & Events
3 Emergency & VIP Access
4 Maintenance Building & Yard / Staff & VIP Parking
5 Proposed Landscape Berms
6 Sensory Zone
7 Water Filling Station
8 Reinforced Sod
9 The Green / Event Lawn / Cricket Field
10 Temporary Bike Parking Location for Events
11 Paved Plaza Area

EVENT PLANNING

EVENT PLANNING
- TRAFFIC MANAGEMENT

The Audley Site is served by a parking lot that has 565 spaces. This is substantially less than the parking that is available at Ajax Downs which has 3,050 spaces. Support to provide alternative modes of travel for the event will be required.

The concept design includes the following traffic management features:

- In addition to the existing pedestrian crossing at Williamson Drive East, a new mid-block crosswalk is recommended. This will allow safe crossing for those who are able to arrive by foot from the neighbouring residential area and connect the Fairground site to the sidewalk that extends along western side of Audley road.
- Ample temporary bicycle storage areas will be provided to take advantage of the bicycle network that runs on Audley and encourage bicycling to the site from further afield.
- Shuttle bus lay-bys on Audley Road (Option 1 on adjacent figure) will separate on-site traffic and help minimize shuttle bus route disruption.
- Alternative shuttle bus stop areas are indicated (Option 2, 3 and 4 on adjacent figure) in the event additional shuttlebus stops are required or the Audley Road overpass gets built.
- Event Vehicles, Operations, Staff, Emergency Vehicles and VIP will have a separate drive access, located at the existing driveway off Audley Road, which will connect to a new vehicular loop around the new Fairground. One-way traffic on this loop will facilitate better traffic control for Event Staff.
- Private vehicles driving to the event could be limited to the Williamson Drive East access. This allows for the possibility of closing Audley Road south of the intersection for traffic management.

- Shuttle Bus Circulation
- Private Vehicle Circulation
- Food Truck / Vendor Circulation
- VIP / Service Circulation
- Pedestrian Crossing
- Bicycle Lane

EVENT PLANNING
- CIRCULATION & ACCESS

A key feature of the concept design is to maintain pedestrian safety while providing functional access for event vehicles. This design separates the circulation of pedestrians and vehicles, limiting their overlap and distributing control point gates around the Fairground, to reduce possibilities of over-crowding and bottlenecks.

Shuttle Bus
Pedestrian Circulation
Vehicular Circulation
Cyclist Circulation

1 Public Access
2 Operations Access - Parks & Events
3 Emergency and VIP Access
4 Maintenance Building & Yard / Staff & VIP Parking
5 Proposed Landscaped Berm
6 Sensory Zone
7 Water Filling Station
8 Reinforced Sod
9 The Green / Event Lawn
10 Temporary Bike Parking for Events
11 Mid-Block Pedestrian Crossing

EVENT PLANNING
- CANADA DAY LAYOUT

The Town of Ajax produces several events throughout the year that range in character and size. The work in this study has been predominantly guided by the largest event, Canada Day, which was chosen as a maximum limit to what the available area of the open space could manage. During a Consultant-Staff workshop, a proposed layout was determined with to consider how the site might be organized.

The following requirements were considered and incorporated into the sample layout:

- A maximum crowd of 20,000 persons with a peak visitor crowd between 15,000 to 18,000 persons
- The use of a Stageline SL100 stage, already owned by the Town
- A second 10' x 10' stage
- Space for 8-10 food trucks
- Space for a variety of vendors
- Space for a quiet or "sensory" area to serve attendees with specific sensitivities to noise and other stimuli
- Space for inflatables (i.e. a bouncy castle)
- A plan for portable restrooms
- Easy access for emergency services

The results of this workshop confirmed that the open field, once re-graded to eliminate the landscape mounds, had adequate space to produce an event of this size. The proposed design concept can accommodate a peak load of 17,800 persons at 1 person / square metre and that other factors such as traffic congestion and attendee transportation will likely be the limiting factor for this site.

1 Info / Main Entrance
2 First Aid
3 Portable Restroom Facilities
4 Main Stage
5 Secondary Stage
6 Inflatables & Attractions
7 Food & Beverage Area
8 Vendor / Sponsor Area
9 Sensory Zone
10 Production Compound
11 Emergency Services Holding Area & Staff Parking

PARK PLANNING

PARK PLANNING - DAILY USE

When not being used for events, the site functions as a beautiful public park. The well-used walking trail which exists today, is supplemented with additional pathways, seating and lighting. The large open lawn can be used for recreational sports and games such as cricket. The Green is large enough to fit a 100m x 118m field. Smaller scale and more intimate park spaces with landscping are located as extension of the perimeter landscape berm and will be activated by picnic tables to create spots for families to gather or individuals to enjoy the shade of the trees.

1 Operations Access
2 Operations Drvieway off Audley Road
3 Maintenance Building & Yard / Staff Parking
4 Proposed Landscape Berm
5 Passive Recreation Zone
6 Water Filling Station
7 The Green / Cricket Field

PARK PLANNING
- CIRCULATION & ACCESS

The new park space will be directly adjacent to the ARC and its existing playground spaces, accessible via new paths and connections to existing ones. The existing walking looping is supplemented with additional pathways, seating and lighting, connecting the outerloop with the open lawn area. Vehicular access will be limited to site operations staff, and will happen along the west of the site as needed. The shuttle bus drop-off, which is made of reinforced meadow grasses, is allowed to grow over, blending into Audley Road's naturalized planting areas and discouraging cars from using this drop-off outside of major events.

Pedestrian Circulation
Vehiclar Circulation

PARK PLANNING - LANDSCAPE

The landscape design focuses on durable, low-maintenance paving and planting materials, that can hold up to the intense activity of a large event and blend into the naturalized aesthetic of the Audley Recreation Centre site.

1 Sod / Meadow Grasses
2 Reinforced sod
3 Naturalized planting
4 Asphalt multi-use trail
5 Engineered mulch
6 Plaza paving
7 Vehicular-grade asphalt

PARK PLANNING - UTILITIES, GRADING & SITE SERVICING

The existing landscape mounds will be re-graded to level the site and to build a new landscape berm which will define the new Farigrounds' perimeter. The new grading will be shaped to minimize the costs associated with importing or exporting fill.

New storm water servicing infrastructure will direct site drainage towards the existing meadow marsh area and storm water management pond, maintaining existing low impact site management strategies.

The domestic water supply will come from an existing watermain at the west side of the site, installed during the ARC Phase 2 construction project. A water meter will be located inside the new Maintenance Building where the water supply for the newly developed site will originate.

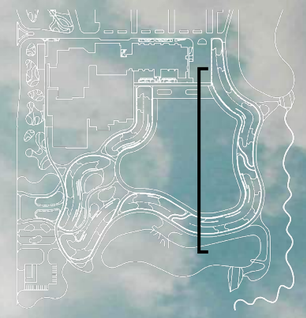
A new sanitary line will be required to serve the Maintenance Building, and will run along Audley Road, connecting to the main near the existing loading area.

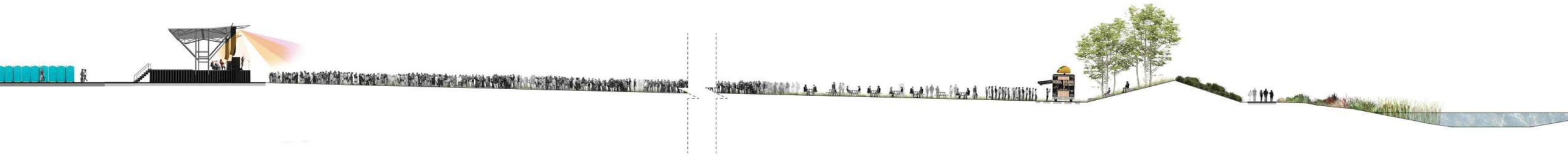
Power for the temporary stage plaza and theinfrastructure loop will come from the existing Garbage Room. Power for the new Maintenance Building and site lighting will be fed from the existing electrical room in the ARC basement.

- New Sanitary Forcemain
- Existing Watermain
- New Watermain
- New Electrical Power Network
- Stormwater Servicing

1 Existing Landscape Mounds - Re-graded
2 Maintenance Access
3 Maintenance Building & Yard / Staff Parking
4 Landscape Berm
5 Passive Recreation Zone
6 Water Filling Station
7 The Green
8 Event Power
9 Irrigation Pond
10 Stormwater Management Pond
11 Meadow Marsh Area

PRELIMINARY SITE SECTION - A





PRELIMINARY SITE SECTION - B

PART 4 - PHASE ONE

- BASE FUNCTIONAL SITE PLAN
- ORDER OF MAGNITUDE PROJECT CONSTRUCTION COSTS

BASE FUNCTIONAL SITE PLAN

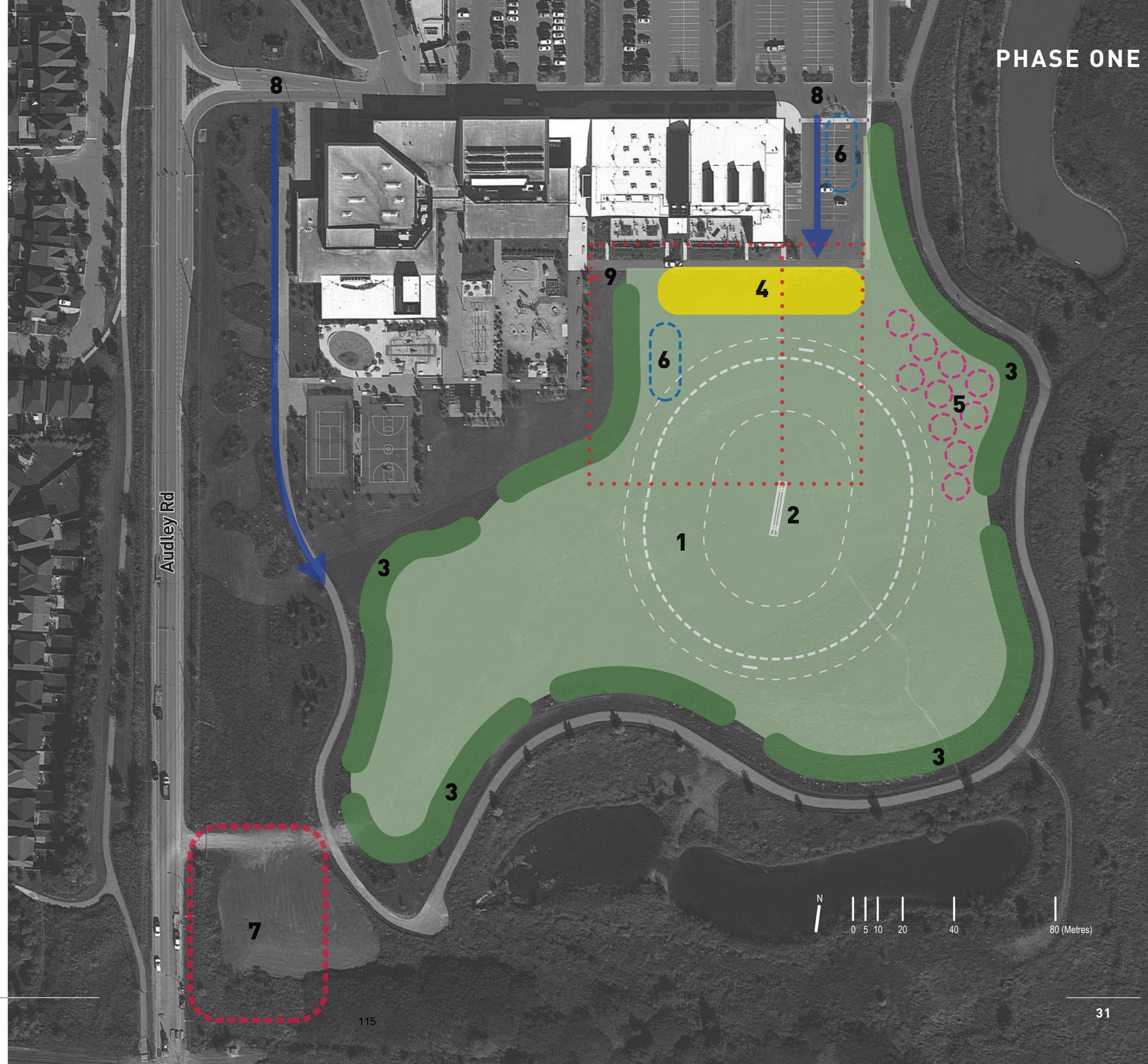
To provide flexibility for funding allocation, a base functional site plan has been devised. This reduced plan establishes the fundamental spatial requirements for hosting a large event of 18,000 people, at peak time, and will leave the possibility for future expansion and upgrades to the site if the need arises or more funding becomes available.

Phase 1 for the new Fairground will include re-grading the existing earth mounds to level the ARC field. Landscape berms at the perimeter will naturally define the event space and the Green. Costs for exporting or importing fill will be minimized with a balanced grading design. Lawn grass cover and new trees on the berms will provide shade and characterize a landscape that will mark the seasons. A cricket field will bring new program to the park, while leaving the field open to accommodate a broad range of recreational activities. Designing multiple purposes for the site will ensure that the Town's investment will have maximum return.

Power for the Fairground will come from a new electrical panel in the existing Garbage Room and will allow plug in from the adjacent outdoor plaza space designed for the temporary concert stage and programs year-round like public gatherings during warmer months and a temporary skate trail during colder months. The extent of new site lighting can be determined once a budget has been set.

Capital cost savings will come from limiting the extent of features shown in the full concept design. The construction market has experienced very high inflation rates over the past few years and costs for materials and labour remain volatile and difficult to predict. It is recommended that the Fairground Schematic Design include options that consider ARC Phase 3, the raising of Audley Road and the extent of scope to facilitate decisions that suit the budget and overall priorities of the Town.

1 The Green
2 Cricket Field
3 Proposed Berm (Relocation of Existing Berm)
4 Multi-Functional Plaza
5 Food and Beverage Area
6 Portable Restroom Facilities
7 Staff / Emergency Vehicle / VIP Area
8 Public Entry
9 Proposed ARC Phase 3 Footprint



Order of Magnitude Cost Estimate for Project Construction*

PHASE ONE

Estimated Hard Cost	$3,534,000
Estimated Soft Cost	$434,000
Total Estimated Project Cost H.S.T. Excluded	**$3,968,000**

* Notes on Order of Magnitude Cost Estimate

The project costs indicated in this report are intended as an order of magnitude based on 2023 rates and subject to a confirmation of the program and scope of the Audley Recreation Centre Fairground Project.

These numbers do not include unknown conditions, development and permit costs, soft costs associated with Town Staff and Event Organziation, design, escalation and construction contingencies.

PART 5 - NEXT STEPS

NEXT STEPS

To move forward with this Concept Design, we would recommend the following next steps during Schematic Design:

- Procure technical analysis of the existing site including a site survey and a geotechnical investigation of the subgrade conditions.
- Confirm the level of design accommodation that will be required for future projects on the site: Audley Road Overpass and ARC Phase 3.
- Obtain feedback from local Authorities Having Jurisdiction including Town Planning, the Central Lake Ontario Conservation Authority, and Emergency Services prior to advancing further to determine any concerns they may have.
- Commission a full traffic management study to confirm recommendations for special event traffic control, transportation and parking requirements.
- Along with the development of the Schematic Design, develop and confirm special event requirements.
- Obtain a Class C cost estimate at the end of Schematic Design to confirm the construction budget.
- Determine metrics that site development will require to meet Town's sustainability goals.
- Engage with the Community for feedback about the concept design.

The work of this study affirms that the Audley Recreation Centre Site is a feasible option for development into a Fairground for the Town. Although challenges such as traffic management for the largest events, the planned raising of Audley Road for the railway overpass and the potential expansion of the Audley Recreation Centre's Phase 3 development will need more study, the concept design demonstrates that the pre-existing conditions contribute to the suitablility of the stie for hosting large scale events. In addition, upgrading the existing open field will transform the underused meadow into a beautiful park with flexible outdoor amenities such as a cricket field, enhancing the community's daily experience. As an investment that contributes to both daily enjoyment and special events for the community, the Audley Recreation Centre Fairground has great potential for becoming a remarkable setting to celebrate together.

APPENDIX A - EVENT PLANNING CONSIDERATIONS

Design Partners

1. INTRODUCTION

The Audley Recreation Centre – Fairgrounds Project is a study commissioned by The Town of Ajax to assess the suitability of using the existing open park space located adjacent to the Audley Recreation Centre as a new urban Fairground. To date, the Town of Ajax has used the Ajax Downs Racetrack, a privately held horse racetrack, for their public events. While that venue has some advantages, the Town of Ajax would prefer to make use of their own space.

Initially opened in 2013, with a second phase opening in 2019, the Audley Recreation Centre is a multi-use sports and recreation facility boasting an aquatic centre, gymnasiums, sports courts, a public library, and a host of other public services. The exterior of the building, specifically to the south, is large open green space. Some of this space had been previously landscaped for use as a park and currently includes a walking path and toboggan hill. To the north of the complex is a large baseball facility with 6 fields.

Design Partners was approached by MJMA and asked to bring its experience in the realm of live event design and management to the design team. In the course of the work, Design Partners, led by Joshua Hind, visited the site on two (2) occasions, once independently and again with the teams from MJMA and the Town of Ajax. Design Partners also discussed the project with colleagues in public events with the City of Toronto and the Ville de Montréal.

2. REQUIREMENTS & CONSTRAINTS

The Town of Ajax internally produces several events throughout the year, though the work in this design phase was predominantly guided by far their largest event, Canada Day. The staff from the Town of Ajax wished to determine if they could produce a Canada Day event at the Audley Recreation Centre which met the following requirements:

- A maximum crowd of 20,000 persons (see Section 4 – Capacity)
- The use of a Stageline SL100 stage (already owned by the Town of Ajax)
- A second 10' x 10' stage
- Space for 8-10 food trucks
- Space for a variety of vendors
- Space for a quiet or "sensory" area to serve attendees with specific sensitivities to noise and other stimuli
- Space for inflatables (i.e., a bouncy castle)
- A plan for portable restrooms
- Easy access for emergency services
- An access plan for event-related vehicles which would not damage the park space

Along with those conditions, the site must also be able to accommodate other types of events, each with their own specific needs. As such, the site will need to be properly services with power, for the main SL100 stage, as well as stages of other configurations, as well as for food trucks, vendors, and other site elements. While Design Partners is providing general comment on the electrical services typically required for an event site, the general specification for power and comment on the electrical appropriateness of the site is by others.

Transportation is another major consideration. The Town of Ajax employs shuttle buses to bring people to and from their current event site at Ajax Downs and anticipates having to do so if events move to the Audley Recreation Centre site. Comment on the appropriateness of the site from the standpoint of transportation is by others.

The overall site design is by MJMA. This report only addresses aspects of the design which are specific to the production of live events on the site.

3. SITE CAPACITY

3.1. REQUIREMENT

The figure of 20,000 persons noted in Section 3 was revised throughout the design process. Initially it was understood to represent a peak crowd load (20,000 persons on the site at one time). This was later revised to 20,000 persons throughout the day, with a peak crowd load of 15,000 – 18,000 persons. The peak load is assumed to occur during the presentation of fireworks, which occurs on the evening of Canada Day.

3.2. DETERMINING APPROPRIATE CROWD SIZE

When determining the appropriateness of a site from a crowd management perspective, we must first make two (2) determinations:

- What is the desired or expected crowd size?
- What is the desired crowd density?

The second consideration, crowd density, is as much an aspect of event design as it is one of public safety. One would not plan a community event with the same crowd density standard used for a concert, especially one planned for an "18+" crowd. Community/Family events require more space, so families with children don't feel crowded, and don't need to worry about losing sight of their children in a busy crowd. Families and those requiring accessibility are also those who may have large, wheeled devices, like strollers, wheelchairs, or other mobility devices. These situations demand more space for movement. By contrast, the "feel of the crowd" is something concert planners and concertgoers often desire, and they must carefully balance the desired experience with safe crowding standards. The Town of Ajax typically plans community events which are attended by families, including families with small children, and which maintain a broad and inclusive accessibility standard.

In the first sample image (**Figure 1**), we see a crowd density of 1 person per square meter. This is the optimal crowd density for a community/family event. As you can see, there is room for movement, even the movement of strollers or mobility devices, between groups of people.

The next image (**Figure 2**) features a crowd of 3 persons per meter and 6 persons per meter, respectively. 3 persons per meter is a safe level of crowding (in ideal conditions with all other safety measures in place), though it becomes apparent why this level of crowding isn't desirable for a community/family event.

The final image (**Figure 3**) shows 6 persons per square meter and is considered extremely dangerous. This condition should not be permitted under any circumstance. The potential for serious problems is readily apparent.

3.3. AUDLEY FAIRGROUNDS

3.3.1. CROWD SIZE

Having established the crowd size (15,000 – 18,000 maximum crowd load) and the desired density (1 person per square meter) and having created a preliminary event layout for the site, it was determined that the proposed Fairground design had an available patron space of 17,800 sq. meters. That figure does not include the landscaped sensory area or the berm which has been proposed to surround the space at delineate the event space from the existing park space.

As such, the preliminary appraisal finds the site is appropriate for the desired crowd size. However, there is little room for future expansion. Should the crowd size for the largest events approach 20,000, other measures for crowd management may be required if the site is to be used.

3.3.2. CROWD FLOW

A constraint of the Audley site is the number of access points. For those driving to the event and using the parking lot, all access will likely be by the east side of the building. For those using shuttle bus services, access (in the proposed design) would be from the southwest corner of the site. Both these access points are wide and should be able to manage the patron flow (considering that it is described to be spread over an entire event day). That said, care should be taken in event site planning to ensure that site elements are placed in such a way as to constrict the flow of patrons through these two (2) access pathways. (**Figure 4**)

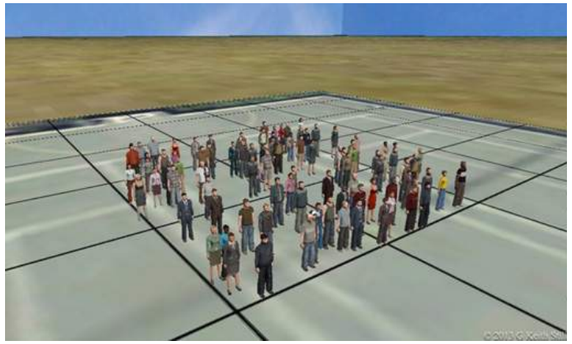


Figure 1: 1 person per square meter (Image by Prof. Dr. G. Keith Still)

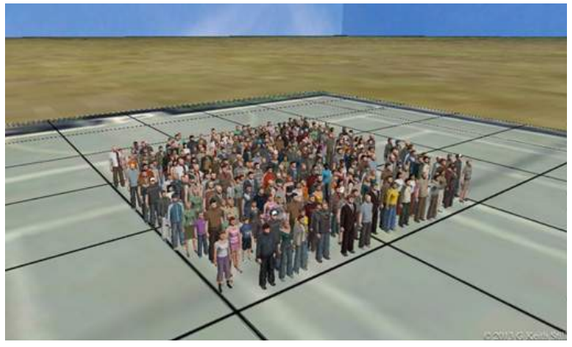


Figure 2: 3 persons per square meter (Image by Prof. Dr. G. Keith Still)

Figure 3: 6 persons per square meter - HIGH RISK (Image by Prof. Dr. G. Keith Still)

4. STAGING

The Town of Ajax owns a Stageline SL100 portable stage. These stages, which fold up in a footprint roughly equivalent to that of a transport truck trailer, can be quickly positioned and set for a performance and include all the overhead rigging supports required for a show. They come in a variety of sizes, with the SL100 being the smallest available size.

Through their design work, MJMA has provided a paved pad on which the SL100 can be set. This will prevent staff from having to drive the stage across the grassy part of the event space. Other, secondary stages can be set up throughout the grassy event space, though is recommended that such stages be located within close proximity to the existing paved walkway which surrounds the space, or to another reinforced surface to prevent damage to the grass caused by vehicle tires.

When setting temporary stages using portable, temporary stage deck systems on any permeable surface, such as grass, it is strongly recommended to first lay a double layer of ¾" plywood under the stage to prevent the stage legs from digging into the dirt. If plywood cannot be placed under the entire footprint of the stage, it should at least be placed under the stage legs, which such footings being at least 12" x 12" in size.

Other options for protecting turf such as "event decking" (a modular system of interlocking plastic tiles which distribute the load of vehicles and other large items and protect the turf beneath) may have to be considered by event planners to protect the integrity of the park space when it is not being used for events.

4.1. AUDIO REINFORCEMENT WITH REGARD TO THE SURROUNDING RESIDENTIAL AREA

Temporary public address speaker systems should be specified in collaboration with a professional rental house, though in general terms the site is well suited to the use of "audio reinforcement" (the common industry term for amplification using a public address system), so long as the audio system and the stage is properly oriented. (**Figure 5**)

In general, audio from the main SL100 stage, or any larger stage, should be directed to the south, toward the railroad tracks. Audio reinforcement systems should never be directed at the residential area on the other side of Audley Road. There are not sufficient topographic features to absorb or block the sound.

Sound systems should avoid the use of wide or ultra-wide dispersion speakers, which a dispersion above 90 degrees. Preliminary sound studies suggest an optimal dispersion pattern is approximate 70 degrees from the stage.

5. ELECTRICAL

5.1. REQUIREMENT

Additional electrical resources are required if the Audley site is to be an ideal, and easily usable event site. In consultation with the end users and other stakeholders, the following general requirements have been identified. This is not meant to be an exhaustive list.

- Mains power for stages and related audio, video, lighting, and staging equipment.
- Food truck power.
- Power "hydrants" distributed throughout the site for use by vendors, site activations, etc.

5.2. STAGE POWER

Stage power should be located near the proposed concrete stage pad. In discussions with staff from the Town of Ajax and with the electrical engineer, two (2) possible solutions were proposed, both of which should be more thoroughly studied in later design phases.

- A series of company switches located in the garage space in the southwest corner of the building, OR...
- An electrical enclosure outside the building, more closely adjacent to the area designated for the portable stage.

In both cases, the minimum recommended requirement based on our current understanding of the maximum event size is three (3) 200A company

Figure 4: Patron Access

Figure 5: Simplified sketch showing audio dispersion

switches. If it's decided to place the stage power in an exterior enclosure, it is also recommended to add a set of 20A service power outlets (NEMA 5-20) to service other common power needs.

5.3. FOOD TRUCK POWER

Food trucks are often responsible for addressing their own electrical needs. As a result, it has become typical practice for food trucks to carry small, gasoline-powered generators. When setting up for a short service on the street, the noise and smell of a small generator can be excused, but over the course of an entire event day, on a site with multiple food trucks present, the impact can be detrimental to the enjoyment of event patrons. Wherever possible, best practices for event site design suggest providing power to food trucks so they don't have to use their generators.

The challenge when committing to providing power for food trucks is there isn't a standard related to electrical systems in food trucks. Many food trucks use a NEMA 14-50 connector (50A, 125/250V), commonly referred to in Canada as a "stove plug". Others use a TT-30 (30A, 125/250V) connector, which is more common to recreational vehicles. And others still simply use household extension cables. While the use of the "stove plug" is the most common, more interaction with the Town of Ajax's preferred vendors is required before deciding on a standard for food truck power delivery at the Audley site.

5.4. POWER "HYDRANTS"

In tandem with considerations of power delivery for food trucks, power delivery is also desired around the periphery of the site for general event needs, such as vendors, site attractions, secondary stages, entertainment, or any other need that would benefit from access to a common household electrical circuit.

The benefit of distributed power is significant, and its often more greatly appreciated by smaller events, which often don't have the logistics and personnel support found on larger events. Readily available power allows for more spontaneous "surprise and delight" attractions, public art, holiday décor, and other uses which otherwise can't happen when power delivery requires the rental of temporary equipment.

The concept of delivering those power through "hydrants", which can be located within light standards, bollards, or any other weathertight enclosure around the site is worth further study involving an electrical engineer.

6. FIREWORKS

6.1. NOTE REGARDING THE DESIGN AND USE OF FIREWORKS DISPLAYS

While Design Partners has considerable experience managing event sites at which fireworks are displayed, Design Partners staff are not licensed pyrotechnicians (as recognized by Natural Resources Canada, Explosive Regulatory Division). As such, commentary on the appropriateness of the site for fireworks display is only general in nature and must be validated with a licensed pyrotechnician. All fireworks display must involve a pyrotechnician who has the required license(s) to purchase, store, transport, and fire pyrotechnic effects.

6.2. REQUIREMENT

The Town of Ajax includes fireworks display in their annual Canada Day event. The size and scale of the display was not exactly defined, though it was assumed, through discussion with staff, that it involves small to medium sized display pyrotechnic effects. (Display effects refers generally to effects which are fired up into the area, as opposed to theatrical effects like bursts of sparks, handheld flame effects, or fire effects.)

6.3. SITE CONSIDERATIONS

The Display Fireworks Manual, as issued by Natural Resources Canada's Explosive Regulatory Division, details the considerations which must be considered when assessing a site for the use of pyrotechnic effects. To summarize for the layperson, the larger the effects used, the greater the distance that must be maintained between the public and the firing location.

6.4. AUDLEY FAIRGROUNDS

Based on very preliminary study, the immediate Audley site (the area to the south of the building which is the primary area of consideration in this study) is not well suited to fireworks display. It does not appear to be possible to reach the minimum distances required for display fireworks without encroaching in the wetland space between the Fairground and the railroad line. While it may be possible to establish a firing position in the wetland space, that introduces another environmental consideration (the flammability of wetland vegetation) which may prove too difficult or costly to mitigate.

However, the baseball facility to the north of the building is large, open, and already well protected with fenced-in areas. Depending on event staff's ability to coordinate with the users of the baseball facility, it might prove an ideal firing position, both for its own qualities and its separation from the Fairground area. (**Figure 6**)

Further study in subsequent phases is required. Such study should include input from a professional licensed pyrotechnician.

7. EMERGENCY SERVICES COORDINATION

Any event plan, regardless of whether it includes a pyrotechnic display, requires coordination with emergency services (fire, paramedics, and police). The advantage to creating a fixed event site, such as a Fairground, is the basics of good incident planning, such as access to fire hydrants, or entry paths for emergency vehicles, can be included in the design of the space. As laws, regulations and bylaws vary from one jurisdiction to another, it's important to consult directly with the local authorities having jurisdiction.

7.1. VEHICLE ACCESS

When planning emergency vehicle access, event planners start with two (2) questions:

- How can I ensure shortest distance between patrons and access to emergency services?
- How can I avoid interaction between people and moving vehicles?

It's typically not safe or reasonable to drive a large fire truck or ambulance into crowded area, even in an emergency. And many sites, such as the Audley site, have topographic features which preclude easy access the centre of the site. A common strategy in these situations, and one that could be employed at the Audley site is to designate holding points around the periphery of the site. Emergency vehicles are staged at these holding points, which must be sited at points with clear vehicle access. Event planners will ensure the site plan allows emergency services personnel to quickly access every area of the site on foot. In the event of an incident, the most common of which is a patron experiencing some form of medical distress, emergency personnel are brought to the incident site on foot, and the patron is then brought to the ambulance (in this case) for transport to hospital.

This strategy of staging points can also be employed for police, though they often develop their own deployments. A good relationship with police is very useful for coordinating the intended site plan with the operational needs of law enforcement.

Fire services, which have the largest, heaviest vehicles, are the most difficult to accommodate on an ad hoc basis on sites where direct vehicle access is not possible. Direct consultation with both fire services and the Office of the Fire Marshal is recommended. The site may require additional fire services infrastructure, if the infrastructure inherent to the Audley Recreation Centre is not sufficient or appropriately located relative to the Fairground site.

7.2. ROAD CLOSURES

If the event plan requires the closure of road which either pass through or around the event site, planners must ensure that a live lane of is always kept clear of obstruction, thereby allowing emergency vehicles to pass, and the closure points must always be staffed with personnel (typically event staff or contract security) who can open the road closure allowing emergency vehicles to pass. (**Figure 7**)

7.3. RULES AND REGULATIONS

Unfortunately for event planners, there are few rules or regulations at any order of government which

Figure 6: Fireworks detail

deal specifically with the production of events. Event planners must develop a good understanding of:

- Local fire codes
- Local building codes
- Ontario Building Code
- Ontario Electrical Safety Code
- Occupational Health and Safety Act and its associated regulations
- Local bylaws

That list is by no means exhaustive. Responsible event planners must carry a wide range of regulator knowledge. The best method for ensuring an event is compliant with all laws, regulations and bylaws is to develop good relationships with the appropriate authorities, such as local emergency services, municipal building inspectors, the Electrical Safety Authority, the Ministry of Labour, and other local representatives of the authorities having jurisdiction.

8. CONCLUSIONS

The Audley Recreation Centre site, with modifications to its landscape design (as proposed by MJMA) and the addition of event-specific resources, as detailed in this document and in supporting documentation by the appropriate engineers and consultants, would serve as a unique, beautiful, and capable event site. The surrounding natural space will provide inspiration for event planners to be more ecologically aware in their approach to design and execution.

The challenges inherent to the site are common to all unpaved event locations. Vehicle access will have to be closely managed to prevent damage to the surface. The use of fireworks will also require additional attention to ensure not only patron safety, but also to protect the surrounding natural space.

Providing site infrastructure, such as the electrical "hydrants" described above, will allow the site to transform more spontaneously from park to event, allowing the space to not only be a space in which events are hosted, but also a space that inspires community gathering.

9. FUTURE DEVELOPMENT

Items to be considering in future development phases of the Audley Fairground site:

- Clarifying site infrastructure requirements and constraints
- Further development of a fireworks plans
- Exploration of other event and non-event uses of the site (i.e., outdoor skating rink) with an eye to their impacts on the infrastructure requirement
- Further study on the best practices for delivery of electrical services to food trucks
- Additional consultation with vendors who've served past Ajax events
- Consultation with emergency services

Figure 7: Emergency vehicle staging points and road access

APPENDIX B - TRAFFIC CONTROL & VEHICULAR SITE CIRCULATION

HDR Consultants

PRELIMINARY TRANSPORTATION CONSIDERATIONS

BACKGROUND

HDR has been retained to assist in the feasibility study of upgrading the Town of Ajax ARC Fairgrounds site to support accommodating major outdoor civic events.

The ARC Fairgrounds site is located south of the Audley Recreation Centre at 1955 Audley Road North. The study will consider the opportunities and requirements for the Town to conduct events and festivals on lands south of the Audley Recreation Centre.

As part of the feasibility assessment, the transportation considerations include:

1. Traffic management planning and special events parking,
2. Active transportation connections, bike corrals, public transit, shuttles, and
3. Vehicular ingress and egress to the event space, and pedestrian flow.

Based on the agreed scope in regard to the above topics, and available information obtained from the City, this memorandum specifically documents our assessment of:

- Internal site circulation to accommodate event vehicles and Town maintenance truck with trailer through the Fairgrounds
- Potential locations for transit shuttle bus lay-by area
- Commentary on how the future grade separation of Audley over the railway tracks south of the Fairgrounds site may impact access and the shuttle bus lay-by area.
- A high-level assessment of parking demands and required shuttle buses to serve a peak event at the Fairgrounds site

This memorandum also provides preliminary recommendations for managing traffic and parking during special events.

FAIRGROUNDS VEHICLE SWEPT PATH ANALYSIS

The Fairgrounds will be a pedestrian-oriented environment. However, maintenance will require vehicles to enter the Fairgrounds. The Town currently uses a pickup truck with a trailer and this was used as the assumed design vehicle for the Fairgrounds swept path analysis. Any event / vendor vehicle will the shorter in length compared to this design vehicle. The dimensions and the outside turning diameter were confirmed with the Town staff and was confirmed to have a 9.5 metres outside turning radius. The selected design vehicle is presented below in **Figure 1**.

The circulation for maintenance vehicles within the Fairgrounds will be in the form of a meandering trail style path that will be wide enough for a Town maintenance truck and will accommodate the truck and trailer combination through all the turns. The Fairground circulation plan is shown in **Figure 2**.

Variations of this circulation plan can accommodate the maintenance vehicle as long as the outside radius of all turns are greater than 9.5 metres. Currently, the turn radii are designed to 19 metres which is more than enough to accommodate the vehicle and should be maintained as it allows for smooth maneuvering through the Fairgrounds. The swept path analysis for the maintenance vehicle in the Fairgrounds is presented in **Figure 3**.

SHUTTLE BUS LAY-BY LOCATION ASSESSMENT

To manage traffic and to reduce congestion surrounding the Fairgrounds, the site will rely on shuttles buses (comprising school buses) to move people in and out of the Fairgrounds. This will reduce overall traffic volumes and will allow for efficient movement of people. This is necessitated by the limited parking supply. Four options were considered including three on-site lay-by locations as well as one off-site lay-by on Audley Road. The options are presented schematically in **Figure 4**.

SHUTTLE LAY-BY OPTION 1 (AUDLEY STREET)

Option 1 involves introducing a shuttle lay-by along Audley Road near the south end of the Fairgrounds lands. This option will allow the shuttles to provide access to the site without entering the Fairgrounds.

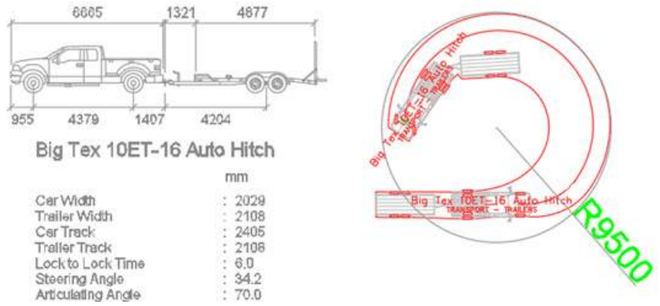


Figure 1: Maintenance Design Vehicle (Pickup Truck with Hitch)

Figure 2: Fairground Circulation Options

For this option, road upgrades must be considered since the cross section of Audley Road is currently rural and does not have a curb and gutter. Pedestrian connections will also need to be constructed to connect the shuttle stop location to the Fairgrounds. A staging area has been assumed in the above concept plan to hold pedestrians waiting to board.

The Audley Road shuttle lay-by location is the preferred option since it provides the best access to the Fairgrounds without conflicting with the recreation centre, reduces mixing of shuttles with vehicles and parking on site, and has a higher potential capacity for shuttles since the lay-by can be extended along Audley Road. At this time, a lay-by capable of handling two buses have been shown on the concept plan. Additional discussions on the number of buses required to serve a peak event is provided later in this report.

The bus lay-by is proposed north of an existing access to the site, which is recommended to serve event operations staff.

Sightlines to the south of this access and the proposed shuttle lay-by location must also be considered, particularly when an overpass is constructed at the rail crossing south of the site.

Audley Road Sightline Assessment (Future Condition with Overpass)

The proposed shuttle bus lay-by is located approximately 75 m north of the existing railroad crossing on Audley Road. We understand that there is a future grade separation that will dictate shifting the shuttle lay-by area and the existing access to the north based on anticipated grades and sightlines. Even after the northerly shift, the driveway and lay-by will still be located on grade as the vertical profile will be raised at the grade separation and will match the existing profile at a location just south of the existing right turns only driveway to the recreation centre.

Sight distances for the proposed bus lay-by were reviewed to confirm if the sightlines are sufficient to allow for safe operations under the current road layout and with the grade separation. The minimum SSD (Stopping Sight Distance) requirements were calculated based on the guidelines provided in the Chapter 2 of the Geometric Design Guide for Canadian Roads (TAC 2017), Table 2.5.2 (Stopping Sight Distance on Level Roadways for Automobiles). For a level road with 70 km/h design speed, the SSD required is 105m. The available distance is more than required as Audley Road is a level and straight road.

Under future conditions with the grade separation over the rail will result in a vehicle traveling northbound in a downgrade. The expected height of the road bridge over the rail will be approximately 7m. If the bridge had to meet back at grade at the existing driveway, the slope would have been steep at 9%. The sight stopping distance required for a 9% downward slope (standard for this type of rail grade separation) is 124m based on the guidelines provided in Chapter 2 of the Geometric Design Guide for Canadian Roads (TAC 2017), Table 2.5.3 (Stopping Sight Distance on Grades). Based on the approximate embankment information provided by the City, we understand that there will be a more gradual slope closer to 3%; however, the proposed lay-by and existing driveway at 75m north, would be approximately 4-5m off the existing ground in which it would be challenging for grading into the site.

The sightline review indicates that left-turns out of this driveway cannot be accommodated due to the limited sightlines resulting from the grade separation crest. Hence in combination with the height issue, it is advisable to shift the proposed bus lay-by and existing driveway (in the future when the rail grade separation is implemented) approximately 75m north of the current proposed location on Audley Road such that approximately 150m distance is available between the lay-by / driveway and railroad to reduce the elevation of the driveway relative to existing ground. At 150m to the north, the shifted driveway would be approximately 3m higher than existing ground, whereas the lay-by area would be approximately 1-2m higher than existing ground.

Police control of the access driveway and closure of Audley Road to public traffic (in the northbound direction) could also facilitate left turns for vehicles exiting the driveway.

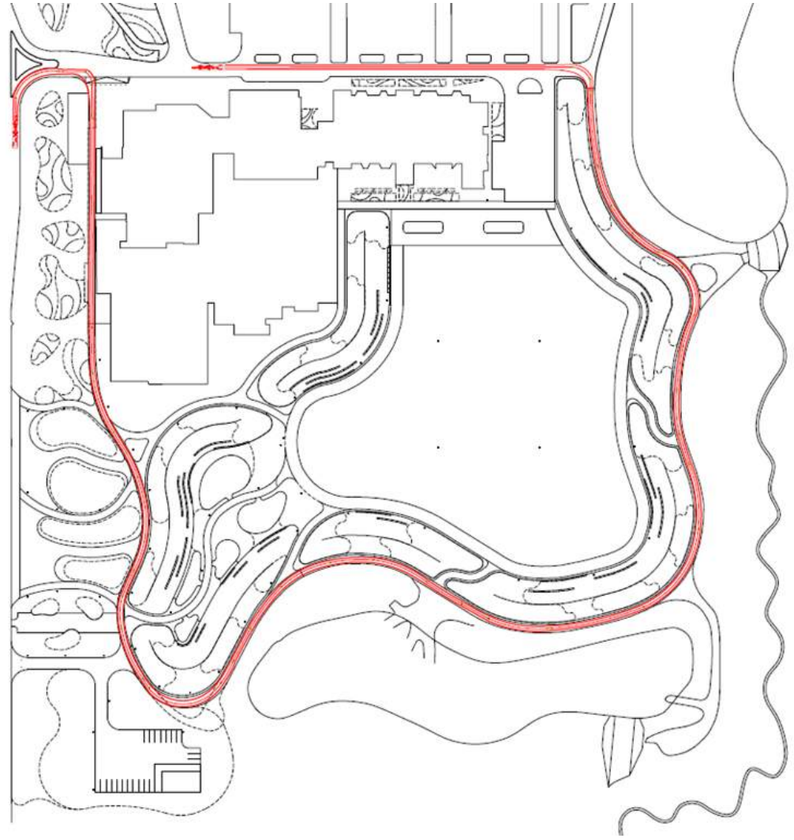


Figure 3: Maintenance Vehicle (Pickup Truck with Hitch) Swept Path

Potential Mid-block Pedestrian Crossing of Audley Road

There are currently limited crossing opportunities for Audley Road and the pedestrian connections in the area are also limited due to the rural cross section of Audley Road which does not include curb, gutter, or sidewalks.

The only crossing opportunity in the vicinity of the site is at Williamson Drive East which is the primary entrance to the existing site. There is also a multi-use path on the west side of Audley Road which runs the full length of the site and connects to the broader pedestrian network but the only formalized pedestrian connection to the site would be via the west side of Audley Road and through the intersection of Audley Road at Williamson Drive East.

Providing an alternative crossing location should be considered as this will reduce the likelihood of pedestrians walking on the east side of Audley Road and will concentrate crossings at a formal crossing location and discourage midblock crossing or 'jay-walking'. The optimal location would be to locate the crossing north of the proposed shuttle lay-by location on Audley Road and where Audley is two lanes wide (that is, south of the right turn lane to the recreation centre driveway) as that would provide an optimal crossing opportunity for pedestrians coming to and from the neighbourhood to the west and avoid conflict with the inbound shuttles and driveway access. It would be ideal to align the crossing with Marriner Crescent for visibility if possible.

SHUTTLE LAY-BY OPTION 2 (WEST OF SKATE PARK)

This option is not the preferred option since it may conflict with vehicle circulation further within the site, will require that all riders walk past the existing building to access the Fairgrounds including crossing the main driveway entrance extending into the site from the right-in/right-out. The right-in/right-out driveway would also require widening to accommodate the buses.

SHUTTLE LAY-BY OPTION 3 (EAST SIDE OF PARKING LOT)

This option is not the preferred option since it may conflict with vehicle circulation further within the site and may necessitate adjustments to curbs within the parking area and will introduce additional congestion into the parking area. This option also requires that all riders walk past the existing building to access the Fairgrounds. The right-in/right-out driveway would also require widening to accommodate the buses.

SHUTTLE LAY-BY OPTION 4 (NW CORNER OF EXISTING BUILDING)

This option is not the preferred option since there is limited capacity for shuttles, it may conflict with inbound movements from Audley Road at the right-in/right-out driveway, it may conflict with vehicle circulation further within the site and access to the loading area for the existing building, and it will require that all riders walk past the existing building to access the Fairgrounds. The right-in/right-out driveway would also require widening to accommodate the buses.

SPECIAL EVENTS

PARKING & SHUTTLE NEEDS ASSESSMENT

Parking needs for the Fairgrounds have been reviewed at a high level using broad assumptions for event attendance, mode shares, and shuttle bus capacity and loading times. For this exercise, the largest crowd size was determined by looking at other events within the Town of Ajax that draw large crowds, and the parking requirements have been determined based on this high attendance peak event scenario.

The largest crowd size expected would be Canada Day where there are up to 25,000 daily attendees and it is assumed that the number of people leaving may be as high as 18,000 people which would occur following the evening fireworks displays. These estimates are based on the Canada Day celebrations at Ajax Downs. If it is assumed that the total number of attendees leaving is 18,000, and that the available parking supply for Ajax Downs is 3,050 spaces with auto occupancy of 2.5 persons per vehicle, then the mode split for those leaving the Ajax Downs Canada Day event is approximately 58% active transportation and shuttles, 17% auto drive, and 25% auto passenger.

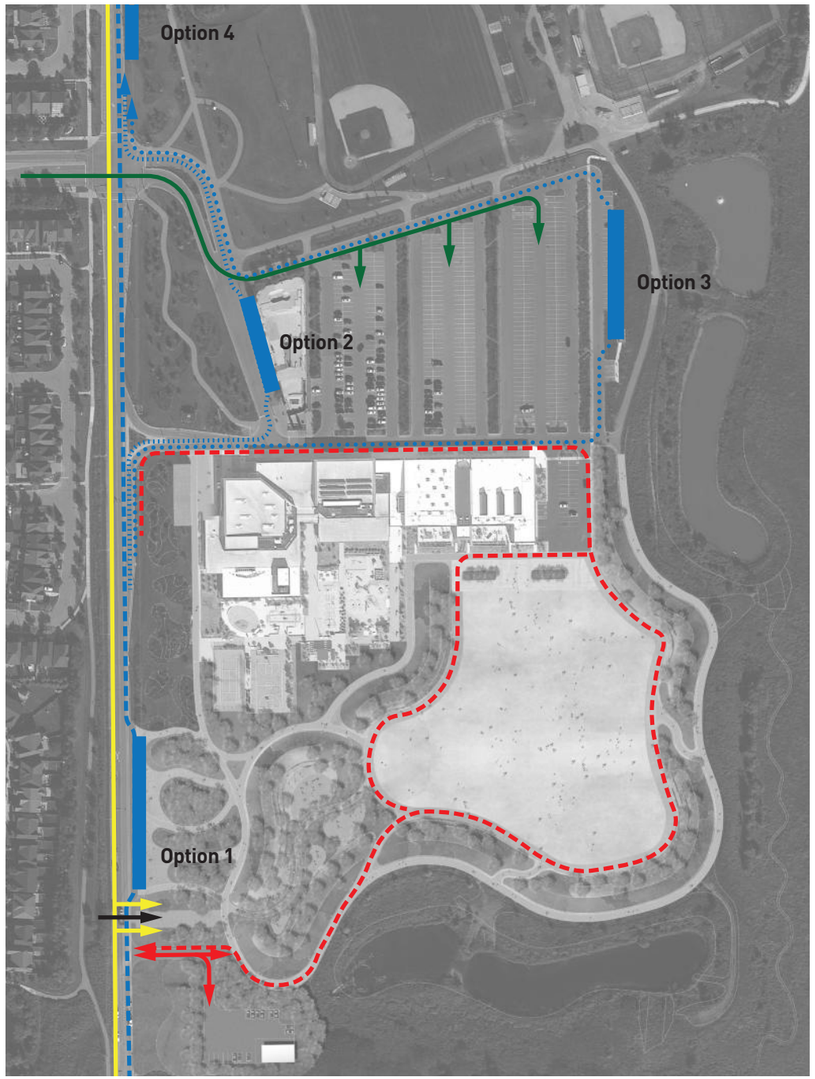


Figure 4: Shuttle Bus Lay-by Location Options

The ARC Fairgrounds will have substantially less parking available. The estimated parking supply for the ARC Fairgrounds will comprise 569 spaces on site, 60 spaces at the nearby Romeo Dallaire Public School, plus 100 on-street parking spaces, for a total available parking supply of 660 spaces. If it is assumed that there will be 2.5 people per vehicle, then the number of people leaving by vehicle will be approximately 1,823. This means that the remainder of the people leaving by non-auto modes will be 16,177.

To accommodate this amount of people, there will be public transit, shuttles, and vehicle pick-up modes available. If we assume 30% of the non-auto trips are active transportation (walk/cycle), a modest 2% for vehicle pickup (given that the area will be congested), and 5% for public transit (based on bus capacity for DRT route 915 Pulse and route 216c running 15-minute headway), then the remaining 10,191 people will have to be served by shuttle buses. The detailed calculations are provided in Attachment A.

The shuttle bus capacity is assumed to be approximately 48 people assuming all sitting space is occupied and no one is standing on a standard school bus with 12 rows of seats. This assumes two people per bench/seat. This means that there will be 220 shuttles required to accommodate 10,515 passengers. If the loading time is assumed to be 3 minutes per bus (including the time needed for the bus to leave and the next bus to arrive and staging the passengers to the bus position), then the time needed to clear the shuttle riders can be calculated based on an assumed number of shuttle lay-bys.

If there are 6 shuttle lay-by spaces provided then the exiting passengers will take approximately 1.77 hours to clear. If there are 8 lay-by spaces provided then the crowd will take 1.33 hours to clear. The detailed calculations assuming 6 lay-by are shown in Attachment A. Six bus lay-by spaces could easily fit between the existing driveway and the existing right turn driveway leading to the Audley Recreation Centre. Approximately 82m of lay-by space would be required based on 12m bus lengths and 2 m clearance in between the buses.

Small changes to the assumptions may result in differences to the estimate time to clear. For example, if the number of shuttle bus riders are assumed to be 50% of the non-auto attendees, then the time to clear is reduced from 1.77 hours down to 1.2 hours. However, given the location of the ARC Fairgrounds, it may be difficult to achieve a 50% non-auto mode share.

The calculations above also assume simultaneous (concurrent) loading of all 6 buses at the lay-by area. If all the buses are destined to the same location then this would be easily achieved since passengers would not need to wayfind to a specific bus/platform and would simply load on to the nearest available bus out of the 6 currently in lay-by. This would constitute the most efficient loading condition.

However, if the buses are destined to multiple locations, then passengers will need to wayfind go the correct shuttle or wait for their desired shuttle to arrive. This will cause inefficiencies in the crowd movement which could result in potential longer wait/loading times compared to the calculations above. Understanding the shuttle destinations and the passenger demand for each shuttle destination would help refine the estimates of shuttle bus requirements.

Considering there are three shuttle routes for Ajax Downs serving passengers from GO parking, Ajax Town Hall and Audley Recreation Centre, we have assumed three routes could also apply to the ARC Fairgrounds as well, except that Ajax Downs would replace the Audley Recreation Centre as a pick-up/drop-off point. With 6 lay-by spaces and 3 routes, there could be 2 shuttle buses per route.

POTENTIAL AUDLEY ROAD CLOSURE AND IMPACTS OF RAIL GRADE SEPARATION ON PROPOSED LAY-BY

Based on the shuttle calculations we estimate that 6 shuttle lay-bys would be required for simultaneous loading to clear the exiting passenger crowd within 2 hours. We believe this to be a reasonable estimate but it relies on consistent shuttle loading and departures.

The analysis assumes that there will be 6 shuttles loading at a time based on the Audley Road frontage available to the buses. Additionally, there is the possibility for there to be other shuttles waiting for an opportunity to load passengers which will be queued along Audley Road.

Since the preferred shuttle location is along Audley Road, the number of potential lay-by stops is limited due to the rail crossing south of the recreation centre along Audley Road and the right-in/right-out driveway to the recreation centre. The distance between the railway and the driveway is approximately 300 metres today.

However, if the grade separation is implemented in the future, the available space along Audley Road is more limited due to the need to shift the existing driveway and proposed lay-by by 75m to the north. Based on the preliminary site plan prepared, 6 buses can be accommodated along Audley between the relocated driveway and the start of the NB right turn lane leading to the right turn driveway to the recreation centre. If more buses arrive and queue along Audley Road, this would suggest that a road closure of the northbound travel lanes could improve safety and operations so that northbound vehicles are not blocked behind the buses on Audley and do not attempt to cross the centreline to pass them.

If a northbound closure is implemented, the closure could be located at Rushworth Drive or at Rossland Road (and only permit local traffic north of Rossland). Only shuttle buses and operations/vendor vehicles would be permitted north of Rushworth Drive. The southbound lane on Audley Road would remain open for exiting vehicles to maximize capacity post peak event.

Northbound shuttle buses leaving the lay-by would proceed north to either Williamson Drive or Taunton Road to make their way west to Salem Road or east to Lakeridge Road and then south towards the three pick-up/drop-off points.

ATTACHMENT A
SHUTTLE LOADING CALCULATIONS

Proxy Event #1 Mode Shares			
If the attendance for Proxy #1 is 18,000 people, then...			
The Non-Auto Mode Share is:	58%	(AT + Shuttles)	(This is not really used since the ARC site will need to have a different mode split with a larger non-auto mode share.)
Auto Driver Mode Share is:	17%		
Auto Passenger Mode Share is:	25%		
	100%		

ARC Statistics			
Anticipated Peak Attendance	18,000	people	(assume the same as Canada Day at Ajax Downs)
ARC Parking Supply	569	spaces	
Romeo Dallaire Public School Parking Supply	60	spaces	
Audley Street (or other street parking)	100	spaces	
Expected Vehicle Occupancy	2.5	ppl/veh	

Shuttle Calculations - Peak Event Exit				
Mode Share: Vehicle Driver + Passenger Attendance	1,823	people		
Mode Share: Non-Auto Persons Attendance	16,177	people		
Mode Share: Passenger Pick up	324	people	2%	(percentage of non-auto trips that are picked up by vehicles after event)
Mode Share: Active Transportation	4,853	people	30%	(percentage of non-auto trips that are walk/cycling trips)
Mode Share: Local Transit - DRT 216c (Harwood/Williamson) and DRT 915 Pulse (Taunton)	809	people	5%	(percentage of non-auto trips that are using local transit - assume 15min headway)
Mode Share: Shuttles (serving GO parking, Ajax Town Hall, Ajax Downs)	10,191	people		
Shuttle Capacity (Sitting only)	48	people		
Shuttle Capacity (Standing if permitted)	0	people		
Shuttle Capacity (fully loaded)	48	people		
Shuttle Bus Loads Required	213	shuttles		
Layby Locations	6	layby		(i.e. simultaneous loading capacity)
Shuttle Loading Time	3	minutes		Based on 30-60 sec for staging passengers to bus, 2-2.5 seconds per passenger boarding,
Shuttle Turnover Per Hour Per Layby	20	shuttles		plus 30 sec cushion before bus moves
Shuttle Turnover Per Hour (Total)	120	shuttles		
Passenger Turnover Per Hour	5760	Persons		
Time to Clear the Passenger Demand	**1.77**	**hours**		

* Entering requirements will be less compared to Peak event exiting requirements

APPENDIX C - GRADING & STORMWATER MANAGEMENT CONSIDERATION

EMC Group Limited

GRADING & STORMWATER MANAGEMENT CONSIDERATIONS

INTRODUCTION

EMC Group Limited has reviewed the design and construction considerations related to Grading and Earthwork, Storm Drainage and Stormwater Management, and Sanitary and Water Servicing and provide a synopsis of how each item should be addressed when considering the development of the open space area into Fairground lands.

GRADING AND EARTHWORKS

The existing ARC open space lands are graded from the existing ARC building towards the south. Drainage from the upper portion is directed to a mid-field cut-off swale that drains towards two DICBs which convey drainage into a lower headwall and ultimately towards existing ditches along the existing pedestrian walkways.

The proposed grading will incorporate naturalized landscape berms along the periphery. The landscape berms will slope up from the existing pedestrian nature trail at a maximum slope of 2H:1V and down to the backside of the proposed reinforced sod road located on the inner side of the Fairgrounds at a slope equal to or less than 4H:1V. The internal slope will be utilized as a pedestrian viewing area where Fairground guests are able to sit on stone benching cascaded up the more gradual interior slope. Walkways will be graded at a maximum 5% longitudinal slope and will have maximum cross-slopes of 2%. The main event stage area will be graded between 1% – 1.5% and will easily direct runoff to designated low points where catch basins will be proposed. The overall Fairground open space area will be graded at a predominantly 2% slope to maintain a reasonably flat area to accommodate a comprehensive arrangement of Fairground amenities. The site will be flat enough to ensure large tents, or temporary amusement park installations could be constructed.

The conceptual grading plan is presented in **Figure 222139-GR1**.

During Phase 2 of the development, the Town created a landscape hill by re-locating a portion of a pre-load fill pile that was originally constructed during Phase 1. The intent of this fill shift was to reduce costs during Phase 2 by reducing the earth export from the site. This approximately 20,000 cu.m. fill mound poses a constraint, as the intent is to limit the volume of export from the site for the Fairground construction. By incorporating berming along the boundary of the Fairground site, the earth cut to fill volumes have been adjusted to reduce the volume of fill to be exported from the site. Based on the conceptual grading review, it appears that most of the existing berm can be utilized on the site. A cut volume of less than 1,000 cu.m. will potentially need to be exported from the site; with an approximate onsite cut of 37,000 cu.m. being placed in fill areas. Topsoil that is stripped from the site shall be stockpiled temporarily; and re-used on the site during fine grading stage at time of landscaping. A small portion of topsoil may have to be exported from the site to offset the areas that will be finished with hard surfaces such as walkways and parking lots.

A conceptual earthwork gradient s presented in **Figure 222139-GR2** and a three-dimensional rendering presented in **Figure 222139-GR3**.

At the time of preparation of the preliminary cut/fill analysis, a recent topographic survey was not made available. The earthwork analysis was prepared by comparing a conceptual proposed grading surface for the new Fairgrounds to an assembled phase 2 design surface – and not an as-constructed surface. At the detailed design stage, it would be prudent that the owner have the area of the proposed Fairgrounds site surveyed and an accurate comparison surface be utilized in the assessment of the grading design and earthworks balancing to ensure a reliable cut/fill estimate is derived for budgeting purposes.

STORM DRAINAGE AND STORMWATER MANAGEMENT

The Fairground is proposed within a 6.48 ha area of the site that drains to the southerly stormwater management pond; and within a 0.77 ha drainage area that drains into the Audley Road ditch.

The Pre-Development drainage plan is presented in **Figure 222139-PRE** and Post-Development in **Figure 222139-POST**.

Area Draining to the Stormwater Management Pond

The current condition of the site area where the Fairgrounds are proposed include a grassed open space area with a landscaped hill to the southeast; a sag depressed landscaped area over drainage structure to the east of the existing building; an asphalt pedestrian walkway spanning the perimeter of the open space; and a drainage system consisting of conveyance swales and ditch inlet style catch basins (DICBs) which ultimately drain into the existing stormwater management facility located at the south limit of the site.

A review of the proposed surface makeup within the 6.58 ha drainage area indicates that the runoff coefficient will slightly decrease to 0.39, from an assigned runoff coefficient of 0.4 which was utilized to size the original stormwater management pond.

The existing overland drainage is captured and conveyed by swales wrapping around the existing landscape hill; convey through culverts, and finally discharged to the south stormwater management pond.

The intention is to maintain the existing drainage pattern while providing efficient drainage solutions to accommodate the proposed Fairground modifications. Drainage collected from single and double catch basins (DCBs) and catch basin maintenance holes (CBMHs) will provide the most efficient method of capture. Swales and ditch inlet catchbasins (DICBs) will also be placed along the toe of all new berms to capture and convey swale drainage towards the proposed or existing storm sewer system. The existing drainage area and runoff contributing to the stormwater management pond will be required to remain unchanged from the parameters outlined through the original SWM pond design.
Should the maintenance building be located on the east side of the drainage line separating drainage directed to the stormwater management pond and Audley Road ditch, then localized stormwater management will be incorporated at the maintenance building to reduce flow to the sewer to pre-development levels.

Proposed berm locations will encroach over the location of a portion of the existing storm sewers and maintenance holes currently constructed within the open space. The existing maintenance holes are to be raised to the new final grade. No re-location of the existing storm sewers beneath the open space are anticipated to be required.

Drainage conveyed through the new proposed sewer system will ultimately discharge into the existing stormwater management pond. A proposed headwall will be constructed towards the north of the stormwater management pond spillway to introduce a new discharge point; and an existing culvert will be upsized to accommodate the upstream drainage area.

Area Draining to Audley Road

The 0.77 ha drainage area directing overland flow towards the Audley Road ditch proposes a runoff coefficient of 0.5, which exceeds the pre-development runoff coefficient of 0.26.

The westerly portion of the site currently drains towards the Audley Road ditch. This drainage condition is proposed to remain unchanged; however proposed ditch inlet catchbasins will capture the additional runoff from the Maintenance building parking lot and Shuttle Bus Drop-off area and will outlet into the existing roadside ditch through a localized stormwater management system.
Development of the bus drop off and maintenance building parking area are proposed within an area reserved to accommodate a grading slope transition from a future Audley Road rail overpass. The bus loop and maintenance building parking lot will require re-location at the time of the overpass construction. The maintenance building itself will be proposed on the east side of the drainage divide; and will be tributary to the SWM Pond. Any increase in peak flow generated by the increased imperviousness introduced by locating the building within this zone will need to be accommodated by establishing localized stormwater attenuation in the form of controlled roof drains, control structures and either aboveground or underground storage.
The bus loop drop off is proposed to have a level surface and lay-by lane adjacent to Audley Road.
The lay-by will be constructed over the existing ditch requiring drainage conveyance by way of a culvert. It

is proposed to extend the culvert from the northerly limit of the bus loop lay-by to the southerly limit of the access driveway.

Construction of the proposed Maintenance Yard and Shuttle Bus Drop-off area located towards the southerly portion of the site will increase the drainage towards Audley Road's existing roadside ditch. With the pre-development area makeup being predominantly pervious, a separate stormwater management system is to be proposed to allow for the site discharge to the Audley Road ditch to remain unchanged. A proposed restriction of flow is necessary to control the discharge from post- to pre-development levels, which will introduce stormwater management storage upstream of the control device. To accomplish this, the proposed parking lot elevation along Audley Road will require grade raising in conjunction with the construction of a retaining wall to account for adequate sewer cover. As paved surfaces accommodating vehicular travel are being proposed within the drainage area out letting to the Audley Road ditch, upstream quality control in the form of an Oil/Grit separator shall be considered to pre-treat stormwater prior to discharge directly to the ditch. To reduce run-off volume to pre-development levels, a form of infiltration for drainage generated from new impervious areas shall also be incorporated. Such infiltration could be in the form of an underground soak away pit collecting clean roof water and filtered runoff from asphalt surfaces; and/or a combination of a roof rainwater soak away pit and permeable pavers to promote filtering and infiltration of surface drainage.

A conceptual plan illustrating the storm sewer improvements is presented in **Figure 222139-STM**.

SANITARY SERVICING

As it relates to wastewater discharge, the predominant generator of wastewater from the Fairgrounds site will be the proposed maintenance building. It is proposed to provide a sanitary service connection to the proposed maintenance building. From review of the depth of the existing sanitary maintenance hole located just south of the Audley Recreation Centre, adjacent to Audley road (i.e. SAN. MH 6A), a gravity connection is not feasible due to the distance and maximum attainable surface elevation at the proposed maintenance building. It is also noted that a sanitary sewer does not exist adjacent to the proposed maintenance building within Audley Road. To facilitate a connection to SAN. MH 6A, a sanitary force main and pump station will be required to service the building. A sanitary pump station complete with a wet well and associated piping and electrical will be proposed at the maintenance building. A force main is to be proposed along the westerly limit of the site adjacent to Audley Road.

WATER SERVICING

An existing 100mm diameter plug had previously been installed to service the future Phase 3 development at the southwest side of the Audley Recreation Centre site. This 100mm watermain extends from the Audley Recreation Centre mechanical room.

The proposed Maintenance Building and Water Bottle Filling Station will require domestic water. It is proposed to extend the existing 100mm diameter PVC watermain from the existing plug to the Maintenance Building and a suitably sized copper or polyethylene water service extended from the maintenance building to the Water Bottle Filling Station. By extending the 50mm dia. line from the new maintenance building, the Town will be able to sub-meter the domestic usage at the maintenance building by installing a separate meter. All water entering the Audley Recreation Centre site is currently metered at the main recreation centre building.

As the owner proceeds to the next stage of the project, it is suggested that the owner investigate with the local fire marshal the requirements related to fire protection where food trucks will be active. Should it be determined that fire protection is necessary in the case of an emergency in the location of the food truck areas, we have laid out a conceptual fire main servicing route in order to facilitate the installation of two fire hydrants. Two new hydrants would provide adequate hydrant coverage to all proposed Fairground areas such as the event lawn, the stage, food and beverage areas, vendor, and sponsor areas. In order to provide access to the hydrants, the construction of the reinforced lawn roadway should consider the structural requirements to support travelling fire trucks to access the remote hydrants. If fire truck access into the Fairgrounds is infeasible, the closest available hydrant is the existing hydrant located at the south end of the existing parking lot which lies adjacent to the ARC Library.

It is understood from the Architect that the maintenance building will be designed and sized in a way that does not require fire protection under the Ontario Building Code.

A conceptual plan illustrating both the sanitary and watermain modifications are presented in **Figure 222139-SS**.

LEGEND:

- DRAINAGE TO SWM POND
- DRAINAGE TO AUDLEY RD
- SLOPE ARROWS
- SINGLE/DOUBLE CATCHBASIN
- SINGLE/DOUBLE CATCHBASIN MANHOLE

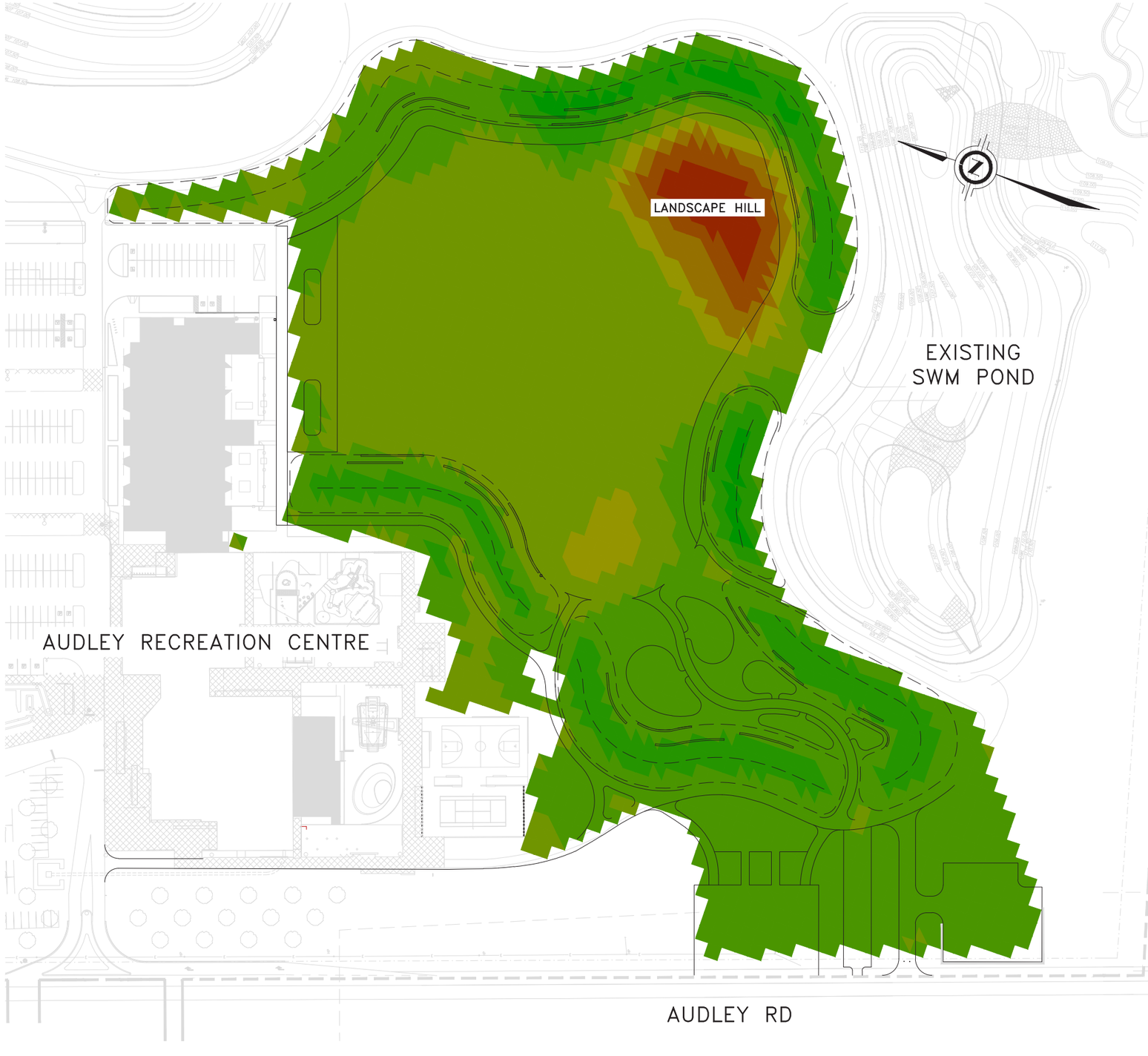
Elevations Table			
Number	Min. Elevation	Max. Elevation	Color
1	109.000	110.000	
2	110.000	111.000	
3	111.000	112.000	
4	112.000	113.000	
5	113.000	114.000	
6	114.000	115.000	
7	115.000	116.000	
8	116.000	117.000	
9	117.000	118.000	

EMC GROUP LIMITED
Engineers Planners Project Managers
Vaughan Ontario (905) 738-3939

CONCEPTUAL GRADING PLAN
FAIRGROUNDS
AUDLEY RECREATION CENTRE

SCALE: 1:1500

DWG. No. **222139-GR1**



Cut and Fill Depth				
#	Min. Elevation	Max. Elevation	Color	volume
1	−10.00	−9.00		0.00
2	−9.00	−8.00		265.35
3	−8.00	−7.00		1053.75
4	−7.00	−6.00		1509.08
5	−6.00	−5.00		1999.66
6	−5.00	−4.00		2567.31
7	−4.00	−3.00		3101.19
8	−3.00	−2.00		3723.96
9	−2.00	−1.00		6904.92
10	−1.00	0.00		15593.66
11	0.00	1.00		20681.60
12	1.00	2.00		9121.74
13	2.00	3.00		4855.55
14	3.00	4.00		1825.48
15	4.00	5.00		221.15
16	5.00	6.00		0.00

PRELIMINARY CUT/FILL SUMMARY

CUT = 36,700.00 cu.m.
FILL = 36,700.00 cu.m.
NET = 0,00 cu.m.

EARTHWORK VOLUMES ARE ESTIMATED BASED ON A PRELIMINARY GRADING CONCEPT AND ARE SUBJECT TO CHANGE

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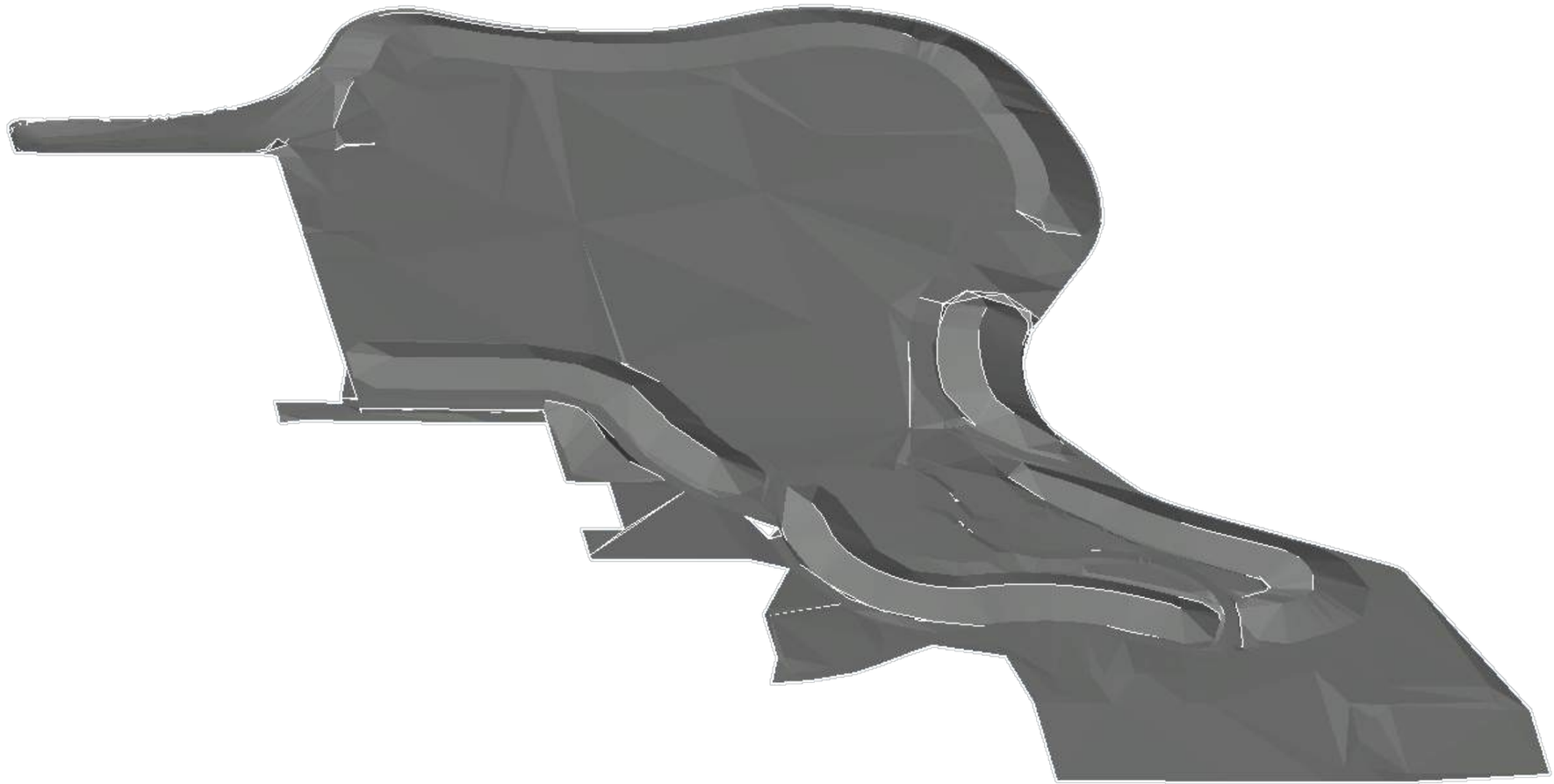
Vaughan Ontario (905) 738−3939

CONCEPTUAL EARTHWORK GRADIENT
FAIRGROUNDS
AUDLEY RECREATION CENTRE

SCALE:	DWG. No.
1:1500	**222139−GR2**



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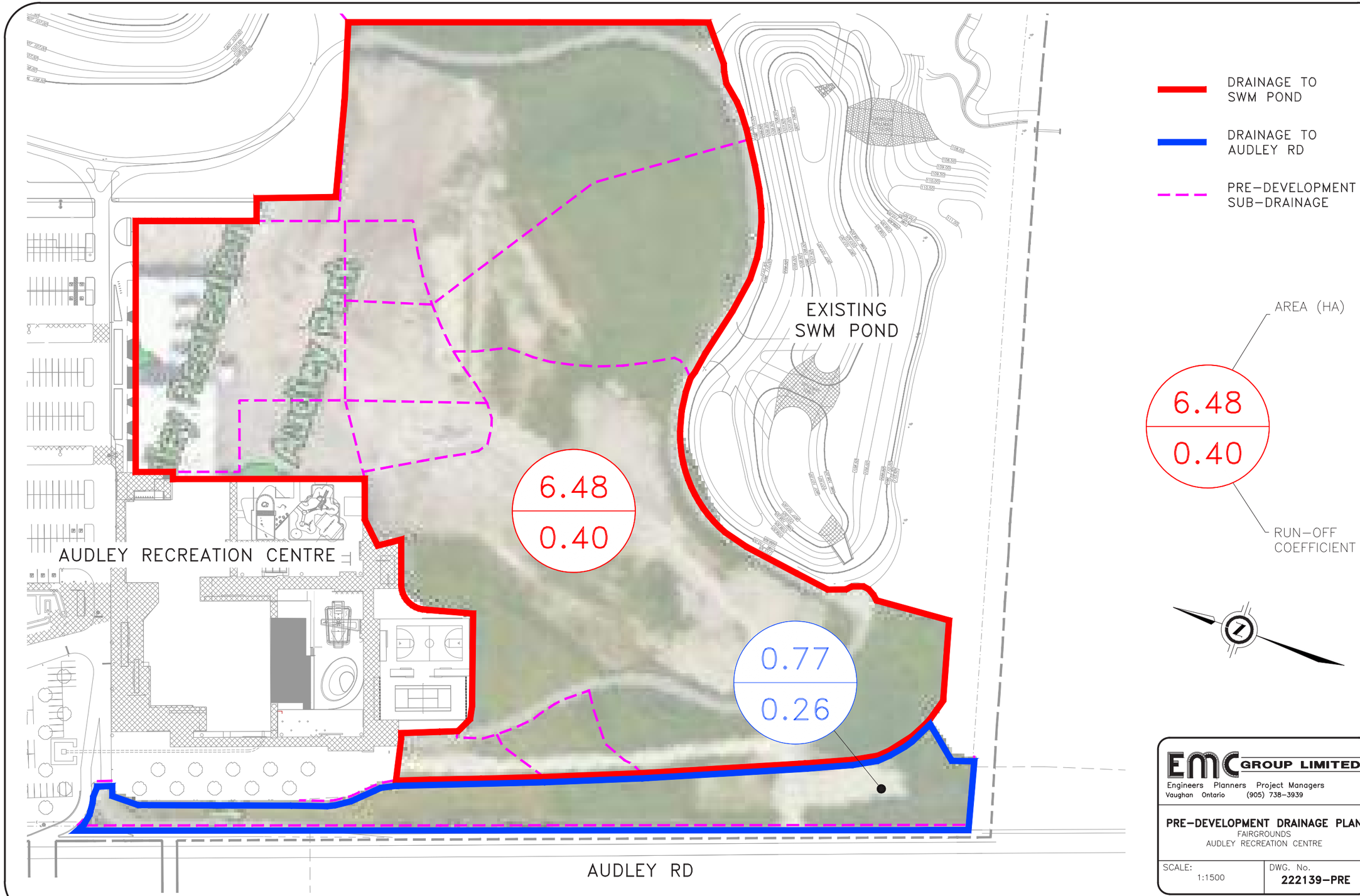
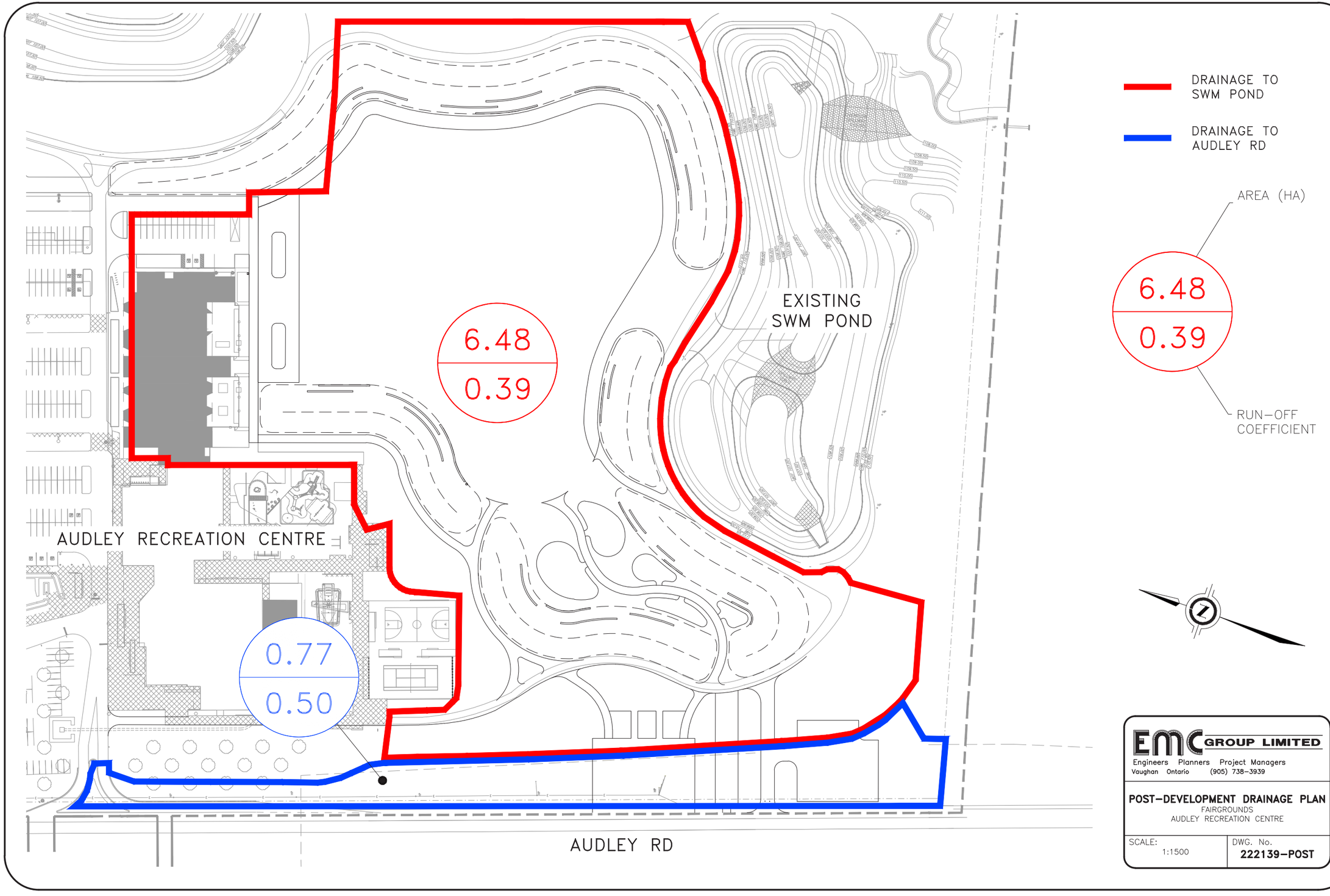
CONCEPTUAL 3D RENDERING
FAIRGROUNDS
AUDLEY RECREATION CENTRE

SCALE:	DWG. No.
N.T.S.	**222139–GR3**


DRAINAGE TO SWM POND
DRAINAGE TO AUDLEY RD
PRE-DEVELOPMENT SUB-DRAINAGE
AREA (HA)
RUN-OFF COEFFICIENT
6.48
0.40
EXISTING SWM POND
AUDLEY RECREATION CENTRE
6.48
0.40
0.77
0.26
AUDLEY RD
EMC GROUP LIMITED
Engineers Planners Project Managers
Vaughan Ontario (905) 738-3939
PRE-DEVELOPMENT DRAINAGE PLAN
FAIRGROUNDS
AUDLEY RECREATION CENTRE
SCALE: 1:1500
DWG. No. 222139-PRE



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POST–DEVELOPMENT DRAINAGE PLAN
FAIRGROUNDS
AUDLEY RECREATION CENTRE

SCALE: 1:1500

DWG. No. **222139–POST**

REMOVE EXISTING SEWER AND HEADWALL RE-USE EX.DCBMH

HEADWALL DISCHARGE TO SWM POND

REMOVE EXISTING SEWER AND HEADWALL. REUSE EX DCBMH

EXISTING SWM POND

AUDLEY RECREATION CENTRE

ORIFICE FLOW CONTROL

OIL/GRIT SEPARATOR

CULVERT DISCHARGE TO DITCH

PROPOSED CULVERT c/w HEADWALLS UNDER BUS DROP-OFF PAD & ACCESS DRIVEWAYS

AUDLEY RD

LEGEND:

PROPOSED STORM SEWER

ST ——— ST EXISTING STORM SEWER

SINGLE/DOUBLE CATCHBASIN

SINGLE/DOUBLE CATCHBASIN MANHOLE

EMC GROUP LIMITED
Engineers Planners Project Managers
Vaughan Ontario (905) 738-3939

CONCEPTUAL STORM SERVICING PLAN
FAIRGROUNDS
AUDLEY RECREATION CENTRE

SCALE: 1:1500

DWG. No. **222139-STM**

HYDRANT & VALVE

REMOVE EXISTING SEWER AND HEADWALL

150mmø PVC FIRE WM

150mmø PVC FIRE WM

HYDRANT & VALVE

EXISTING SWM POND

REMOVE EXISTING SEWER AND HEADWALL

WATER BOTTLE FILL STATION

AUDLEY RECREATION CENTRE

50mmø CU DOM. WM

EXISTING HYDRANT TO BE REMOVED

CONNECT TO EXISTING PLUG

100mmø PVC DOM. WM

SANITARY MH PUMP STATION

MAINTENANCE BUILDING

SAN FORCEMAIN via HDD

EXISTING SAN MH

AUDLEY RD

LEGEND:

- PROPOSED SANITARY SEWER
- PROPOSED WATERMAIN
- W —— W EXISTING WATERMAIN
- — — — EXISTING WATERMAIN
- PROPOSED STORM SEWER
- ST —— ST EXISTING STORM SEWER
- Ⓜ PROPOSED WATER METER INSIDE MAINTENANCE BUILDING

EMC GROUP LIMITED
Engineers Planners Project Managers
Vaughan Ontario (905) 738-3939

CONCEPTUAL SANITARY AND WATER SERVICING PLAN
FAIRGROUNDS
AUDLEY RECREATION CENTRE

SCALE: 1:1500

DWG. No. **222139-SS**

APPENDIX D - MECHANICAL & ELECTRICAL CONSIDERATIONS

Smith + Anderson

ELECTRICAL

1.1 GENERAL

1.1.1. The new Fairgrounds will be electrically serviced from the existing Audley Recreation Centre (ARC) distribution equipment.

1.1.2. As part of the previous ARC Phase 2 design, empty conduits were capped outside at the south end of the building, just south of the Garbage Room. These conduits were planned for the future Phase 3 expansion of the building. Below is a summary of the conduits installed:

- 3 x 78mm conduits for Normal Power (from Penthouse Library Electrical Room).
- 1 x 78mm conduit for Emergency Power (from Penthouse Library Electrical Room).
- 1 x 53mm conduit for Fire Alarm (from Penthouse Library Electrical Room).
- 2 x 103mm conduit for Communications (from Garbage Rooms Telecom Room).
- 1 x 53mm conduit for Security (from Garbage Rooms Telecom Room).

1.1.3. All of the normal power requirements for the new Fairgrounds will be provided using the 3 x 78mm conduits running back to the PH Library Electrical Room. Any new distribution equipment will also be installed within this existing Electrical Room.

1.2. MULTI-FUNCTION PLAZA

1.2.1. Power for the temporary stage in the Multi-Function Plaza area will be provided via 3 new Company Switches, each rated at 200A, 120/208V, 3PH, 4W. The Company Switches will be located on the south exterior wall of the Garbage Room, housed within a lockable weatherproof enclosure. A new 225 kVA, 600:120/208V distribution transformer and 800A, 120/208V, 3PH, 4W Power Panel will be provided in the PH Library Electrical Room that will sub-feed the 3 new Company Switches.

1.2.2. A new permanent 3m long copper ground rod shall also be provided adjacent to the future Stage location and provided with a lockage inspection pit. The Ground Rod shall be provided with mechanical lugs for the temporary grounding of the mobile Stage.

1.2.3. A new 200A, 600V, 3PH, 3W disconnect c/w Cam-lock connections will also be provided next to the Company Switches. This will be used for the portable chiller for the temporary ice mats/skating trail system. This power connection will be fed from the existing 600V distribution panel (DP-PC) located in the Penthouse Library Electrical Room.

1.3. INFRASTRUCTURE LOOP AND PEDESTRIAN PATH

1.3.1. The inner Infrastructure Loop will be provided with 40-60 ft high light poles mounted with multiple LED floodlights to illuminate the main Event Lawn space. The LED floodlights will be provided with visors and snoots to control light spill and be aimed below the horizontal plane to avoid light pollution.

1.3.2. 50A, 120/208V, 3-pole, 4-wire pre-wired receptacles in a rainproof enclosure will also be located around the Infrastructure Loop, spaced approximately every 300 ft. This plug connection will be used to connect portable power distribution boxes for events requiring power out in the Lawn space such as Food Trucks.

1.3.3. For the outer Pedestrian Path, new 6m high LED lighting will be provided to match the existing cylindrical light standards installed around the site.

1.4. MAINTENANCE BUILDING

1.4.1. A new 60A, 120/208V, 1PH, 3W loadcentre panel will be provided in the new Maintenance Building and will be electrically serviced from the existing Audley Recreation Centre. The new electrical feed will originate from the existing basement Main Electrical Room (panel RP-BC).

1.4.2. New LED strip lighting and convenience receptacles will be provided in the Maintenance building and fed from the new 60A loadcentre panel.

MECHANICAL

1.1. GENERAL

1.1.1. The Fairgrounds will make use of an underground water service provided for future phases at the southwest corner of the site rather than extending from the existing recreation building which would involve renovations to both the interior of the building and impacts to existing exterior site elements to the south of the recreation centre. The water service will route into the new Maintenance Building on the southwest corner of the site where it will be metered and then exit the building below grade to the site connection points.

1.2. MULTI-FUNCTION PLAZA

1.3.1. A 25mm [1 in.] water supply will be provided at the back of the main stage and another near the sensory zone (locations indicated on the Architectural Plans) for water filling stations.

1.2.2. A 25mm [1 in.] water supply for ice production will be provided adjacent to the Multi-function Loop/outdoor skating surface.

1.2.3. Water service piping for the above connections will come from a new incoming water service entering the Maintenance Building. All piping will route below the frost line and each connection shall be complete with non-freeze ground hydrant flush with finished grade with access box, hinged cover and drain port.

1.3 MAINTENANCE BUILDING

1.3.1. A new 100mm [4 in.] incoming domestic cold water service complete with backflow preventor and water meter will be provided in the new Maintenance Building from the existing water line on site.

1.3.2. A new sanitary line connected to municipal service will be required to service the new building.

1.3.3. New floor drains and fire extinguishers will be provided in the Maintenance Building; should a climate-controlled staff room complete with washroom facility be provided, cooling/heating for staff areas will be provided through a heat-pump style variable refrigerant flow (VRF) system complete with energy recovery ventilator for outdoor air supply and supplemented with perimeter electric heating as required. Plumbing services will be provided to any washroom or kitchen sinks; domestic hot water will be generated by an electric hot water heater.

END OF DOCUMENT

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Armi De Francia, MCIP, RPP Active Transportation Coordinator
Report #:	PDS-2023-12
Subject:	**2023 Transportation Demand Management Plan Update**
Ward(s):	All
Date of Meeting:	April 11, 2023
Reference:	2015 Transportation Demand Management Plan 2019 Integrated Transportation Master Plan 1022311 Transportation Demand Management Plan Update Staff Contract Award Report December 16, 2021 2022-2026 Strategic Plan

Recommendations:

1. **That the report entitled "2023 Transportation Demand Management Plan Update" for the Town of Ajax be received for information, and**

2. **That Council endorse the Transportation Demand Management Plan for the Town of Ajax prepared by WSP Canada, dated March 2023.**

1.0 Background:

Residents need more transportation options. Transportation Demand Management (TDM) is about responding to this need using demand-side solutions. TDM is the application of strategies that encourage greater use of sustainable transportation modes and trip decision-making that reduces, combines, or shortens vehicle trips. Specifically, it encourages transportation options such as:

- Walking (including the use of mobility aid devices)
- Cycling (including the use of adaptive bikes or trikes for people with disabilities)
- Scooters (including adaptive scooters)
- E-bikes, e-scooters, and other micromobility devices
- Transit
- Ridesharing
- Telecommuting (remote work)

A number of policies support the update of the Transportation Demand Management Plan:

- Section 4 of the Official Plan includes a goal to "Support Transportation Demand Management (TDM) initiatives that promote alternative modes of transportation and increase transit ridership, walking and cycling."
- The Town's first Transportation Demand Management Plan (TDM) in 2015 contained development guidelines, bicycle parking guidelines, and demand-side actions for enabling the use of various sustainable transportation modes. It recommended an update every five years.
- In 2019, Council approved the Integrated Transportation Master Plan (ITMP) which focuses on the transportation network and set a vision to:

  "…Enhance the sustainable and multi-modal transportation system for the Town of Ajax by planning, designing, constructing, and maintaining a community and transportation network that supports pedestrians, cyclists, transit, and vehicular traffic. The Town's transportation system will be accessible to all users and all modes of transportation".
- In 2022, the Town approved the Ajax Green Standard which includes active transportation and pedestrian friendliness as one of the impact sub-categories for reducing greenhouse gases and working towards clean air.

Since 2015, the Town has completed 7 actions. These actions initiated a track record for engaging residents and guiding development processes.

Historically, TDM plans have been designed to focus on the needs of residents working a 9-5 office job, seeking ways to alleviate vehicle-based commuting pressures at peak travel times. The 2023 TDM Plan Update uses a less conventional yet holistic approach that builds on existing strengths and assets with guidance and input from residents, community leaders, and agency stakeholders.

With the proliferation of hybrid work and the increasing interest of advancing transportation equity considerations, the Ajax TDM Plan considers people who:

- work irregular schedules
- take trips that occur at unconventional times,
- are dependent on other mobility options as their primary means of travel
- take non-work-related trips such as trips to grocery stores/food banks, doctors' appointments, and places of worship.

In December 2021, the Town contracted WSP Canada to develop the TDM Plan Update. Actions within the 2023 TDM Plan Update work towards a vision of mobility choice that meets the needs and abilities of residents and visitors through a shift to sustainable transportation. Given staff capacity and intention for more holistic engagement with residents and stakeholders, the TDM Update was completed in 2023.

The purpose of this report is to provide an overview of the 2023 TDM Plan Update and to seek Council's endorsement of the proposed actions and guidelines, which will be future capital projects that will be consistent through the future budget deliberations.

2.0 Discussion:

2.1 What is Transportation Demand Management?

Transportation Demand Management (TDM) is the application of policies, programs, services, and products for the purpose of reducing the overall demand for single-occupant vehicle use and increasing the demand for sustainable transportation options.

TDM does not include supply-side solutions. It does not include increasing the frequency of transit, building new active transportation infrastructure, new transit routes, or changes to the road network. Supply side solutions fall within the scope of other plans (such as the Integrated Transportation Master Plan), some of which are implemented by other jurisdictions or agencies (such as transit plans by Durham Region Transit or Metrolinx).

Relying on infrastructure improvements and transit frequency alone is not enough. Below are types of demand-side solutions which respond to challenges that cannot be addressed through infrastructure improvements alone:

Strategy Type	Value
Technology	Provides updates to changes in routes, construction, weather, and other factors that may disrupt the trip.
Policies	Increases amenities addressing needs that may arise during or at the end of a trip. Amenities include bicycle parking, pedestrian walkways, charging stations for pedal assist bicycles, or lighting.
Communications (Outreach, education, and marketing)	It is less likely that people will change their travel behaviours if they are unaware of other transportation options available as well as actions and resources that can make these alternatives more convenient.
Incentives	Motivates people to try options that are outside of their routines and comfort zones. Incentives include pre-loaded transit passes, or points or discounts at businesses or other destinations for walking or biking.

2.2 What is the scope of the 2023 TDM Update?

The 2023 TDM Plan Update includes:

- A report summarizing best practices and results from engaging residents, community leaders, staff, and other stakeholders with lessons and considerations from the Novel COVID-19 pandemic.
- Updated Development Guidelines that consider the Ajax Green Standard and the incoming Zoning By-law update.
- Updated Bicycle Parking Guidelines that consider a wider variety of bicycles (cargo bikes, bicycles with strollers, and adaptive bicycles for people with disabilities) and provides greater clarity on requirements for a larger diversity of land uses.
- Actions for the next five years that involve expanding the Town's capacity, strengthening partnerships, integrating with other Town policies and practices, and working towards transportation equity.

2.3 What is the 2023 TDM Update trying to achieve?

Like other municipalities across Ontario, the Town of Ajax has a legacy of a transportation system designed to prioritize the movement of automobiles. This has limited the mobility of residents such as children, youth, elders, shift workers and people with disabilities. Many

community leaders, service providers, and residents have expressed numerous ways in which car-centric design has not been working for them and in many cases, has created barriers to access essential services, programs, and opportunities. The Town's efforts to improve conditions for sustainable transportation help address barriers resulting from car-centric design.

With the Town's growing population, a transportation system that prioritizes cars and maintains current travel patterns has many risks, including but not limited to the following:

- significant congestion on local roads;
- expensive expansions to roadway capacity;
- increases in pedestrian deaths and serious injury;
- increased barriers (typically distance and financial) to accessing job opportunities, appointments for necessary services or programs, etc.;
- increased pressure on families and caregivers to provide rides for people in their lives who cannot drive; and
- isolation among people who cannot drive or who do not have access to a vehicle or driver.

Not all residents can drive, and not everyone has people in their lives who have the time or means to drive them to the places they need to go. By examining the 2021 age demographics of the Town alone (ages 0-14, ages 70+), approximately 30% of the Town's population are unable to drive. This figure alone does not account for the numbers of people with disabilities, illnesses, or other conditions that can affect their ability to drive who do not fall within these age ranges.

Improving mobility choice makes it easier for children, seniors, people with disabilities and residents on low income to access their community independently. Providing more convenient options for pedestrians, cyclists, transit users, and people using other modes of transportation expands the number of choices residents have available to get around Ajax.

Therefore, the 2023 TDM Plan works towards a vision of mobility choice that meets the needs and abilities of residents and visitors through a shift to sustainable transportation.

To achieve this vision, the goals of the 2023 TDM Update are to:

1. **Improve Mobility Choice:** Provide opportunities and access to alternatives modes of travel, including transit, active transportation, ridesharing, and micromobility to minimize dependency on single-occupant automobile use.

2. **Build a Culture of Sustainable Transportation:** Where active transportation, transit, and ridesharing are recognized and celebrated as accessible, feasible, safe options.

3. **Collaborate with Stakeholders and Community Leaders:** Strengthen partnerships on a range of TDM initiatives locally and regionally to leverage the knowledge, skills and connections of others and increase impact.

4. **Improve Transportation Equity:** Expand mobility choice where car-centric design of the past has created barriers for disproportionately impacted communities to access essential services, programs and opportunities.

5. **Measure and Communicate the Impacts of TDM:** Evaluate the TDM initiatives and communicate the outcomes to staff, Council and the public to build support for continued investment.

2.4 What has the Town accomplished since the previous TDM?

The Town completed 7 actions from the previous TDM Plan between 2015-2022:

- Create Sustainable Transportation Awareness, Education and Safety Campaign Material
- Conduct Community Outreach Events
- Conduct Public Awareness Campaigns
- Provide Guidance on TDM-friendly Site Design
- Prepare Wayfinding and Street Furniture Strategy
- Provide Active Transportation Amenities
- Update Walkable and Bikeable Ajax: The Ajax Pedestrian and Bicycle Master Plan.

Through experience implementing the 2015 TDM Plan, staff have learned that for a TDM Plan to be effective, it must propose actions that have at least one of the following conditions:

- Have strong interest and support from multiple partners if the idea initiated from the Town.
- Build on similar past efforts that were delivered by Town staff, including efforts implemented by different departments or teams.
- Build on an idea or existing efforts initiated by community organizations that are well trusted and respected in the community and that are willing to collaborate with the Town.

2.5 What Community Strengths and Assets can help achieve the goals of TDM?

a) Mobility of care network

To overcome mobility barriers, many community organizations, faith groups and social support networks in Ajax have created both formal and informal tools to support people facing a variety of mobility constraints.

Mobility of Care refers to trips that are vital for a person's needs and can include trips to medical appointments, grocery stores, places of worship or a food bank. These trips provide for physical, social and spiritual needs in Ajax, and are vital to the ongoing health of its population. In Ajax, Mobility of Care programs manifest in a variety of formats such as door-to-door specialized transit ((Kids/Seniors on Wheels; Community Car Durham), carpooling arrangements), or by bringing essential services directly to clients (Doctors on Wheels). The wide variety of Mobility of Care programs by community organizations are delivered with compassion and a deliberate focus on improving the wellbeing of others. To expand mobility options available to residents, it is necessary for the Town to support the strengths of this community-led mobility of care network and replicate a compassionate approach that focuses on the wellbeing of others.

b) School-led efforts and enthusiasm for sustainability and mental health

Since 2010, a number of teachers and school staff employees have led efforts encouraging active school travel. Roughly 30% of Ajax elementary schools participate in the Town's Active School Travel (AST) events each year since 2016, the highest track record in Durham Region.

Notable activities encouraging active school travel championed by teachers and school staff are:

- Two Ajax schools won the Metrolinx Smart Commute Bike to School Week Awards in 2016 and 2018
- One school in Ward 2 implemented a student-led Walking School Bus for multiple years prior to the pandemic. Although a number of parents complimented the Walking School Bus program, school staff noted challenges with the capacity of student leaders during

key sports seasons. In 2019, this school organized a 8km group walk to school starting from Rotary Park.
- Three South Ajax schools organized a group bike ride in 2022 and regularly organize annual bike to school activities.

Further, growing interest in mental health and well-being in the education sector and the benefits of AST for self-regulation provide further opportunities for sustaining or increasing this enthusiasm at the school and school board level.

2.6 What are the proposed actions?

a) Foundations

Action	Steps to Implementation
A1 – Secure TDM funding and resources	A1.1 Review all existing and potential TDM funding sources (including Regional, Provincial, and Federal sources where applicable), including 1% of capital project funding for TDM, and Development Charges as part of next DC By-law update. A1.2 Secure funding and staff resources needed to implement the Plan.
A2 – Create TDM Communications Plan	A2.1 Market research to identify target audiences. A2.2 Evaluate current marketing and communication materials. A2.3 Create communications plan that identifies key messages, campaigns, tactics and partners. Incorporate lessons learned from past campaigns like #GetAjaxMoving as well as behavioural science insights. It can be a living document that gets updated as new lessons are learned and focus areas change.
A3 – Prepare Annual TDM Summary Report	A3.1 Create annual report from 2023-2027.

b) Partnerships

Action	Steps to Implementation
B1 – Leverage the Ajax Partnership Fund to encourage micro-grants to support and encourage TDM-related community projects.	B1.1 Update the Ajax Partnership Fund program criteria to encourage micro-grants to support and encourage TDM-related community projects. B1.2 Promote the program to known community partners who may be interested. B1.3 Issue funding grants, and work with grant recipients to implement programs and report on results.
B2 – Build Capacity Through Community Partnerships. Potential partner organizations can include: Businesses, Boards of Trade,	B2.1 Implement Ajax-led video campaigns partnering with community leaders and other partners promoting the benefits of sustainable transportation and building capacity among residents, with potential support from the Region. B2.2 Prepare a TDM Community Champions Toolkit to support community partners for awareness, education and marketing campaigns.

Action	Steps to Implementation
Places of Worship, Resident Associations, Community Development Council Durham, Durham Region Cycling Coalition, Accessibility Advisory Committee.	B2.3 Town of Ajax staff will collaborate with the Region to run a capacity building workshop to leverage and support community partners for awareness, education and marketing campaigns, including honoraria for community champions.
B3 – Collaborate with Durham Smart Mobility.	B3.1 Work with the Region to grow number of participating businesses in Ajax by using existing program resources, working with the Region to adapt or create new marketing materials, securing Smart Commute workplace champions, and establishing relationships between key TDM contacts at the Town, Region and Workplaces. B3.2 Work with the Region to identify topics for Workplace champion training/capacity building workshops. B3.3 Explore opportunities to work with the Region of Durham to coordinate Ajax-specific projects/campaigns at each member workplaces based on their needs and opportunities. **Cycle Durham:** B3.4 Increase participation in Bike Month in Ajax through community events, marketing campaigns using existing Bike Month supports from the Region. B3.5 Explore ways for the Region of Durham to support Ajax community/neighbourhood bike rides series. **Active School Travel** B3.6 Work with the Region to promote the Town's Active School Travel toolkit and explore ways to for the Region to support the Town's efforts in building capacity among Youth.
B4 – Transit Marketing and Promotion: Coordinate communications efforts to promote existing transit services.	B4.1 Durham Region Transit - Work together on marketing campaigns, engagement activities and incentive programs to increase transit use and promote new and existing transit services. B4.2 Metrolinx – Ajax GO Station- Work together on marketing campaigns, engagement activities and incentive programs to increase transit use and promote new and existing transit services.
B5 – Carshare	B5.1 Explore partnerships with carshare providers on the feasibility of carshare services in Ajax, and implement a pilot project. Carshare companies can include: • Communauto • Enterprise CarShare • Zipcar • Turo Also consider a community carshare model.

Action	Steps to Implementation
B6 – Shared Micromobility Pilot	B6.1 Consult with staff and the public on the feasibility of permitting and accommodating for shared e-bicycles and e-scooters. B6.2 If there is a desire to permit either shared e-bicycles or e-scooters or both, establish partnerships with vendors to implement a shared micromobility pilot program.

c) TDM Integration with Town Policies and Practices

Action	Steps to Implementation
C1 – Complete Streets	C.1.1 Develop a Complete Streets Design Manual specifically for the Town of Ajax that serves as a guide to staff for planning, designing and implementing multi-modal supportive transportation corridors, including adopting a Multi-Modal Level of Service (MMLOS) evaluation framework for transportation planning.
C2 – Bicycle Parking Implementation Plan	C2.1 Establish a bike parking request program to add bike parking in areas of need (through business requests and within the public right-of-way). C2.2 Proactively install short- and long-term bicycle parking within Town right-of-way near employment, retail, transit, and other key Town destinations. C2.3 Convert the spacing standards in the updated Ajax Bicycle Design Guidelines into typical Ajax Standards (AS-300 or AS-900 series), so that developers can select the type of bike parking style, the site scenario, and attach one of them into their site plan submission details.
C3 – Parking Policy Reform	C3.1 Provide more flexibility with parking minimums as part of next zoning by-law update, drawing upon best practices.
C4 – TDM in New Developments	C4.1 Develop a pilot project for residents in new developments to better understand their transportation options at the time of occupancy. Identify the barriers to a behaviour, construct a program to overcome these barriers, implement the program across the community, and evaluate the effectiveness of the pilot project. C4.2 Review lessons learned from pilot projects and scale it up to a program. Review the different governance models for such a program, such as delivery though a Transportation Management Association (TMA). C4.3 Encourage developers to provide transit passes to new residents or with rental/office agreements with a guideline of one month fare per pass.
C5 – Coordination with the Integrated Transportation Master Plan (ITMP) Update	C5.1 Combine the TDM Plan and IMTP into one Master Plan as part of the next update cycle to bring TDM policies to the forefront of transportation planning policy and actions. C5.2 Contribute to the refinement, finalization and adoption of Durham Region's TDM Development Guidelines for a coordinated TDM approach in development planning.

d) Transportation Equity

Action	Steps to Implementation
D1 – Inclusive Media Library	D1.1 Create a repository of photos promoting sustainable transportation that reflect the diversity and built environment of Ajax (will be done in 2023 as part of the Ontario Community Changemakers Grant). D1.2 Ensure all Town staff are aware and have access to the media library for use in Town reports, publications and communications. D1.3 Continue to expand the media library with more visual assets (images and video) and consider accepting submissions from community organizations and the general public.
D2 – Build Youth Capacity in Equity-Deserving Communities.	D2.1 Develop a plan for launching the Town of Ajax Active School Travel Advocacy Toolkit. D2.2 Facilitate a youth-based capacity-building workshop for improving walking, rolling and transit in equity-deserving communities. Provide honoraria to support community champion involvement.
D3 – Engage Equity-Deserving Communities.	D3.1 Implement one or more bike events targeting underrepresented groups. (This will be done in 2023 as part of a project funded by the Ontario Community Changemakers Grant). D3.2 Implement a driver safety awareness campaign centering pedestrians with disabilities. (This will be done in 2023 as part of a project funded by the Ontario Community Changemakers Grant). D3.3 Create a resident assessment panel to bring voice to the concerns of residents and ensure that a broad diversity of opinions are considered as the TDM Plan moves forward.
D4 – Investigate the feasibility of creating a bike library.	D4.1 Create a business case for the creation of a Bike Library, preferably at the libraries and other Town Facilities, as feasible. D4.2 Review possible sites for bike storage and review administrative capacity for lending bicycles. D4.3 Seek funding for procuring bicycles and accessories.

Financial Implications:

There are no financial implications as a result of the recommendations of this report.

The funding requirements associated with the implementation of 2023 TDM Plan Update will be included in the Town’s future capital and operating budgets for Council deliberation and consideration.

4.0 Communication Issues:

Public engagement methods prioritized advancing transportation equity by engaging community organizations and groups who have traditionally been left out of the TDM policy development process through a combination of in-person and virtual events and activities.

Engagement methods were informed by:

- The **International Association of Public Participation (IAP2) process** emphasizes the importance of a tailored Community Engagement Strategy based on the understanding, commitment, and contribution of each unique group of stakeholders.
- Principles of the **Dignity Infused Community Engagement (DICE)**, which focuses on building trust and respect within marginalized communities by building coalitions, valuing existing work in those communities, building capacity, and thoroughly vetting recommendations to ensure they provide an environment of dignity.

Method	# of people engaged	Key Findings
IMO page and survey	• 910 page visitors • 252 survey respondents	Resident priorities that can be addressed through TDM: • Improving secure bicycle storage across Ajax; • Making cycling wayfinding signage and maps more accessible; and • Support bicycle skills and education programs. **Supply-side** Resident priorities for transit and active transportation • Building more bicycle lanes to connect employment, commercial and recreation areas to residential neighbourhoods; and • Improving cycling safety at intersections. Durham Region Transit • Increasing the frequency of buses; • Improving route directness; • Improving connections to GO Transit service; • Extending transit coverage area; and • Adding more bus stops across existing routes.
Social Climate Analysis: Interviews with Community Leaders	10 community leaders engaged: • Durham Region Transit • Ajax Masjid Quba	Communities across Ajax have established community-based transportation systems and programming designed to help residents, particularly those who do not own personal vehicles, get around easier.

Method	# of people engaged	Key Findings
	• Southside Worship • Christian Faith Outreach Centre (2 representatives engaged) • Kids/Seniors on Wheels • The Lighthouse Retreat • Ajax/Pickering Hospital Foundation • Community Care Durham • Womxn of Colour Durham Collective	There is active and ongoing collaboration between many of the community groups: • The Christian Faith Outreach Centre has worked with Seniors on Wheels to ensure that aging populations can access church provided services. • The South Side Worship has partnered with Care Mongers to provide foodbank delivers to residents. Community leaders identified a need for • Mobility options for workers (such as hospital support staff and personal support workers), seniors, and vulnerable residents during non-traditional commuting hours. • Increased communication when changing bus stops, routes, and service frequency. Moving stops and reducing frequency has created barriers for seniors, volunteers, and other residents in accessing to services such as food banks and support services for people with dementia. • A holistic list of all transportation service providers, regardless of whether they are formal or informal services. Despite many organizations working together with one another, there is still a lack of familiarity with what options are available for residents throughout the Town.
Pop Up Events with postcards containing the QR code directing to the survey and physical copies of the survey in large font	Engaged over 200 residents while promoting the IMO survey at the following locations: • August 11, 2022: - Lighthouse Retreat • September 23, 2022: - Masjid Quba during afternoon prayer	Many of the results are captured in the survey results. Notable insight gained from the pop-up booths include the following: • There are Recreation staff who work late night shifts without access to a vehicle. • There is high traffic at Masjid Quba as well as many residents who share their support networks (i.e. people willing to carpool).

Method	# of people engaged	Key Findings
	- McLean Community Centre - Ajax GO Station	
Virtual Open House: January 16, 2023	15 attendees	Improvements that would make it easier for residents to walk, bike, or take transit more often: Biking • Secure bike storage, including bike lockers at the Ajax GO Station • Maintenance tools • Safe connectivity • Cargo bike storage • Wayfinding on trails E-bikes and e-scooters • Charging stations • E-bike subsidy for everyone Other improvemenents • Paved transit stops with shelters • Improvements to traffic and pedestrian signals Metrics to define success • Busy trails • Youth retention, not moving to cities for more transportation options • People of all ages, races, and abilities biking • Modeshare and culture normalizing walking, cycling and transit Other concerns brought up: • Supply side solutions are also important: removing a bus route creates significant barriers for people who are blind and who do not live in an area designated for On-Demand service • Some residents in support of e-bikes and e-scooters, others not in favour
Agency Stakeholder Meetings: • **October 12, 2022**	Internal stakeholders present: • Transportation Services • Environment and Climate Change	High Impact Actions on the Supply side: • Improving walking and cycling access to the Ajax GO Station. • Providing active transportation amenities.

Method	# of people engaged	Key Findings
• **December 8, 2022**	• Policy Planning • Development Planning • Economic Development • Finance • Infrastructure and Asset Management • Facility Management • Road Operations and Maintenance • Ajax Library • Communications • Public and Strategic Affairs External stakeholders present: • Durham Region Transit • Metrolinx Stations Planning Team • Durham Region Planning and Economic Development • Durham District School Board • Conseil scolaire catholique MonAvenir	Measures that can be addressed through TDM Update: 1. Encouraging Active School Travel, preferably through a designated full-time staff member who creates AST programs and partnerships across the Region (if there is interest, budget, and resources at the school boards). 2. Enhancing partnerships and collaboration: - Support workplace TDM champions. - Leverage community groups and their community networks. - Strongly encourage development applicants to implement TDM measures. 3. Integrating TDM into policies, processes, and programs. 4. Coordinating with developers and property managers to distribute TDM information to residents encouraging sustainable transportation.

5.0 Relationship to the Strategic Plan:

A new 2023 to 2026 Strategic Plan is currently in development.

6.0 Conclusion:

The vision of the 2023 TDM Update is mobility choice that meets the needs and abilities of residents and visitors through a shift to sustainable transportation.

The Town of Ajax is home to an increasingly diverse population with different identities, lifestyles, talents, capacities, household compositions, needs, and aspirations. It needs a transportation system that allows residents to access local services, programs, resources, activities, and opportunities regardless of whether they are able to drive a car.

Like other municipalities across Ontario, the Town of Ajax has maintained a transportation system prioritizing the movement of automobiles for decades. This placed significant financial costs for residents to access job opportunities and necessary services, creating isolation among people who cannot drive or who do not have access to a vehicle or driver. Further, designing streets in favour of the automobile has led to increases in pedestrian deaths and serious injury while increasing costs to expand and maintain roadway capacity.

Experience in implementing the 2015 TDM Plan, the process of engaging residents, as well as awareness of health and equity concerns that became further pronounced during the Novel COVID-19 pandemic reveal that more transportation options are needed beyond the 9-5 commute.

In developing the 2023 TDM Update, the Town of Ajax used a more holistic approach to mobility that considers a wide variety of trip purposes in addition to trips to/from work outside of the 9-5 shift. This approach involved greater collaboration with community leaders and stakeholders and uses existing strengths and assets in the community and at the Town.

The 2023 TDM Plan Update includes:

- An improved understanding of resident needs and best practices with consideration to trends from the COVID-19 pandemic.
- Updated Development Guidelines that are better integrated with new and incoming guidelines and by-law update.
- Updated Bicycle Parking Guidelines with guidance on accommodating for cargo bikes, bicycles with strollers, and adaptive bicycles for people with disabilities while clarifying requirements for a larger diversity of land uses.
- Actions for the next five years that involve building foundations, strengthening partnerships, integrating with other Town policies and practices, and working towards transportation equity.

Transportation staff will continue to revisit and update the Transportation Demand Management Plan every 5 years and will report back to Council at each update. The next update is anticipated to take place in 2028.

Attachments:

ATT-1: 2023 Ajax TDM Plan Update Draft/Final Report

Prepared by:

Armi De Francia, MCIP, RPP – Active Transportation Coordinator

Submitted by:

Geoff Romanowski, CPT, MCIP, RPP – Director of Planning and Development Services

Approved by:

Shane Baker – Chief Administrative Officer

Town of Ajax

Transportation Demand Management Plan

March 2023 (Draft)

#GetAjaxMoving

This document is available in alternative accessible formats upon request by contacting 905-619-2529, ext. 3347, or email accessibility@ajax.ca

TOWN OF AJAX
TRANSPORTATION DEMAND MANAGEMENT PLAN

MARCH 2023

PREPARED BY:
WSP Canada Inc.

PROJECT TEAM
Armi De Francia, Project Manager
Dhaval Pandya
Ayca Kocakusak Sezgin
Cameron Richardson

CONSULTANT TEAM
Shawn Smith, Project Manager
Pegah Tootoonchian
Enya Huang
Justin Jones
Kyle Dow
Tony De Crescenzo

#GetAjaxMoving

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1 INTRODUCTION

A New Direction for Ajax

Vision

Goals

Organization

1.1 | A New Direction for Ajax

The Town's Transportation Demand Management (TDM) Plan was first launched in 2015 to help alleviate traffic congestion and encourage the use of more sustainable travel modes. With the benefit of 7 years of experience implementing the 2015 TDM Plan, Town staff have a strong understanding of what measures have an impact and what measures often fail to influence change. There has also been an increased need for more transportation options. The Transportation Demand Management Plan Update responds to this need using demand-side solutions.

Driving is not an option for up to 30% of Ajax's population. Children, seniors, people with disabilities and people who cannot afford a car still need to get to everyday destinations in Ajax. By making efforts today to support mobility choice in Ajax, the Town is building a more resilient transportation system that allows all residents to enjoy the quality of life and amenities that the Town has to offer. The Town recognizes that not all residents can drive, and not everyone has people in their lives who have the time or means to drive them to the places they need to go. To help improve access for residents of all ages and abilities, the Town will continue to expand access to safe, convenient and reliable mobility options.

FIGURE 1.1: Street space taken up by 60 people by car, bus, and bike. (Source: City of Muenster Planning Office, August 2021)

If residents are able to drive and have access to a vehicle, residents can still choose to drive. The TDM Plan update simply aims to make it easier for residents to choose other forms of mobility to get around Town. It recognizes that with the Town's growing population, a transportation system that maintains current travel patterns will result in either significant congestion on local roads or expensive expansions to roadway capacity, neither of which will support the Town's long-term objective of creating inclusive, thriving communities.

The Town's overall goals for transportation, as expressed through this update and the Integrated Transportation Master Plan, reflect a desire to make it easier for residents who

cannot drive or who choose not to drive to move through the community safely and conveniently. By improving mobility choice, the Town makes it easier for children, seniors, people with disabilities and people whose self-reported incomes were in the lowest income bracket to access their community independently. Providing more convenient options for pedestrians, cyclists, transit users, and people using other modes of transportation expands the number of choices residents have available to get around Ajax.

This Plan provides a blueprint for change, outlining programs, policies, and partnerships that can help realize a more balanced, efficient and equitable transportation system.

1.2 | Vision

Mobility choice that meets the needs and abilities of residents and visitors through a shift to sustainable transportation.

1.3 | Goals

The TDM Plan is meant to achieve the following goals:

1. **Improve Mobility Choice:** provide opportunities and access to alternative modes of travel, including transit, active transportation, ridesharing, and micromobility to minimize dependency on single-occupant automobile use. It is important to ensure that the Town is providing an equitable and efficient transportation system that supports the travel demands of residents to and from key destinations.
2. **Build a Culture of Sustainable Transportation:** change travel behaviour and create an environment that celebrates and recognizes active transportation, transit, and ridesharing as accessible, feasible, safe options.
3. **Collaborate with stakeholders and community leaders:** strengthen partnerships on a range of TDM initiatives locally and regionally to leverage the knowledge, skills and connections of others and increase impact.

4. **Improve transportation equity:** expand mobility choice where car-centric design of the past has created barriers for disproportionately impacted communities to access essential services, programs and opportunities.
5. **Measure and communicate the impacts of TDM:** evaluate the TDM initiatives and communicate the outcomes to staff, Council and the public to build support for continued investment.

1.4 | Organization

The TDM Plan is structured into the following chapters:

- **Chapter 1** provides an introduction to the TDM Plan.
- **Chapter 2** provides an overview of TDM and the steps to implementing a TDM strategy.
- **Chapter 3** outlines the policy framework for TDM implementation in the Town of Ajax.
- **Chapter 4** provides an overview of the methodology used in developing the plan including jurisdictional scan, background review and engagement program undertaken.

- **Chapter 5** summarizes lessons from best practices, including strategies on sourcing funding, alternative work arrangements, and integrating TDM in new developments.
- **Chapter 6** summarizes current travel trends that influence TDM opportunities in the Town.
- **Chapter 7** identifies the TDM opportunities and initiatives which could help address the Town's transportation challenges.
- **Chapter 8** provides TDM strengths and accomplishments at the community and Town scales, including mobility of care network, school efforts, and completed TDM actions.

- **Chapter 9** presents the Action Plan of recommended TDM interventions that the Town will seek to implement over the next five years.

The report also contains the following appendices:

- **Appendix A: Engagement Summary and Public Survey Findings** summarizes the results of the public survey conducted in developing the TDM Plan.
- **Appendix B: Bicycle Parking Design Guidelines 2022 Update** provides planners, developers and property owners with information to support the design, installation and management of high quality bicycle parking facilities within the Town.
- **Appendix C: Supplement to Durham TDM Development Guidelines.** Supplemental checklist for Town of Ajax for the planning and design of new neighbourhoods and development sites.
- **Appendix D: TDM Best Practices Review.** Summary of jurisdictional scan to inform the TDM Plan Update.

2 WHAT IS TRANSPORTATION DEMAND MANAGEMENT?

Defining TDM

Steps to TDM Implementation

Potential Impact of a Comprehensive TDM Program

2.1 | Defining TDM

TDM is defined as the application of strategies and policies to influence travel behaviour, with the objective of both reducing overall demand, especially from single-occupant vehicle use, and seeking to influence when and where this demand occurs. TDM concentrates on changing the travel behaviour to encourage greater use of sustainable modes of transportation and trip decision-making that reduces, combines, or shortens vehicle trips. Therefore, it can be considered either in tandem with, or distinct from, infrastructure or service improvements. **Figure 2.1** illustrates the different ways to influence travel decisions through TDM.

There are numerous TDM strategies to influence travel decisions, typically summarized into four categories:

- **Options** – Expand the range of viable transportation options through awareness and education campaigns, apps that provide travel information or connect people to existing services, and programs that offer new ways to travel.
- **Incentives** – Provide incentives to change travel mode, time or destination.
- **Accessibility** – Improve land use accessibility to facilitate greater use of more sustainable transportation modes.
- **Policy** – Introduce and/or reform transportation policies and programs to foster travel by more sustainable modes.

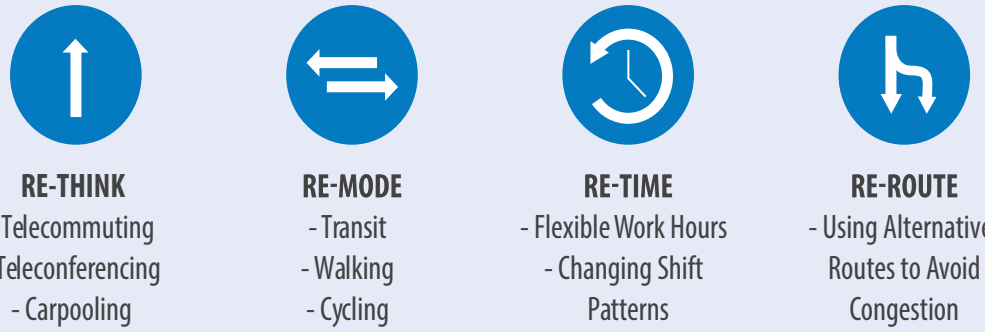


FIGURE 2.1: Applying TDM to Influence Travel Decisions

TDM Includes Demand-Side Solutions

- Technology-based devices to help people use transit, ridesharing, walking, biking, and work from home. This improves the convenience of these options while also notifying users of challenges to help them prepare and adjust.
- Policies that help create an environment that more naturally encourages the use of sustainable transportation modes. This could include requiring or encouraging developers to provide amenities and features such as of bicycle parking, pedestrian walkways, charging stations for electric mobility aid devices or e-bicycles or e-scooters, or lighting.
- Outreach, education and marketing campaigns to help people better understand their transportation options and use the infrastructure in place more effectively, and change social norms.
- Incentives to "nudge" people to other modes instead of driving alone to reach their destinations. Examples include pre-loaded transit passes, points or discounts for walking or biking.

TDM Does Not Include Supply-Side Solutions

- Providing more frequent transit service.
- Building new cycle infrastructure.
- Changes to the road network.

Transit

Public bus and rail services hold significant potential for reducing the number of people driving to work and other destinations. The integrated electronic fare system of the Greater Toronto and Hamilton Area's (GTHA) transit network (the Presto Card) and accompanying marketing offers a great opportunity to see a step-change in transit use as well as bundle with other transportation services.

Typical benefits of using transit include:

- Enables people to use their time more effectively to work or relax.
- Makes more efficient use of fuel compared to single-occupant vehicles.
- Reduces personal vehicle ownership/operating costs.
- Reduces risk and economic cost of traffic collisions.
- Reduce greenhouse gas emissions and improves air quality.

Walking and Cycling

Walking is the simplest and most sustainable form of transportation and includes people who use mobility aid devices. It carries zero cost, is versatile and is impervious to congestion or delay on the roads. Except in the most extreme temperatures, walking is a viable means of travel for all short trips (subject to the health and mobility of the individual).

In the past, cycling was overlooked as a significant travel mode within the Ajax transportation network, primarily due to the local climate, the need for the municipality to set infrastructure and spending priorities with limited funds, and development designs that did not envision separate or dedicated bicycle facilities. However, in recent years, there has been significant progress in developing cycling facilities and services, resulting in increasing interest across the Town. Cycling is cheaper, healthier, and at times, quicker than driving, especially over shorter distances and during

FIGURE 2.2: Different Types of Bicycles considered in Town of Ajax's Bicycle Parking Guidelines

peak hours. Cycling can be used by all ages and abilities including the use of adaptive bikes for people with disabilities.

Across the globe, governments and societies are increasingly recognizing the valuable contribution that walking and cycling can make in tackling congestion, reducing journey time delays and improving air quality. Other benefits of walking and cycling include:

- Very cost efficient forms of transportation and require only modest expenditures (relatively) on footwear, protective equipment and bicycles.
- Potentially saves time for commuters when it is congested on roadways.
- Potential to replace many of the shortest auto trips.
- The healthiest ways to travel. They improve fitness and personal health, and increase opportunities for social interaction, all of which provide wider societal benefits. It is statistically proven that people who have healthier lifestyles have lower absentee rates from work.

Carpooling and Staff Shuttles

Carpooling and staff shuttles represent two forms of ridesharing. In addition to its ability to reduce traffic congestion, carpooling creates significant reductions in fuel consumption for private automobiles and diminishes costs associated with delays.

Trips that can be easily accommodated by carpooling are typically those that are undertaken most often and with a pattern of routine. This means the best journeys for carpooling are:

- School trips.
- Commuting trips.
- Regular trips to shops, the gym, places of worship or other commitments.

Carpooling is most effective when it is undertaken on a company-wide or office-wide/specific location basis, with formal monitoring and website support to "match" appropriate people. Although there are some start-up costs, it should largely be a self-sustaining system once fully operational.

Smart Commute has a ride-matching tool that can be used by Ajax employees and businesses at *www.smartcommute.ca*.

Staff shuttles are an effective way of transporting large numbers of people between fixed sites on a fixed rotation. They are most effective at catering to staff that work fixed hours/shift patterns and are commonly used to address transportation demands at the world's busiest airports, manufacturing facilities, power plants and other large scale, single site developments.

Shuttle bus services are most appropriate for larger employment bases where many people can be transported at once, and where the investment in vehicles is offset by the benefits achieved.

Typical benefits of carpooling and staff shuttles include:

- Fuel cost savings.
- Reduced parking requirements for employers.
- Reduced vehicle emissions, which means reduced pollution and improved air quality.
- Reduction of stress levels when using alternative modes of travel instead of driving a single-occupancy vehicle.
- Fewer vehicles on the roads, which reduces congestion and delays and helps the economy to prosper.
- Reductions in localized congestion and unnecessary mileage caused by drivers searching for parking.
- Easy to set up and appropriate for all organizations, large or small.
- Travel benefits such as use of high-occupancy vehicle lanes and preferential parking.
- Some sites offer additional benefits to staff, such as free car washes or enabling staff to "clock in" as they board the bus.

Remote and Flexible Working

The ability to change the way people work– whether by changing the hours they travel ("Re-Time") or enabling a certain level of remote working ("Re-think") – is a very effective way of tackling congestion. After all, there is no more effective way of reducing the number of autos on the road than by reducing the number of trips made in them.

Compressed Work Week Example

If a company currently employs 1,000 people full time and 800 of them drive to work from Monday to Friday, this equates to 8,000 individual auto trips a week. If the company launched a flexible working policy that enabled each staff member to work compressed hours, where employees could fulfil their contractual requirement to work 40 hours per week (or the appropriate standard maximum) by working nine (9) days over a two-week period, and taking the remaining day off, this would reduce the number of auto trips made by 800 per week (10%).

For employees based in offices that do not require regular "face-time" with customers or co-workers, working remotely or making small changes to the hours worked can have a real impact. In today's world, remote and flexible working is easier and more cost efficient than ever. Provided staff have the tools they need to do their job effectively – which usually only requires a laptop, a phone and internet access – then they can be logged in almost anywhere. Of course, there is no substitute for on-site presence for certain types of work, and such a policy is not suitable for some sectors of employment such as shift work, and customer service for instance.

In most instances, remote and flexible working arrangements:

- Support a healthy work/life balance.
- Reduce the number of vehicles on the road during the busiest times.
- Provide fuel cost and vehicle depreciation savings.
- Reduce congestion and delay and help the economy to prosper since there are fewer vehicles on the roads.
- May help in retaining high performing staff as a form of benefit.

While supply-side solutions, such as increasing the frequency of transit and

expanding active transportation networks, are essential to shifting travel behaviours, demand-side solutions are also necessary. Focusing on supply-side solutions is incomplete. People need to be aware of the routes and resources available to them. TDM addresses gaps that cannot be addressed by supply-side solution, such as policy focusing on sustainable transportation modes, technology, and education or marketing programs. These create better understanding around access to transportation options. The Town's TDM Plan focuses on the "less-necessary" vehicle trips and taking action to support a shift to the alternatives modes available for use.

2.2 | Steps to TDM Implementations

There may be several steps between a particular TDM policy or program and its desired outcomes, as illustrated in **Figure 2.3**. Although many TDM strategies have modest impacts, only affecting a few percent of total trips, their impacts are cumulative and synergistic (total impacts are larger than the sum of individual impacts). As such, a comprehensive TDM program can often affect a significant portion of total travel and provide large total benefits. It is therefore important to plan and evaluate integrated TDM programs rather than individual strategies.

The TDM Plan that follows provides types of integrated, comprehensive strategies required to elicit the desired change in travel behaviour over time. The full effect and benefits of the measures conceived and recommended today may not be felt in the short or immediate term, but will set the Town on a path of evolutionary change necessary for the community to address transportation demand while at the same time becoming more sustainable in the future.

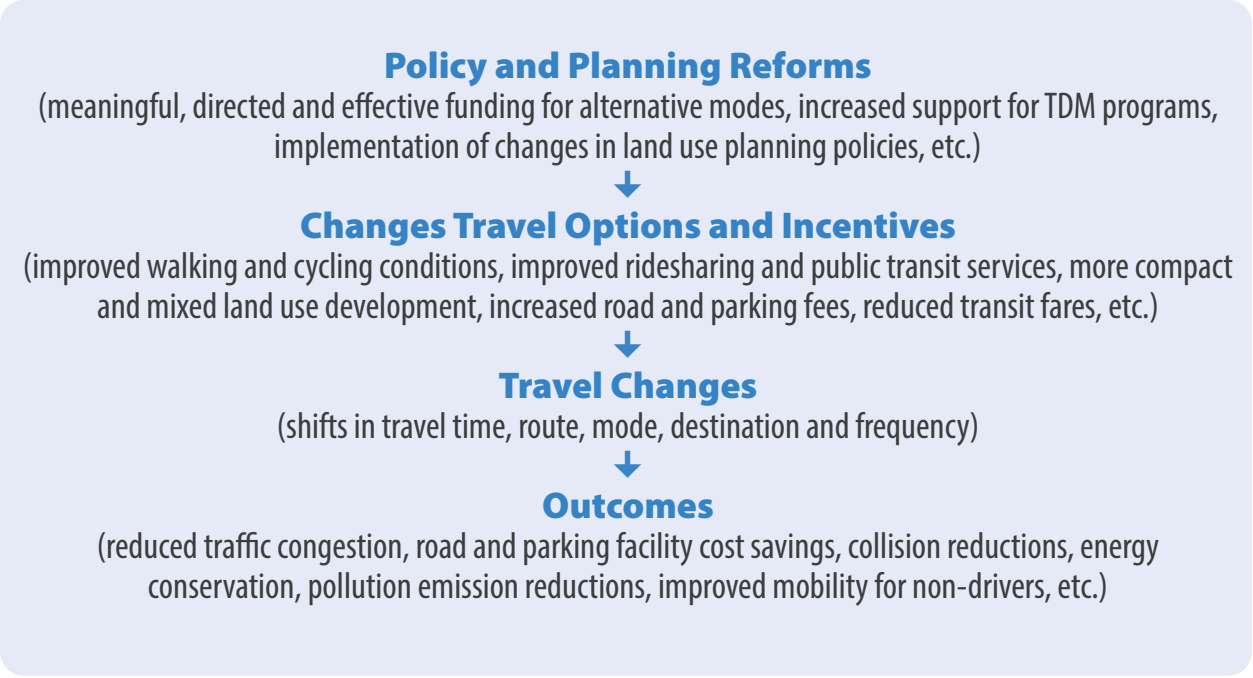


FIGURE 2.3: Steps to TDM Implementation (Source: Victoria Transportation Policy Institute Website, accessed on 2015-04-16 at *http://www.vtpi.org/tdm/tdm12.htm*)

2.3 | Potential Impact of a Comprehensive TDM Program

As previously noted, the objective of a TDM program is to maximize the movement of people rather than vehicles. Many TDM strategies are directed at reducing auto travel and, in particular, single-occupant vehicle trips.

The potential impact of TDM strategies and measures is difficult to quantify and is dependent upon a number of variables including but not limited to:

- Availability and quality of transit services and active transportation facilities.
- Community structure.
- Current TDM strategies and measures in place within the community.
- Overall community awareness and buy-in of TDM benefits.

The research shows anecdotal evidence of the success of various TDM strategies and measures. **Table 2.1** denotes the auto trip reduction potential of different TDM measures based on the experiences of other jurisdictions across North America. As seen in the table, the results of the research suggest that a reduction in automobile trips can be anticipated when a municipality has established a well-planned and coordinated TDM Plan.

To be noted as part of the results, the ridesharing TDM measure may reflect potential overlap with transit services. This overlap however, does not take away from the importance around the availability of transport options beyond the automobile.

TABLE 2.1: Potential Reduction in Automobile Trips from TDM Measures (Exclusive of Transit)
(Source: Region of Waterloo "Travel Wise" Strategies Overview, Regional Transportation Master Plan)

TDM Measure	Short Trips (<10km)		Long Trips (>10km)	
	Short to Medium Term (within 10 years)	Long Term (beyond 10 years)	Short to Medium Term (within 10 years)	Long Term (beyond 10 years)
Improved Land Use and Transportation Integration	1%	2.5%	1%	2.5%
Ridesharing (numbers reflects potential for overlap with transit	2%	2%	3%	3%
Walking/Cycling	4%	5%-8%	Minimal	
Telecommuting	0.5%	0.5%	1%	1.5%

Note: The COVID-19 pandemic has had unprecedented effects on how people travel in the Town of Ajax. The implementation timeline of actions within this plan are dependent on the length of the COVID-19 pandemic and the recovery process.

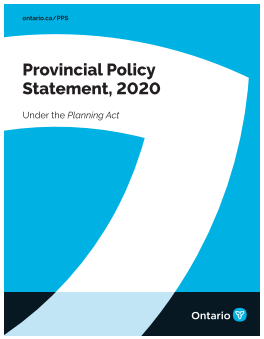
3 POLICY FRAMEWORK

Province of Ontario
Durham Region
Town of Ajax
Policy Summary

This chapter outlines the strategic policy framework for the implementation of TDM in Ajax and provides the basis of the TDM Plan.

3.1 | Province of Ontario

Provincial Policy Statement (2020)



The Provincial Policy Statement, 2020 (PPS) provides policy direction on matters of provincial interest related to land use planning and development. The policies of the PPS, provide a framework for a comprehensive, integrated, place-based and long-term planning that supports and integrated the principles of strong communities, clean and healthy environments, and economic growth.

As part of the Transportation Systems policy framework, Section 1.6.7, the PPS articulates the following reference to Transportation Demand Management for municipalities across Ontario:

Efficient use should be made of existing and planned infrastructure, including through the use of transportation demand management strategies, where feasible.

Places to Grow: Growth Plan for the Greater Golden Horseshoe (2020)

The Places to Grow: Growth Plan for the Greater Golden Horseshoe (GGH) (Growth Plan, 2020) provides policy objectives to guide the planning and development of an integrated and efficient transportation that supports economic development and high quality of life.

The Growth Plan includes policies for infrastructure to support growth. Part of this includes equipping municipalities with tools to develop and implement transportation demand management policies in their Official Plans and other Planning documents. The scope of Transportation Demand Management policies in the Growth Plan is to reduce reliance on single-occupant vehicle use.

The Transportation Demand Management policies stemming from the Growth Plan are focused on:

1) Reducing trip distance and time

2) Increasing the modal share of alternatives to the automobile

3) Prioritizing active transportation, transit, and goods movement over the use of single-occupant vehicles

4) Expanding infrastructure to support active transportation, and,

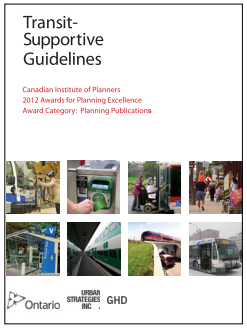
5) Considering the needs of major trip generators

2041 Regional Transportation Plan (2018)

The 2041 Regional Transportation Plan (RTP) outlines a framework for governments and transit agencies in the Greater Toronto and Hamilton Area (GTHA) to continue improving traveller experience in the region in combination with strategies such as active transportation connections, transportation demand management, climate change resiliency, and intelligent transportation systems. The transit strategies and priorities highlighted in the RTP are developed against

transit ridership growth and accessibility to employment. The ultimate goal of the RTP is to integrate seamless into transportation systems across the GTHA, offer enhanced mobility choices, improving access to reliable and frequent service, and embedding more affordability into transit and reducing car-dependency.

Transit Supportive Guidelines (2012)



The Ministry of Transportation (MTO) has developed a suite of policies and guidelines that encourage the promotion of sustainable transportation across the Province.

The Ministry's Transit Supportive Guidelines, published in 2012, were developed to provide "tools and strategies to create a more compact land use pattern supportive of walking and cycling, optimize the effectiveness of existing or future transit services, retain ridership and better target transit service. In mid-size or larger cities, the guideline will help to better utilize existing infrastructure, grow ridership and manage urban growth in a more transit-supportive manner." Some examples of specific policies are listed below:

Article 2.5.1 (Parking Management):

- Require large developments, institutions and employers to submit TDM strategies as a component of the site plan approval process.
- Encourage existing land uses to implement TDM strategies by showcasing local best practice examples.
- Reduce parking allocations once TDM is in effect.

Article 3.4.2 (Access for Cyclists):

- Develop community support by engaging local bicycle groups, students, public organizations and environmental groups.
- Build support by increasing active transportation and development of TDM.

Guideline 3.5.3 (Targeting Ridership Through Partnerships):

- Provide transit routes to meet the needs of employees.
- Implement workplace TDM measures.

Such measures are repeated throughout the document, and are intended to apply to education, recreation, commercial and health facilities as well as workplace and residential locations.

#CycleON 2.0 Action Plan (2017)

Action Plan 2.0 is the second wave of initiatives that are intended to achieve #CycleON: Ontario's Cycling Strategy (2013). #CycleON is a 20-year plan, promoting cycling across Ontario through designing more cycle-friendly environments.

Action Plan 2.0 guides efforts across ministries, strategies, and initiatives to ensure collaboration towards coordinated and complementary cycling infrastructure. The intent of this Plan is to shift provincial initiatives to build on the success of the Action Plan 1.0 with the following overarching priorities:

1. Design Healthy, active and prosperous communities.

2. Improve Cycling Infrastructure.

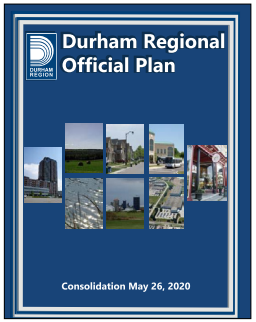
3. Make Highways and Streets Safer.

4. Promote Cycling Awareness and Behavioural Shifts.

5. Increase Cycling Tourism Opportunities.

3.2 | Durham Region

Durham Regional Official Plan (2020)



The Durham Regional Official Plan is a comprehensive policy framework for land-use decision making processes and priorities of the Region. The Regional Official Plan sets the context for detailed planning and policy for the local municipalities, including the Town of Ajax.

Section 11 of the Durham Regional Official Plan (Transportation System) highlights a series of goals and policies including:

1. Providing a transportation system that is integrated, safe, efficient, reliables for all users and modes.

2. Offering a variety of mobility choices for all Durham residents.

More specifically, Policy Section 11.3.16 is for Transportation Demand Management, supporting the development of a comprehensive TDM program to reduce single-occupant vehicle dependency by promoting alternatives modes of transportation (transit, carpooling, cycling and walking, and alternative working arrangements).

Durham Transportation Master Plan (2017)

The Durham Transportation Master Plan (TMP) is a multi-modal strategic plan that defines the policies, programs, and infrastructure required to manage future growth and the anticipated transportation demands. With 60% of morning peak trips anticipated to remain within the Region, a change in travel patterns will require an emphasis on transportation improvements focused on shorter and localized trip patterns. The TMP includes seven key directions, one of which is to Promote Sustainable Travel Choices with TDM measures being fostered in planning approval processes.

Durham Region Transit: The Route Ahead – Service Strategy 2022-2025

Durham Region's goal in the next three years is to adapt to the changing needs of customers by delivering efficient and effective mobility solutions, with a sustainable approach. The Route Ahead encompasses five key pillars, including:

1. Availability: accessing transit when needing to travel, with service available in all areas of the Region.
2. Innovation: adapting flexible service delivery models to customer needs and travel patterns, leveraging technological solutions.
3. Reliability: transit priority measures that focus on improving reliability while reducing travel time. Adjusting service to new travel patterns post-pandemic.
4. Safety: improving customer environment on-board vehicles, at stops, and in stations.
5. Growth: providing transit-first

solutions to new communities as population and employment grows in the region, supporting economic development.

To continue supporting initiatives focused on creating compact and transit-friendly communities, the Route Ahead is focused on integrating broader strategic plans such as Durham Region's Strategic Plan, the Transportation Master Plan, the Metrolinx RTP, and the Greater Golden Horseshoe Transportation Plan. As the ridership market continues to experience changes due to the Covid-19 pandemic, structural changes in travel patterns is required to support ridership recover and continued growth to 2025.

The Regional Cycling Plan (2021)

The Regional Cycling Plan 2021 provides Durham Region and the area municipalities with the tools and guidance needed to support with the planning, design, implementation, and promotion of a connected, safe, and accessible cycling network.

Durham Region's TMP Update introduced the Primary Cycling Network (PCN) concept, with the hopes of achieving connectivity to area municipalities with on- and off-road cycling networks.

To be a leader in healthy community development, safe road connections, improved personal mobility, economic development, and reduced carbon emissions, the 2021 Cycling Plan is a blueprint for the next 10+ years to guide cycling infrastructure planning and design, programming, and policy development.

The Plan has also been coordinated to align with and support some of the Region's Strategic Goals identified in the 2020-2024 Strategic Plan. They include: expanding sustainable and active transportation, promoting economic prosperity by enhancing transportation networks to better connect people and move goods effectively, and optimize resources and partnerships to achieve service excellence.

3.3 | Town of Ajax

Town of Ajax Official Plan (2022)

The Town of Ajax Official Plan (OP) is a policy framework for managing land use planning and aiding development within the Town. The Ajax OP establishes principles, goals, and policies that govern long-term growth, in accordance to with provincial policy statements and conformity with provincial plans. The 2022 OP establishes the context for future urban structure and managing the built environment with policies focused on sustainable growth, preservation of natural heritage features, employment opportunities, and fiscal resiliency. The Ajax OP should be read in conjunction with Durham Region's Official Plan.

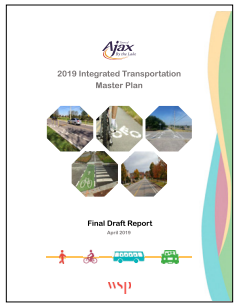
Section 4.0 of the OP includes several goals, objectives, and policies that support TDM, including:

- Supporting TDM initiatives that promote alternative modes of transportation

and increase transit ridership as well as active modes as a means of achieving vibrant streets. Policies in support of this goal include supporting initiatives that encourage developers and employers to utilize TDM strategies such as carpooling, vanpooling, and preferred parking spaces for participants.

- To establish a road network that provides safe, efficient, and reliable movement of people and goods using a variety of transportation modes. This includes encouraging the Province and Durham Region to develop High Occupancy Vehicle (HOV) lanes along major freeways and associated links to support transit, vanpooling, carpooling, and other TDM measures.

Integrated Transportation Master Plan (2019)



The Town of Ajax's Integrated Transportation Master Plan (ITMP) is a multi-modal planning tool that defines a comprehensive mobility network, addressing growth in the Town with infrastructure recommendations focused on a combination of active transportation, transit, and automobile. The ITMP also encompasses an approach focused on strengthening Transportation Demand Management with a wide-range of policies, programs and services that encourage increased reliability on sustainable travel behaviour. The ITMP addresses the supply side, which benefits TDM.

In this ITMP, it is recommended the Town take a three-pronged approach to changing travel behaviour, including: infrastructure, promotion, and normalization. The three-pronged approach includes short-, medium-, and long-term policy recommendations in support of TDM change.

3.4 | Policy Summary

Sustainable transportation is a recurring theme in most policies and strategies outlined in Provincial, Regional, and Municipal frameworks. The policy documents reviewed highlight the importance of TDM playing a vital part in addressing the growing issues of single occupant vehicle dependency and the associated challenges around traffic congestion, air pollution, road safety, and inactive lifestyle. The policy frameworks noted in this chapter include goals and objectives that can be leveraged to support the Town of Ajax's goals of shifting travel behaviour and incentivizing alternative modes of travel.

4 METHODOLOGY

Jurisdictional Scan

Review of 2016 TTS Data

Engagement

- Social Climate Analysis
- InMyOpinion Webpage and Survey and Promotion
- Agency Meetings
- Virtual Public Open House

This chapter outlines the methodology used to develop the Transportation Demand Management Plan Update. Chapters 5-8 provide the results on methods conducted.

4.1 | Jurisdictional Scan

The purpose of the jurisdictional scan is to learn about and discuss measures that can be adapted to thrive within the context of Ajax's communities. This jurisdictional scan reviews ten case studies which were selected for their similar geographical and municipal make-up (single-tier versus two-tier) to the Town, and some for their innovative approaches that could serve to be valuable consideration upon which the Town's TDM Plan can be updated. The jurisdictions range throughout Canada and internationally.

The case studies reviewed were:

1. Metrolinx Transportation Demand Management Background Paper
2. Region of Waterloo Transportation Demand Management Program Plan
3. City of Kitchener Integrated Transportation Master Plan
4. Peel Region Sustainable Transportation Strategy
5. City of Brampton TDM
6. City of Ottawa TDM Strategy
7. City of Whitehorse TDM Plan
8. City of Coquitlam
9. Arlington Country Commuter Services TDM Plan
10. San Francisco TDM Plan

The results are discussed in **Chapter 5** and **Appendix D**.

4.2 | Review of 2016 TTS Data

Data from the 2016 Transportation Tomorrow Survey (TTS) was analyzed to provide a "snapshot" of current travel behaviour within the Town of Ajax, and a baseline for bench-marking and measuring future progress in TDM implementation.

The TTS is a comprehensive travel survey undertaken in the Greater Golden Horseshoe (GGH). Durham Region is encompassed within the GGH and the survey is conducted every five years that collects detailed demographic information and a travel diary (for persons over the age of 11) for all members of a surveyed household for an entire weekday. The data is then expanded based on population counts by age from the Statistics Canada Census to represent the total population of the survey area, thereby providing a precise match in population totals between the Census and TTS at the municipal level.

The results are discussed in **Chapter 6**, Current Travel Trends.

4.3 | Engagement

Meaningfully engaging and collaborating with residents of Ajax is essential to ensuring the final TDM Plan and its measures are supported by the community. Public engagement methods prioritized advancing transportation equity by engaging community organizations and groups who have traditionally been left out of the TDM policy development process through a combination of in-person and virtual events and activities. The Project Team also engaged key stakeholders and community leaders for perspectives on implementation.

Engagement methods were informed by:

- The **International Association of Public Participation (IAP2) process** emphasizes the importance of a tailored Community Engagement Strategy based on the understanding, commitment, and contribution of each unique group of stakeholders.

- Principles of the **Dignity Infused Community Engagement (DICE)**, which focuses on building trust and respect within marginalized communities by building coalitions, valuing existing work in those communities, building capacity, and thoroughly vetting recommendations to ensure they provide an environment of dignity.

Key audiences consulted and collaborated with throughout the project included Town staff, transit agencies, school boards, local businesses, active transit and cycling committees, age-friendly and community organizations, and seniors' organizations, as well as the general public. Engagement activities included online surveys, focus groups, pop-up events, and an open house. The resulting TDM plan update aims to address transportation barriers and improve mobility for all members of the community, with a focus on advancing transportation equity.

Social Climate Analysis

The project team conducted a Social Climate Analysis (SCA) to inform the development of the TDM Plan update. A Social Climate Analysis (SCA) is a way of building trust and genuine partnerships within a community. Rather than seeking to "improve" a community from the top down, a Social Climate Analysis identifies the critical social infrastructure that already exists within a community, focusing on building relationships and capacity to support the community's ability to fulfill its potential by naming, mitigating, and working toward the removal of systemic barriers that impede a community's ability to thrive.

InMyOpinion Webpage and Survey and Promotion

A digital platform was developed to promote and engage the general public on the development of the TDM Plan update. It included a survey with questions on travel behaviours and demographics to identify trends and ensure that responses reflected the Town's demographics.

The platform was launched in June 2022

and closed in late September. The platform and survey were promoted on the Town's social media as well as through e-mails to community leaders who contributed to the Social Climate Analysis.

Additionally, three members of the Project Team from both the Town of Ajax and WSP Canada conducted a series of in-person engagement and promotion for the TDM Plan update community survey. The Project Team members distributed physical copies of the survey (with large print versions available for accessibility) and post-cards with a QR code and a link to the digital version of the survey. During this session, the Project Team also provided prizes featuring #GetAjaxMoving branding to incentivize uptake of the survey and build awareness around the TDM campaign across the Town. In total, the Project Team engaged with over 230 residents throughout the day at 4 different venues, boosting the uptake of the survey and increasing awareness around the TDM Plan update.

The platform received the following uptake:

- 910 visitors to the page
- 252 survey respondents
- 318 visitors who viewed multiple pages or accessed files shared on the webpage, raising awareness behind the project and its purpose.

The majority of the 252 survey responses were representative of the Town of Ajax's demographic profile, albeit with a slight overrepresentation of white and male respondents. The 2022 TDM survey asked similar questions as a community survey that was conducted to produce the 2015 TDM Plan, allowing results and trends to be compared between 2015, 2022, and the 2016 census where applicable.

Because the 2022 survey was built with the SCA results in mind, the number of survey respondents were far more representative of groups who did not have access to a private vehicle as their primary mode of commuting. This ensures that the results obtained are more representative of those most dependent on sustainable transportation modes that ensures the recommendations are applicable to a broad range of residents, not just those working within the structure of a 9-5 routine. The resulting TDM Plan update will aim to address these transportation barriers and improve mobility for all members of the community, with a focus on advancing transportation equity.

Agency Meetings

During the development of the TDM Plan update, technical agency representatives were asked to provide their priorities, perspective, and feedback on proposed measures for the plan. Agencies included representatives from the Town of Ajax, Metrolinx, Durham Region Transit (DRT), the Durham District School Board, and other local stakeholder organizations. Representatives were convened in two agency meetings. The first meeting focused on obtaining ideas for desired TDM measures guided by results from the Social Climate Analysis and considerations to past short-term and medium-term TDM measures. The second meeting focused on obtaining feedback on proposed actions.

Virtual Public Open House

The Project Team hosted a Virtual Public House on Monday, January 16, 2023 from 7-8:30PM. The event was promoted on the Town's social media channels as well as direct e-mails to community groups involved in the Social Climate Analysis and previous active transportation activities. Fourteen people attended. During the event, the project team explained the scope of the project, shared results from the Social Climate Analysis and surveys, and provided opportunities for residents to ask questions and share their ideas and concerns.

More details of the Engagement and Consultation activities can be found in **Appendix A**.

5 Lessons from Best Practices

Understanding the Stakeholders

Assigning Responsibility Where Authority Exists

Funding Sources and Measuring and Monitoring Reports

Alternative Work Arrangements

Integrating TDM in New Developments

The case studies reviewed provided a broad sample of best practices in Transportation Demand Management in the GTHA and beyond. Below are the key themes that emerged:

5.1 | Understanding the Stakeholders

A holistic change to modal shift requires the involvement and coordination of many stakeholders. The Metrolinx case study provides a useful overview of some of those organizations including all levels of government, municipal transportation and planning departments, schools and school boards, developers, employers, and community organizations. The City of Kitchener example demonstrates the importance of engagement between both local groups and the Region as a means of meeting TDM targets and goals in the long-term. Any TDM strategy should consider how to involve these stakeholders and work with them to achieve mutually beneficial outcomes (such as encouraging less parking in a new development, in exchange for providing discounted transit passes to residents/workers where service is provided, as seen with the City of Coquitlam). The York Region case study touches on a specific stakeholder collaboration, targeting active school travel amongst students. Part of the coordination involved here is providing infrastructure and materials that support safe and accessible school commute, that in turn reduces single-occupant trips during school pick-up and drop-off periods.

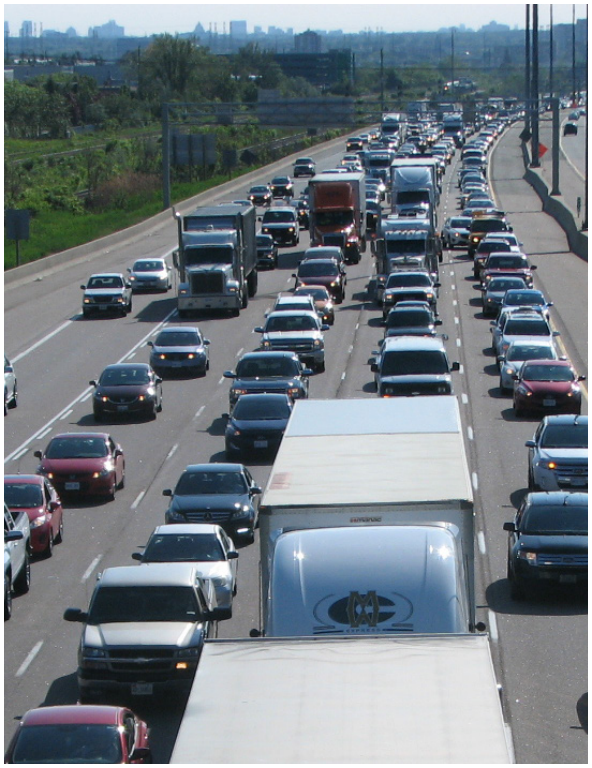
5.2 | Assigning Responsibility Where Authority Exists

While a third-party can be relied upon for providing effective TDM programming ("soft" measures), these organizations have limited ability to implement "hard" measures such as policy changes or infrastructure-related initiatives. The Peel case study assigns responsibility and actions to the groups with the greatest ability to make change; for example, by calling upon transportation staff to implement a multimodal level of service (MMLOS) evaluation framework for new roadway projects. TDM must be core to an organization's operation to be successful.

5.3 | Funding Sources and Measuring and Monitoring Reports

Regular evaluation and reporting are needed to ensure that the value of various TDM measures are understood and communicated to the public, as noted through the City of Coquitlam, San Francisco, and Arlington case studies. When tangible results are produced, as in the case of Arlington, they can be used to justify funding increases or taking on a broader mandate. Without results, TDM risks being perceived as a "nice-to-have" that can be cut during a tight budget cycle. Measuring and monitoring tools are also critical to updating and/or introducing new measures as opportunities become available for communities. San Francisco evaluates some TDM initiatives by analyzing vehicle miles travelled, green house gas emissions, and reduction of single occupancy vehicles (SOV).

It is also important that TDM be recognized as a long-term, value-adding initiative for achieving sustainable mode share



goals, rather than a series of marketing initiatives. The Arlington case study shows the value of robust funding requirements, such as government-funded programs, and evaluation tools that assess the impacts of TDM services to see a positive return on investment, in contrast to Waterloo's public-private partnership which relies more on short-term funding. Similarly, the City of Whitehorse identifies specific funding sources, such as gas tax funds, as a means to allocate financial means towards maximizing existing infrastructure with opportunities for transit and active transportation.

5.4 | Alternative Work Arrangements

Most impactful to travel demand strategies is reducing or eliminating the need for travel altogether. Implementing alternative work arrangements can yield outcome-oriented productivity amongst employees, increasing both the social and economic benefits for both the employer and employee. Both the York Region and City of Whitehorse case studies speak to measures that qualify as alternative travel and work arrangements, including teleworking or even flexible work schedules outside peak working hours to spread out travel demand over time.

5.5 | Integrating TDM in New Developments

Establishing a method for quantifying the TDM benefits of a project and incorporating an evaluation process into planning documentation. Regulating evaluation ensures that projects are meeting the minimum TDM threshold recognized by the municipality. The City of San Francisco uses monitoring and evaluation techniques related to development projects that range from reporting practices, incorporating data tracking, identifying the effectiveness of TDM measures, and informing developers. The City of Kitchener outlines strategies to integrate TDM into site planning and development approval processes, by using a TDM checklist. For instance, that applications for new developments must include a TDM Plan as a component of transportation impact studies. The City of Brampton and the City of Coquitlam are also examples where minimum thresholds and development application tools, such as a TDM checklist, are used to evaluate elements such as parking requirements to minimize dependency on single-occupant vehicles and instead encourage use of sustainable modes with alternative incentives such as bicycle parking thresholds and subsidized transit passes.

More details from the jurisdictional scan can be found in **Appendix D**.

6 CURRENT TRAVEL TRENDS

Vehicle Ownership
Average Daily Trips
Trip Purpose
Travel Mode
Overall Trends
COVID 19 Trend: Interrupted Travel

This section provides a snapshot of current travel behaviour within the Town of Ajax and an update for benchmarking and measuring future progress in TDM implementation. This analysis is primarily based on the results from the 2016 Transportation Tomorrow Survey (TTS) with insight from the IMO survey results. At the time that this report was written, the most up-to-date data available from the Transportation Tomorrow Survey was from 2016, before the COVID-19 pandemic.

Several household and person attributes collected through the TTS provide valuable insight into current trends of interest from a TDM perspective. These include:

6.1 | Vehicle Ownership

Vehicle ownership has been shown to correlate closely with auto trip making. **Figure 6.1** shows average vehicle ownership trends in 2016 for the Town of Ajax, Durham Region and the GTHA.

6.2 | Average Daily Trips

The rate and magnitude of trip making provide an indication of the opportunity for TDM penetration. **Table 6.1** shows some key trip making statistics for the Town of Ajax, the Region of Durham and the GTHA.

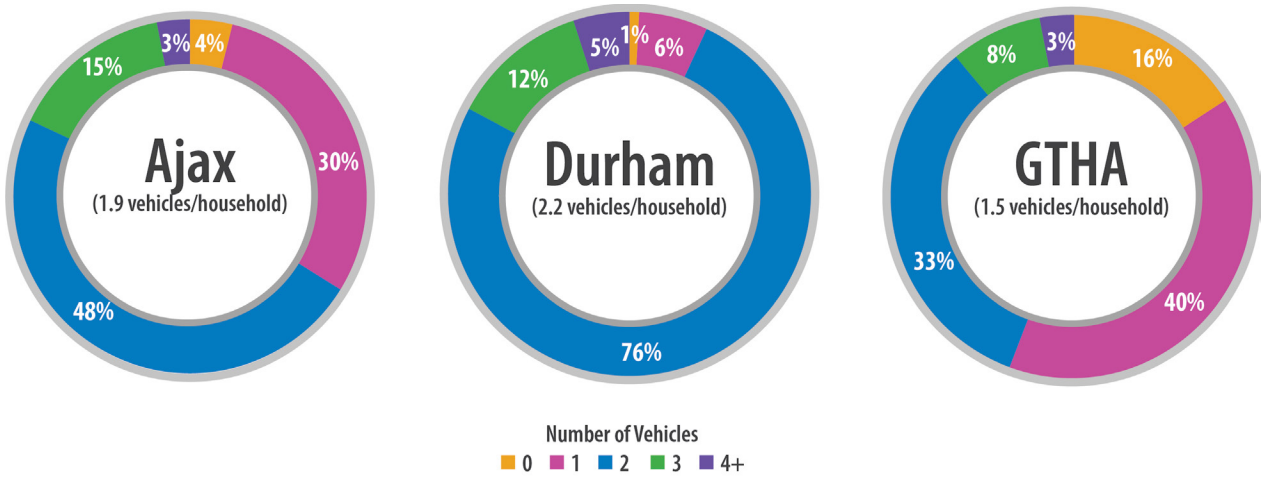


FIGURE 6.1: Household Vehicle Ownership (2016)

6.3 | Trip Purpose

It is important to understand the purpose of the trip, as certain trip types, in particular Home-Based to Work (HB-W) and Home-Based to School (HB-S), have greater potential for TDM measures than others, such as Home-Based to Discretionary (HB-D) and Non Home-Based to (N-HB). **Figure 6.2** summarizes the purpose of trips made by Town of Ajax, Durham Region and GTHA residents daily and during the morning peak period (6:00 to 9:00 AM).

TABLE 6.1: Daily Trips Made by Residents (2016)

Area	Average Daily Trips per Person	Total Number of Trips	% of Trips Between 6 and 9 AM
Ajax	2.0	235,336	25.7%
Durham	2.0	1, 279, 237	24.3%
GTHA	1.9	13, 260, 736	26.5%

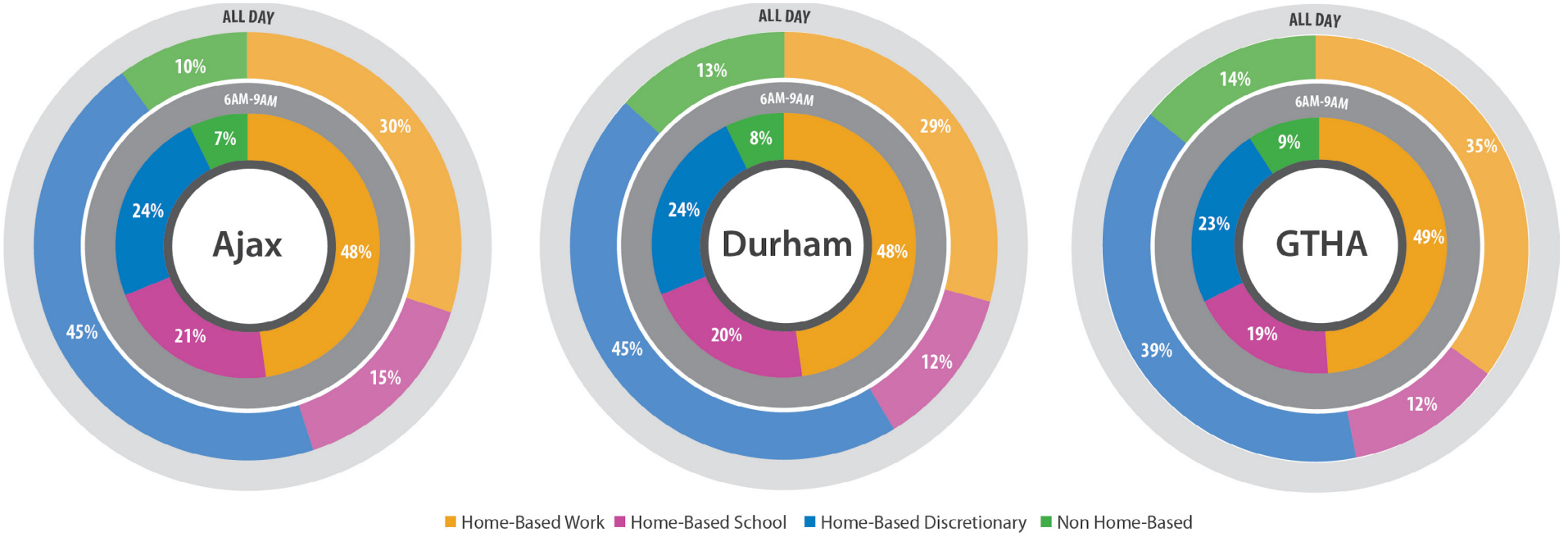


FIGURE 6.2: Trip Purpose for Trips Made by Town of Ajax Residents (2016)

6.4 | Travel Mode

Data on travel mode provides an indication of the extent of travel by modes other than the single-occupant vehicle, and the potential market for TDM. **Figure 6.3** depicts the modes of travel selected by Town of Ajax, Durham Region and GTHA residents for daily and morning peak period trips in Ajax.

The survey also found that the most common modes of transportation were driving alone, carpooling, and public transit, with the main barriers to using active transportation being a lack of infrastructure and safety concerns.

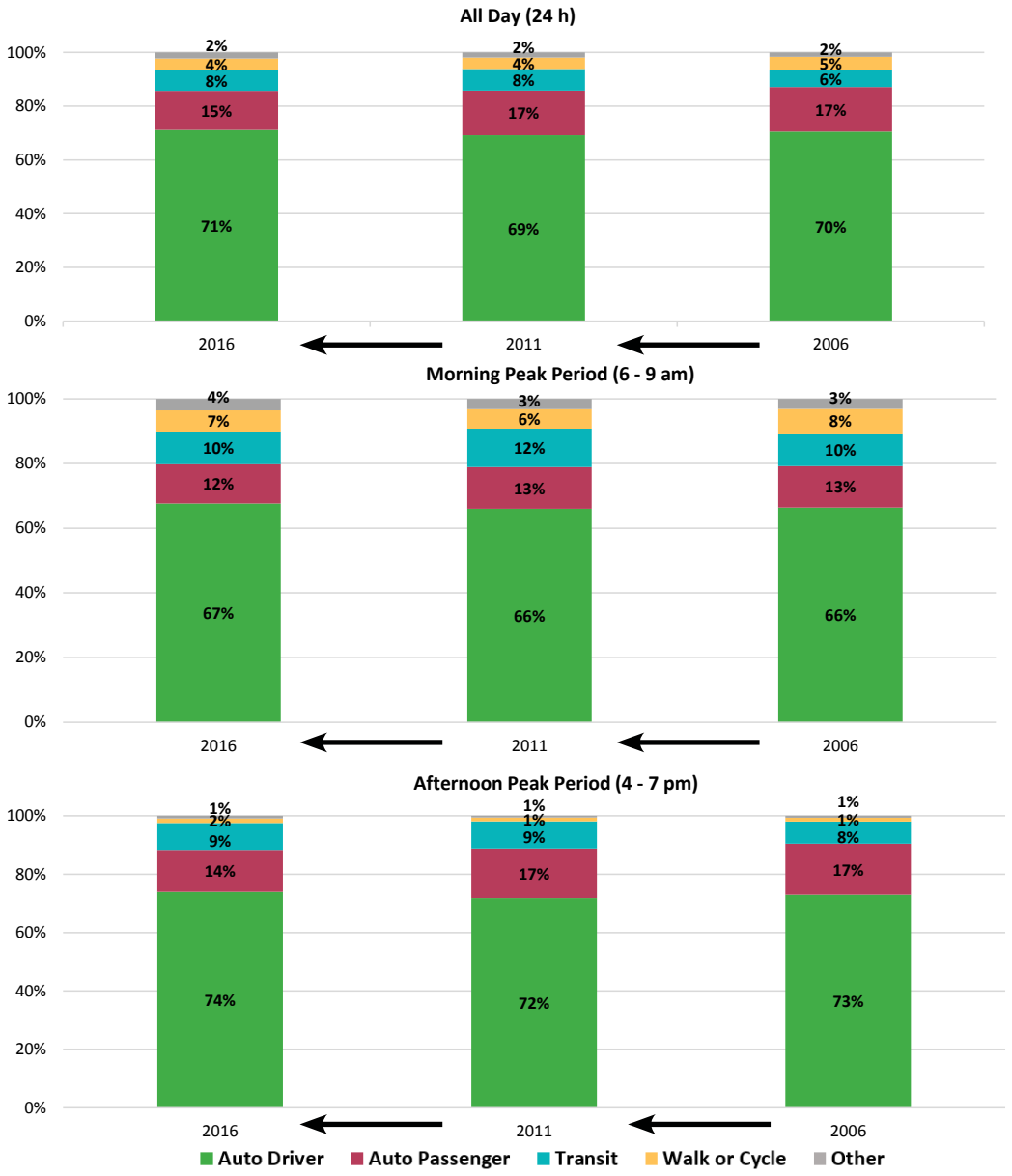


FIGURE 6.3: Modal Split (Trips) by Period of Day in Ajax (2006-2016)

#GetAjaxMoving

6.5 | Overall Trends

Vehicle Ownership

- Vehicle ownership is higher in Ajax (1.9 vehicles per household) than in the GTHA as a whole (1.5 vehicles per household), indicating a higher level of auto dependency than average.
- One notable trend observed in the 2022 survey is an overall decline in the number of vehicles owned per household in Ajax. The survey revealed significant increases in the percentages of zero and one-car households (46% in 2022 vs 31% in 2015), with a corresponding decline in the percentage of households with two or more vehicles (54% in 2022 vs 70% in 2015).

Trips Made by Ajax Residents

- Ajax residents (aged 11+) make an average of 2.0 daily trips, similar to the GTHA as a whole.
- About 26% of trips made by Ajax residents occur in the morning peak period. There is a higher concentration of peak period trips than in the GTHA as a whole indicating a stronger commuting profile.
- Most trips are made by the private automobile.
- Transit use in Ajax is significantly less than the GTHA as a whole, which may reflect the level of service in Ajax. The exception is during the peak period, when use of GO Transit is higher in Ajax (9%) than the wider average.
- Walking and cycling travel has a low mode share (4% of all trips), but there is an opportunity to convert short car trips to active transportation.

Trips Made to Ajax

- On a daily basis, approximately 235,300 trips are made to Ajax. About 26% occur during the morning peak period.
- The vast majority of trips made to Ajax are by private automobile.

Remote Work

- The increase in those able to work from home and the decline in resident commute times demonstrates that the workforce has shifted across the Town.
- However, despite these shifts, there was an increase in single-occupancy vehicle usage with many residents citing convenience as the primary reason why they continue to drive.

COVID 19 Trend: Interrupted Travel

The Covid-19 pandemic has been a strong disruptor to travel patterns that has previously been considered to be relatively unchangeable. With a widespread adoption of working from home, where the option has been available to employees, remote schooling, and remote services such as doctors' appointments have shown how much travel demand can be affected. Since 2020, some of the changes in travel trends have been temporary in nature, while others are more permanent. While the Town of Ajax works towards understanding the full extent of changes caused by the Covid-19 pandemic through the upcoming Transportation Tomorrow Survey (TTS) results in 2023, there are some insights that shed light on changed behaviour moving forward into the foreseeable future.

The University of Toronto conducted a Covid-19 survey, similar to the TTS, that looked at the ways Covid-19 influenced household interrupted travel (COVHITS). The survey was completed in three cycles with summary reports disrupted in January, October, and December 2021. The survey is a travel demand dataset in the Greater Golden Horseshoe (GGH). In summary, a "new normal" has emerged with respect to travel patterns, which has yet to unfold in its entirety. The pandemic not only resulted in a disruption in daily travel, but triggered new trends such as telecommuting, e-shopping, online social activities, and virtual essential services. Urban passenger mobility has dropped significantly since the beginning of the pandemic, with immobility (no trips per day) reaching 58% in Fall 2021, versus the 22% pre-pandemic. Although urban passenger mobility has declined, the surveys point to modal share of driving in the AM peak period to be at 56%, which is similar

to levels pre-pandemic (55%). On a plus side, 14% of trips generated by residents in the study area were by mode of active transportation, which is a 5% increase from pre-pandemic times.

As this "new normal" continues to evolve throughout the Greater Golden Horseshoe, including the Town of Ajax, this TDM Plan update is an opportunity to build upon the strides that have been made in travel behaviour changes since the start of the pandemic. By leveraging and actively implementing the action items detailed in Chapter 8 of this document, the Town can continue to provide programs and advocate for infrastructure improvements that favour alternatives modes of travel to the single-occupant vehicle.

Sources:

Covid-19 Household's interrupted Travel Schedules (COVHITS) Survey Report, 2021

7 Challenges and Priorities

Challenges

Challenges That Can Be Addressed Through TDM

Priorities

This section discusses the barriers to using sustainable transportation modes based on results from the Social Climate Analysis (SCA), survey results, and agency and stakeholder meetings in addition to previous knowledge. The engagement demonstrated that Town residents are open to change. However, alternatives must be safe, convenient and easily accessible. While much of the concerns raised in the community engagement processes pertain to supply-side challenges, residents also brought up challenges that can be addressed through TDM.

7.1 | Challenges

A persistent theme to emerge in discussions with community leaders was a strong recognition of the mobility deficiencies that exist throughout the Town for those who do not have direct access to a personal vehicle and rely on public transit or active transportation. The interview process revealed that transportation needs were not being met for workers, seniors, and vulnerable residents during non-traditional commuting hours.

Supply-Side Barriers

While these elements are out of scope for this project, they should be noted and should continue to inform infrastructure and transit planning at both the Town and Regional level.

Land Use and Transportation System

Previous zoning allowed for mostly lower density development, resulting in higher walking distances between residential and commercial or employment uses. Much of the Town of Ajax developed considerably in the 1970s and 80s, at a time when the private auto prevailed as the dominant transportation mode in Canada (and elsewhere), and when developments were being constructed to accommodate the private auto.

There was also significant development during the early 2000's which improved by providing more sidewalks and nearby schools, and to a certain extent, retail and employment uses. However, much of this development continued using lower densities, which makes transit systems less efficient and more difficult to access.

Ajax and the immediate area have a very strong historical and economic link with the automotive industry, with a local presence for Volkswagen Canada (in Ajax), General Motors (in nearby Oshawa) and BMW. The manufacturing plants associated with these companies require proximate highway access, further promoting the ease of auto travel. While the presence of two highway interchanges fulfills this need and creates access from the rest of the Greater Golden Horseshoe by car, it creates challenges to providing north/south connections for active transportation.

As seen in the Policy Framework in Chapter 3, there is now policy support in the Town's Official Plan to create more walkable communities, build active transportation facilities, and implement TDM measures.

Active Transportation Facilities

Physical infrastructure changes such as the addition of protected cycling facilities or new sidewalks are addressed by the Town's Integrated Transportation Master Plan, but the importance of those changes emerged as a key theme multiple times during this project.

New sidewalk connections and improved winter maintenance on sidewalks were identified as priorities, in addition to paving more trails to improve accessibility. Adding more mid-block crossings, pedestrian controlled intersections, and lighting were all highlighted as key enhancements to improve walkability across the town.

Transit Frequency and Routes

The SCA and public survey both demonstrated a desire for both local and regional transit improvements. Community members identified a need to improve transportation frequency and increase the number of stops

within the existing system. There were also community leaders and residents who expressed that the removal of stops and changing the eligibility for accessible transit created significant barriers to accessing food banks, programs for people with dementia, and the Ajax GO Station.

7.2 | Challenges That Can Be Addressed Through TDM

Communications and Transparency

The most consistent theme that emerged through the engagement for the TDM Plan update was the central role that communication needs to play in travel behaviour change.

The Town is already making significant changes in the design and operation of its streets – with new sidewalks, multi-use paths and cycling facilities being added each year. Many residents and community leaders reported feeling unaware of how these changes might impact them, highlighting the need for the Town to work closely with local champions to promote the potential for mobility choice within their networks.

Community members also indicated that throughout the pandemic there was a lack of transparency from local and regional transportation providers when it came to transit service reductions, route changes, and moving stops. The lack of transparency and integration of community feedback negatively affected users who were reliant on the service.

Suggestions for enhanced communication also included making public transportation more predictable, easier to navigate and adding more transparency behind route and scheduling changes. For example, integrating real-time bus tracking allows residents to accurately predict when their bus will arrive and whether it has been delayed. Additionally, a number of comments were made regarding an abrupt route change or termination of bus stops throughout the DRT system. Residents expressed a desire to have DRT operate with a higher degree of transparency and consultation when adjusting transportation route and service levels.

Bicycle Parking and Active Transportation Amenities

Residents also brought up the need for secure bicycle storage in the surveys results and the Public Open House. There were comments asking for weather protected bicycle storage at the Ajax GO Station, schools, and other destinations. Currently, the Town has approximately 1967 bicycle parking spaces, many of which can be found on the interactive trail map at ajax.ca/bike.

Residents also brought up improving wayfinding signage as a priority. The 2015 TDM Plan included developing a Wayfinding Strategy in its actions. The Wayfinding Strategy was completed in 2020 and is currently being implemented.

7.3 | Priorities

The 2022 survey asked respondents to identify how to improve the uptake of public transit, where residents identified that frequency, reliability, and accessibility were their top priorities. To improve active transportation uptake, residents identified that there is a need for more active transportation infrastructure and education.

Table 7.1 illustrates priorities recommended by residents based on need and recommendations from stakeholders based on past implementation experience and existing plans and resources. Further information on these priorities and recommendations can be found in the survey results and the agency meeting results.

TABLE 7.1: Priorities Recommended by Residents and Stakeholders

	Resident Priorities	Stakeholder Recommendations
Supply-Side	**Priorities for Transit and Active Transportation** 1. Building more bicycle lanes to connect employment, commercial and recreation areas to residential neighbourhoods; 2. Improving cycling safety at intersections; **Durham Region Transit (DRT)** 1. Increasing the frequency of buses; 2. Improving route directness; 3. Improve connections to GO Transit service; 4. Extend transit coverage area; and 5. Add more bus stops across existing routes.	**High Impact Actions** 1. Improving walking and cycling access to the Ajax GO Station 2. Providing active transportation amenities
Demand-Side (measures that can be addressed through TDM)	1. Improving secure bicycle storage across Ajax; 2. Making cycling wayfinding signage and maps more accessible; 3. Support bicycle skills and education programs; and 4. Improve the awareness of transit options and how to use them.	1. Encouraging Active School Travel, preferably through a designated full-time staff member who creates AST programs and partnerships across the Region 2. Enhancing partnerships and collaboration • Support workplace TDM champions, • Leverage community groups and their community networks • Strongly encourage development applicants to implement TDM measures 3. Integration of TDM into policies, processes, and programs 4. Coordinate with developers and property managers to distribute TDM information to residents encouraging sustainable transportation.

8 Strengths and Accomplishments at the Community and Town Levels

Mobility of Care Network

School-led Efforts and Enthusiasm for Sustainability and Mental Health

Completed TDM Efforts

In identifying feasible actions for the TDM Plan update, it is highly beneficial to leverage existing strengths and accomplishments at the community and Town levels. Further, it is important that future actions do not burden community-level resources that help with community safety and care.

8.1 | Mobility of Care Network

Through the interview series in the Social Climate Analysis, the project team identified that community organizations across Ajax have established community-based transportation systems and programming designed to help residents, particularly those who do not own personal vehicles, get around. These programs manifest in a variety of formats such as door-to-door specialized transit (Kids/Seniors on Wheels; Community Car Durham), carpooling arrangements, or by bringing essential services directly to clients (Doctors on Wheels).

The strengths of the community emerged in response to existing transportation gaps in the community that were then exacerbated by the COVID-19 pandemic. Each organization leader was eager to make travel and access to services easier for clients who are frequently left out of the fold as a priority group for improving mobility. Our team also identified that there is coordination between different community groups and community transportation providers that allows for a greater reach to connect community members to essential services and work.

8.2 | School-led Efforts and Enthusiasm for Sustainability and Mental Health

Since 2010, a number of teachers and school staff employees have led efforts encouraging active school travel (AST). In Durham Region, Ajax has maintained a track record of having the highest number of schools registered for events celebrating AST. These events are Walktober, Winter Walk Day, and Bike to School Week. Roughly 30% of Ajax elementary schools participate in the Town's Active School Travel events each year since 2016.

Below are notable activities encouraging active school travel championed by teachers and school staff:

- Two Ajax schools won the Metrolinx Smart Commute Bike to School Week Awards in 2016 and 2018
- One school in Ward 2 implemented a student-led Walking School Bus for multiple years prior to the pandemic. Although a number of parents complimented the Walking School Bus program, school staff noted challenges with the capacity of student leaders during key sports seasons. In 2019, this school organized a 8km group walk to school starting from Rotary Park.
- Three South Ajax schools organized a group bike ride in 2022 and regularly organize annual bike to school activities.

Further, the interest in sustainability and mental health among school boards can be leveraged to encourage active school travel. All Ajax elementary schools within the Durham Catholic District School Board are Certified Eco-Schools. Growing interest in mental health and well-being in the education sector and the benefits of AST for self-regulation provide further opportunities for sustaining or increasing this enthusiasm at the school and school board level.

8.3 | Completed TDM Efforts

Table 8.1 outlines actions completed from the 2015 TDM Plan.

TABLE 8.1: Actions Completed from the 2015 TDM Plan

Recommendation	Accomplishments
S.2 Create Sustainable Transportation Awareness, Education and Safety Campaign Material	Toolkits on sustainable transportation options can be found at https://ajax.ca/GetAjaxMoving Growing list of Youtube active transportation videos as part of the following series: • Explore Ajax by Bike • Back to School TOA • WinterTOA
S.3 Conduct Community Outreach Events	Average of 3 in-person community events each year, 1 in-person event each year from 2020-2021, 1-2 virtual events with attendees from 2020-2021. These events engage approximately 1000 residents annually.
S.4 Conduct Public Awareness Campaigns	Over 3 campaigns per year since 2016. Ongoing campaigns Major Partners: • Durham Planning and Economic Development • Durham Public Health • Durham Catholic District School Board
S.6 Provide Guidance on TDM-friendly Site Design	Planning and Development Services regularly requests development applicants to include TDM measures.
S.7 Prepare Wayfinding and Street Furniture Strategy	4 bike routes identified. 14 signs installed in 2020. Currently being implemented over a 15 year horizon.
S.10 Provide Active Transportation Amenities	Each school provided with a bike rack. Bike racks provided to Ontario By Bike bicycle-friendly businesses, where feasible. Five bicycle repair stations installed in 2021.
S.12 Update Walkable and Bikeable Ajax: The Ajax Pedestrian and Bicycle Master Plan	The Pedestrian and Bicycle Master Plan is now the Integrated Transportation Master Plan (ITMP). Staff have implemented a number of facilities outlined in the ITMP to date and are continuing to expand the network.

9 ACTION PLAN

Foundations

Partnerships

TDM Integration with Town Policies and Practices

Transportation Equity

Table 9.1: Summary of Actions

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A2	Create TDM Communications Plan
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B3	Durham Region TDM Working Group
B4	Transit Marketing and Promotion
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C	**TDM Integration with Town Policies and Practices**
C1	Complete Streets
C2	Bicycle Parking Implementation Plan
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C5	Coordination with the Integrated Transportation Master Plan (ITMP) Update
D	**Transportation Equity**
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D3	Build Youth Capacity in Equity Deserving Communities
D4	Investigate the Feasibility of Creating a Bike Library
D5	Engage Equity-Deserving Communities

The actions are categories into the following four theme areas:

Foundations: The important first steps to set the new TDM plan in motion for long-term success.

Partnerships: Collaborate and strengthen partnerships on a range of TDM initiatives locally and regionally to leverage the knowledge, skills and connections of others and increase impact.

Integration with Town Policies and Practices: Embed TDM in site design and street improvements to promote greater uptake of sustainable transportation .

Transportation Equity: Expand mobility choice where car-centric design of the past has created barriers for disproportionately impacted communities to access essential services, programs and opportunities.

9.1 | A) Foundations

ACTION A1 – Secure TDM Funding and Resources

Description:	Secure the funding and staff resources needed to deliver the TDM Action Plan from Q1 2023 to Q4 2027 (5 years)
Rationale:	A long-term source of funding supports the longevity and lasting impact of TDM strategies and has greater potential to influence more sustainable and efficient travel to achieve the Town's transportation needs.
Steps to Implementation:	**A1.1** Review all existing and potential TDM funding sources (including Regional, Provincial, and Federal sources where applicable), including 1% of capital project funding for TDM, and Development Charges as part of next DC By-law update. **A1.2** Secure funding and staff resources needed to implement the Plan.
Lead:	Town (Planning & Development)
Support:	Durham Region
Estimated Budget Needs:	2023 - $0 2024 - $218,500 2025 - $276,000 2026 - $275,500 2027 - $230,000 Total - $1,000,000
Performance Indicators:	Annual funding amount

ACTION A2 – Create TDM Communications Plan

Description:	A communication plan is a tool to help marketing or public relations teams align their communications and produce cohesive messaging. It will draw upon the Town of Ajax's current marketing and communications materials and incorporate lessons learned from past campaigns.
Rationale:	A strategic and coordinated approach is needed for a Transportation Demand Management plan to be successful, with key messages, campaigns, tactics and partners identified. Segmentation and understanding of target markets are keys to success in behaviour change initiatives, and focused research is often crucial to developing such an understanding. Surveys at intervals enable an understanding of trends as they develop and can reveal developments in the public's awareness and understanding of key issues, programs and specific initiatives..
Steps to Implementation:	**A2.1** Market research to identify target audiences. **A2.2** Evaluate current marketing and communication materials. **A2.3** Create communications plan that identifies key messages, campaigns, tactics and partners. Incorporate lessons learned from past campaigns like #GetAjaxMoving as well as behavioural science insights. It can be a living document that gets updated as new lessons are learned and focus areas change.
Lead:	Town (Corporate Communications, Planning & Development)
Support:	All Town Departments
Estimated Budget Needs:	Need to expand staff resources to ensure adequate capacity.
Performance Indicators:	• Number of marketing and outreach impressions • Number of activities (events held/resources distributed)
Examples:	• City of Vancouver, BC - Active Transportation Promotion and Enabling Plan https://vancouver.ca/files/cov/active-transportation-promotion-and-enabling-full-plan.pdf • Cycling Scotland – Communication tool for community groups https://www.cycling.scot/mediaLibrary/other/english/7371.pdf • Association of Commuter Transportation (ACT) TDM messaging guidelines and templates https://www.actweb.org/tdm-messaging

ACTION A3 – Prepare Annual TDM Summary Report

Description:	A summary report on TDM highlights of accomplishments and status of action plan, including use of infographics and results from program monitoring and evaluation.
Rationale:	An annual report demonstrates Ajax as a committed leader in the field of TDM, build supports for greater partnership involvement, and provides transparency on return on investment.
Steps to Implementation:	Create annual report from 2023-2027.
Lead:	Town (Planning & Development)
Support:	Town (Corporate Communications)
Estimated Budget Needs:	Need to expand staff resources to ensure adequate capacity.
Performance Indicators:	• Number of impressions, likes, shares • Qualitative feedback • Sustainable transportation mode share (Census, TTS data)
Examples:	• Atlanta, Georgia - https://atlantaregional.org/transportation-mobility/transportation-planning/transportation-demand-management-annual-report/ • Region of Durham - https://www.durham.ca/en/living-here/resources/Smart-Mobility-Durham-2021-Snapshot.pdf • Mobility Lab Infographics Catalogue - https://mobilitylab.org/infographics-catalog/

9.2 | B) Partnerships

ACTION B1 – TDM Community Fund

Description:	Leverage the Ajax Partnership Fund to encourage micro-grants to support and encourage TDM-related community projects.
Rationale:	The strengths of the community have emerged in response to existing transportation gaps. A Social Climate Analysis completed as part of this study identified that community organizations across Ajax have established community-based transportation systems and programming designed to help residents, particularly those who do not own personal vehicles, get around. These programs manifest in a variety of formats such as door-to-door specialized transit (Kids/Seniors on Wheels; Community Car Durham, carpooling arrangements), or by bringing essential services directly to clients (Doctors on Wheels). The initiative of creating a TDM Community Fund will support these types of community projects as well as develop other community partnership through pilot projects such as vanpools or cycling education, that would have otherwise not been possible.
Steps to Implementation:	**B1.1** Update the Ajax Partnership Fund program criteria to encourage micro-grants to support and encourage TDM-related community projects. **B1.2** Promote the program to known community partners who may be interested. **B1.3** Issue funding grants, and work with grant recipients to implement programs and report on results.
Lead:	Town (Planning & Development)
Support:	Town (Corporate Communications)
Estimated Budget Needs:	2023 - $0 2024 - $10,000 2025 - $10,000 2026 - $10,000 2027 - $10,000 Total - $40,000
Performance Indicators:	• Impacts of projects (number of participants, satisfaction of participants, shift to sustainable modes)
Examples:	• The Colorado Department of Transportation (CDOT) Office of Innovative Mobility (OIM) has two TDM funding opportunities under the OIM Grant Program. The OIM Grant Program includes TDM Seed Funding Grants and TDM Innovation Grants. https://www.codot.gov/programs/innovativemobility/grants/overview • Greater Sudbury, ON - https://www.greatersudbury.ca/city-hall/news-and-public-notices/2021/city-launches-transportation-demand-management-community-grant-program/ • Sacramento Area Council of Governments - https://www.sacog.org/sites/main/files/file-attachments/sacog_tdm_innovations_grant_guidelines_1718.pdf?1495474010

ACTION B2 – Build Capacity Through Community Partnerships

Description:	Identify, support and train community partners to help with TDM awareness, education and marketing campaigns, including providing honoraria to support their participation.
Rationale:	Having a healthy, vibrant city is everybody's business. There are existing community groups that can help build a culture of sustainable transportation use by educating residents on their travel options and changing social norms so that walking, cycling, rolling, transit use, ridesharing and working remotely are encouraged and celebrated.
Steps to Implementation:	**B2.1** Implement Ajax-led video campaigns partnering with community leaders and other partners promoting the benefits of sustainable transportation and building capacity among residents, with potential support from the Region. **B2.2** Prepare a TDM Community Champions Toolkit to support community partners for awareness, education and marketing campaigns. **B2.3** Town of Ajax staff will collaborate with the Region to run a capacity building workshop to leverage and support community partners for awareness, education and marketing campaigns, including honoraria for community champions. Potential partner organizations can include: Businesses, Boards of Trade, Places of Worship, Resident Associations, Community Development Council Durham, Durham Region Cycling Coalition, Accessibility Advisory Committee.
Lead:	Durham Region (Smart Mobility, Public Health, Active Transportation Committee)
Support:	Town (Planning & Development)
Estimated Budget Needs:	2023 - $0 2024 - $5,000 2025 - $5,000 2026 - $5,000 2027 - $5,000 Total - $20,000
Performance Indicators:	• Number of participants / partners • Number of activities (events held, resources distributed) • Videos: # of marketing and outreach impressions
Examples:	• Centre for Scientific Collaboration and Community Engagement – Community champions programs - https://www.cscce.org/resources/community-champions-programs/ • Mobility Lab, The Switch video series - https://www.youtube.com/playlist?list=PLvxkEFOhOXMZZhr8w8MYxfE9SZzGwVECe

ACTION B3 – Durham Region TDM Working Group

Description:	An inter-municipal working group within Durham Region to collaborate and coordinate on TDM initiatives.
Rationale:	Smart Mobility Durham is the umbrella for the Region's Transportation Demand Management (TDM) programs. These programs reduce the number of single-occupancy vehicle trips by making it easier and more attractive for travellers to use active and sustainable modes of transportation such as walking, cycling and transit. A regional working group can help strengthen and grow these existing programs within the Town of Ajax.
Steps to Implementation:	**Smart Commute Durham:** **B3.1** Work with the Region to grow number of participating businesses in Ajax by using existing program resources, working with the Region to adapt or create new marketing materials, securing Smart Commute workplace champions, and establishing relationships between key TDM contacts at the Town, Region and Workplaces. **B3.2** Work with the Region to identify topics for Workplace champion training/capacity building workshops. **B3.3** Explore opportunities to work with the Region of Durham to coordinate Ajax-specific projects/campaigns at each member workplaces based on their needs and opportunities. **Cycle Durham:** **B3.4** Increase participation in Bike Month in Ajax through community events, marketing campaigns using existing Bike Month supports from the Region. **B3.5** Explore ways for the Region of Durham to support Ajax community/neighbourhood bike rides series. **Active and Safe Routes to School** **B3.6** Work with the Region to promote the Town's Active School Travel toolkit and explore ways to for the Region to support the Town's efforts in building capacity among Youth.
Lead:	Durham Region (Smart Mobility)
Support:	Town (Planning and Development)
Estimated Budget Needs:	2023 - $0 2024 - $7,500 2025 - $10,000 2026 - $12,500 2027 - $15,000 Total - $45,000

ACTION B3 – Durham Region TDM Working Group

Performance Indicators:

Smart Commute:

- Number of employees participating
- Shift to sustainable transportation (surveyed annually)

Cycle Durham:

- Participation in Bike Month
- Number of events held/resources distributed
- Satisfaction of participants

Active and Safe Routes to School:

- Number of schools participating in school travel planning
- Shift to sustainable transportation

Examples:

- Region of Waterloo
- York Region
- Metrolinx

ACTION B4 – Transit Marketing and Promotion

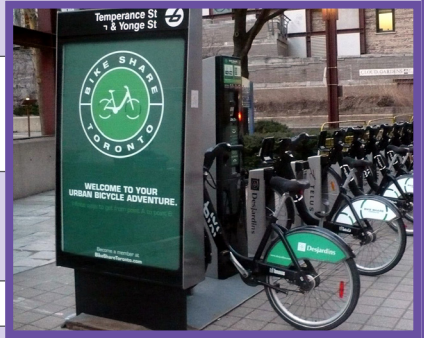
Description:	Coordinate communications efforts to promote existing transit services.
Rationale:	Marketing and promotional activities are an important supplement to the supply side of transit, as they help people understand their travel options and overcome barriers to using the transit system. Increased transit ridership can lead to increased service levels over the long-term.
Steps to Implementation:	**B4.1** Durham Region Transit - Work together on marketing campaigns, engagement activities and incentive programs to increase transit use and promote new and existing transit services. **B4.2** Metrolinx – Ajax GO Station- Work together on marketing campaigns, engagement activities and incentive programs to increase transit use and promote new and existing transit services.
Lead:	Durham Region Transit, Metrolinx
Support:	Town (Planning & Development)
Estimated Budget Needs:	2023 - $0 2024 - $5,000 2025 - $5,000 2026 - $5,000 2027 - $5,000 Total - $20,000
Performance Indicators:	• Number of marketing and outreach impressions • Transit mode share
Examples:	• GO Transit - https://www.gotransit.com/en/travelling-with-us/promotions-and-events • Durham Region Transit - https://www.durhamregiontransit.com/en/index.aspx • TDM for Transit Agencies - https://nacto.org/event/tdm-for-dot-transit-agency-practitioners/

ACTION B5 – Carshare

Description:	Establish partnerships with carshare providers to make carshare services available in Ajax. Carshare is designed to be convenient for people who want to rent cars for short periods of times (a few hours) and only pay for their usage, and therefore eliminate the need to own a car.
Rationale:	Carsharing allows people to take transit, walk or cycle for some of their trips while giving them access to a fleet of vehicles, only when one is needed. Vehicles are available on demand without the ongoing charges and hassles of owning, maintaining and parking (in some cases). Research shows that each carshare vehicle takes 15 personally-owned vehicles off the road.
Steps to Implementation:	**B5.1** Explore partnerships with carshare providers on the feasibility of carshare services in Ajax, and implement a pilot project. Carshare companies can include: • Communauto • Enterprise CarShare • Zipcar • Turo Also consider a community carshare model.
Lead:	Town (Planning & Development), Carshare companies
Support:	Durham Region
Estimated Budget Needs:	2023 - $0 2024 - $0 2025 - $0 2026 - $0 2027 - $0 Total - $0
Performance Indicators:	• Number of carshare vehicles available • Number of carshare memberships • Number of trips by carshare
Examples:	• Region of Waterloo, https://www.regionofwaterloo.ca/en/exploring-the-region/carsharing.aspx • BC Climate Action Toolkit – Community Car Share, https://toolkit.bc.ca/tool/community-car-share/ • Kootenay, BC, https://www.carsharecoop.ca/home-page/

ACTION B6 – Shared Micromobility Pilot

Description:	Review the feasibility implementing a shared micromobility pilot project.
Rationale:	E-scooters and e-bikes have the potential to replace car trips for short trips, and provide more transportation options for those who cannot or chose not to drive.
Steps to Implementation:	**B6.1** Consult with staff and the public on the feasibility of permitting and accommodating for shared e-bicycles and e-scooters. **B6.2** If there is a desire to permit either shared e-bicycles or e-scooters or both, establish partnerships with vendors to implement a shared micromobility pilot program.
Lead:	Town (Planning & Development)
Support:	Durham Region, Durham Region Transit
Estimated Budget Needs:	2023 - $0 2024 - $50,000 2025 - $50,000 2026 - $0 2027 - $0 Total - $100,000 Note: Total budget is for public engagement and communication, additional signage, delineating e-mobility parking areas, evaluation and monitoring. Assumes capital and operating costs of pilot project are covered by vendors.
Performance Indicators:	• Number of trips • User feedback • Cost of program
Examples:	• Calgary, AB, https://engage.calgary.ca/scootershare • Mississauga, ON, https://yoursay.mississauga.ca/micro-mobility • Grand Junction, CO, USA, https://www.gjcity.org/1228/Shared-Micromobility-Pilot-Study • Santa Monica, CA, USA, https://www.smgov.net/uploadedFiles/Departments/PCD/Transportation/SantaMonicaSharedMobilityEvaluation_Final_110419.pdf



9.3 | C) TDM INTEGRATION WITH TOWN POLICIES AND PRACTICES

ACTION C1 – Complete Streets

Description:	Complete streets are streets for all modes and users of all abilities. Focused on a higher standard of design for the common good of the public, Complete Streets have the potential to enhance the Town's quality of life by promoting transportation options, improving public health and safety, enhancing community livability and economic vitality, and promoting environmental sustainability.
Rationale:	There is an opportunity to operationalize the Complete Streets policy detailed in the Integrated Transportation Master Plan to provide attractive streets while simultaneously supporting improved multi-modal operations as way of reducing dependency on cars. A Complete Streets Design Manual for the Town of Ajax can serve as a multi-modal tool for implementing the better, safer more liveable streets.
Steps to Implementation:	**C1.1** Develop a Complete Streets Design Manual specifically for the Town of Ajax that serves as a guide to staff for planning, designing and implementing multi-modal supportive transportation corridors, including adopting a Multi-Modal Level of Service (MMLOS) evaluation framework for transportation planning.
Lead:	Town (Planning & Development)
Support:	All Town Departments, Durham Region
Estimated Budget Needs:	2023 - $0 2024 - $0 2025 - $50,000 2026 - $50,000 2027 - $0 Total - $100,000
Performance Indicators:	• Multi-modal Level of Service (MMLOS)
Examples:	Several municipalities across Ontario, and globally, have adopted robust Complete Street policies to help realize their community's vision for safe, connected, and multi-modal transportation network. • City of London, Ontario - https://london.ca/sites/default/files/2020-09/Complete%20Streets%20Design%20Manual.pdf • City of Hamilton (Ontario) - https://www.hamilton.ca/sites/default/files/2023-01/pedpolicies-street-complete-streets-design-guidelines.pdf • Niagara Region (Ontario) - https://www.niagararegion.ca/2041/pdf/tmp-complete-streets-design-guidelines.pdf

ACTION C2 – Bicycle Parking Implementation Plan

Description:	Developing a comprehensive active transportation network goes beyond implementing a network focused specifically on the infrastructure. A robust implementation plan focusing on programs and policies to foster a cycling-supportive environment is also needed. Bicycle parking provides a safe and convenient way for people to secure their bicycle at destinations. It incentivizes people to take their bike for shorter distanced trips, with the added benefit of supporting decreased automobile use and parking needs as well as supporting local businesses.
Rationale:	The implementation plan will focus on policies and initiatives that support trip-end facilities including short- and long-term bicycle parking. It should also focus on how to best establish connections between modes such as bicycle parking at transit stations. The Town's Bicycle Parking Design Guidelines have been updated to reflect best practices and support the installation of bicycle parking at key destinations.
Steps to Implementation:	**C2.1** Establish a bike parking request program to add bike parking in areas of need (through business requests and within the public right-of-way). **C2.2** Proactively install short- and long-term bicycle parking within Town right-of-way near employment, retail, transit, and other key Town destinations. **C2.3** Convert the spacing standards in the updated Ajax Bicycle Design Guidelines into typical Ajax Standards (AS-300 or AS-900 series), so that developers can select the type of bike parking style, the site scenario, and attach one of them into their site plan submission details.
Lead:	Town (Planning & Development)
Support:	Town (Operations & Environmental Services)
Estimated Budget Needs:	2023 - $0 2024 - $10,000 2025 - $10,000 2026 - $10,000 2027 - $10,000 Total - $40,000
Performance Indicators:	• Number of bike parking spaces added • Bike rack utilization
Examples:	• Thunder Bay, ON - https://www.thunderbay.ca/en/city-services/bike-parking.aspx#:~:text=The%20Bike%20Racks%20for%20Business%20program%20is%20a,for%20Business%20include%3A%20Free%20site%20visit%20and%20consultation • Mississauga, ON - https://www.mississaugabikes.ca/bicycleparking/ • Hamilton, ON - https://www.hamilton.ca/home-neighbourhood/getting-around/biking-cyclists/bike-parking-request-form • Victoria, BC - https://www.victoria.ca/assets/Community/Cycling/Bicycle%20Parking%20Request%20Info.pdf

ACTION C3 – Flexibility in Parking Policy

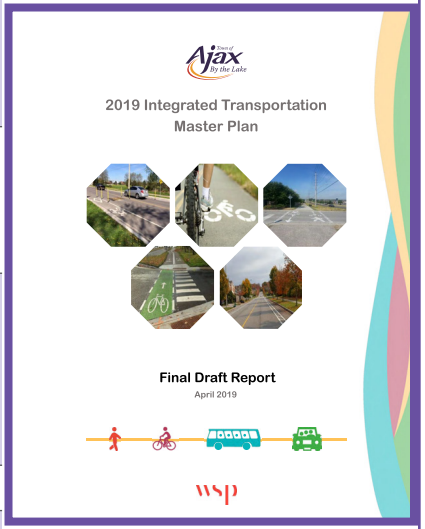
Description:	This action focuses on providing more flexibility with parking mandates along corridors to encourage more creativity and discussion on transportation solutions when development applications come in.
Rationale:	By moving away from prescribing parking minimums at the Town of Ajax, it opens opportunities for introducing TDM measures as part of incoming development applications. This has the potential to minimize the impacts of parking costs on society such as housing affordability, dependency on automobiles and creating car centric places.
Steps to Implementation:	**C3.1** Provide more flexibility with parking minimums as part of next zoning by-law update, drawing upon best practices. Focusing on the quality of life outcomes and shifting travel behaviours that stem from implementing a Parking Policy reform, some municipalities across Canada have been able to shift towards planning cities that are more walkable, affordable, and support small businesses. Examples of such places include the City of Ottawa, Ontario and the City of Edmonton, Alberta.
Lead:	Town (Planning & Development)
Support:	Community Groups
Estimated Budget Needs:	No cost.
Performance Indicators:	n/a
Examples:	• Kingston, Ontario - https://www.cityofkingston.ca/documents/10180/18914138/Projects_Bylaw_ParkingDiscussionPaper.pdf/53ccdf4f-4b32-cd80-395b-90f3fb69258d?t=1622570454241 • Ottawa, Ontario - https://ottawa.ca/en/transit-oriented-development-guidelines/development-guidelines#section-5-vehicles-parking • Edmonton, Alberta - https://www.edmonton.ca/city_government/urban_planning_and_design/comprehensive-parking-review

ACTION C4 – TDM in New Developments

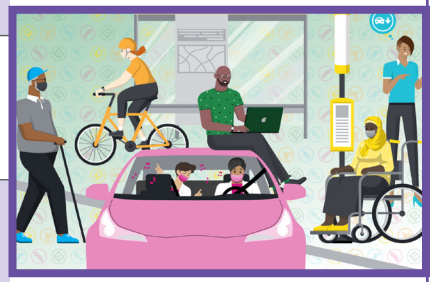
Description:	A community-based social marketing program along with individualized trip-planning is designed to help residents make informed choices that will improve their travel, making it easier to get around.
Rationale:	Increasingly, those who develop and deliver programs to promote health and sustainability are turning to community-based social marketing for assistance. Community-based social marketing emphasizes direct contact among community members and the removal of structural barriers, since research suggests that such approaches are often most likely to bring about behaviour change. In addition, we are particularly likely to change our habits during periods of transition and change, such as when we move homes. Providing nudges at opportune moments can also increase the likelihood of behaviour change.
Steps to Implementation:	**C4.1** Develop a pilot project for residents in new developments to better understand their transportation options at the time of occupancy. Identify the barriers to a behaviour, construct a program to overcome these barriers, implement the program across the community, and evaluate the effectiveness of the pilot project. **C4.2** Review lessons learned from pilot projects and scale it up to a program. Review the different governance models for such a program, such as delivery though a Transportation Management Association (TMA). **C4.3** Encourage developers to provide transit passes to new residents or with rental/office agreements with a guideline of 1 month fare per pass.
Lead:	Durham Region
Support:	Town Departments, Durham Region Transit
Estimated Budget Needs:	2023 - $0 2024 - $0 2025 - $0 2026 - $45,000 2027 - $45,000 Total - $90,000
Performance Indicators:	• Number of participants • Satisfaction of participants • Mode shift to sustainable transportation
Examples:	• King County, WA, USA - https://altago.com/projects/king-county-tdm/ • York Region - York Region's TDM program for new and growing communities, MyTrip, started as a pilot project in 2017. Local community ambassadors were hired and trained to provide individualized trip planning information to their community. Incentives included Presto cards for new residents to give transit a try. - https://www.york.ca/transportation/transportation-demand-management

ACTION C5 – Coordination with the Integrated Transportation Master Plan (ITMP) Update

Description:	Coordinating the ITMP update to integrate the Transportation Demand Management Plan does not require developing a new planning process for the Town of Ajax. Instead, it is intended to focus on identifying how the programs, policies, and initiatives noted in this TDM Plan can play a stronger presence in strategic directions carried out under the Master Plan. This also includes adopting of the forthcoming Durham Region TDM Development Guidelines.
Rationale:	Coordination and integration is an opportunity for the Town of Ajax to bring TDM to the forefront in its transportation planning efforts. This eliminates the need for a separate plan to implement measures that promote shifting travel behaviour. Since the Town's objectives are to promote more sustainable modes of travel, reduce automobile dependency, and create an interconnected and multi-modal network, combining the two plans will help embed TDM into its actions.
Steps to Implementation:	**C5.1** Combine the TDM Plan and IMTP into one Master Plan as part of the next update cycle to bring TDM policies to the forefront of transportation planning policy and actions. **C5.2** Contribute to the refinement, finalization and adoption of Durham Region's TDM Development Guidelines for a coordinated TDM approach in development planning.
Lead:	Town (Planning & Development)
Support:	All Other Town Departments, Durham Region
Estimated Budget Needs:	2023 - $0 2024 - $0 2025 - $0 2026 - $0 2027 - $0 Total - $0
Performance Indicators:	• Average score of developments
Examples:	• City of Pickering, ON • York Region, ON • Peel Region, ON



ACTION D1 – Inclusive Media Library



Description:	Create an image/video library that is representative of Ajax residents.
Rationale:	Images convey an organization's values and its intended or desired audience. Because the Town of Ajax community is diverse, images that reflect that diversity are more accurate and will connect more with the intended audience. When people do not see themselves in images, they feel that the measures being discussed are not ' for them'. A more inclusive set of images will help to make all of Ajax's diverse communities feel welcomed in the discussion about mobility.
Steps to Implementation:	**D1.1** Create a repository of photos promoting sustainable transportation that reflect the diversity and built environment of Ajax (will be done in 2023 as part of the Ontario Community Changemakers Grant). **D1.2** Ensure all Town staff have are aware and have access to the media library for use in Town reports, publications and communications. **D1.3** Continue to expand the media library with more visual assets (images and video) and consider accepting submissions from community organizations and the general public.
Lead:	Town (Corporate Communications, Planning & Development)
Support:	All Town Departments
Estimated Budget Needs:	2023 - $0 2024 - $10,000 2025 - $10,000 2026 - $10,000 2027 - $10,000 Total - $40,000 (models and photographer or videographer)
Performance Indicators:	• Number of images that reflect the diversity of Ajax • Evaluation of images used in Town communications reflect the diversity of Ajax
Examples:	• SmartCommute Toronto created graphics that include people of colour, people with disabilities and gender non-conforming people for their promotional materials - https://smartcommutetoronto.ca/mysmartcommute/ • Wheels for Well-Being (UK) campaign for more images of disabled people cycling - https://wheelsforwellbeing.org.uk/campaigning/invisible-cyclists/ • Waka Kotahi (NZ) photo library of everyday New Zealanders cycling in a variety of settings - https://www.nzta.govt.nz/walking-cycling-and-public-transport/cycling/cycling-photo-library/ • Associated General Contractors of America (AGCOA) Construction Diversity Image Library showcases diversity in the construction industry - https://www.agc.org/construction-diversity-image-library

ACTION D2 – Investigate the Feasibility of Participating in an E-cargo Bike Pilot

Description:	Electric Assist Bicycles, or E-Bikes, are human-powered bicycles with an assisting electric motor. In Ontario, all E-Bikes must have functional pedals, and are speed-limited to 32km/hr. To operate an e-bike in Ontario, a person must be 16 years of age or older and is required to wear a bicycle helmet. However, a driver's licence, registration or insurance are not required. E-Bikes provide reliable, efficient and inexpensive transportation for short and medium trips. They expand the availability of cycling to people with mobility impairments or physical limitations, make longer trips easier and increase the cargo carrying abilities of any bike. Research has shown that e-bikes are an effective replacement for most car trips in urban and suburban environments, especially in communities that are well-served by local and regional transit like Ajax.
Rationale:	Despite their myriad benefits, however, the relatively high purchase price of high-quality e-bikes puts e-bike ownership out of reach for the very communities that would most benefit from having access to one. It is recommended that the Town of Ajax review the considerations for an E-bikes program such as where they are permitted, mechanisms to implement, and what a pilot project would look like.
Steps to Implementation:	**D2.1** Review the feasibility of participating in the Ontario Cargo E-bike Pilot Program with input from Fire and Emergency Services, and Operations.

ACTION D2 – Investigate the Feasibility of Participating in an E-Cargo Bike Pilot

Lead:	Town (Planning & Development)
Support:	Service delivery agencies Local E-Bike retailers Region of Durham
Estimated Budget Needs:	2023 - $0 2024 - $2,000 2025 - $2,000 2026 - $2,000 2027 - $2,000 Total - $8,000
Performance Indicators:	
Examples:	• Denver's E-Bike Rebate Program - (https://www.denvergov.org/Government/Agencies-Departments-Offices/Agencies-Departments-Offices-Directory/Climate-Action-Sustainability-Resiliency/Sustainable-Transportation/Electric-Bikes-E-Bikes-Rebates) has resulted in participants shifting an average of 34 trips and over 40km travelled each week (https://www.smartcitiesdive.com/news/denver-ebike-rebate-vouchers-climate-action-mike-salisbury/641856/) • CalBike e-bike Affordability Program - https://www.calbike.org/bike_purchase_incentives/ • District of Saanich E-Bike Incentive Pilot Program - https://www.saanich.ca/EN/main/community/sustainable-saanich/climate-change/programs-rebates/e-bike-incentives.html • Tracker detailing e-bike incentives across North America - https://docs.google.com/spreadsheets/d/1C-sYcwLrQFsr8r2A6RiAP2RwGsBNwr1BK0F_HJvCsVU/edit#gid=0

ACTION D3 – Build Youth Capacity in Equity Deserving Communities

Description:	Building upon the recently released Town of Ajax Active School Travel Toolkit, a active school travel advocacy and placemaking resource centring the insights of young people. Host capacity-building workshops at local schools, with a focus on middle school and high school students, to identify the needs within each school community, and develop a plan to address those needs with youth at the forefront. Over time, expand capacity-building workshops to all schools, creating tailored, school-specific solutions across Town to encourage more active travel by youth.
Rationale:	Many municipalities and active school travel organizations are working to encourage active school travel such as walking, cycling, rolling and skateboarding. However, youth rarely have an opportunity to shape the priorities, key messages and engagement initiatives related to these programs. This approach is not consistent with the disability justice mantra, "Nothing about us without us." While adults such as parents/caregivers, educators and city staff play an important role in increasing active school travel, it is important to centre the lived experiences and intelligence of young people who will ultimately live with the social, environmental and health impacts related to this important issue. Working with youth to understand their specific concerns and needs means that the solutions proposed will meet the needs of the target audience, increasing the likelihood of program success.
Steps to Implementation:	**D3.1** Develop a plan for launching the Town of Ajax Active School Travel Advocacy Toolkit. **D3.2** Facilitate a youth-based capacity-building workshop for improving walking, rolling and transit in equity-deserving communities. Provide honoraria to support community champion involvement. **D3.3** Create a resident assessment panel to bring voice to the concerns of residents and ensure that a broad diversity of opinions are considered as the TDM Plan moves forward.

ACTION D3 – Build Youth Capacity in Equity Deserving Communities

Lead:	Region of Durham
Support:	School Boards, Town (Planning & Development)
Estimated Budget Needs:	2023 - $0 2024 - $2,500 2025 - $2,500 2026 - $2,500 2027 - $2,500 Total - $12,500
Performance Indicators:	• Number of youth involved • Number of Schools that have hosted a capacity-building workshop • Number of activities (events held/resources distributed) • Positive stories from youth
Examples:	• York Region Transit YO! (Youth On-board) Ambassador Program - https://forms.yrt.ca/YO-Program-registration# • Ontario Active School Travel success stories - https://ontarioactiveschooltravel.ca/student-leader-talks-about-his-high-schools-booming-cycling-culture/ • Youth Bike Summit - https://youthbikesummit.org/

ACTION D4 – Investigate the Feasibility of Creating a Bike Library

Description:	A Bike Library involves lending bikes, often with the use of a Library card. A bike library is different from Bike Share. Bike share, typically delivered by a private company, involves paying to use a bicycle for a set amount of time. In contrast, Bike Libraries lend bikes either for free or with a refundable deposit fee to pay for potential damage or replacement.
Rationale:	To increase access to bicycles, a bike library may be a more appropriate initial step for Ajax since it can be implemented locally using existing resources and additional funding sources without additional efforts to attract bike share companies. It also reduces the risk of technical difficulties associated with docking bicycles and are more user-friendly for people who do not have a smartphone or credit card. Currently, the Ajax Public Library lends bike locks with three at each location. The bike locks are frequently being borrowed.
Steps to Implementation:	**D4.1** Create a business case for the creation of a Bike Library, preferably at the libraries and other Town Facilities, as feasible. **D4.2** Review possible sites for bike storage and review administrative capacity for lending bicycles. **D4.3** Seek funding for procuring bicycles and accessories.
Lead:	Town (Transportation)
Support:	Ajax Public Library, Operations, Recreation
Estimated Budget Needs:	2023 - $0 2024 - $30,000 for 3 sets of 10 bikes (including helmets, lights, bells, and locks) 2025 - $6,000 (maintenance) 2026 - $6,000 (maintenance) 2027 - $8000 (maintenance + helmet replacement) Total - $50,000
Performance Indicators:	• Number of rentals annually
Examples:	• Markham Cycles Bike Lending Library - https://markhampubliclibrary.ca/bicycle-lending-library/ • Bike Brampton Bike Library - https://bikebrampton.ca/different-spokes/brampton-bike-hub/

ACTION D5 – Engage Equity-Deserving Communities

Description:	Engage equity-deserving communities in transportation options to expand mobility choice through supply-side solutions.
Rationale:	Car-centric design of the past has created barriers for disproportionately impacted communities to access essential services, programs and opportunities. Reliable and affordable transportation is critical for accessing educational opportunities, jobs, housing, health care, and other services that enable economic prosperity, physical and emotional well-being, and social mobility. Conversely, a lack of transportation access often further exacerbates structural inequities in health and wealth in equity-deserving communities.
Steps to Implementation:	**D5.1** Implement one or more bike events targeting underrepresented groups. (This will be done in 2023 as part of a project funded by the Ontario Community Changemakers Grant). **D5.2** Implement a driver safety awareness campaign centering pedestrians with disabilities. (This will be done in 2023 as part of a project funded by the Ontario Community Changemakers Grant). **D5.3** Create and invite residents and community leaders to sign up for updates on upcoming engagement opportunities for supply-side projects.
Lead:	Town (Planning & Development)
Support:	Town (Corporate Communications)
Estimated Budget Needs:	2023 - $0 2024 - $2,500 2025 - $2,500 2026 - $2,500 2027 - $2,500 Total - $10,000
Performance Indicators:	• Satisfaction of participants • Number of impressions • Number of residents signed up for supply-side project updates.
Examples:	• City of Toronto - https://uttri.utoronto.ca/news/planning-for-equity-in-torontos-active-transportation-infrastructure-maharaj/ • City of Hamilton - https://www.hamilton.ca/people-programs/equity-diversity-inclusion • Seattle, WA, USA Transportation Equity Program - https://www.seattle.gov/transportation/projects-and-programs/programs/transportation-equity-program • Austin, TX, USA Walk, Bike Roll – 12 community ambassadors representing Austin's most vulnerable populations were hired to collect feedback from their communities. The ambassadors were compensated, had flexible hours and were involved in decision-making. - https://storymaps.arcgis.com/stories/ceaaa82f2db543b5bb3ef760a5d601b9

In April 2019, the Ajax Integrated Transportation Master Plan (ITMP) was completed. It is a multi-modal planning tool for short, medium, and long-term goals that seek to improve different modes of transportation in the Town to the year 2031. The ITMP provided a comprehensive detailed data collection framework as a blueprint to monitor progress. The data collection framework with key indicators can help inform the overall effectiveness of TDM initiatives that have been implemented. **Table 9.2** shows the multi-modal data collection framework with key indicators from the Ajax ITMP.

Table 9.2: Key Multi-modal Transportation Indicators

#	Mode	Indicator	Unit	Data Source	Frequency
1	Active Transportation	Total kilometres of on-road and off-road cycling facilities	Km	Town of Ajax Durham Region	Every 2 years
2	Active Transportation	Total kilometres of new sidewalks	Km	Town of Ajax	Every 2 years
3	Active Transportation	Number of collisions or accidents to pedestrians or cyclists	Unit Frequency	Durham Region Police OPP	Every year
4	Active Transportation/ Transit	Number of existing and new bicycle end-trip facilities (bike parking, bike share, bus units with bike racks)	Unit Frequency	Town of Ajax	Every year
5	Transit	Number of kilometres of existing and new transit routes (Transit coverage)	Km	Durham Region Transit	Every 2 years
6	Transit	Ridership	Ridership	Durham Region Transit GO Transit	Every year
7	Transit	Effective kilometres travelled by transit units	Km	Durham Region Transit GO transit	Every year
8	Transit	PKI – Passenger-fares per effective kilometre index	Index (Pax/km)	Durham Region Transit GO Transit	Every year
9	Car	Private vehicle ownership per 1000 inhabitants	Index (registered vehicles / 1000 people)	MTO/Census Canada (population)	Every 5 years
10	Car	Number of collisions or accidents (motorists)	Unit Frequency	Durham Region Police OPP	Every year
11	Car	Total lane kilometres of new, repaved or newly-treated roads	Lane km	Town of Ajax Durham Region	Every 3 years
12	Car	Daily Vehicle Kilometres Travelled (VKT)	Km	Town of Ajax (through Durham Region traffic model)	Every 5 years
13	Car	Daily Vehicle Hours Travelled (VHT)	Hours	Town of Ajax (through Durham Region traffic model)	Every 5 years
14	Car	Screenline analysis (volume/capacity)	A.M. peak volume / capacity	Town of Ajax (through Durham Region traffic model)	Every 5 years
15	All modes	Modal split (commuting)	Percentage of trips	Town of Ajax (through Durham Region traffic model)	Every 5 years
16	All modes	Number of daily trips	Trips	Transportation for Tomorrow Survey (TTS)	Every 5 years

Town of Ajax Transportation Demand Management Plan

Appendices

#GetAjaxMoving

MEMO

TO: Armi De Francia - Town of Ajax

FROM: Shawn Smith, Justin Jones - WSP

SUBJECT: Ajax TDM Plan Update | Engagement Summary

DATE: **March 16, 2023**

Background

To inform the development of Transportation Demand Management Plan update, the Project Team conducted several engagement activities with the public and key technical stakeholders throughout the community. Public engagement methods prioritized advancing transportation equity by engaging community organizations and groups who have traditionally been left out of the TDM policy development process. Meaningfully engaging and collaborating with the community of Ajax is essential to ensuring the final TDM plan and its measures are accepted by the community and are implemented. Community and stakeholder engagement was conducted both in-person and remotely to engage a broad range of residents.

Engagement Objectives

This project was developed using International Association of Public Participation (IAP2) process and practices, as illustrated in **Figure 1** below. The IAP2 Process outlines the preparation, management, and evolution of engagement tactics based on a spectrum of involvement tailored to the wants and needs of the anticipated or desired audiences. There are five levels of commitment, which are known as the IAP2 Spectrum of Public Participation.

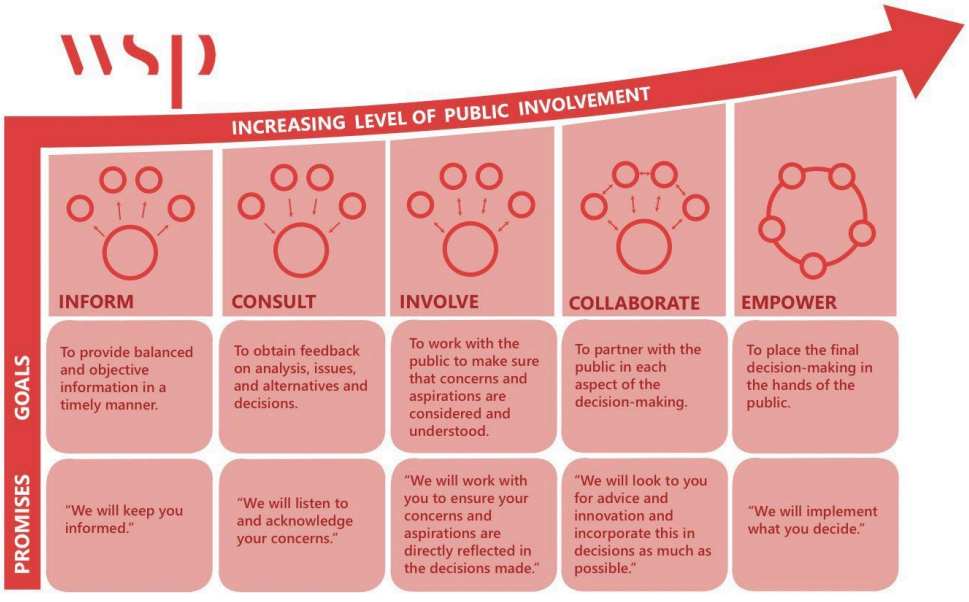


Figure 1. IAP2 Spectrum of Audience Involvement

The amount of information sharing, gathering and integration increases as you "move up" the spectrum. The intent is to recognize that not all stakeholders will have the same level of participation in the project or need the same amount of information to inform their involvement. The IAP2 approach emphasizes the importance of a Community Engagement Strategy which is tailored to the understanding, commitment and contribution of each of the unique groups. By identifying the stakeholders early in the study process the project team will be able to anticipate, identify, plan for and communicate the expectations based on the intended audience.

For the Ajax TDM Plan update, the Project Team focused on providing a high level of involvement and engagement for all residents and stakeholders within Ajax. Engagement activities for the project achieved the level of "involve" or higher for each stakeholder group, including residents of Ajax.

In addition to the IAP2 Approach, the Project Team adopted the principles of Dignity Infused Community Engagement (DICE). Created by Thrivance Group CEO Dr. Destiny Thomas, the DICE Method centres the lived experiences of communities that have been marginalized in the creation of engagement and outcomes. The DICE method focuses on building trust and respect within communities by:

1. Building coalitions
2. Valuing the work that is already being done in those communities
3. Building capacity
4. Thoroughly vetting recommendations to ensure that what is being proposed provides an environment of dignity for those who are at the highest risk of harm

At the start of the engagement process, the Project Team identified key audiences that were deemed essential be consulted and collaborated with throughout the project to achieve actionable TDM measures. The identified audiences included but was not limited to:

- Town of Ajax Staff;
- Durham Regional Transit and Metrolinx;
- Durham District Schoolboard, Durham Catholic District Schoolboard and Conseil Scolaire Catholique Mon Avenir;
- Local businesses;
- Durham Active Transit Committee;
- Durham Region Cycling Coalition;
- The Age Friendly Ajax Sub Committee;
- Community Organizations;
- Seniors Organizations; and
- Members of the public.

Engagement Approach

Throughout the Spring and Summer of 2022, the WSP Project Team worked with the Town of Ajax to facilitate several engagement activities with the identified audiences listed above to inform the development of the Ajax TDM Plan update. These activities were completed to gain an understanding for the success and shortcomings of the existing TDM plan, what contemporary priorities are for the update, and what communities are most susceptible to transportation barriers within the Town. The

following activities were conducted to help inform the development of TDM measures and to help inform the development of the recommended measures in the Ajax TDM Plan update.

Social Climate Analysis

A Social Climate Analysis (SCA) is a way of building trust and genuine partnerships within a community. Rather than seeking to "improve" a community from the top down, a Social Climate Analysis identifies the critical social infrastructure that already exists within a community, focusing on building relationships and capacity to support the community's ability to fulfill its potential by naming, mitigating, and working toward the removal of systemic barriers that impede a community's ability to thrive. A SCA is developed in three stages that includes

1. listening and observing – going to where residents are already gathering to develop a baseline understanding of community conditions,
2. engaging in discussions with community leaders to confirm observational findings, and;
3. providing space for community leaders to co-create solutions.

The scope of this project allowed the first two phases to be conducted, whereas the third phase will be an on-going process following the completion of the project.

The goals of the Ajax Social Climate Analysis were to:

- Identify the needs of residents who do not have a high degree of mobility choice by making deliberate outreach efforts to residents who have not been heard in previous transportation planning processes;
- Understand the existing social networks within Ajax that support mobility;
- Highlight areas where the Town and its partners can build capacity within community groups to support ongoing mobility programs; and
- Chart a path forward where the Town and its institutional partners can honour the work being done by community groups while building trust and new capacity in those communities.

By striving to achieve each of these goals, the proposed measures in the Ajax TDM Plan update will also play a role in the third stage of the social climate analysis, providing agencies and community groups with the resources and capacity to develop and enhance mobility choice in their communities. Additionally, the proposed measures will not only work for people with a high degree of mobility choice and 'traditional' commuting patterns, but will also better serve those with low mobility choice whose access to amenities have been limited by processes centered around the movement of single-occupancy vehicles.

Due to the time constraints of the project, the first phase of the SCA was expedited by working with Town staff from the Recreation, Culture and Community Development and Transportation departments to identify community leaders who could provide insight into the project. Additional research from the Project Team identified 10 community leaders who were subsequently interviewed to identify transportation successes, challenges, and barriers experienced by their respective communities. The interviews took place in early Spring with community advocates, community support and senior care organizations, faith leaders, and regional service providers.

#GetAjaxMoving Webpage

A digital platform was developed to promote and engage the general public on the development of the TDM Plan update. The platform was launched in June 2022 and closed in late September. Throughout the lifecycle of the website, there were 910 visitors to the page, 233 of which participated in a virtual engagement activity. Another 318 visitors looked at multiple pages or accessed files shared on the webpage, raising awareness behind the project.

InMyOpinion Survey

Between June 28 and September 25, 2022, the Town of Ajax ran a TDM community survey hosted on the InMyOpinion Ajax webpage with the intent of identifying current travel patterns, transportation mode choice, and gaps within the transportation system throughout the Town. The survey was promoted online by the Town of Ajax through social media platforms, in print media from local and regional outlets, and in-person. Town of Ajax staff and members of the Project Team took survey promotion directly to the public at the Ajax GO Transit station during a Friday morning rush hour, mid-day at the Ajax Community Centre and in the afternoon at a Mosque in Ajax to promote the survey to Zuhr attendees (afternoon prayer).

In-person survey promotion involved Town Staff and Project Team members handing out leaflets that had information regarding the TDM Survey and were offered #GetAjaxMoving souvenirs for free to help incentivise survey uptake. In total, 233 surveys were completed.

Agency Meetings

The project team convened a series of agency meetings to present findings from the social climate analysis and to receive feedback on the proposed TDM measures for Ajax. On October 12, 2022, the first agency meeting was hosted with members of the Town of Ajax planning, engineering and transportation staff alongside regional transportation authorities such Metrolinx and DRT. To begin the session the Project Team distributed a Mentimeter poll asking each respective agency to identify what TDM measures they would like to see included, not included, and what their overall goals and objectives were for the update. After soliciting feedback, our team then presented the SCA and the resulting Thrivance Narrative to the committee, and subsequently asked the group to the same series of questions at the start of the session. The purpose of this exercise was to provide each attendee with the opportunity to revaluate their initially proposed measures through a transportation equity lens.

A second Agency Meeting was held remotely on December 8, 2022 with the intent of soliciting feedback on the proposed TDM measures that the Project Team and Town developed. During this meeting, 17 representatives from public agencies attend including those from the Town of Ajax, DRT, the Durham District Schoolboard, Durham Region, Metrolinx and the Ajax Library. The session included a presentation by the Project Team where attendees were invited to share feedback as recommendations were presented. The open conversation format allowed for the identification of potential partnerships, and created opportunities for the project team to verify the overall direction of the project and the recommendations contained in the TDM Plan update.

In-Person Engagement

On September 23, 2022 3 members of the Project Team from both the Town of Ajax and WSP Canada conducted a series of in-person engagement and promotion for the TDM Plan Update community survey. The Project Team members distributed physical copies of the survey and post-cards with a QR

code and a link to the digital version of the survey. During this session the Project Team also provided giveaways including umbrellas, water bottles, bike lights, and tote bags featuring #GetAjaxMoving branding to incentivize uptake of the survey and build awareness around the TDM campaign across the Town.

Figure 2: Left: Town of Ajax and WSP staff promoting the TDM survey at the Ajax GO Transit Station. Right: Frontside of the promotional postcard distributed at in-person engagement events.

The first session was held at the Ajax GO Transit Station from 07:00 – 10:00 to capture morning commuters in the survey. During this morning session over 100 commuters were provided with promotional postcards and giveaways promoting the TDM survey. The second location was the McLean Community Centre from 10:30 – 11:30 to capture seniors, parents, and staff members within the survey results. Finally, the Project Team was invited by the Islamic Society of Ajax to promote the TDM survey at two Jumma (Friday afternoon prayer) sessions at the Ajax Masjid Quba from 12:30 – 2:00. The Islamic Society of Ajax allowed the Project Team to setup a booth outside of the Masjid Quba where the team was able to interact with attendees prior-to and following each Jumma session. Overall, the Project Team engaged with over 100 attendees between the two sessions.

Public Open House

A virtual Public Open House was held in the evening on Monday, January 16, 2023. It provided an opportunity for the team to show residents the impact of their survey/interview contributions, receive additional public feedback, and provided transparency about the project scope and recommendations. There were 23 people who originally registered and 14 that attended. A public recording of the public open house is available at https://youtu.be/ZtsWp6C-7tA.

What was said?

Social Climate Analysis

Strengths and Assets in Ajax

Through our interview series our team identified that community organizations across Ajax have established community-based transportation systems and programming designed to help residents, particularly those who do not own personal vehicles, get around. These programs manifest in a variety of formats such as door-to-door specialized transit (Kids/Seniors on Wheels; Community Car Durham), carpooling arrangements), or by bringing essential services directly to clients (Doctors on Wheels).

A persistent theme to emerge in discussions with community leaders was that there was a strong recognition of the mobility deficiencies that exist throughout the Town for those who do not have direct access to a personal vehicle and rely on public or active transportation modes. Each organization leader was eager to make travel and access to services easier for clients who are frequently left out of the fold as a priority group for improving mobility. Our team also identified that there is coordination between different community groups and community transportation providers that allows for a greater reach to connect community members to essential services and work.

The strengths of the community emerged in response to existing transportation gaps in the community that were then exacerbated by the COVID-19 pandemic. The interview process revealed that transportation needs were not being met for workers, seniors, and vulnerable residents during non-traditional commuting hours. In addition to these gaps in mobility, community members also indicated that throughout the pandemic there was a lack of transparency from local and regional transportation providers when it came to transit service reductions, route changes, and moving stops. The lack of transparency and integration of community feedback negatively affected users who were reliant on the service.

InMyOpinion Survey

With a total of 252 responses collected, the online survey hosted on the Town's In My Opinion Ajax Engagement Platform provides an important snapshot of the current mobility needs of the community. While the survey does not meet the standard of being statistically representative of the Town's population as a whole, there are still important insights to be gleaned from the responses. The project team gathered the following information and public priorities for TDM measures through the public survey:

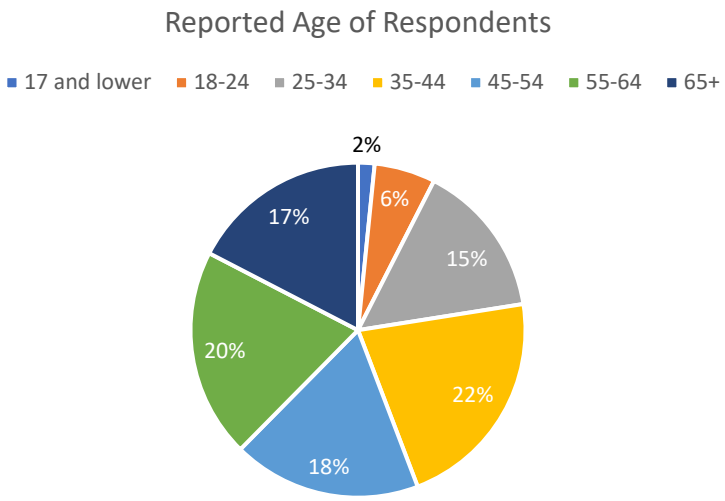


Figure 3: Age of 2022 TDM survey participants.

The results from the survey showed that the respondents were largely representative of the Town as a whole. Responses from across different age groups and ethnic groups showed that participation was not limited to a particular demographic group. It should be noted that male identifying respondents were overrepresented in the responses, accounting for 56.6% of response compared to females (42.9%) and non-binary and two-spirited at 0.4% each, respectfully.

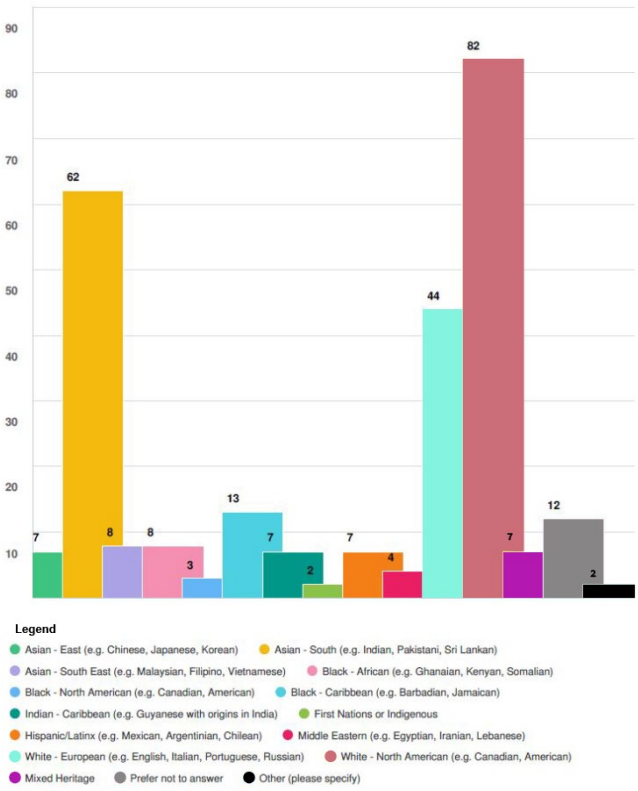


Figure 4: Cultural heritage of survey participants.

Survey respondents also largely reflected the cultural diversity of the Town, although White-identifying respondents were slightly over-represented among participants.

15.9% of respondents also indicated that they live with either a mental or physical condition that reduces the amount or kinds of activities they can do in a given day, whereas 6.9% used an assistive device to help with their mobility.

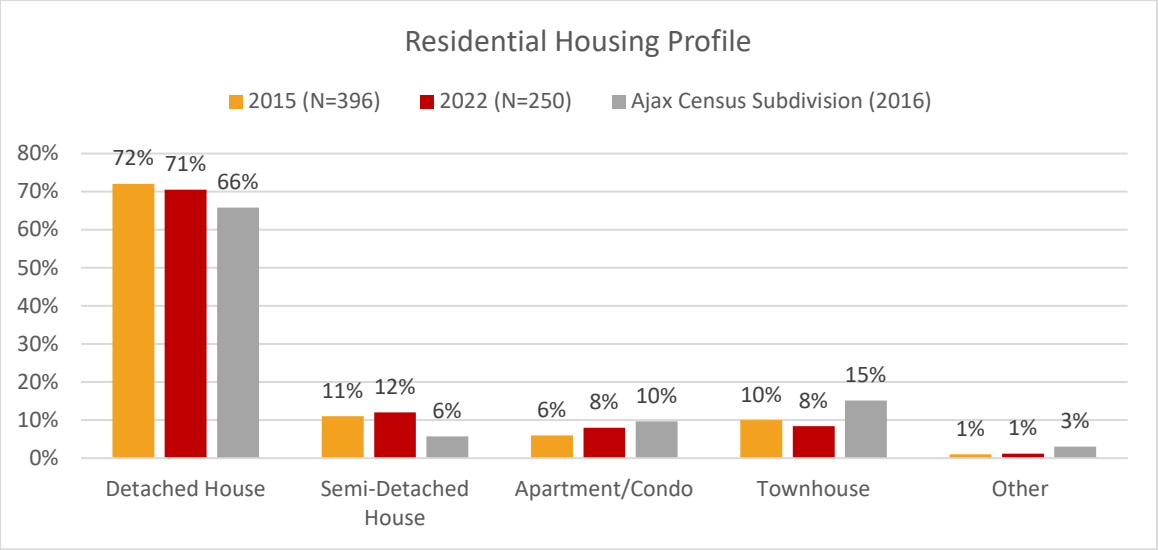


Figure 5: Comparison of the housing profile between the 2015 TDM Survey, 2022 TDM Survey, and Statistics Canada 2016 data.

While the Survey was not designed to be a statistically representative sample of the Town, some of the data available does demonstrate that the survey did collect responses that, on the whole, reflect the Town's overall demographic makeup. In particular, the results relating to household makeup and ownership demonstrate that the survey's respondents were largely consistent with what would be expected based on Statistics Canada data. The exception is that the survey underrepresented 1 resident households, although the response rates among this demographic were an improvement from the 2015 survey (Figure 6).

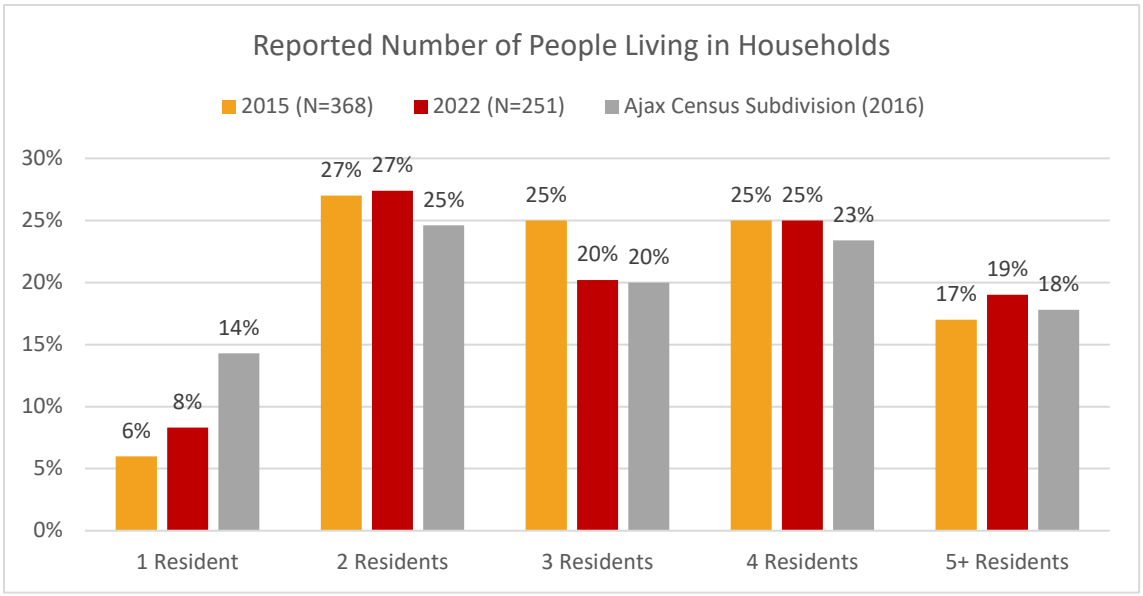


Figure 6: Comparison of the reported number of people residing in respondents' households between the 2015 TDM Survey, 2022 TDM Survey, and Statistics Canada 2016 data.

Households across Ajax predominantly comprise of multi-dwelling single-detached homes. However, the number of renters and single dwelling residents appears to be steadily increasing, aligning with regional trends. Community members who participated in the survey reported a high level of home ownership, notably the polled group captured a higher portion of renters compared to the Town percentage reported in the 2016 census (Figure 7).

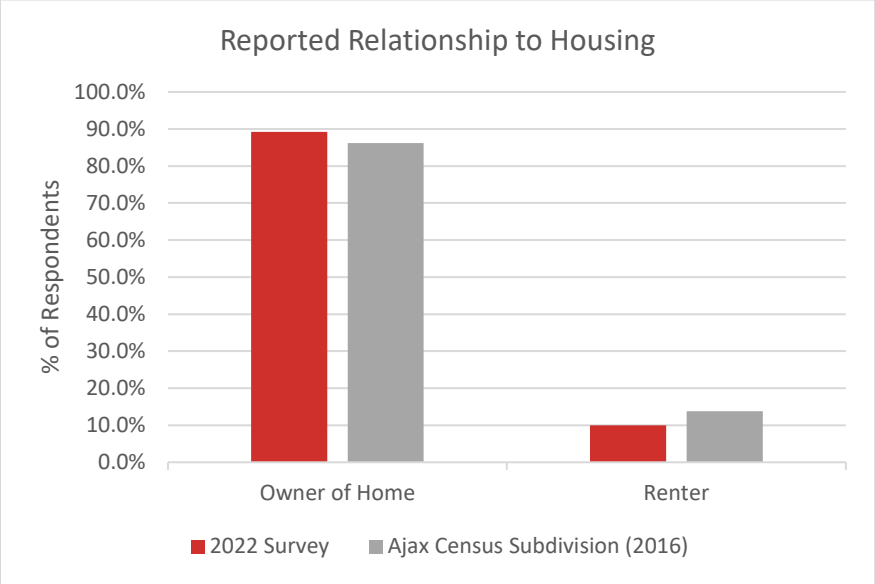


Figure 7: Comparison of residents' relationship to housing between the 2022 TDM Survey and the 2016 Census results.

Travel and Commuting Profile

One notable trend observed in the 2022 survey is an overall decline in the number of vehicles owned per household in Ajax. The survey revealed significant increases in the percentages of zero and one-car households, with a corresponding decline in the percentage of households with two or more vehicles. (Figure 8).

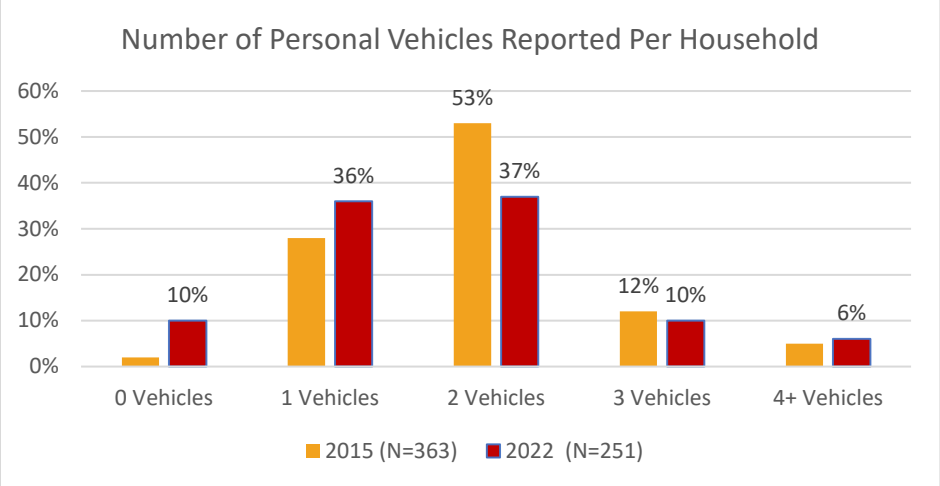


Figure 8: Reported number of vehicles per household between 2015 and 2022.

Commuting behaviours among residents for whom remote work is an option (Approximately 40% of respondents overall) exhibited perhaps the most significant shift compared to the 2015 Survey. For residents that can work remotely, working one day a week or more in a remote environment has become the norm, with 78% of respondents indicating that they work from home at least one day a week if they are able. (Figure 9).

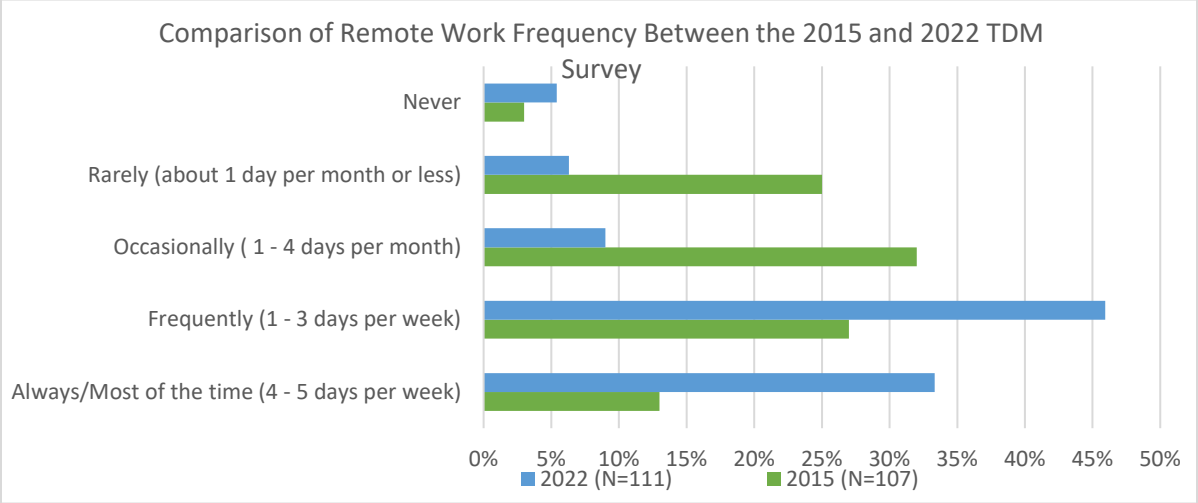


Figure 9: Ability to work remotely and frequency between 2022 and 2015.

Figure 10 sheds further light on how remote work has changed in Ajax, with far fewer respondents indicating that onsite work is a requirement of their job, or that remote work is not permitted. For many residents, the 'push' factors of commuting to work have been reduced; there is more flexibility in how work is performed. The 'pull' factors, which are more around the in-person experience of going into a defined workspace, have become more important as barriers to working remotely have been removed over the course of the past 3 years.

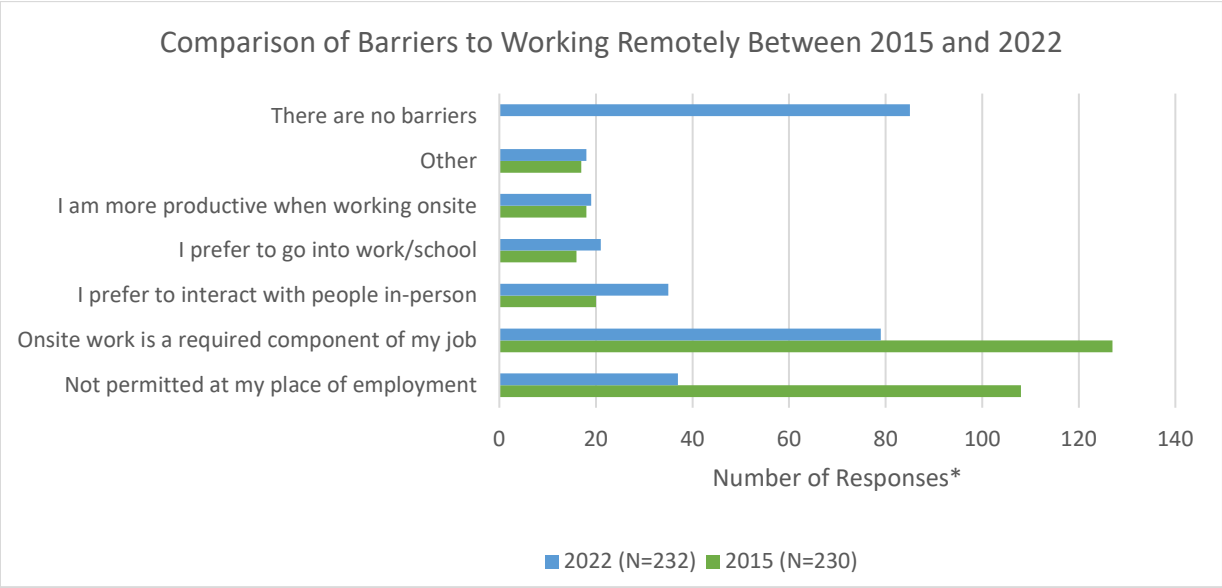


*Figure 10: Comparisons of the barriers to remote work between 2015 and 2022. *Note that respondents were allowed to select more than one barrier leading to a more responses than survey participants.*

When assessing where residents work most frequently there is an observable trend that more residents are remaining in Ajax for work. Respondents reported working closer to home, with higher rates of respondents remaining in Ajax to work each day, either at a primary place of work or when working from home. The sharpest decline in reported work location was among respondents who work in the GTA, but outside of Durham Region and those who work outside of the GTA, demonstrating a sizeable shift in employee behaviours and preferences. (Figure 11)

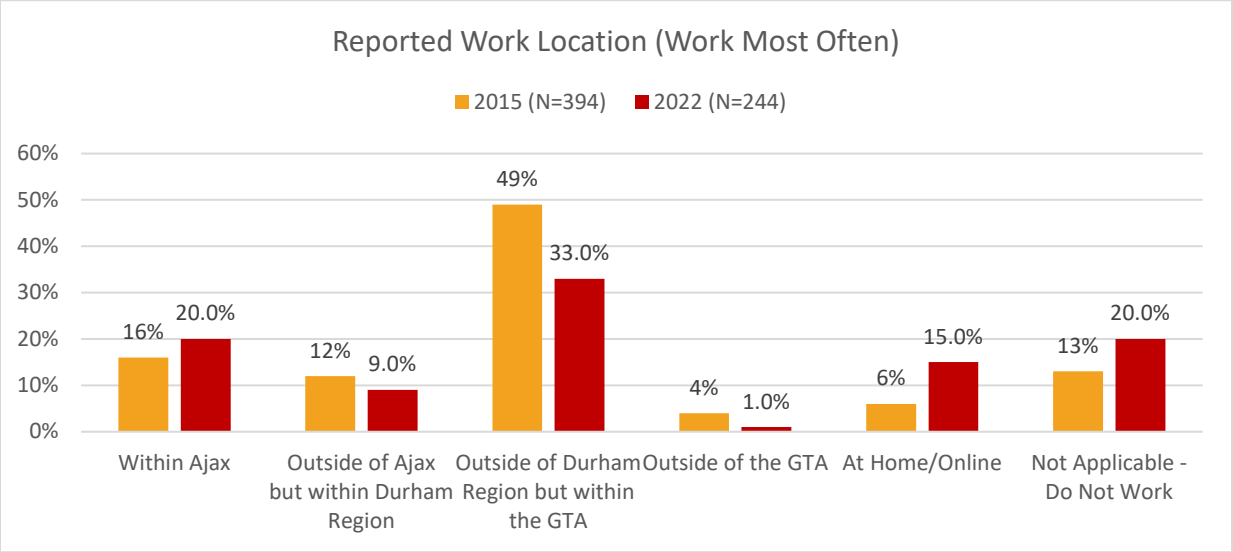


Figure 11: Reported work location between 2015 and 2022.

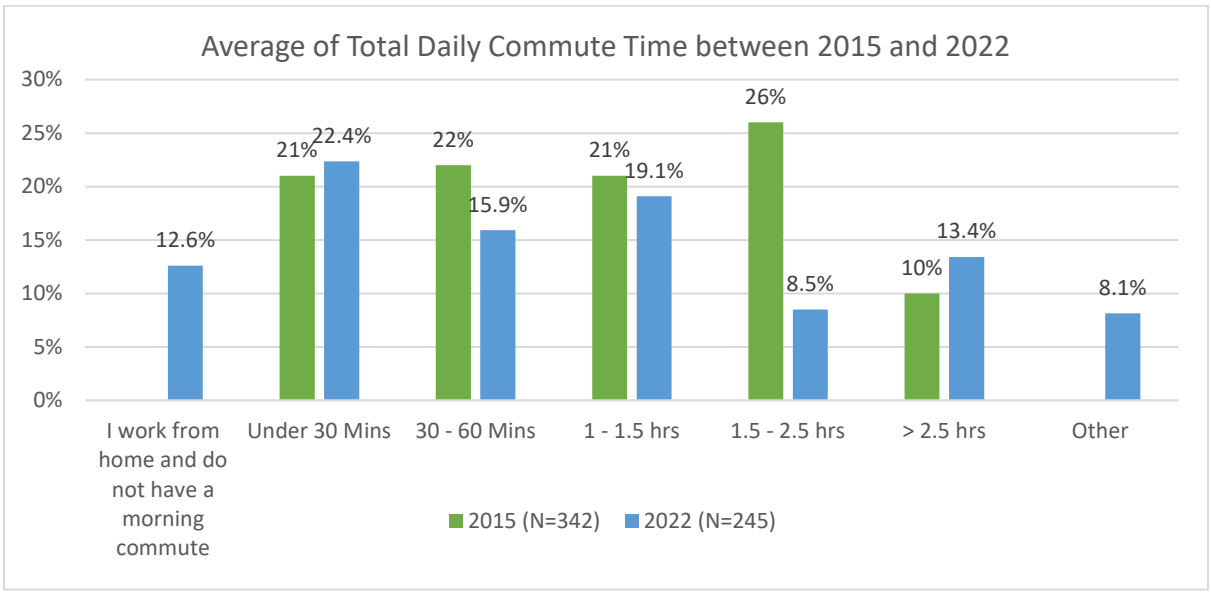
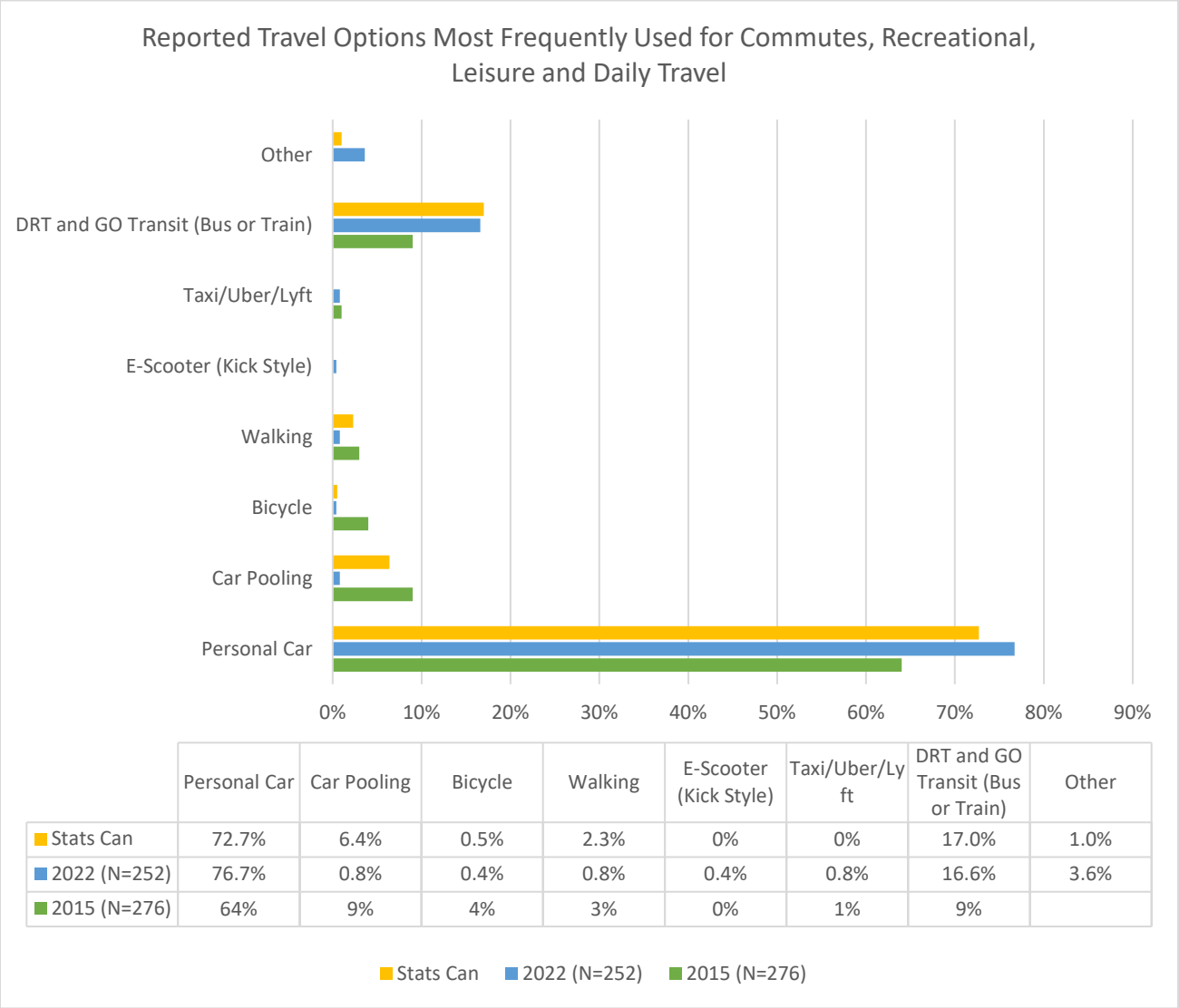


Figure 12: Comparison of average daily commutes between 2015 and 2022.

The final major difference in residential commuting patterns was found when asking residents what the average cumulative time they spend commuting in a day. From respondents there was a sharp decline in those who spend 1.5 – 2.5 hours commuting in day that appears to have been predominantly reallocated to those working remotely or having an under 30-minute commute. Interestingly, among

those surveyed in the 2022 study there were more individuals who reported having a commute longer than 2.5 hours in a day. Finally. Those who reported “other” worked in the transportation sector and did not indicate their cumulative average commute time.

Mode Choice



	Personal Car	Car Pooling	Bicycle	Walking	E-Scooter (Kick Style)	Taxi/Uber/Lyft	DRT and GO Transit (Bus or Train)	Other
Stats Can	72.7%	6.4%	0.5%	2.3%	0%	0%	17.0%	1.0%
2022 (N=252)	76.7%	0.8%	0.4%	0.8%	0.4%	0.8%	16.6%	3.6%
2015 (N=276)	64%	9%	4%	3%	0%	1%	9%	

Figure 13: Comparison of reported mode choice between the 2015, 2022 TDM Surveys and the 2016 Census.

When asking residents to identify their mode choice, it was evident that the use of single-occupancy vehicles are by far the most popular choice for travel of all purposes across Ajax with 76.7% of respondents indicating that to be the case. Following this was the selection of DRT and GO Transit services that saw an increase in usage between those surveyed in the 2022 and 2015 survey. The sampled population for the 2022 survey aligned with the Statistics Canada 2016 data except for the personal car option that was 4% higher. The increase in personal car usage and decrease in DRT and GO Transit could be a lasting trend observed during the pandemic, however it could also be a statistical error in this case. Interestingly, the 2022 survey did not capture as diverse of a range of mobility choices

as the 2015 survey, noting a sharp decline in respondents who carpool or use active transportation modes.

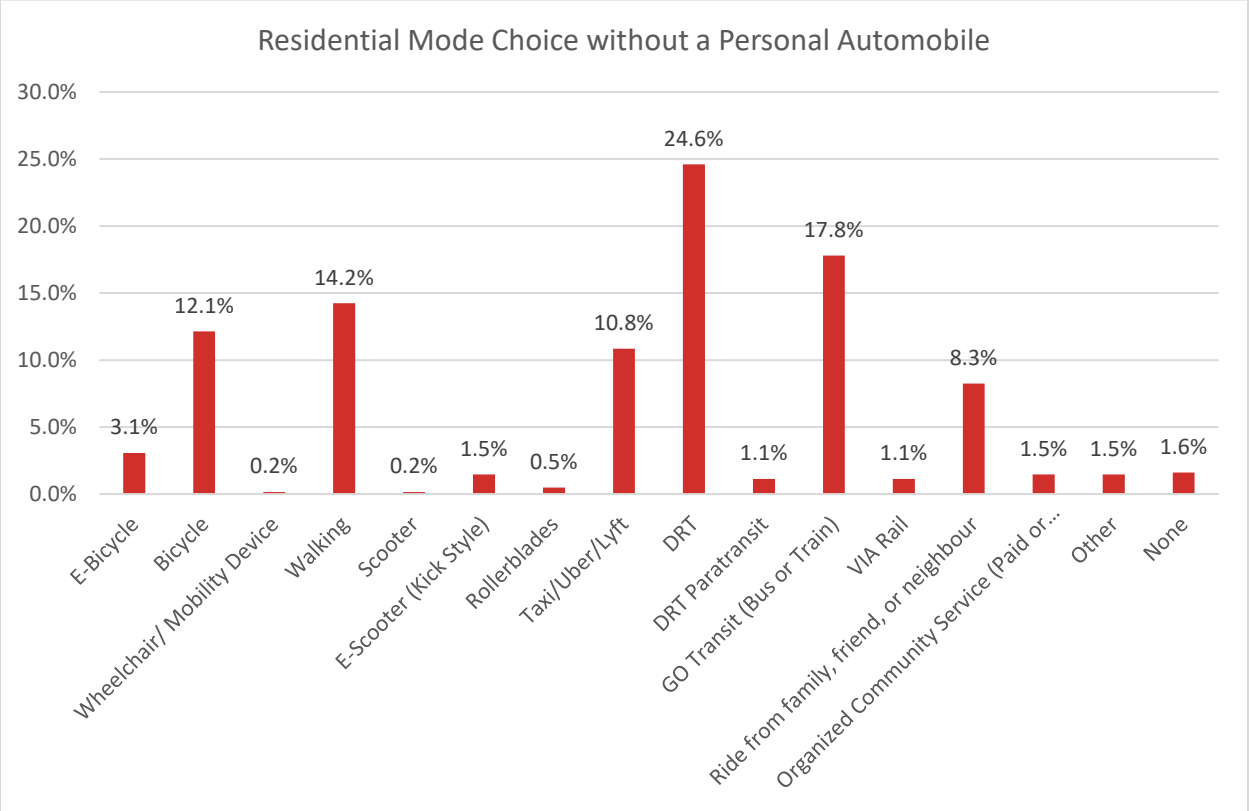


Figure 14: Residents were asked how they would travel without the use of a personal automobile.

When asked how residents would travel if the use of a personal car was not an option, most respondents answered that they would rely on using public transportation such as DRT, GO Transit and VIA Rail services. Notably, walking, cycling, the use of ride sharing services and carpooling were all substantially higher than their current usage levels, demonstrating a potential appeal of switching to sustainable and shared transportation modes among Ajax residents.

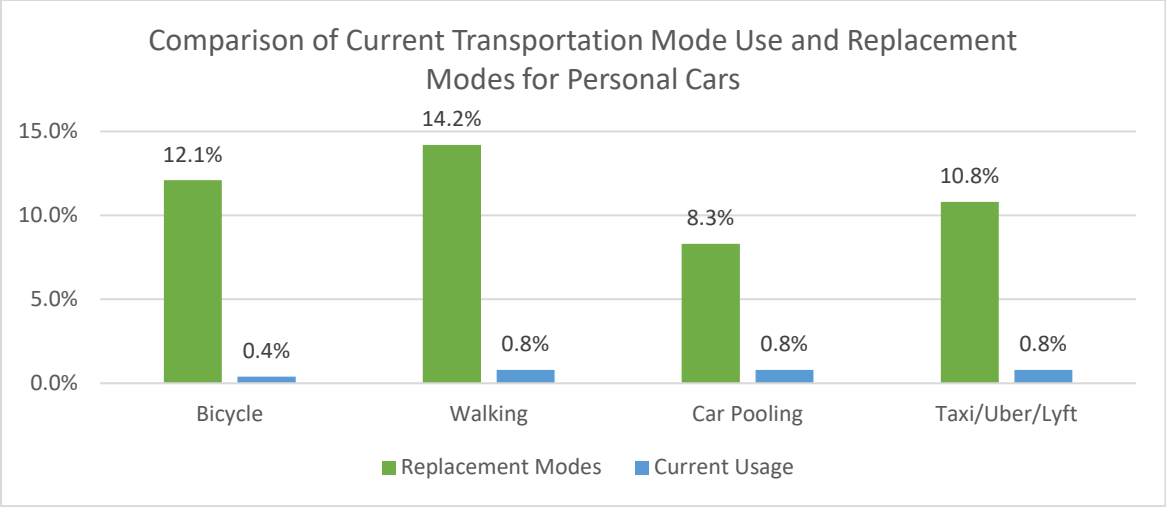


Figure 15: Comparison of current transportation modes and replacement modes for personal cars.

When asked how the Town of Ajax can make more sustainable modes of transportation more comfortable to use many residents indicated that improving transit frequency and routing options were a top priority. The most frequent response had to do with the increase in time it takes to use sustainable transportation modes over personal cars for commuting and travel across the Town, meaning improving directness and frequency of service was a top priority for residents.

Improving the Viability of Sustainable Transportation Modes

Residential Priorities to make Cycling a Viable Transportation Option

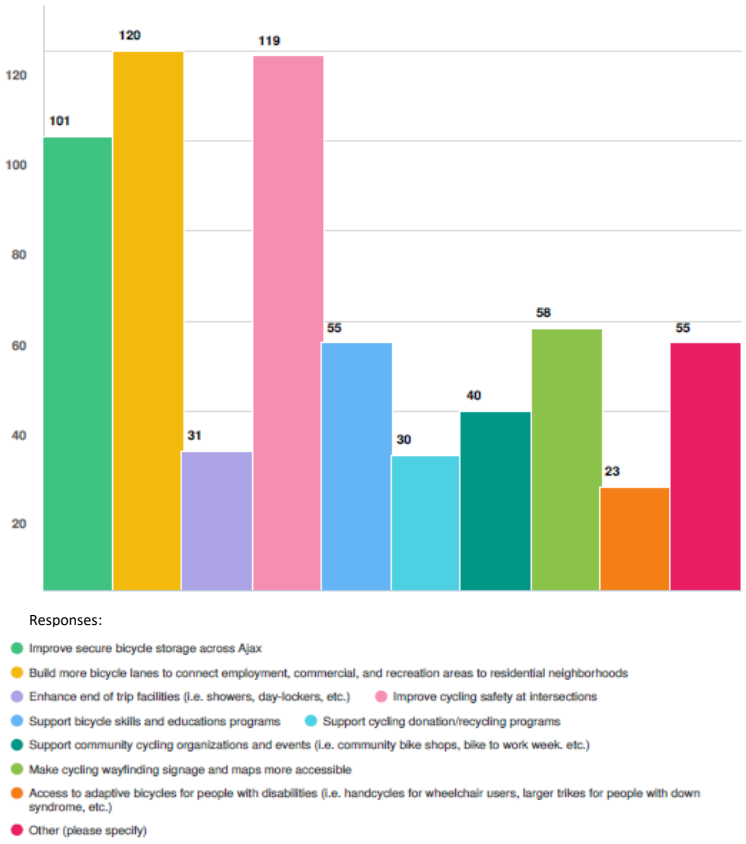


Figure 16: Residential priorities to make cycling a viable transportation option in Ajax

Residents were asked to rank their top 5 priorities for improving cycling, DRT, and walk/rolling as viable forms of transportation within Ajax. The top responses for cycling included:

1. Building more bicycle lanes to connect employment, commercial and recreation areas to residential neighbourhoods;
2. Improving cycling safety at intersections;
3. Improving secure bicycle storage across Ajax;
4. Making cycling wayfinding signage and maps more accessible; and
5. Support bicycle skills and education programs.

Themes that emerged from the “other” category included:

- Many residents indicating that cycling is not a viable mode of transportation in Ajax due to the distances people need to commute and the long winter months;
- Residence seeking barriers and protected cycling paths and trails;
- Adding a bikeshare/rental system in Ajax; and
- Improving crossings and connections across Highway 401.

Residential Priorities to make Public Transit a Viable Transportation Option

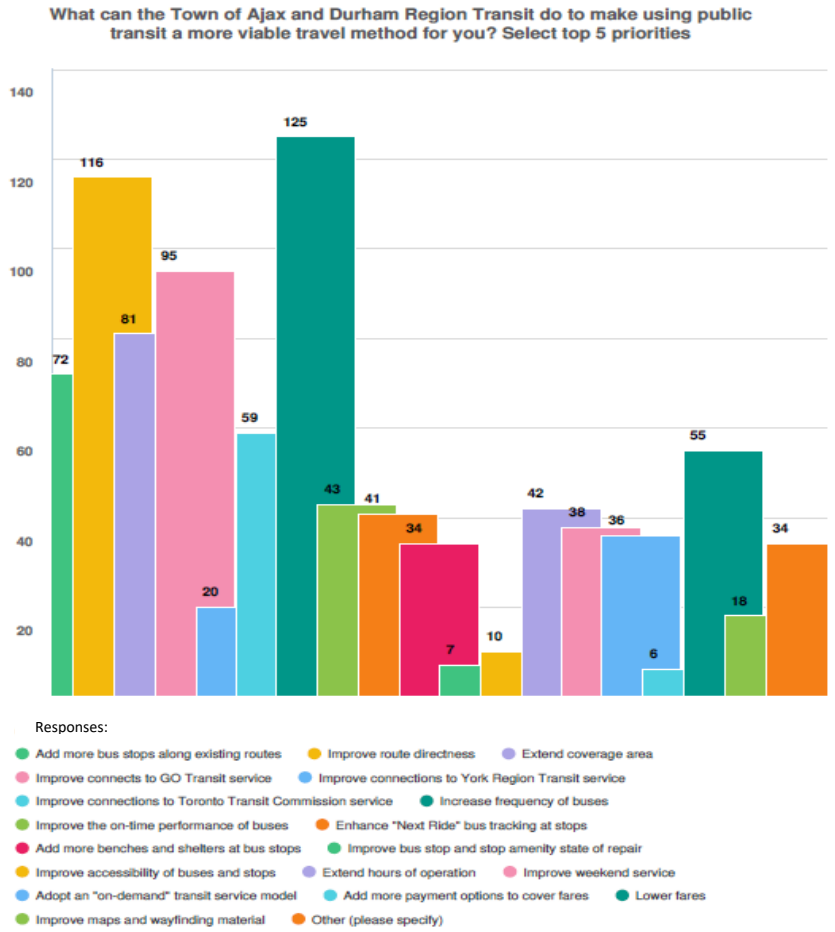


Figure 17: Residential priorities to make public transit a viable transportation option in Ajax

The top 5 community priorities for improving DRT included:

1. Increasing the frequency of buses;
2. Improving route directness;
3. Improve connections to GO Transit service;
4. Extend transit coverage area; and
5. Add more bus stops across existing routes.

Other themes to emerge from the assessment included:

- Reinstating routes that were removed during the pandemic;
- Improve transit priority at intersections and along road rights-of-way;
- Introduce real-time bus tracking information to improve predictability; and
- Improve wayfinding and navigation of the system.

Residential Priorities to make Walking a Viable Transportation Option

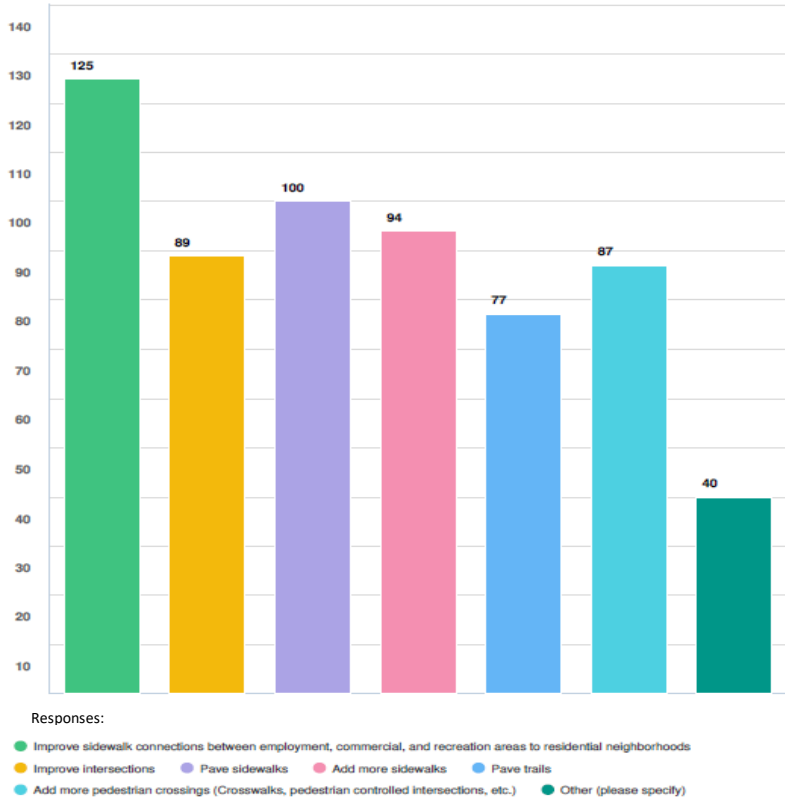


Figure 18: Residential priorities to make walking/rolling a viable transportation option in Ajax

The final question in the InMyOpinion survey asked residents how walking/rolling could be improved as a viable mode of transportation within the Town. Respondents indicated that their top 5 priorities were:

1. Improving sidewalk connections between employment, commercial and recreation areas to neighbourhoods;
2. Pave sidewalks across the Town;
3. Add additional sidewalks;
4. Improve safety and accessibility at intersections; and
5. Add more mid-block crossings and pedestrian controlled intersections.

Other themes identified included:

- Better snow clearing and winter maintenance;
- Improve lighting along roads and trail network;
- Host cycling and walking education and etiquette programs to reduce conflicts between users.

Agency Meeting

Agency Meeting #1

Agency Priorities for the TDM Plan

At the start of the meeting, agency representatives were asked to rank their knowledge of the Town of Ajax's existing TDM measures on a scale of 1 – 8 with 1 being little to no knowledge and 8 being a very high degree of knowledge. Respondents averaged a score of 4 (50%) with a general familiarity of the 2015 TDM plan but not being well versed in its policies.

When asked what each agency would most like to see as a result of the program, responses included:

- Introducing bike share programs in the Town;
- Develop incentive-based programs to encourage a shift to sustainable transportation modes;
- Prioritize the creation of an active transportation network; and
- Improve transit reliability.

When asked what measures agency representatives would least like to see in the TDM Plan update attendees identified that they do not want the plan to:

- Prioritize moving cars over people and continue emphasizing the reliance on private vehicles;
- Have unrealistic expectations, actions, and unfeasible measures for the Town to implement;
- Propose incentive programs that cannot be funded long-term; and
- Include measures from the 2015 Plan that were not implemented or are not currently in progress.

Following these two questions the project team presented the findings from the Social Climate Analysis and subsequently asked the agency representatives the exact same series of questions but asked their responses to consider transportation equity. The results for what measures attendees most wanted to see in the plan included:

- Ensuring that the walking/rolling network is up to AODA standards;
- Improving the safety and reliability of using public transit and active transportation options; and
- Introducing a social climate analysis toolkit for practitioners and improving the relationships and collaboration opportunities between governing agencies and community groups/leaders.

With the Social Climate Analysis results in mind, agency representatives indicated that TDM Plan update measures that they would least like to see included:

- Delivering an update with no resources or consistent funding to implement; and
- Transportation equity goals that are too broad and unclear.

Perceived Level of Effort/Impact for Short and Medium-term Measures

The Project Team introduced the draft list of proposed short and medium-term TDM measures and asked attendees to rank their results. Respondents indicated that each of the six proposed measures would have a high-impact but also would require a high level of effort/complexity to implement.

Figure 19: Short-term TDM Measure Impact/Effort Matrix

The highest-level impact program indicated by the group was improving walking and cycling access to the Ajax GO Station followed by providing active transportation amenities.

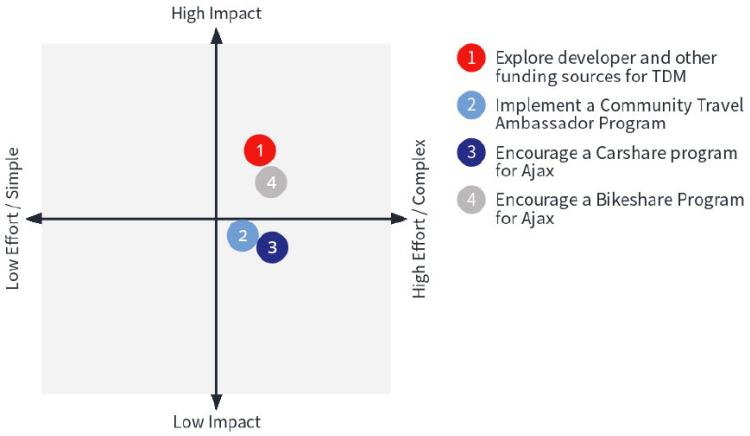


Figure 20: Medium-term TDM Measure Impact/Effort Matrix

Agency members also identified each of the four proposed measures as requiring a high-level of effort or complexity to implement, however, two programs were identified as having a high level of impact relative to the level of effort required to implement. This included exploring developer and other funding sources to support TDM measures and implementing a community travel ambassador program.

Agency Meeting #2

During the second Agency Meeting the group presented the proposed TDM Plan update measures and sought feedback on implementation support and challenges from each group. Key takeaways from the meeting included:

Active School Travel

- Improving conditions around school zones from September – June to mitigate peak travel periods;

- Dedicating a full-time staff member to prioritize creating active school travel programs and partnerships across the Town and Region; and
- Piloting or funding a “free transit for students” program to encourage the use of active and sustainable transportation early-on.

Partnerships and Collaboration

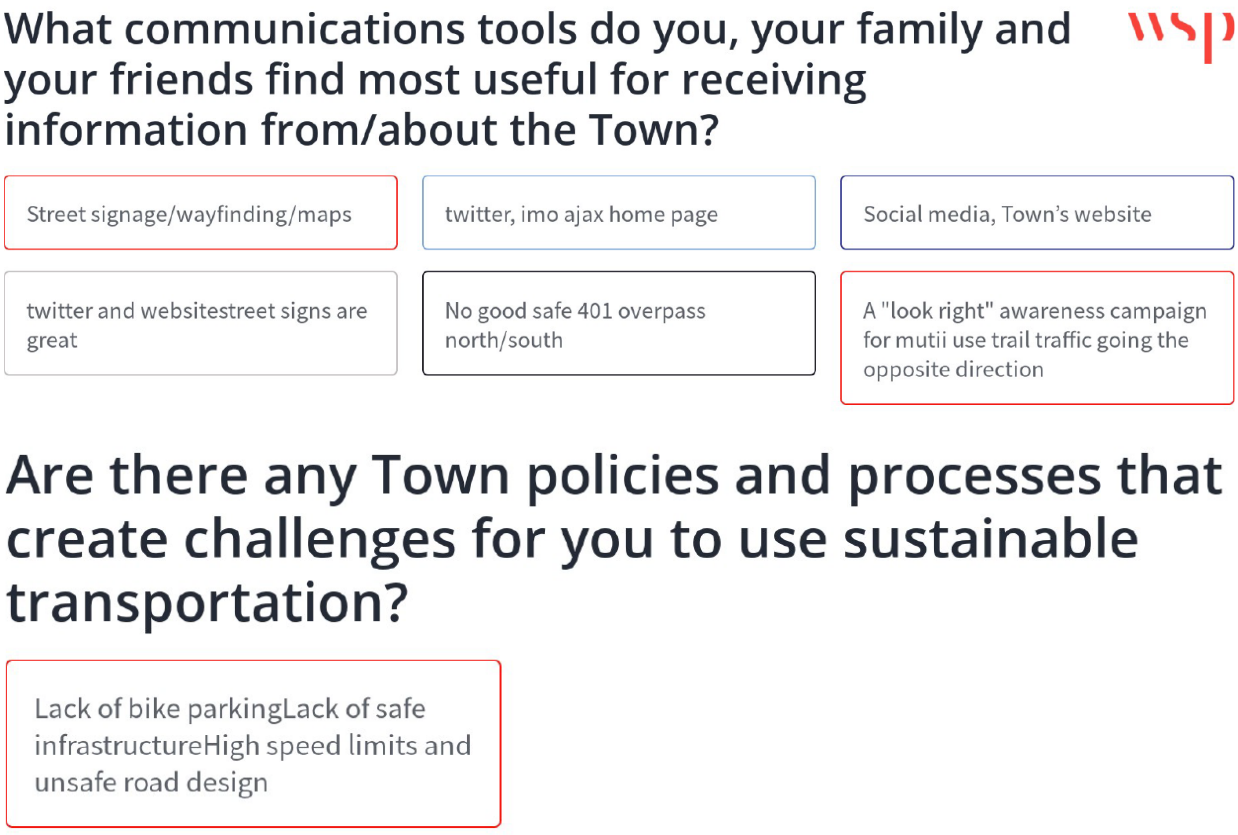
- Develop programs that encourage and support workplace TDM champions;
- Leverage existing community groups and their communication networks to promote TDM measures; and
- Embed TDM measures in new development projects. Ensure that developers are aware and support the TDM plan.

Integrate TDM into the Town Policies, Processes and Programs

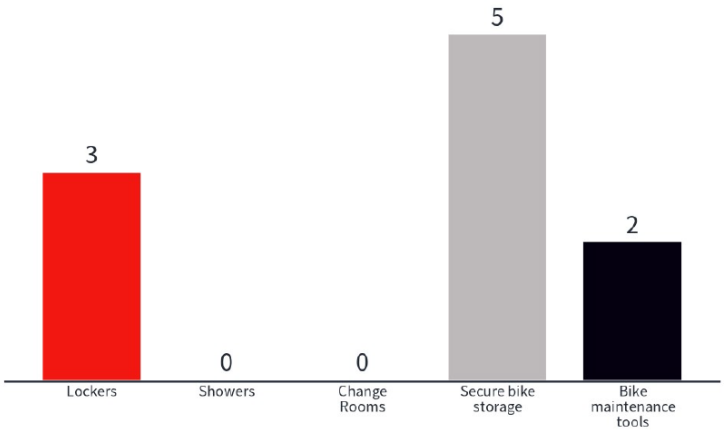
- Coordinate with developers and property mangers to distribute TDM information to new residents encouraging the use of sustainable transportation;
- Soften language around removing parking minimums.

Public Open House

A menti tool was used to collect feedback at the Jan 16, 2023 Public Open House:



What kinds of facilities would make it easier for you to walk, bike or take transit more often?



What other facilities would make it easier for you to walk, bike or take transit more often?

- safe connectivity for cycling will only improve TDM
- Charging stations for mobility scooters and E bikes
- Cargo biek storage
- Bike racks at shops
- Wayfinding on trails (ie to Kerrison rd on Carruther creek trail)
- E-bike subsidy for everyone, not just low income
- Sheltered bike storage, differentiation between temporary and long-term bike storage
- Allow small scooters/ e bikes to push through (safely) while doing business
- secure, covered bike parking. Better traffic/pedestrian signals. Paved transit stops with shelters
- Grocery stores
- Secure locker at Go Station.

Where would you like to see bike parking installed in Ajax? Be as specific as possible.

- Rio-Can Durham Centre
- J Clarke/ND track
- Sobey's
- Everywhere, more covered bike parking around schools, around every commerical/employment place, and in parks and near community facilities
- Sheltered parking on Rossland/Salem plaza
- South Ajax
- All rec centres, or do they have already?
- The North Sobeys has a huge indoor area for carts that would work
- Yes

What types of people would you like to see represented in Ajax's image library? Be as specific as possible.

- The non-tour-de-france cyclists, regular people in sweats on mountain bikes
- Elementary and high school students wearing helmets
- Yes, love that too!
- Working-class/tradespeople
- As diverse as Ajax
- Larger folks.

What groups or organizations could the Town partner with to deliver bike events to underserved groups in Town?

- Lake ridge health(?)
- Ajax Municipal Housing
- Riding clubs teach how to ride safe on toad

What other metrics would you suggest to define success for the TDM Plan Update?

- Busy trails
- People of all ages/colours/abilities cycling
- High modeshare for walking, cycling, and transit usage, as well as cycling and transit being seen more as the norm culturally
- Youth retention, not moving to cities for more transportation options

Other feedback:

- Hard for those with disabilities or seniors to use transit to get places such as from Pickering Village to GO Station.

- Trail system is fantastic, well maintained including in winter. Cycling facilities to community centres, GO Train, Town Hall, libraries, schools can avoid the need to get a second or third car
- Cycling access North-South over Hwy 401 is a challenge. ITMP has a 30+ year network proposed, but these crossings need coordination with Region, are big-ticket items and require a lot of time to implement; safest connections are currently Harwood, Church.
- Pedestrians feel very vulnerable at intersections.
- E-scooters can provide another transportation option and best marketing is getting people to try them. Town is looking into shared e-scooter services
- Suggests database of safest ways to get places

What was heard?

Supply-side improvements

Much of the feedback received focused on aspects of the mobility system that are outside the scope of a TDM Strategy Update. Physical infrastructure changes such as the addition of protected cycling facilities or new sidewalks are addressed by the Town's Integrated Transportation Master Plan, but the importance of those changes emerged as a key theme multiple times during this project.

- The SCA and public survey both demonstrated a desire for both local and regional transit improvements. Community members identified a need to improve transportation frequency and increase the number of stops within the existing system.
- Community priorities for improving cycling included adding bicycle lanes, protected intersections, and improving wayfinding signage across the Town.
- New sidewalk connections and improved winter maintenance on sidewalks were identified as priorities, in addition to paving more trails to improve accessibility.
- adding more mid-block crossings, pedestrian controlled intersections, and lighting were all highlighted as key enhancements to improve walkability across the town.

While these elements are out of scope for this project, they should be noted and should continue to inform infrastructure and transit planning at both the Town and Regional level.

Communications and Transparency

The most consistent theme that emerged through the engagement for the TDM Plan update was the central role that communication needs to play in behaviour change. The Town is already making significant changes in the design and operation of its streets – with new sidewalks, multi-use paths and cycling facilities being added each year. Many residents and community leaders reported feeling unaware of how these changes might impact them, highlighting the need for the Town to work closely with local champions to promote the potential for mobility choice within their networks.

Suggestions for enhanced communication also included making public transportation more predictable, easier to navigate and adding more transparency behind route and scheduling changes. For example, integrating real-time bus tracking allows residents to accurately predict when their bus will arrive and whether it has been delayed. Additionally, a number of comments were made regarding an abrupt route change or termination of bus stops throughout the DRT system. Residents expressed a desire to have DRT operate with a higher degree of transparency and consultation when adjusting transportation route and service levels.

Impact of Remote Work on Traditional Commuting Patterns

The proliferation of remote work has had a sizeable impact on the commuting habits of Ajax residents. Notably the increase in those able to work from home and the decline in resident commute times demonstrates that the workforce has shifted across the Town. Despite these shifts, however, there was an increase in single-occupancy vehicle usage with many residents citing convenience as the primary reason why they continue to drive. The engagement demonstrated that Town residents are open to change, however, if alternatives are safe, convenient and easily accessible.

Policy, Resources and Partnerships

Finally, it was identified that Town policies relating to TDM should be strengthened or updated to reflect the Town's experience with requiring TDM measures as part of the development application process. With the benefit of 7 years of experience implementing the 2015 TDM Plan, Town Staff have a strong understanding of what measures have an impact and what measures often fail to influence change. There was a strong desire among stakeholders and partners to see this update reflect those learnings, and to reduce the number of recommendations while focusing on measures that will have the highest likelihood of success. The addition of new resources at the Town to support these measures was a consistent theme of consultations, as was the importance of partnering with both Durham Region and community organizations within Ajax to make the TDM Strategy Update a success.

MEMO

TO: Armi De Francia - Town of Ajax

FROM: Shawn Smith, Justin Jones, Kyle Dow - WSP

SUBJECT: **Town of Ajax TDM Plan Update: Social Climate Analysis**

DATE: **January 26, 2023**

INTRODUCTION

The Town of Ajax recognizes that mobility choice is a key determinant of health, quality of life and connection to community. As the Town becomes more diverse and mobility patterns shift away from the traditional peak hour commute that has been the focus of previous development, providing options that go beyond the use of a single-occupant vehicle will support the Town's broader objectives of becoming a better place to live, work and play. The goal of the Transportation Demand Management Plan (TDM) Update is to provide all residents of Ajax with more options for how they get around, helping to reduce the number of cars on the road and creating a more efficient and equitable transportation system that moves more people in the same amount of space.

To create a more equitable and sustainable transportation system, the TDM Plan Update will focus on how the Town can meaningfully increase levels of:

- transit use (including local, regional and accessible transit offerings)
- walking (including the use of mobility aid devices)
- biking (including the use of adaptive bikes or trikes for people with disabilities)
- rolling (including skateboarding and rollerblading)
- scooters (including adaptive scooters)
- e-bikes, e-scooters, and other micromobility devices (lightweight vehicles that can operate safely in urban areas)

The TDM Plan will also incorporate emerging trends related to telework, flexible commuting options and ridesharing, all of which have the potential to reduce peak-hour travel demand and improve mobility across the Town.

2015 TDM PLAN

The Town's existing TDM Plan was adopted in 2015, and largely reflected existing best practices and understandings of TDM strategies at the time. The significant shifts in mobility patterns and a growing recognition of the ways in which race, class, income, ability, age, gender and a host of other characteristics impact how communities experience mobility have necessitated this Update to the Plan.

Historically, TDM plans have been designed to focus on the needs of residents working a 'traditional' 9-5 job, seeking ways to alleviate vehicle-based commuting pressures at peak travel times. With the proliferation of hybrid work and the increasing interest of advancing transportation equity considerations WSP Canada Inc has reframed the focus of the Ajax TDM Plan to account for those who work irregular schedules, whose trips occur at unconventional times, or who are dependent on other mobility options as their primary means of travel and non-work-related trips such as trips to food banks, doctors' appointments, and places of worship. In developing the TDM Plan to address the needs of these populations, it is necessary to develop a strong understanding of what barriers and challenges these residents face, what techniques they are already deploying to overcome those barriers, and what social networks the Town and its partners can support to build up additional capacity to move more residents throughout the Town. The Social Climate Analysis component of the Ajax TDM Plan update seeks to understand and what mobility choices, options, gaps, and programs community organizations have developed in response to what is available throughout the Town of Ajax. Interviewing community organizations who have an ongoing relationship with some of the residents who experience the highest degree of mobility poverty helps the project team to better understand the needs of the community, the existing measures being taken to fill those needs and the opportunities for the Town of Ajax to provide supports to build capacity within these organizations as they look to expand the mobility choice of their communities.

SOCIAL CLIMATE ANALYSIS – PURPOSE AND GOALS

WHAT IS A SOCIAL CLIMATE ANALYSIS?

A Social Climate Analysis (SCA) is a way of building trust and genuine partnerships within a community. Rather than seeking to "improve" a community from the top down, a Social Climate Analysis identifies the critical social infrastructure that already exists within a community, focusing on building relationships and capacity to support the community's ability to fulfill its potential by naming and mitigating, and working towards removing some of the systemic barriers that impede a community's ability to thrive. A Social Climate Analysis is a process developed by the Thrivance Group, led by Dr. Destiny Thomas, and is grounded in the framework of critical race theory, which encourages practitioners to consider the numerous ways that systemic oppression impacts a community's ability to achieve self-determination. A Social Climate Analysis is an exercise in care and understanding – one that examines how communities have needs that are going unmet, how the intersectional identities of community members affect how they experience mobility and how the often-invisible communities of care show a community's strength.

For the TDM Plan Update, the purpose of the Social Climate Analysis is to develop a stronger understanding of the networks of care that already exist in Ajax – to identify and name these community assets in a way that supports their preservation. The goal is not to have the Town of Ajax and its institutional partners absorb and adopt the existing services being delivered by community groups, but rather for governmental agencies to realize where their services need support to be sustained, and to work collaboratively with community partners to co-create solutions to improve the mobility and the quality of life for all residents in Ajax.

STEPS FOR A SOCIAL CLIMATE ANALYSIS

A SCA is developed through a three-stage process which is centered on building understanding, capacity and trust within the community. This document represents the culmination of Stages one and two of the SCA Process, but it should be noted that Stage 3 should be seen as an ongoing process of collaboration, capacity building and co-creation of solutions.

The first stage involves in-situ observations of the community's assets by the project team. This involves listening and absorbing oral histories, observing community activities and identifying community leaders and elders who hold knowledge about the community. This stage is important to gain a foundational understanding of the community's assets, which are then confirmed and refined in later stages of the SCA process. Given the project timeframe, these observations involved reviewing past engagement processes conducted by Transportation and other departments. Notably, many community leaders were identified based on past experiences working on events and committees primarily led by the Town's Recreation, Culture, and Community Development Department. Additionally, the project team researched service providers and online neighbourhood groups.

Stage two of the SCA is focused on engaging in discussions with community members and leaders to refine and confirm what was already observed during the first stage. During this stage, the project team gains additional insights into the needs of the community, establishing the foundations for moving into Stage three.

Stage three provides space for the community leaders to co-create solutions to the problems identified during the first two stages. During this Stage, the SCA is put into practice, providing project staff with an opportunity to learn what solutions are preferred by the communities they wish to serve, and how the engagement of the community leaders can support the Town's overall goals of creating a more sustainable and equitable transportation system.

CAPACITY BUILDING VERSUS AGENCY SUBSTITUTION

It is important to note that the purpose of the SCA for the Ajax TDM Plan is not to enable the Town of Ajax or Durham Region Transit (DRT) to formalize and adopt nor compete with existing informal/community transportation systems within the Town. Rather it is an opportunity to acknowledge the existing services that are being delivered and realize where improvements can be made within each of the governmental bodies. If The Town of Ajax and DRT began offering the same services that local organizations have developed, then it does not recognize the value of the work and the sacrifice individuals have made to improve the wellbeing of residents within the town. Relocating services from community organizations which have close relationships with their communities to the Town or the Region risks increasing the distance between those with a need and those providing a service. The relationships held by these community organizations should be valued as a significant asset, and the Town and its partners should work with these groups to co-create a path forward where locally delivered services benefit from the support and collaboration of the Town and the Region's ongoing programs.

AJAX SCA GOALS

The goals of the SCA for the Ajax TDM Plan include:

- Identifying the needs of residents who do not have a high degree of mobility choice by making deliberate outreach efforts to residents who have not been heard in previous transportation planning processes
- Understanding the existing social networks within Ajax that support mobility
- Highlighting areas where the Town and its partners can build capacity within community groups to support ongoing mobility programs
- Charting a path forward where the Town and its institutional partners can honour the work being done by community groups while building trust and new capacity in those communities

It is important to note that this TDM Plan update will still serve those whose mobility patterns follow the 'traditional' commute pattern. These people tend to have a high degree of mobility choice, and a traditional TDM Plan places an emphasis on encouraging people to use that choice to integrate sustainable mobility into their regular travel. Where the Ajax TDM Plan Update will differ from those traditional TDM approaches, however, is in recognizing how people with a low degree of mobility choice are disproportionately harmed by transportation systems that prioritize single occupant vehicles and making recommendations to better serve those residents' needs.

COMMUNITY SNAPSHOT

AJAX DEMOGRAPHICS AND POPULATION

The Town of Ajax has historically been a commuter community for the Greater Toronto Area (GTA). Suburbanization trends throughout the latter half of the 20th century made the town increasingly popular to settle in among young professionals working 9 – 5 jobs across Durham Region and the GTA. The Town's demographic profile has shifted rapidly since 2001, with significant growth in the Town's South Asian, Black, Filipino and Chinese populations compared to the Town's previous settlement patterns, where it was a community with a large majority of white residents. Many people of different ethnicities move to Ajax from other parts of the GTHA as well as from different parts of the world. Since 1981, over 35,000 immigrants have made their home in Ajax. As these more recent Canadians make the Town their home and begin to raise the next generation of residents, the Town is rapidly diversifying and is now one of Ontario's most diverse communities, with over 56% of its population identifying as a member of a visible minority group as of the 2016 census (Figure 1).

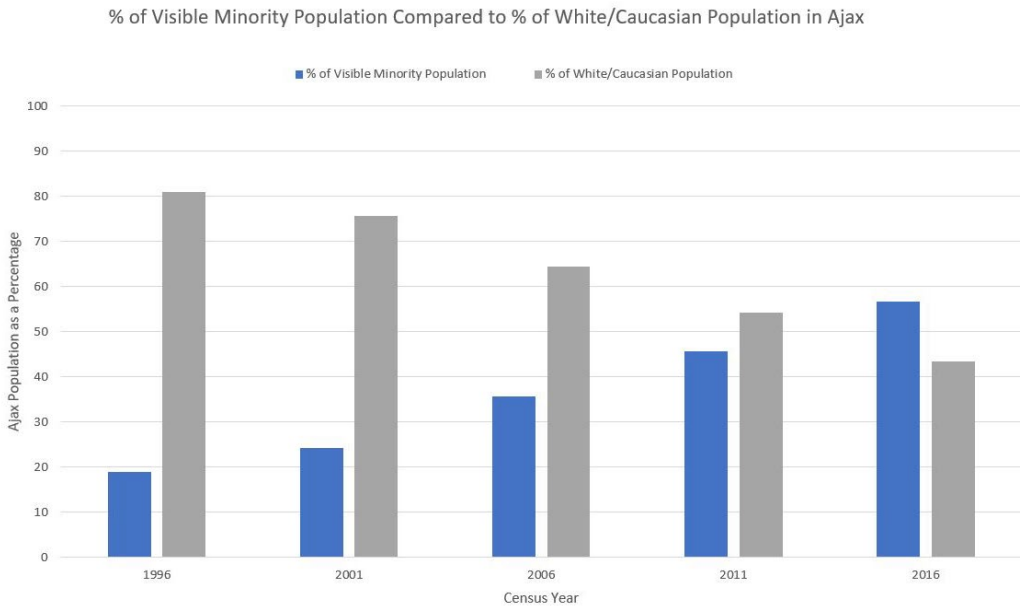


Figure 1: Demographic change of visible minority population in Ajax between 1996 and 2016.

Today Ajax is home to approximately 126,000 residents, the majority of whom now identify as a member of a visible minority. The largest visible minority communities in Ajax are South Asian (24,895 residents), Black (19,860 residents), Filipino (5,685 residents) and Chinese (3,400). There are also significant West Asian, Arab, Latin American and East Asian communities in the town (Figure 2).

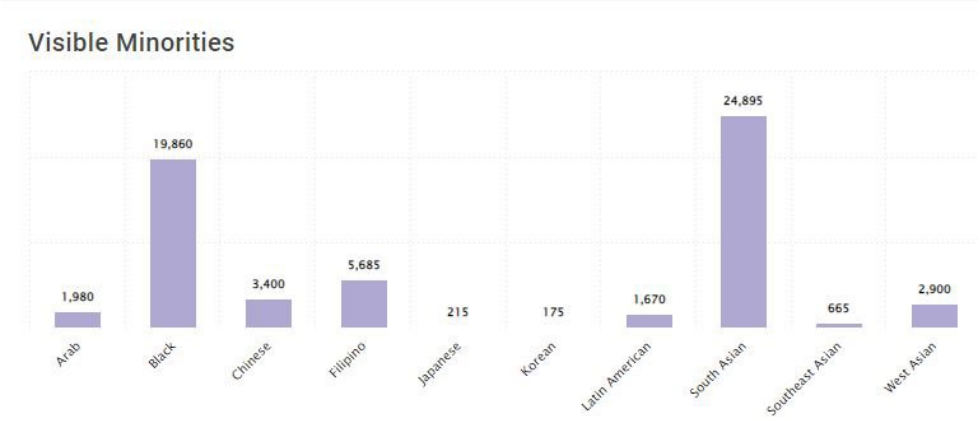


Figure 2: Breakdown of ethnic origins in Ajax (2016).

As of 2016, the Town of Ajax has a median age of 37, a full 4 years younger than the average age in Ontario (41). While the Town is generally younger than many communities in Ontario, there is a growing number of older adults in Town, with the percentage of residents over the age of 65 growing from 8.7% in 2011 to 11% in 2016 In developing its Age-Friendly Community Plan, the Town of Ajax engaged residents to better understand the mobility needs of seniors and residents approaching their senior years. To ensure that the TDM Plan considers the needs the Town's aging population, the Project Team incorporated learnings from the development of the Age-Friendly Community Plan. Further, the Project Team also engaged many of the community leaders in the Age-Friendly Steering Committee.

MOBILITY PATTERNS

As the Town of Ajax's demographic profile has changed, so has the employment and mobility patterns of the town. Historically Ajax was a community with a high proportion of commuters who travelled out of Town for work every day – either travelling into adjacent municipalities in Durham Region or elsewhere in the GTHA. In the 2015 TDM Plan, community surveys showed that nearly two thirds of the working population left Durham Region for work every day (Town of Ajax, 2015, p. 26). With the proliferation of remote work resulting from the COVID-19 pandemic and an increase in employment opportunities within Ajax and Durham Region, the Town is experiencing a dramatic shift in the historic mobility patterns. Peak hour commutes into the GTA are declining, work schedules are more flexible or require irregular work schedules and more people are choosing a community where they can live, work and play A car-centric transportation system that serves commuters working outside of Ajax within the 9-5 workday is increasingly becoming ineffective for meeting residents' needs. While many residents still work and study outside of Ajax, residents are becoming less reliant on daily commutes to and from the Greater Toronto Area, opening up new opportunities for the Town to use mobility as a means of connecting people to their community in more meaningful ways.

The majority of residents in Ajax still use a car as their primary mobility choice, but the results of the survey and the interviews conducted as part of this analysis show both the myriad ways that the reliance on automobile transportation is leaving some residents behind. The 2016 Census found that **nearly one in four households** in Ajax had an annual household income of less than $60,000 a year. The Canadian Automobile Association's *Driving Cost Calculator* states that the average annual cost of one vehicle is over $9,400, taking into account insurance, maintenance,

monthly car payments, fuel, and depreciation costs. For households earning $60,000 a year or less, the average cost of a vehicle represents nearly 1/6 of their income – a significant financial burden to be borne by those who can least afford it. For these households, having access to lower cost mobility options including walking, cycling, micromobility and public transit frees up much needed resources in their household budgets and expands their access to jobs, services and recreational activities within the Town and beyond.

The 2022 community survey highlights the ways that mobility patterns are shifting in Ajax, with a dramatic increase in zero and one-car households reported in the 2022 survey compared to the 2015 survey. The *2015 Transportation Demand Management Plan* found that 70% of households in Ajax had 2 or more vehicles, a percentage that dropped to 53% in the 2022 survey. Whether this is a reflection of the rising costs of housing and mobility or a reflection of the Town's ongoing efforts to promote more sustainable and efficient forms of mobility is unclear, but what is certain is that the residents of Ajax are shifting their mobility patterns already, and that the Town is in a position to support that ongoing shift to create a healthier, more prosperous and more connected community.

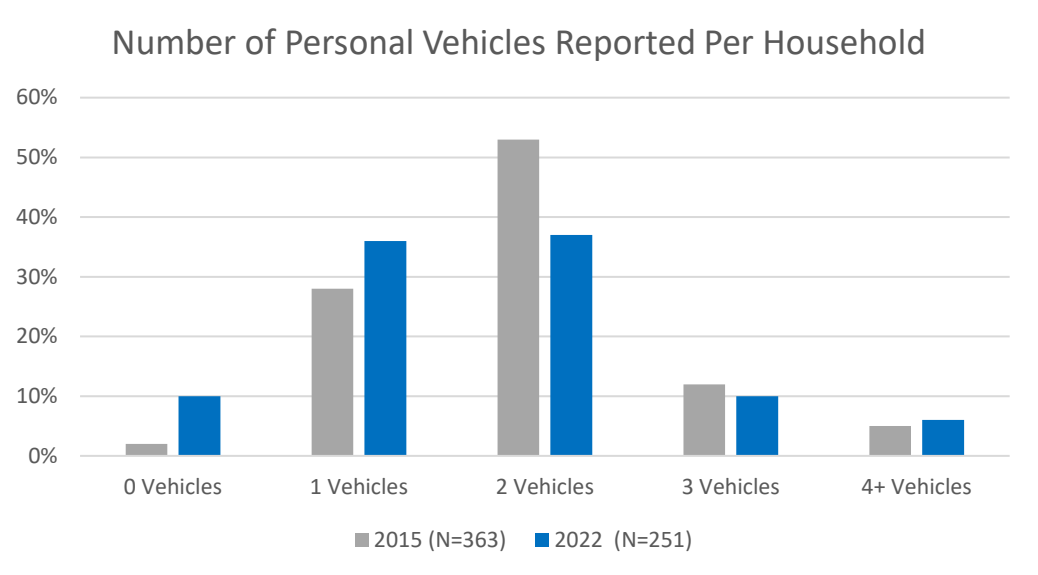


Figure 3: Reported number of vehicles per household between the 2015 and 2022 TDM community surveys.

The Town of Ajax has been investing in sustainable mobility options for several years, creating new physical and social infrastructure to expand mobility choice in Town. Despite these investments, however, there remains a significant population in Town who are not able to access their daily needs. To support these residents, community organizations have stepped up to support their community members, creating a number of formal and informal services to increase mobility for populations made the most vulnerable by the existing transportation system. The remainder of this Social Climate Analysis aims to document those efforts and highlight the communities of care that exist to address those mobility gaps, especially for those residents for whom an automobile is not a viable option.

HEARING DIRECTLY FROM COMMUNITY LEADERS

A key tenet of a Social Climate Analysis is talking with the community leaders – the people who are providing services, connecting with people in their community, and organizing efforts to support the wellbeing of their neighbours. For this project we talked with groups all across Ajax who interact with some of the Town's residents who lack mobility choice, and whose networks have been adversely impacted by previous transportation planning decisions.

REGIONAL SERVICE PROVIDERS

DURHAM REGION TRANSIT (DRT)

DRT partakes in the Smart Commute program across the Greater Toronto Hamilton Area. The transportation agency actively works with employers, organizations and individuals to seek ways of diversifying transportation modes across the Town of Ajax and Durham Region. The goal is to help organizations shift to more sustainable transportation modes and to incentivize travel during off-peak times. The transit agency has adopted Social Equity Guidelines that facilitates the integration transportation equity measures within all aspects of the service providers operations. The guidelines have facilitated an increase in night-bus service, a improving accessibility across the system, and piloting new projects to enhance mobility such as on-demand transit. DRT also assists in sustainable school travel programming across Ajax by collaborating with the city and local school boards.

FAITH LEADERS

AJAX MASJID QUBA

The Masjid Quba is a place of faith and community gathering in Ajax. The organization operates prayers 5-times per day with approximately 200 people attending each session. The Islamic Society of Ajax also provides and assists in programming to care for youth, seniors, and newcomers to the town. Organizers of the Masjid Quba noted that many mobility barriers exist for those who do not drive or own a personal auto-mobile, particularly senior residents seeking to access prayers throughout the day. The Mosque organizers noted that seniors rely on family members for rides to and from the Mosque, but that for residents in Ajax that work outside the home these transportation demands can be burdensome. The organizer stated that the Ajax Masjid Quba has tried to organize taxi and specialized bus services in the past, however the Mosque has shifted to coordinating with volunteers to fill service gaps.

The Masjid Quba organizer noted the positive impact of dedicated cycling and pedestrian infrastructure, noting that there has been a major increase in the number of residents using bicycles, ebikes, and e-scooters to access prayers throughout the day, however this usage dissipates during the winter months due to inconsistent maintenance and other weather concerns on Town and Regional roads and Trails.

> Interview Quote:
>
> "Right now there's a trend for [using} ebikes. Like everybody is going towards that technology because it's very easy to handle, it's very easy to drive and it's cheap. So we should have something for the seniors, especially because the ebike thing which really helps them and their mobility"

THE SOUTHSIDE WORSHIP

Southside Worship is a church that operates a foodbank that clients can access either on-site or by a request for delivery. The Southside Worship partners directly with mental health and long-term care homes to donate food to seniors and mental health clients who are in need of assistance. Southside Worship conducts approximately 500 deliveries and pick-ups each month. Deliveries are organized by Southside Worship and are carried out by church volunteers and by the community group CareMongers. Current challenges experienced by the Southside Worship include a lack of delivery drivers during peak travel times, a trend that has been exacerbated by recent changes in DRT routing that removed the ability of clients to easily access the food bank on their own.

Interview Quotes:

"We used to have a bus that stopped right outside our door[that] is no longer here… We no longer have that direct access to transit. They need to walk quite a bit from Harwood to get here. So, people try to arrange rides to get here."

"We have people that take taxis every other week. They come to the foodbank, and then they have to take a taxi. … there is no other way because they can't carry it all home and [they] have some mobility issues. So they will walk here and then take a cab home."

CHRISTIAN FAITH OUTREACH CENTRE

The Christian Faith Outreach Centre is a well-connected community church that operates a food bank, emergency shelter, and acts as a community hub within Ajax. The organization helps coordinate mobility options for its clientele to access these services, whether it is arranging rides using taxis, public transit, or coordinating rides with Kids/Seniors on Wheels. The Christian Faith Outreach Centre's clientele has experienced barriers accessing services due to reductions of DRT bus stops and operating hours, particularly during weekends, limiting access to Sunday Service. The organization has coordinated the use of taxis, Kids/Seniors on Wheels, and volunteer drivers to address these service gaps for their clientele.

Interview Quotes:

"The knowledge that it's there, what is the transportation choices, what's available, how to access it, especially if I'm a senior. I met a senior or you know they called us and they moved recently, moved into the area and they and that's why I asked you about the wheel trans cause. Actually I didn't know myself that they didn't have access to move around"

"We have the GO bus. We have like the local transit as well. But it is not easy for people, I think, to get around."

"it seems like transportation has been designed for really those who are commuting back and forth to work, but with no consideration. Maybe for others that are not. And then, you know, we think workplaces are nine to five. But I've seen the evolution of workplaces that are beyond that. You know, they have a lot of different shift environments that people are working in and even overnight. So where is it transportation in terms of, you know, those type of services for late evening as well as overnight or early morning?"

"I look at it from the shelter perspective, you know, to have somebody taking a taxi everywhere that they need to go. I think that's not good. I think that's an expensive venture. There's a lot of vehicles on the road, so there has to be a way that we can reduce that ability for moving people around."

COMMUNITY SUPPORT AND SENIOR CARE ORGANIZATIONS

KIDS/SENIORS ON WHEELS

Kids on Wheels is a private transportation company that began operating in 2011 to help alleviate the transportation needs of working parents who are not able to drive their children to and from school, daycare programs, sports, and recreational activities. The company recognized mobility gaps for school-aged children who lived within 1.6 kilometers of a school and were not being serviced by the Durham Student Transportation Services or buses from other school boards to take them to and from school. The company recognized that parents were not always available to walk their younger children to and from school and began working with local neighborhoods to coordinate rides to help parents in need. Following the success of coordinating rides to schools and recreational programs, Kids on Wheels expanded their operations to include providing services to seniors. Seniors on Wheels, a division of Kids on Wheels, provides personalized door-to-door transportation services to seniors that includes trips to hospitals and medical services, community events, personal shopping, and connecting to their personal social networks. The service is partnered with Community Care Durham and actively connects seniors with the organization. Kids on Wheels operates 3 school buses and 5 company cars around the clock to improve mobility options for youth and seniors in Ajax.

Seniors on Wheels noted that much of their clientele has issues using Durham Region Transit services due to an unfamiliarity with the system and a lack of personalized door-to-door services offered to senior residents, especially those with disabilities. Kids on Wheels and Seniors on wheels keeps a polaroid picture of clientele with Alzheimer's, Dementia, and other at-risk individuals to ensure that they do not get lost and have assistance getting to and from their destinations. The organization emphasizes the importance of their drivers requiring a high level of compassion to be able to work for the organization.

Interview Quote:

"City Transportation services for the seniors is very hard. I can tell for seniors, it's very hard for them especially the ones I have, dementia or Alzheimer's or can't walk. So seniors need special care. It's very hard for them to [take] public transportation or … an uber/taxi and when they can hardly walk out of the house."

"I can tell you this much, all of our drivers have a compassionate heart. Not only for seniors, but for children as well. So when they use our service we're not just another part of an organization. So when the unsheltered come in [to use the service] they feel more free to speak … so they become our friends and we know them by name. We know [our clients] because it's usually the same drivers, the same places they're going to. They need the compassion."

THE LIGHTHOUSE RETREAT

Since 2013, the Lighthouse Retreat has provided seniors care for residents and works with families to help to slow the onset of Alzheimer's for clients who have been diagnosed with the disease. The

Lighthouse Retreat has an extensive network of personal support workers that provide service not only at the Lighthouse Retreat, but are also sent to conduct one-on-one in-home support for clients, supporting physician's office coordinators, and providing direct support to local shelters and other long-term care facilities.

Since personal support workers often work hours outside of the standard 9-5 schedule, they have noticed inconsistency with early morning and late evening DRT service, noting that service frequency is the largest barrier. For clients, the Lighthouse Retreat actively works with DRT's specialized transit service to provide pick-up and drop-off to seniors and residents seeking shelter access. However, the Lighthouse Retreat noted that following the pandemic the pick-up window for specialized transit exceeds 30-minutes for seniors and those requiring mobility aid. To respond to the long delays, the Lighthouse Retreat has partnered and used the services of Seniors on Wheels and have had staff volunteer to drive clients as a way to address the pick-up and drop-off window gap.

> Interview Quotes:
>
> "I think first and foremost for from a DRT perspective, it is just being able to fine tune what they currently have. If we could go back to that pre pandemic schedule that would be great. That does help to lighten the anxiety for our seniors which is so huge for the population that we're serving, especially the ones who use … are already, you know, dealing with social isolation and depression and anxiety issues and we don't need to further add stressors."
>
> "Also, I see it with the clients in particular …, it's just the frequency of the busses and the routes [available] means they have to wait a long time. [For staff] it adds an exorbitant amount of time on their work day because they have to wait so long for in between buses as well as some of the routes."

AJAX PICKERING HOSPITAL FOUNDATION

The Ajax Pickering Hospital Foundation currently operates a service called Doctors on Wheels that consists of three doctors who provide in-home and remote healthcare services to clients. This reduces the burden of accessing the hospital for clients and removes stresses from local specialized transportation services and reduces the likelihood of a patient in non-critical condition calling an ambulance.

The Ajax Pickering Hospital Foundation and its staff predominantly drive to work, however hospital staff noted that personal support workers often receive pick-up and drop-off services either from family, friends, or DRT services. The demand for parking at the Ajax Pickering Hospital has increased so much that an additional parking garage is currently under construction. In the interim, the Hospital has responded to the parking garage being at capacity by operating a school bus to transport workers from an overflow parking facility to the hospital and back. Because of the nature of shift work for many of the hospital workers, the Ajax Pickering Hospital Foundation indicated that the current DRT scheduling structure does not meet their staff needs. As a result of lowered bus frequency early in the morning, mid-day, and in the evening, several staff members rely on car-pooling, ride-sharing, and taxi services to complete their trips.

> Interview Quotes:
>
> "We do have doctors on wheels … [clients] can just call up request a doctor that goes directly to your house, which is really convenient. That eliminates having [the need] to call a family member or depend on wheel trans showing up and not"

COMMUNITY CARE DURHAM

Community Care Durham provides transportation and mobility coordination for seniors, residents requiring mobility support, and mental health support across Durham Region. Community Care Durham helps to connect residents to grocery stores, doctors' appointments, and leisure activities across the region. The service allows personal support workers to join their clients along for their trips as well. The organization has a fleet of 10 vans with several outfitted with a rear hydraulic lift. Service is predominantly offered between 9 – 5 but several drivers are on-call for morning and evening service.

Community Care Durham drivers are not only there to drive their clients to appointments, they are all specialized to assist them to and from the vehicle to their required service. Community Care Durham charges residents at a rate of 45 cents per kilometer, however the organization actively seeks subsidies, especially for longer trips or vulnerable populations.

COMMUNITY ADVOCATES

WOMXN OF COLOUR DURHAM COLLECTIVE

The Womxn of Colour Durham Collective is a youth-led community organization that seeks to create spaces and elevate the voices of Black, Indigenous, and other self-identifying womxn within Durham Region. The organization provides networking opportunities, safe spaces for discussion, resources, and cultivates projects, workshops, and panels that are designed to create change within the community and beyond. Ultimately, the intent is to create a space for knowledge-sharing, leadership, healing and creativity.

> Among the barriers mentioned by Womxn of Colour Durham Collective, notable challenges affecting young racialized residents in Ajax include on-time performance and frequency of DRT busses, the perceived necessity of vehicle ownership, and the need for more transportation options within high-density racialized areas.
>
> Interview Quotes:
>
> "No transit in terms of buses is a difficult one, especially with when you look at really high density areas around like Westney, Kingston Rd where you find at least a lot of racialized folks waiting for long hours for buses that are already packed and perhaps just driving by them."
>
> "I'm sure you've heard a lot around costs, but I know that's something that I imagine is only going to continue to rise, but that's one of the bigger conversations I've had [with people]. You know what? Of course I can't own a car, but then … transit is also very expensive. I can't purchase a bike like, how can I get around aside from walking? And if I'm experiencing challenges that make walking difficult for me, then I just end up at home. One thing we've noticed is that folks are very isolated within Durham, very siloed."
>
> "Of course, I've seen shelters that are heated now, which is also great because that's another barrier. Folks are waiting outside in the cold to get around. And for some folks

who just can't do it. So they end up going home. They just decide to not to go about them, their trips and that sense too."

WHAT WE LEARNED

EXISTING STRENGTHS AND ASSETS IN AJAX

Communities across Ajax have established community-based transportation systems and programming designed to help residents, particularly those who do not own personal vehicles, get around easier. These programs manifest in a variety of formats such as: door-to-door specialized transit (Kids/Seniors on Wheels; Community Car Durham), carpooling arrangements (Christian Faith Outreach Centre; Masjid Quba), or by bringing essential services directly to clients (Doctors on Wheels; Community Care Durham; Lighthouse Retreat). There was a strong recognition of the mobility deficiencies that exist throughout the Town for those who do not have direct access to a personal vehicle and rely on public or active transportation modes. Each organization leader was eager to make travel and access to services easier for clients who are frequently excluded out of the fold as a priority group for improving mobility.

In addition to individual organization's efforts, the interviews revealed that there is active and ongoing collaboration between many of the community groups. For example, The Christian Faith Outreach Centre will work with Seniors on Wheels to ensure that aging populations can access church provided services with ease. The South Side Worship will partner with a community group called Care Mongers to provide foodbank delivers to residents. When services cannot be coordinated between the organizations, the community leaders interviewed provided assurance that their staff, volunteers, and members go out of their way to provide carpooled rides, pick-ups, and drop-offs to those unable to move around the Town, highlighting the positive impact a tightknit community can have on improving transportation outcomes for those in need.

Finally, each of the community partners interviewed throughout this process demonstrated an eagerness to help promote and support the development of the TDM Plan update. Organizers at the Lighthouse Retreat offered to distribute and walk-through a community engagement survey with seniors. The Masjid Quba offered for members of the project team to promote the TDM Plan and survey at a Friday congregational prayer. Community leaders helped to identify and reach out to other community leaders and organizations to aid in the development of the SCA, expanding the reach of the project.

COMMUNITY NEEDS

The strengths of the community emerged in response to existing transportation gaps in the community that were then exacerbated by the COVID-19 pandemic. The interview process revealed that transportation needs were not being met for workers, seniors, and vulnerable residents during non-traditional commuting hours. For example, the Pickering Hospital Foundation noted that most of their personal support worker staff is scheduled through shift work meaning that their hours are far more inconsistent than 9-5 office workers. In some cases, workers will finish their shift at odd hours in the afternoon or late at night where municipally operated transportation services are infrequent.

Another need identified was the lack of transparency when changing bus stops, routes, and service frequency. Community leaders identified a need for increased communication when changing bus stops, routes, and service frequency. While changes may be required due to factors affecting the resources and capacity of transit service providers, greater transparency on these factors and notices of changes would be beneficial.

Organizations such as the Southside Worship, Community Care Durham, and the Lighthouse Retreat indicated that bus stops had been either moved or had service reduced without any notice from the DRT. Moving stops and reducing frequency has reduced access to these service providers for seniors and volunteers who rely on public transit, leading these organizations to address the gaps on their own accord.

Community leaders also identified the need for a holistic list of all transportation service providers, regardless of whether they are formal or informal services. Despite many organizations working together with one another, there is still a lack of familiarity with what options are available for residents throughout the Town.

THRIVANCE NARRATIVE

A deliverable from the Social Climate Analysis is a Thrivance Narrative for the Town of Ajax. A Thrivance Narrative is a community-developed demographic description that is centred on how the community has been impacted by previous decisions, with a focus on the community's strengths and its response to the needs of its members. For the Town of Ajax, the following Thrivance Narrative has been developed:

The Town of Ajax is home to an increasingly diverse population with different identities, lifestyles, talents, capacities, household compositions, needs, and aspirations. While many residents travel outside of Ajax for work or other purposes, a growing number of residents are choosing to live and work in the Town each passing year. There is also a variety of services, programs, events, resources, and activities within Ajax provided by the Town, community groups, places of worship, institutions, businesses, residents, and other formal and informal providers. The Town needs a transportation system that allows residents to access local services, programs, resources, activities, and opportunities regardless of whether they are able to drive a car.

Similar to other municipalities across Ontario, the Town of Ajax has a legacy of a transportation system designed to prioritize the movement of automobiles. This has limited the mobility of residents such as children, youth, elders, shift workers and people with disabilities. Many community leaders, service providers, and residents have expressed numerous ways in which car-centric design has not been working for them and in many cases, has created barriers to access essential services, programs, and opportunities.

To overcome these mobility barriers, Ajax's community organizations, faith groups and social support networks have created both formal and informal tools to support those whose access to mobility choice has been limited. The community's support for its elders, for people experiencing poverty, and for people with disabilities clearly demonstrates the ways that the people of Ajax honour the needs of their neighbours, creating numerous tools to expand the Mobility of Care within the community.

Mobility of Care refers to trips that are vital for a person's needs and can include trips to medical appointments, grocery stores, places of worship or a food bank. These trips provide for physical, social and spiritual needs in Ajax, and are vital to the ongoing health of its population. The wide variety of Mobility of Care programs delivered by community organizations in Town shows the empathy and love that community members have for their neighbours. These programs are delivered with compassion and a deliberate focus on improving the wellbeing of others, an approach that should be both supported and replicated as the Town aims to expand the mobility choice available to residents.

The community leaders, residents, and knowledge keepers in Ajax are eager to provide their feedback to improve the Town's mobility options. This willingness to engage and provide feedback should not be taken for granted – it is vital for the building of trust that when a community speaks, its institutions take the time to listen and respond in kind. The connections that have been made between Town Staff and community organizers as part of this project have the potential to improve mobility throughout the Town, but only if the Town continues to value the input provided and respond to the needs of those residents who experience the least amount of mobility choice.

Ajax's increasing diversity is the foundation upon which the Town's mobility future will be built. The Town of Ajax will apply a more holistic approach to mobility that considers a wide variety of trip purposes in addition to trips to/from work outside of the 9-5 shift. With increasing population, economic opportunities, workplaces allowing employees to work from home, and resident and community-led activities, the Town has an opportunity to meaningfully connect with its communities, building upon the work being done by community organizations to expand mobility choice to all residents of Ajax.

Transportation Demand Management Plan Update

Town of Ajax

Public Open House
Monday January 16th, 2023

#GetAjaxMoving

Land Acknowledgment

Town of Ajax

I would like to begin this event by acknowledging that the land on which we gather is situated within the traditional and treaty territory of the Mississaugas. More specifically, the Mississauga's of Scugog Island First Nation, signatories of the Gunshot Treaty of 1788 and the Williams Treaties of 1923. This land is, and will continue to be, home to the Indigenous Peoples. Let us acknowledge the mistakes and traumas of the past through authenticity and support truth and reconciliation. Let us engage and celebrate Indigenous communities by being leaders of action in acknowledging the United Nations Declaration on the Rights of Indigenous Peoples and the Truth and Reconciliation Commission's recommendations towards truth and reconciliation. Let us keep these principles close, as we continue towards truth and reconciliation and as we move forward with kindness and respect as a community.

Town of Ajax By the Lake

MISSISSAUGAS OF SCUGOG ISLAND FIRST NATION

This meeting is being recorded for note-taking purposes

Project Team Introductions

Armi De Francia, Project Manager

Dhaval Pandya, Manager, Transportation Planning

Shawn Smith, Project Manager

Justin Jones, Engagement Lead

Kyle Dow, Engagement Support

Meeting Purpose

- Introduce the study goals, engagement completed so far & present initial recommendations
- Gather feedback, input and ideas to improve the Transportation Demand Management Plan

Code of Conduct

Be Patient:

Virtual meetings don't always run as smoothly as planned

Be Brief:

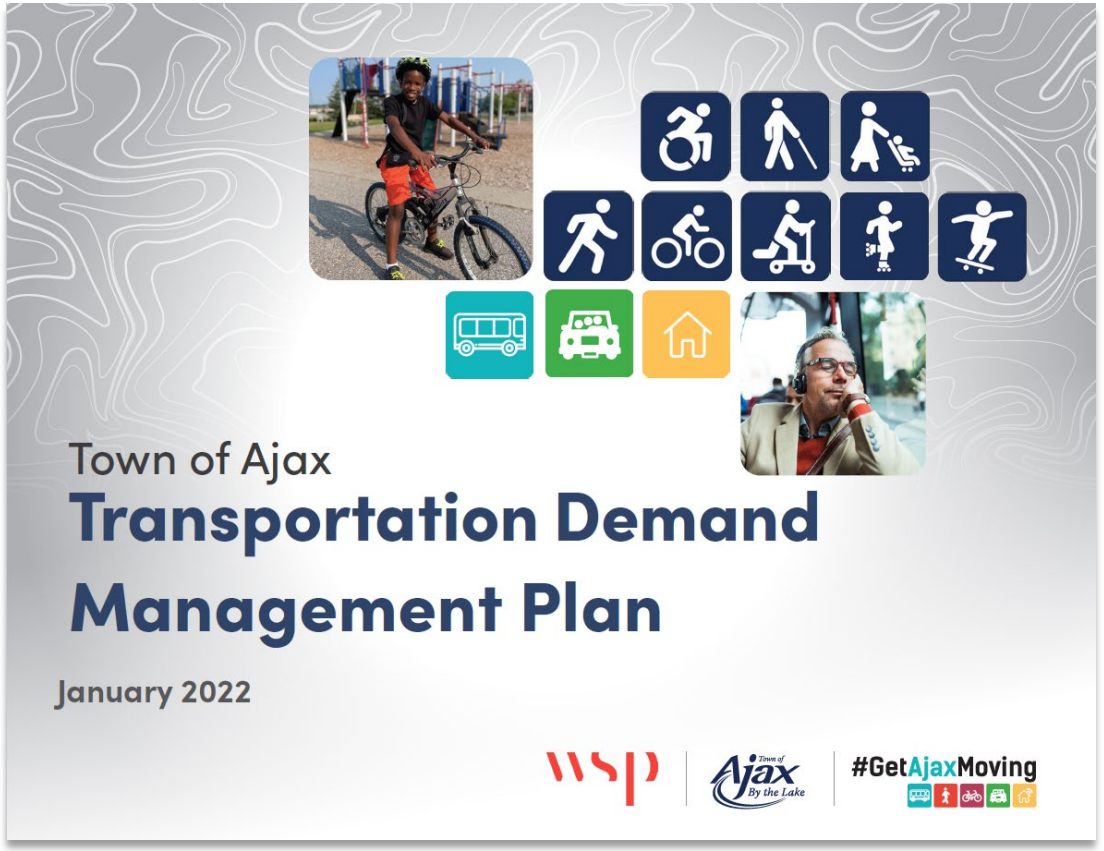
Limit yourself to one question or comment when called on to speak to give others opportunities to speak

Be Respectful:

The Town of Ajax is an inclusive public organization. Discriminatory, prejudicial or hateful comments and questions will not be tolerated and you will be removed from the meeting

Agenda

1. Project Overview	15’
2. Recommendations	15’
3. Participant Feedback Exercise	20’
4. Q&A	20’
5. Wrap up and Next Steps	5’



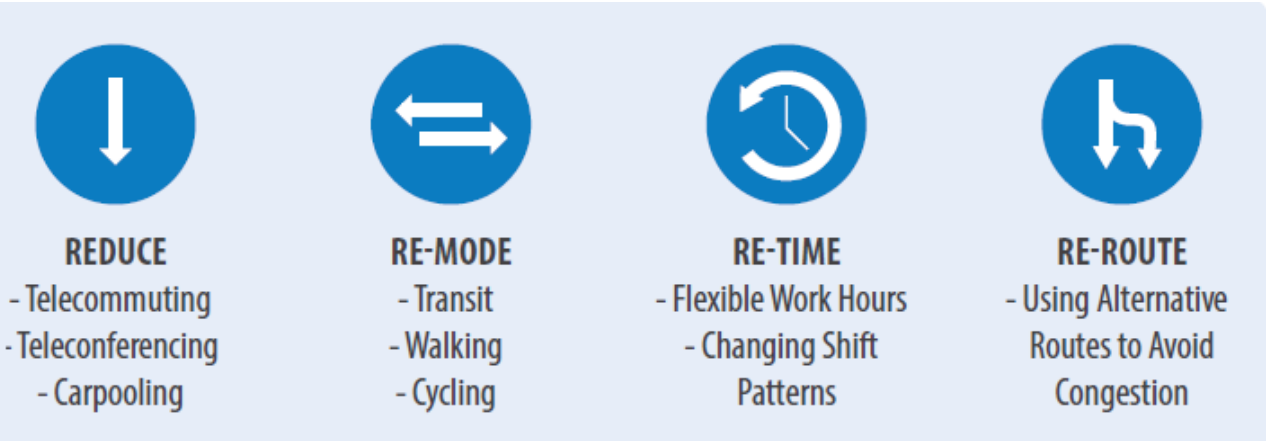
Project Overview

#GetAjaxMoving

What is TDM?

What is a Transportation Demand Management Plan (TDM)?

TDM is the use of policies, programs, services and products to influence whether, why, when, where and how people travel



The Ajax TDM Plan

What it is

Includes demand-side solutions:

- **Technology-based services**
  Ex: Ridematching app, trip planning resources
- **Policies**
  Ex: Site design, parking reduction
- **Outreach, education and marketing campaigns**
  Ex: Transit promotion, bike to work day
- **Incentives** to “nudge” people to other modes for trips that don’t have to be drive alone
  Ex: Try transit for free, e-bike subsidy

What it is not

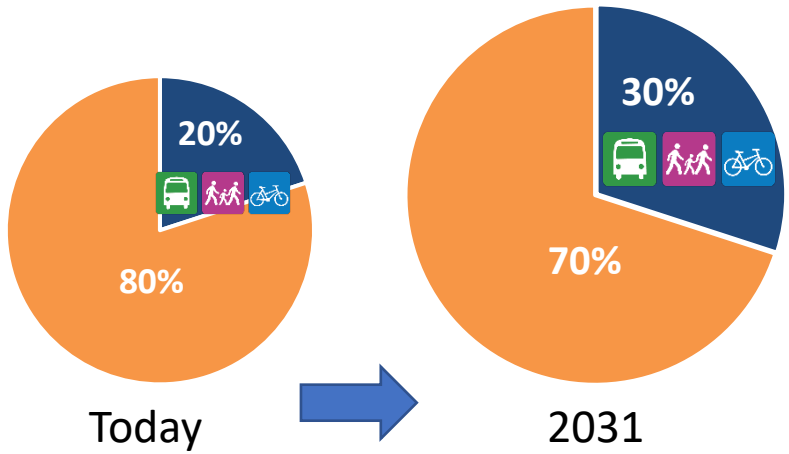
Does <u>not</u> include supply-side solutions:

- providing more frequent transit service
- building new cycle infrastructure
- changes to the road network

There are separate projects being done by different jurisdictions to address these supply side solutions.

Why TDM?

Increase morning peak period trips made by transit, cycling & walking from 20% today to 30% by 2031



Over next 20 years, the Town faces several challenges:

Rapid Growth
Traffic Impacts
Changing Demographics
Downtown Uptown
Auto-Oriented Costs
Health Concerns

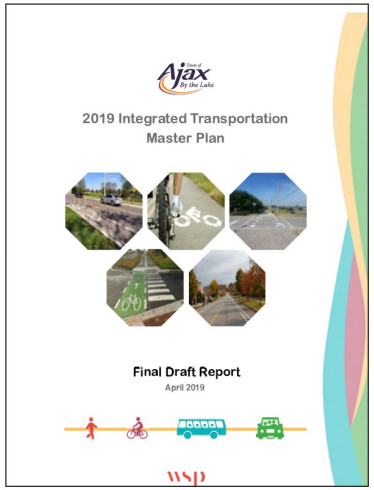
Why Update the 2015 TDM Plan?

- Build upon Town's experience and lessons learned in implementing TDM programs since 2015 Plan
- Capture new trends such as changing demographics and impacts in travel patterns due to COVID-19
- Reflect a desire to make it easier for residents to choose other forms of mobility than driving to get around Town
  - Improve mobility choice using "demand side" solutions
  - Particularly focus on children, seniors, people with disabilities and low-income residents to access their community independently

Project Goals and Objectives

Vision

Mobility choice that meet the needs and abilities of residents and visitors through a shift to sustainable transportation



Alignment with other Town Initiatives

2022 Goals & Objectives

1. Encourage a shift to sustainable modes
2. Build a culture that celebrates and recognizes the value of walking, cycling, rolling, transit and ridesharing
3. Collaborate
4. Improve transportation equity
5. Measure and communicate

TDM Plan Structure

Mobility choice that meet the needs and abilities of residents and visitors through a shift to sustainable transportation

1. Encourage sustainable modes
2. Build a culture of sustainable transportation
3. Collaborate with stakeholders and community leaders
4. Improve transportation equity
5. Measure and communicate the impacts of TDM

1. TDM Foundations
2. Partnerships & Collaboration
3. Integrate TDM into Town Policies, Processes and Programs
4. Transportation Equity

Action Plan

Evaluation Plan

Engagement Process

In My Opinion Digital Survey

Survey collected transportation patterns from community members between July and September 2022.

In-person Survey Promotion

Connected with over 200 community members and boosted survey uptake.

Agency Meeting Series

Visioning & goal workshop

Preliminary TDM measures & partnerships

Social Climate Analysis – Definition

A Social Climate Analysis (SCA) is a way of **building trust** and **genuine partnerships** within a community.

Rather than seeking to "improve" a community from the top down, an SCA identifies the **critical social infrastructure** that already exists within a community, focusing on building relationships and capacity to **support the community's ability to fulfill its potential** by naming and removing some of the systemic barriers that impede a community's ability to thrive.

Social Climate Analysis – In Practice

Identified...

Formal and informal community networks.

Interviewed...

10 Community leaders from equity deserving community organizations.

Engaged…

With commuters, seniors, youth and communities that have been historically unaccounted for; and

Utilized community networks to increase project engagement and feedback.

What Did We Hear?

If you can't drive, it's difficult to participate

Essential trips like getting to the grocery store, to medical appointments or to spiritual gatherings become a challenge.

The Community cares for its own

Formal and informal services connect residents to the services they need – e.g. Seniors on Wheels, Community Care Mongers, Doctors on Wheels.

Small changes have big impacts

Negative: DRT Schedule changes

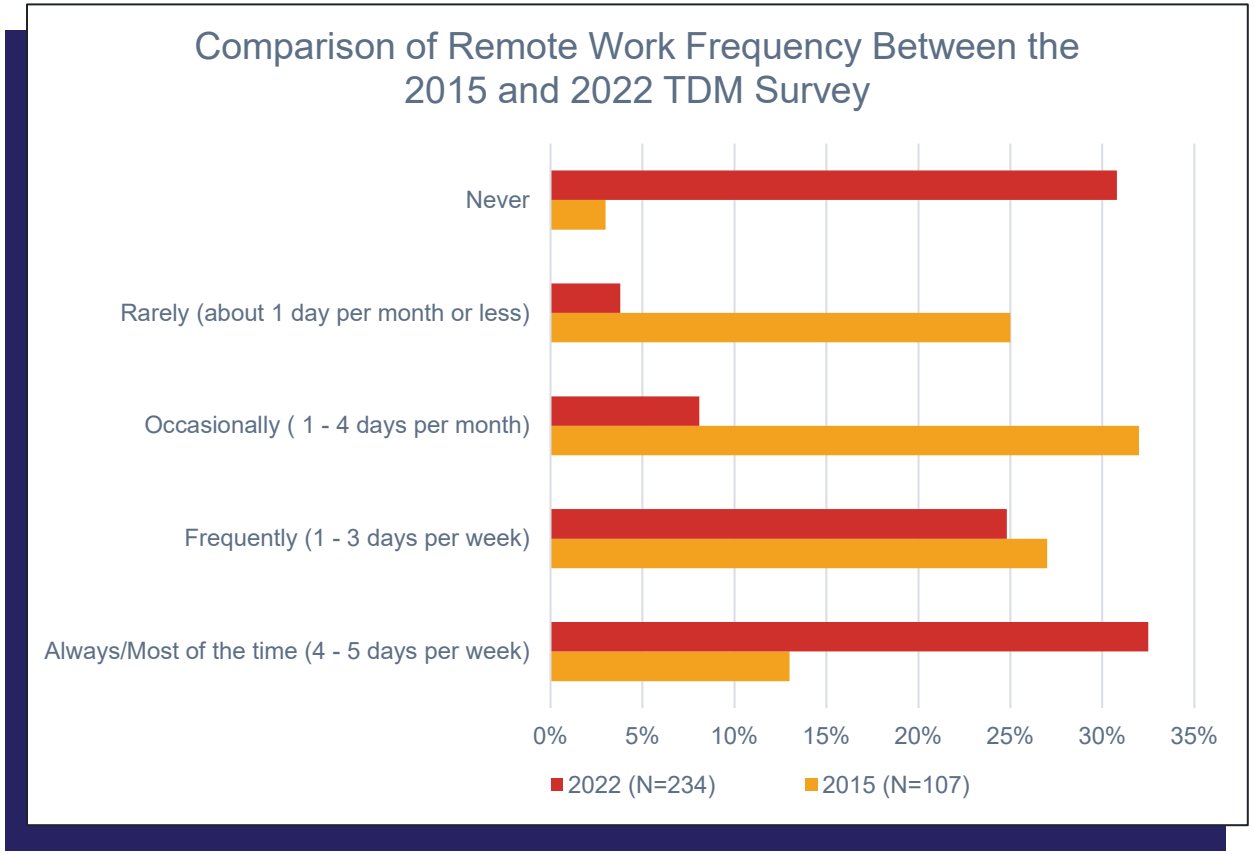
Positive: new trails on Lower Harwood

Retrieved from seniorsonwheels.ca/gallery

Building on the 2015 TDM Plan

Since 2015…

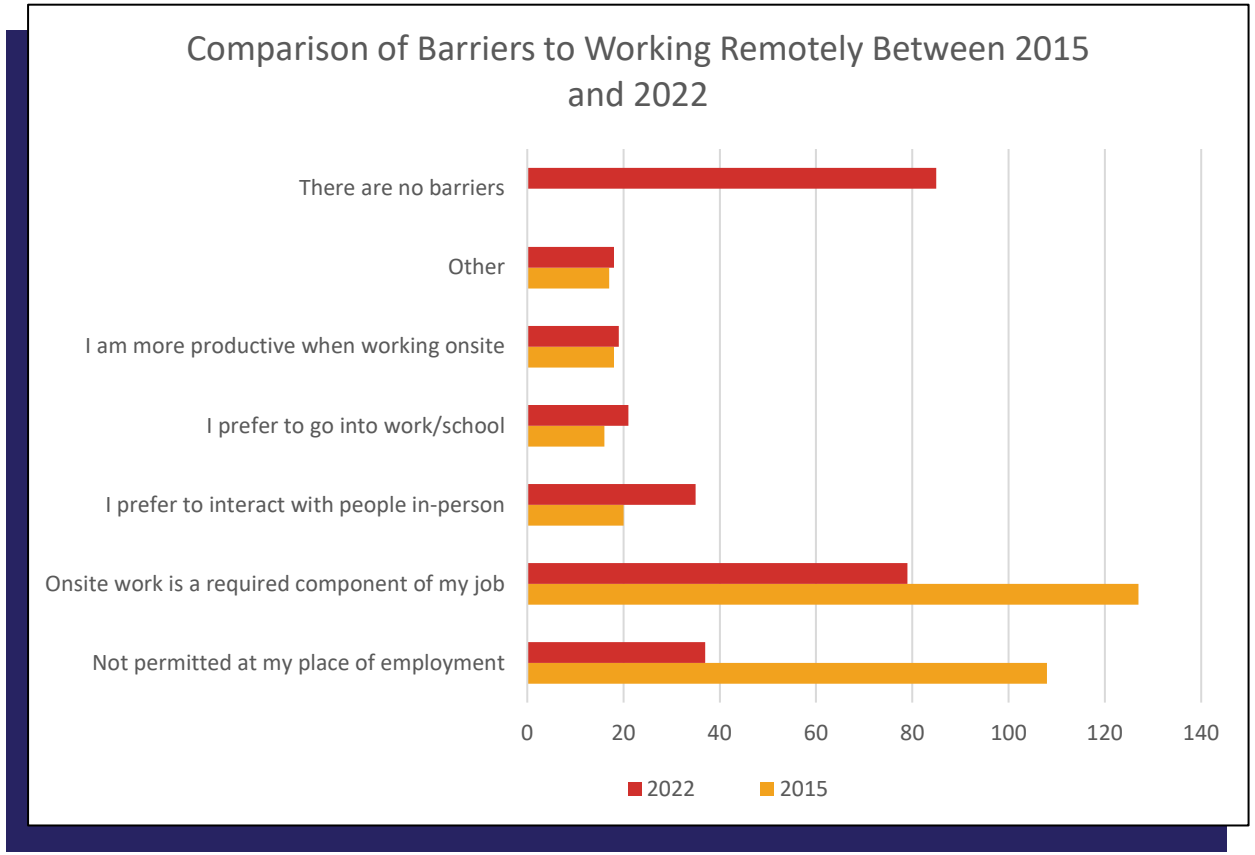
- Ajax's population has increased and has become more diverse;
- Workplaces have embraced remote work; and
- Transportation measures that address climate change and social equity challenges have gained more public support.



Highlights of Survey Results

In 2022…

- The number of households owning more than 1 vehicle declined by 17% ;
- Barriers to remote work for many positions have been removed; and
- Average time spent commuting has declined with over 50% of respondents spending less than 60 minutes on their commute.



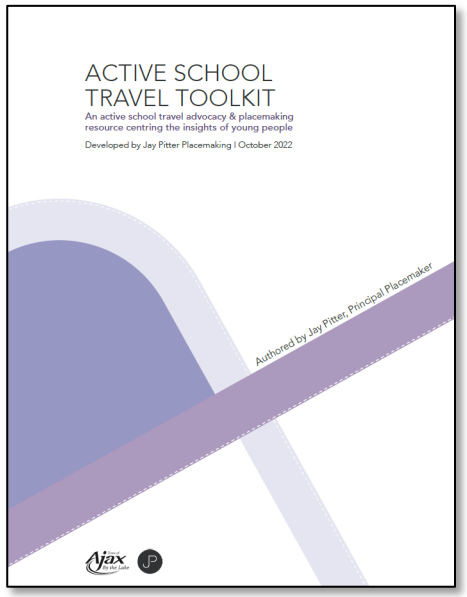
Preliminary Recommendations

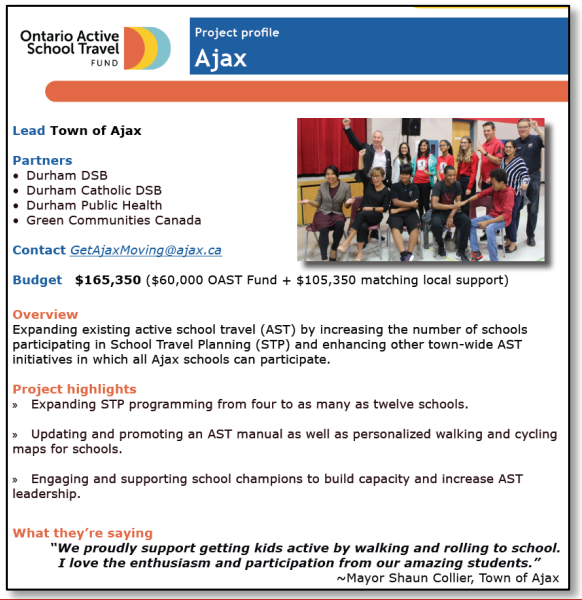
#GetAjaxMoving

A. TDM Foundations

	Measure
A1	Secure funding and resources need to deliver TDM Action Plan from 2023 to 2028
A2	Dedicated staff resource for Active and Sustainable School Travel
A3	Create TDM communications strategy
A4	Prepare annual TDM impact report

A2 – Expand STP Programming and build upon previous school travel successes and lessons learned





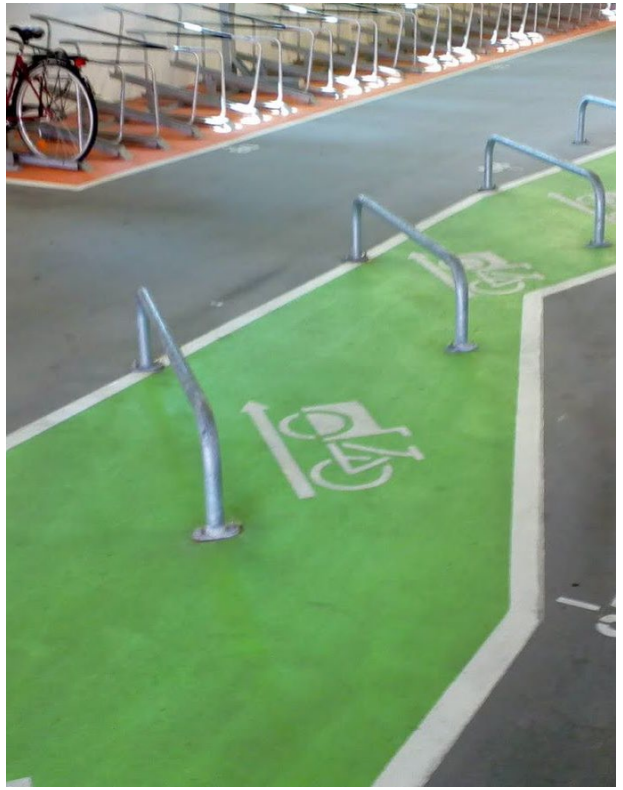
B. Partnerships and Collaboration

	Measure
B1	**Leverage Ajax Partnership Fund for TDM-related Community Projects**
B2	**Capacity Building Workshops with Community Partners**
B3	**Regional TDM Working Group**
B4	**Transit Marketing Campaigns and Incentives**
B5	**Explore Carshare Feasibility**
B6	**Shared Micromobility (e-scooter, bikeshare) Pilot**

Smart Mobility Durham's TDM programs include:

C. Integrate TDM into Ajax Policies, Programs & Processes

	Measure
C1	**Develop Complete Streets Design Guidelines**
C2	**Bike Parking Implementation Program**
C3	**Parking Policy Reforms**
C4	**TDM Programs for New Developments**
C5	**Integrate TDM into next Transportation Master Plan**
C6	**Adopt Durham Region's TDM Development Guidelines**



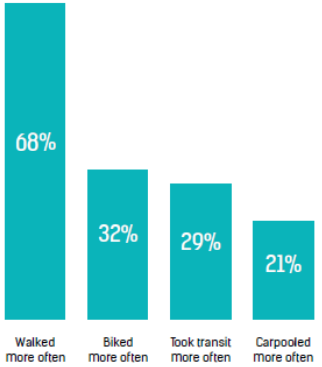
D. Transportation Equity

	Measure
D1	**Inclusive Resource/Image Library Promoting Sustainable Transportation**
D2	**E-bike subsidy program for low-income families**
D3	**Build youth capacity in equity deserving communities**
D4	**Pilot a "mobility wallet" for transit poverty areas**
D5	**Engage equity deserving communities in transportation options programs**

Source: Zygg Ebikes

Evaluation and Monitoring

"What gets measured, gets improved"



Metrics

- Trips shifted to sustainable transportation modes
- Marketing and outreach impressions
- Quantity of activities (events held/resources distributed/number of participants)
- Satisfaction of participants
- Involvement of equity-deserving communities
- Comments from participants

Impact of #GetAjaxMoving Campaign 2016-2018

Participant Feedback

#GetAjaxMoving

Participant Feedback

Go to

www.menti.com

And enter Code

4770 3793

Q&A Session

#GetAjaxMoving

Next Steps

- Review feedback from today's Public Open House
- Continue to develop and refine 2023 TDM Plan
- Present to Town Council in April 2023
- Form Resident Assessment Panel

Open Discussion

Ajax Seniors Friendship Club

Children mapping their routes to school

Families sharing their thoughts on sustainable transportation options in Ajax.

Taste Ajax 2016

TRAILfest

Armi De Francia

Active Transportation Coordinator, Town of Ajax

Armi.DeFrancia@Ajax.ca

Shawn Smith

Senior Project Manager, WSP

Shawn.Smith@wsp.com

Thank you

wsp.com

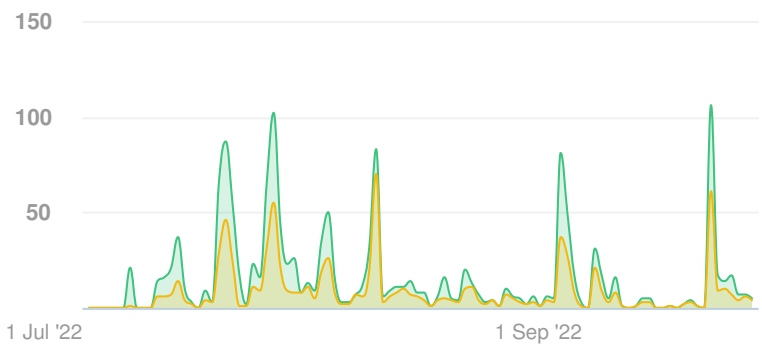
Project Report

01 May 2022 - 29 September 2022

In My Opinion

Transportation Demand Management Plan

Visitors Summary



Highlights

TOTAL VISITS	MAX VISITORS PER DAY	NEW REGISTRATIONS
910	70	1

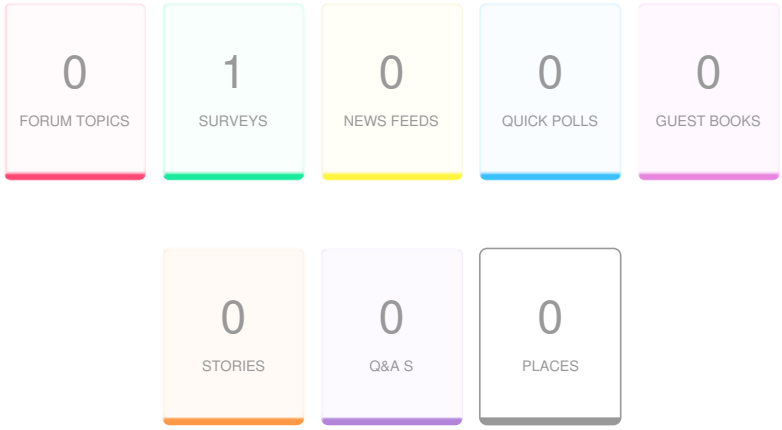
ENGAGED VISITORS	INFORMED VISITORS	AWARE VISITORS
233	318	676

Aware Participants	676
Aware Actions Performed	Participants
Visited a Project or Tool Page	676

Informed Participants	318
Informed Actions Performed	Participants
Viewed a video	0
Viewed a photo	0
Downloaded a document	0
Visited the Key Dates page	11
Visited an FAQ list Page	0
Visited Instagram Page	0
Visited Multiple Project Pages	80
Contributed to a tool (engaged)	233

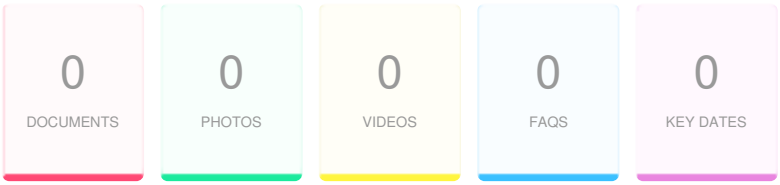
Engaged Participants	233		
Engaged Actions Performed	Registered	Unverified	Anonymous
Contributed on Forums	0	0	0
Participated in Surveys	2	0	231
Contributed to Newsfeeds	0	0	0
Participated in Quick Polls	0	0	0
Posted on Guestbooks	0	0	0
Contributed to Stories	0	0	0
Asked Questions	0	0	0
Placed Pins on Places	0	0	0
Contributed to Ideas	0	0	0

ENGAGEMENT TOOLS SUMMARY



Tool Type	Engagement Tool Name	Tool Status	Visitors	Contributors		
				Registered	Unverified	Anonymous
Survey Tool	Ajax TDM Survey	Draft	338	2	0	231

INFORMATION WIDGET SUMMARY



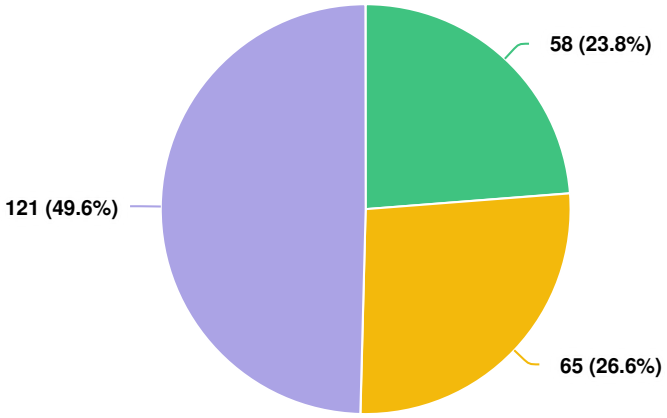
Widget Type	Engagement Tool Name	Visitors	Views/Downloads
Key Dates	Key Date	11	11

ENGAGEMENT TOOL: SURVEY TOOL

Ajax TDM Survey

Visitors 338	Contributors 233	CONTRIBUTIONS 252

What ward do you live in?



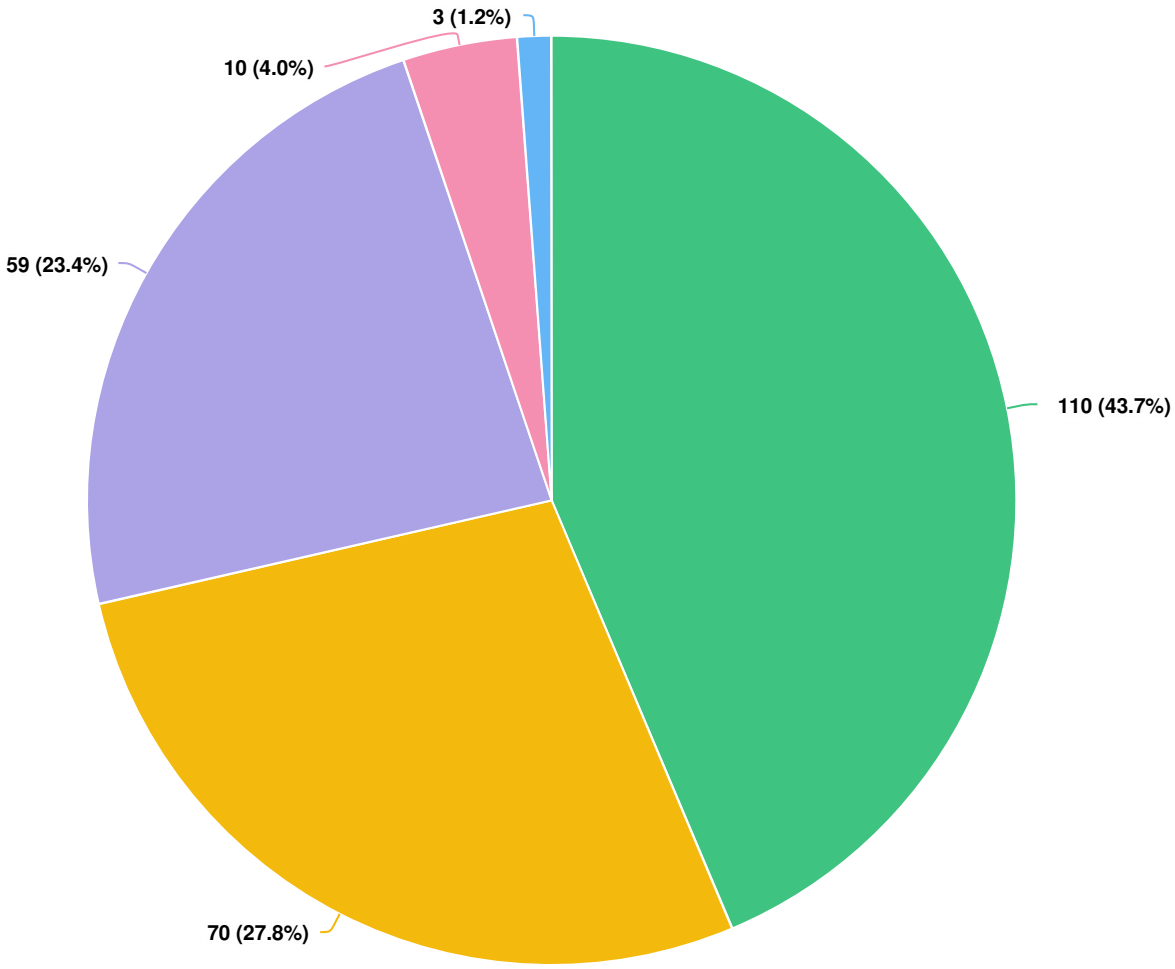
Question options

Ward 1 Ward 2 Ward 3

Optional question (244 response(s), 8 skipped)

Question type: Radio Button Question

What are the first three letters/digits of your postal code? [Optional]



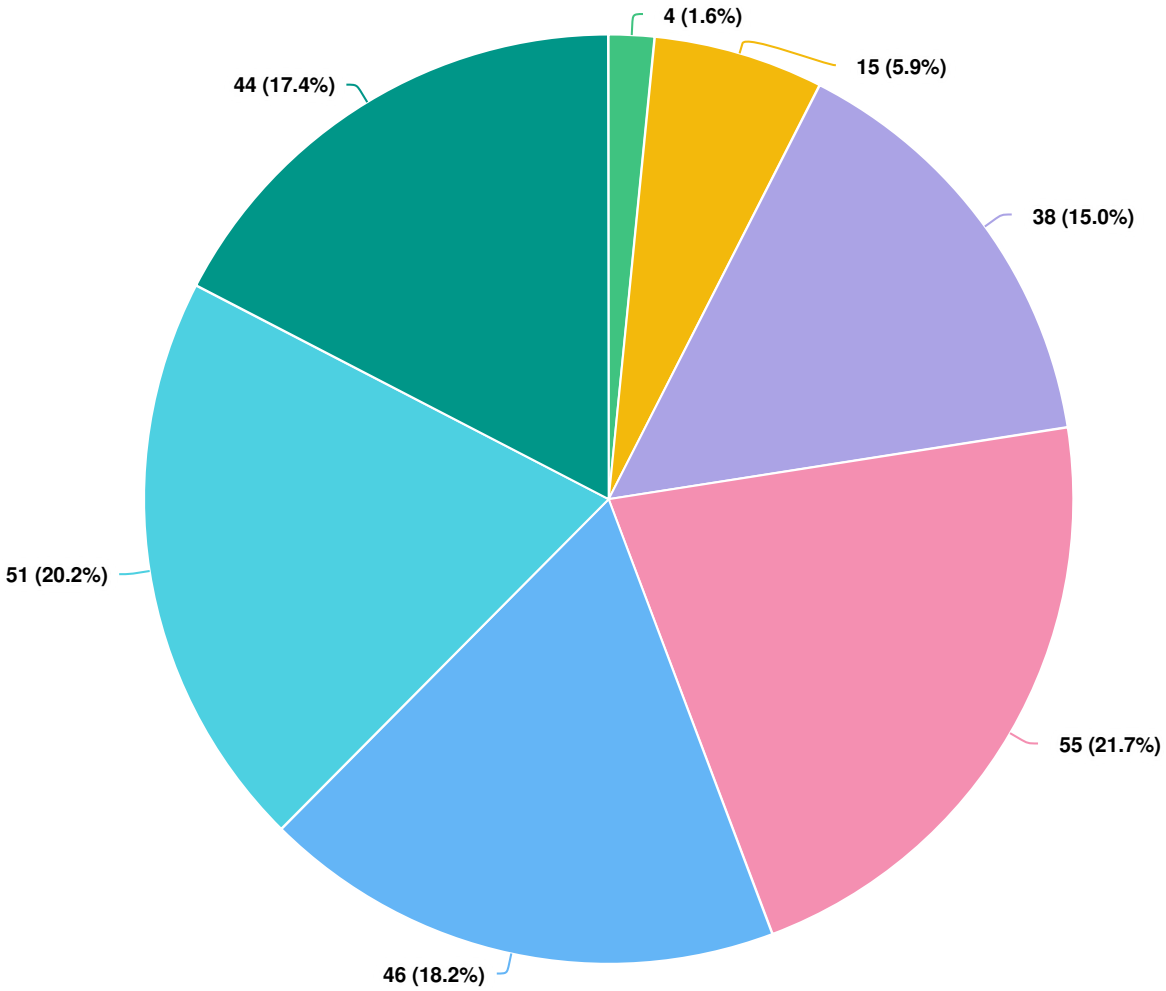
Question options

L1S L1T L1Z Prefer not to say Other (please specify)

Optional question (251 response(s), 1 skipped)

Question type: Radio Button Question

What is your age range?



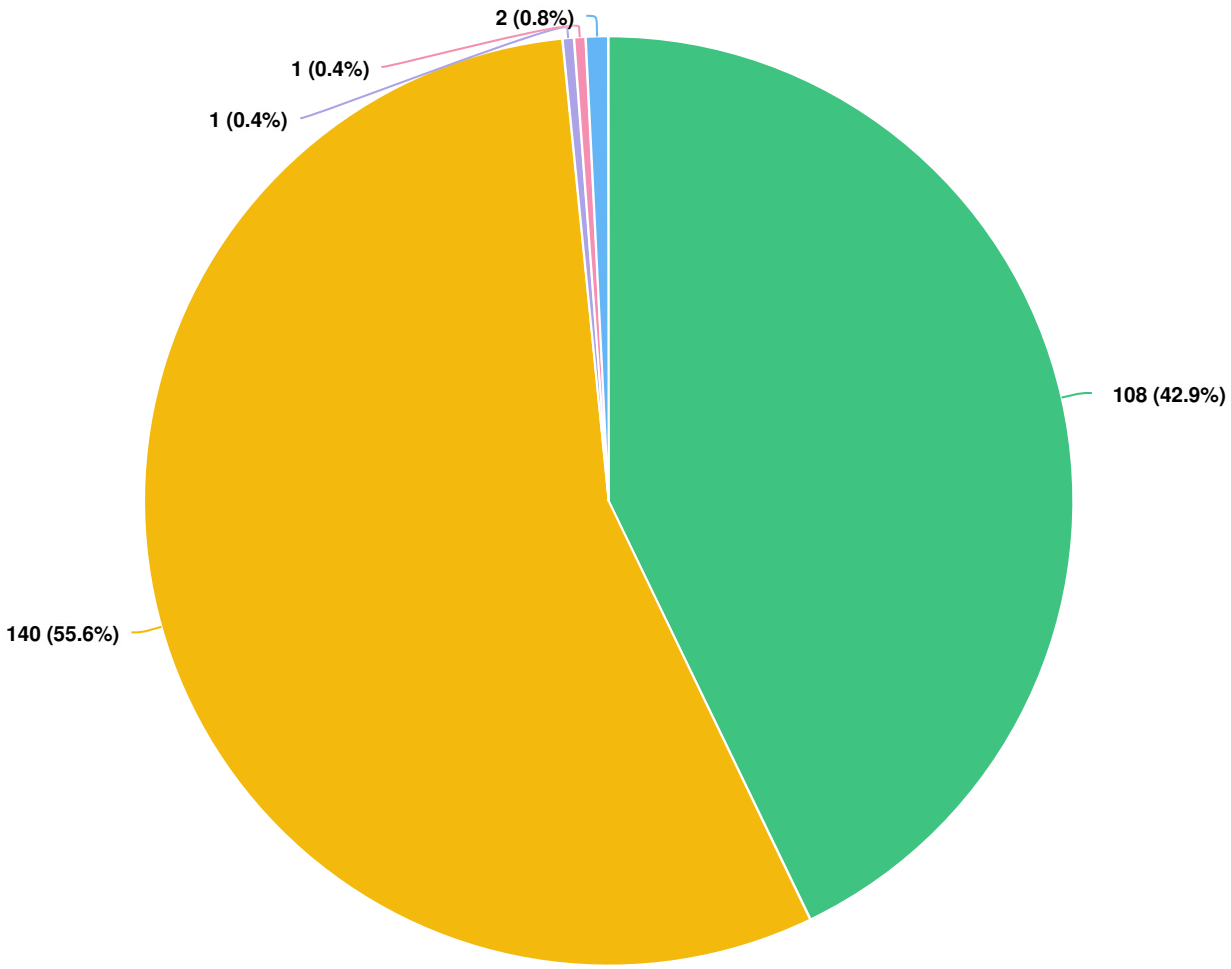
Question options

17 and lower | 18-24 | 25-34 | 35-44 | 45-54 | 55-64 | 65+

Optional question (252 response(s), 0 skipped)

Question type: Radio Button Question

Gender: How do you identify?



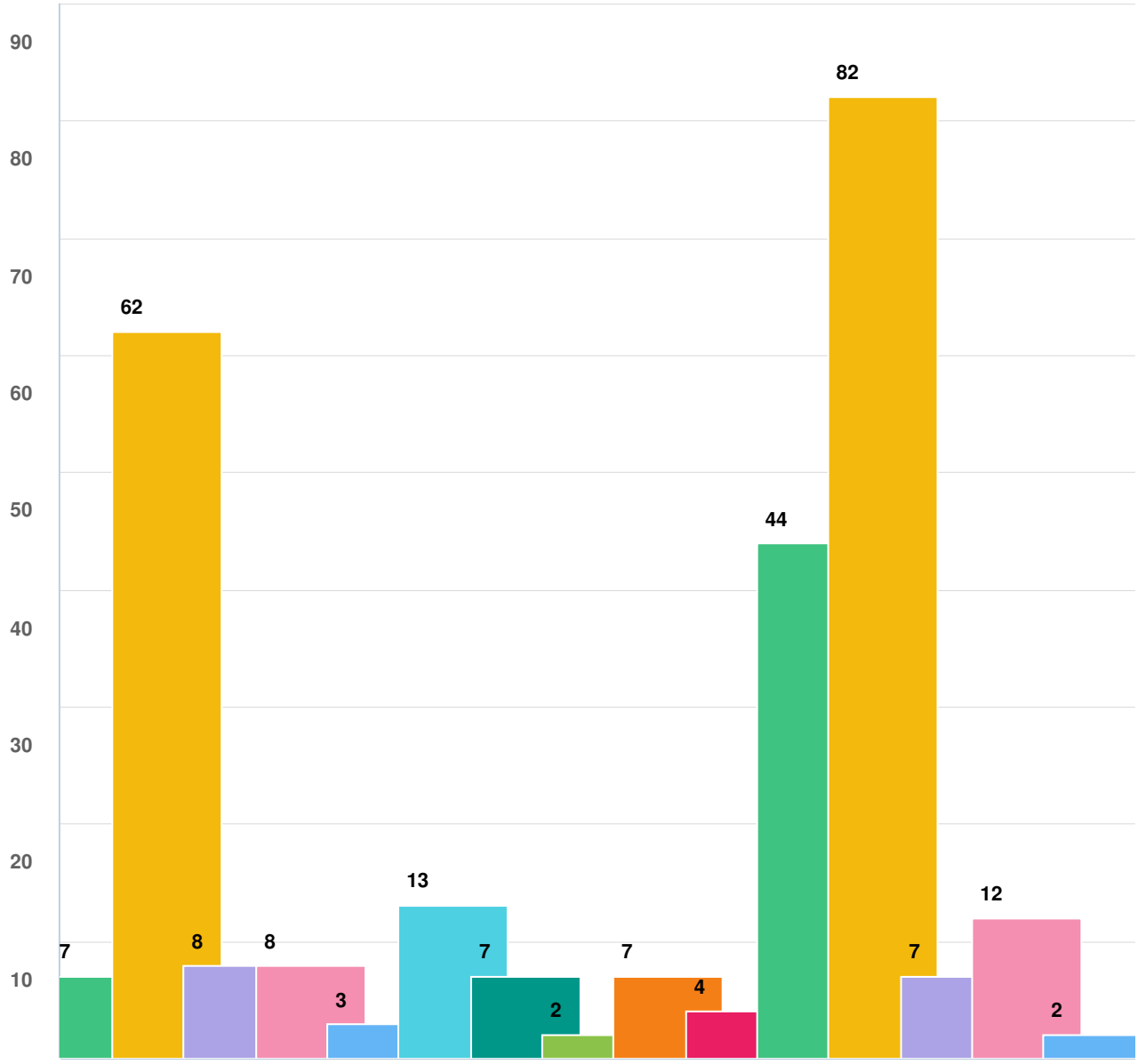
Question options

Female Male Non-binary Two Spirited Prefer not to answer

Optional question (251 response(s), 1 skipped)

Question type: Radio Button Question

This question helps us see if the survey results represent the Town's cultural diversity. We understand that structural racism impacts our experiences with safety. Your answer to this question will put us in a better position to address gaps in und...



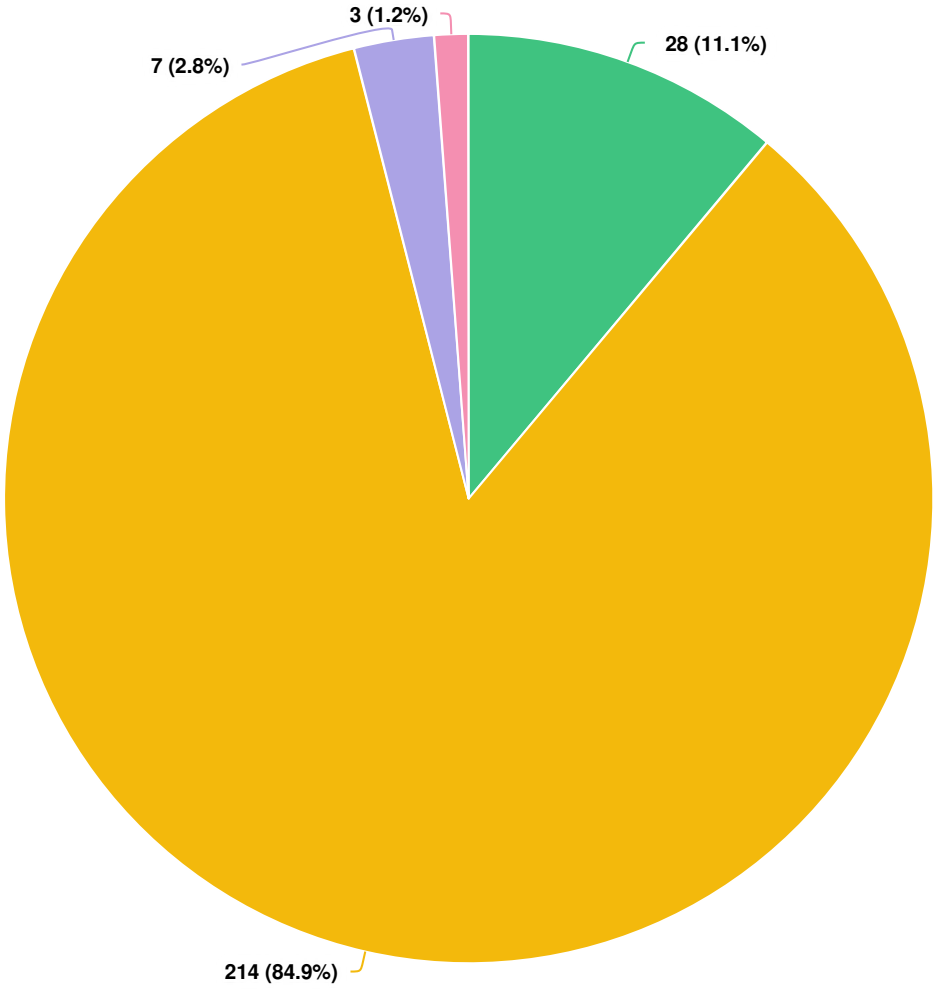
Question options

- Asian - East (e.g. Chinese, Japanese, Korean)
- Asian - South (e.g. Indian, Pakistani, Sri Lankan)
- Asian - South East (e.g. Malaysian, Filipino, Vietnamese)
- Black - African (e.g. Ghanaian, Kenyan, Somalian)
- Black - North American (e.g. Canadian, American)
- Black - Caribbean (e.g. Barbadian, Jamaican)
- Indian - Caribbean (e.g. Guyanese with origins in India)
- First Nations or Indigenous
- Hispanic/Latinx (e.g. Mexican, Argentinian, Chilean)
- Middle Eastern (e.g. Egyptian, Iranian, Lebanese)
- White - European (e.g. English, Italian, Portuguese, Russian)
- White - North American (e.g. Canadian, American)
- Mixed Heritage
- Prefer not to answer
- Other (please specify)

Optional question (251 response(s), 1 skipped)

Question type: Checkbox Question

Do you have difficulty seeing, even with the use of glasses or contact lenses?



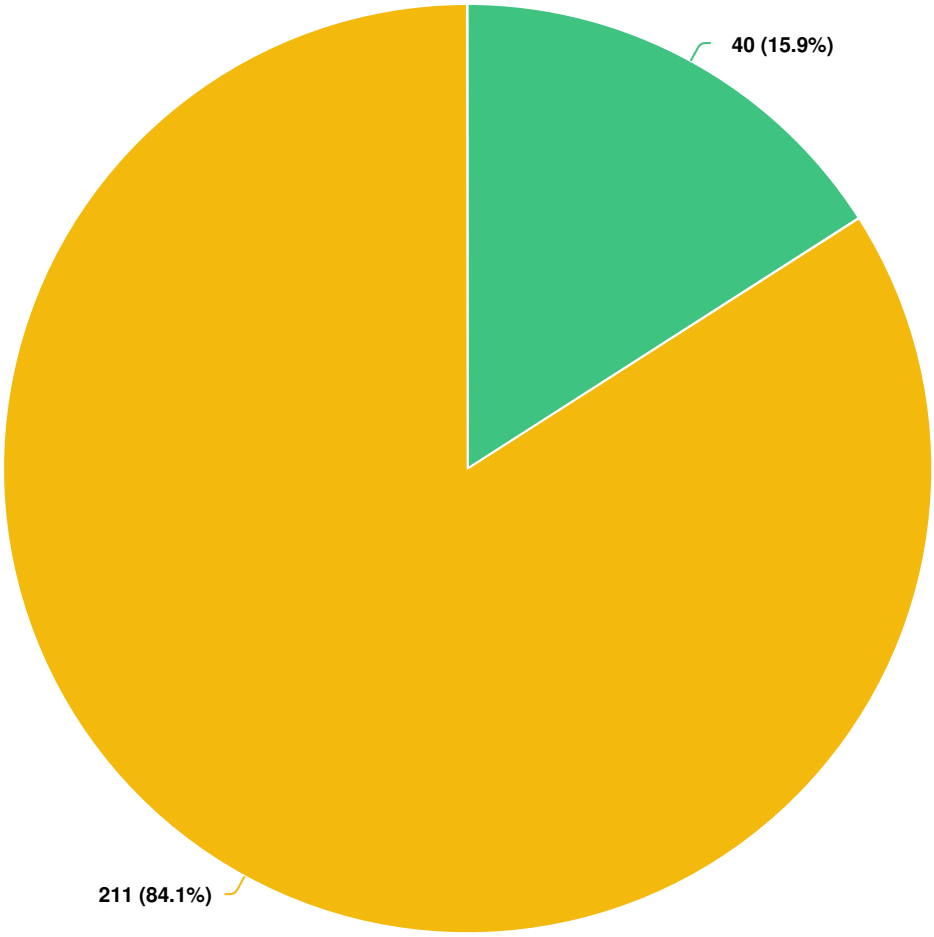
Question options

Yes No I prefer not to answer Other (please specify)

Optional question (251 response(s), 1 skipped)

Question type: Radio Button Question

Do you live with a physical or mental condition or health problem that reduces the amount or kind of activity you can do?



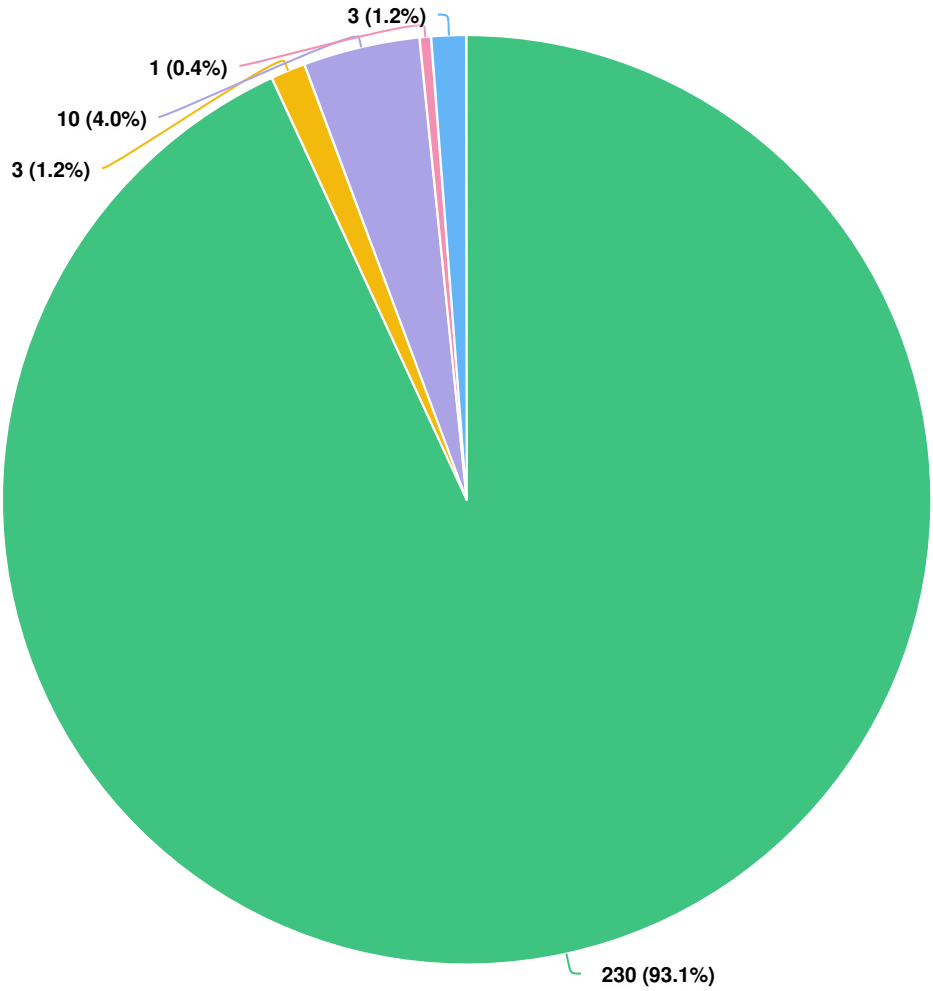
Question options

Yes No

Optional question (250 response(s), 2 skipped)

Question type: Radio Button Question

Do you use an assistive device?



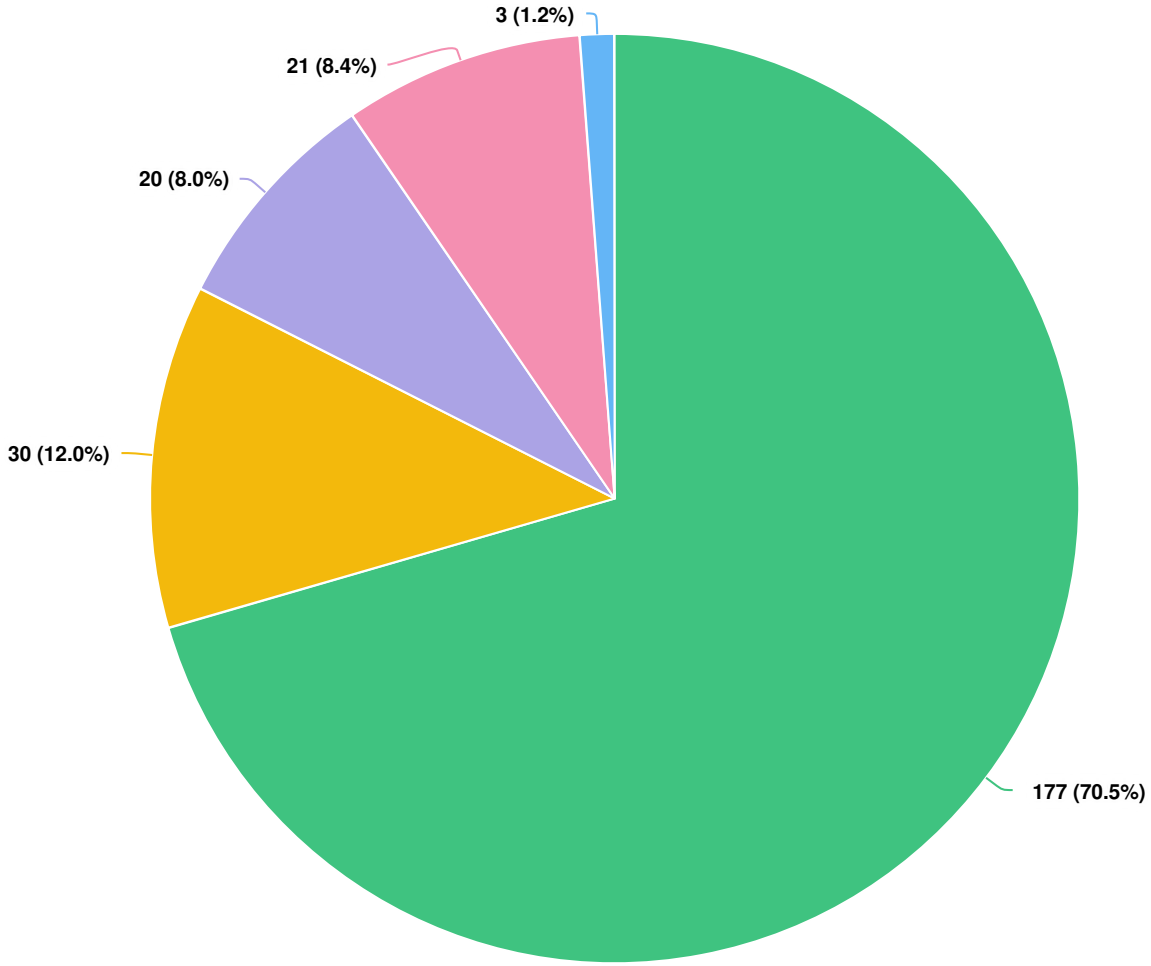
Question options

No Walker Cane Service Animal Other (please specify)

Optional question (246 response(s), 6 skipped)

Question type: Radio Button Question

What type of housing unit best describes your living situation?



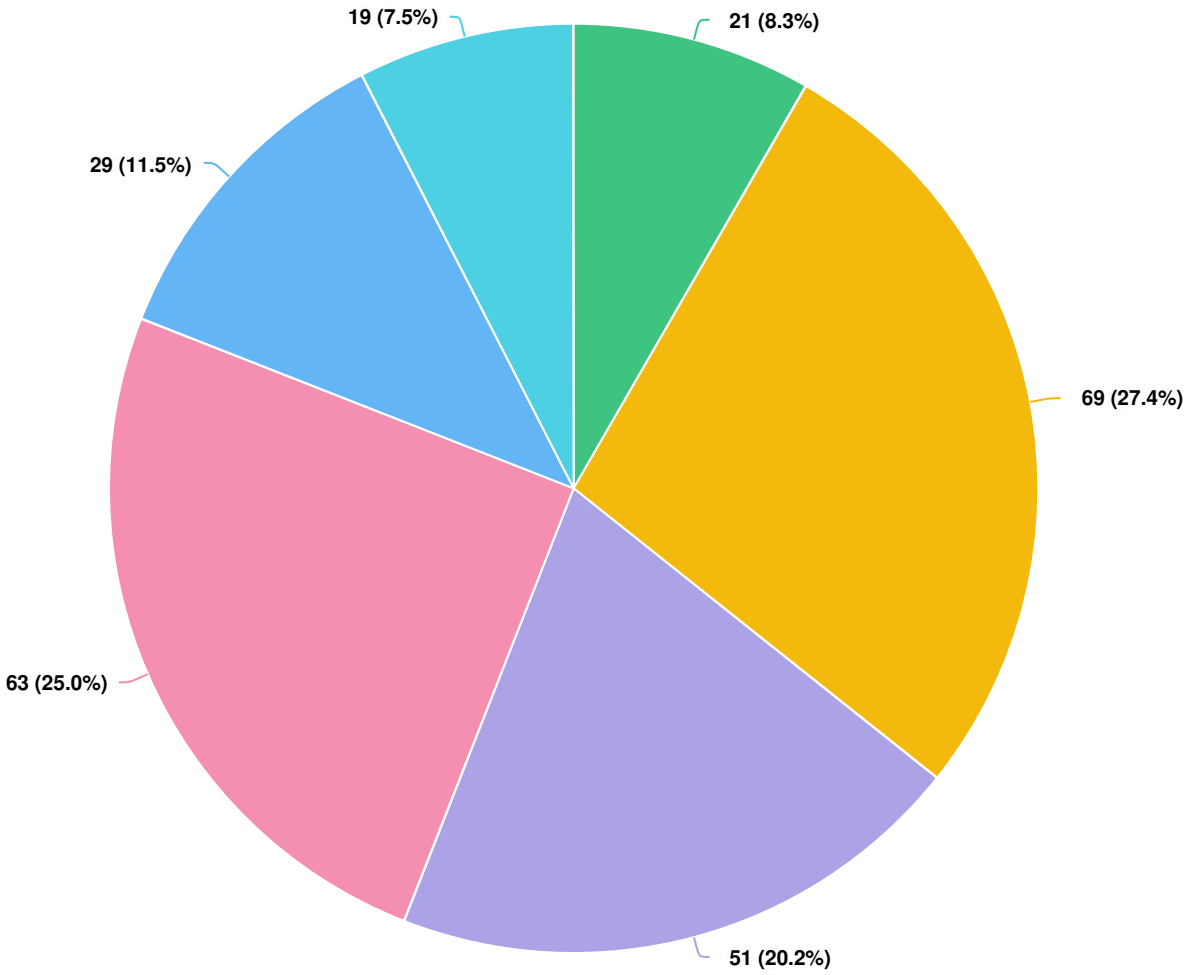
Question options

Detached house | Semi-detached house | Apartment/Condo | Townhouse | Other (please specify)

Optional question (250 response(s), 2 skipped)

Question type: Radio Button Question

How many people live in your household?



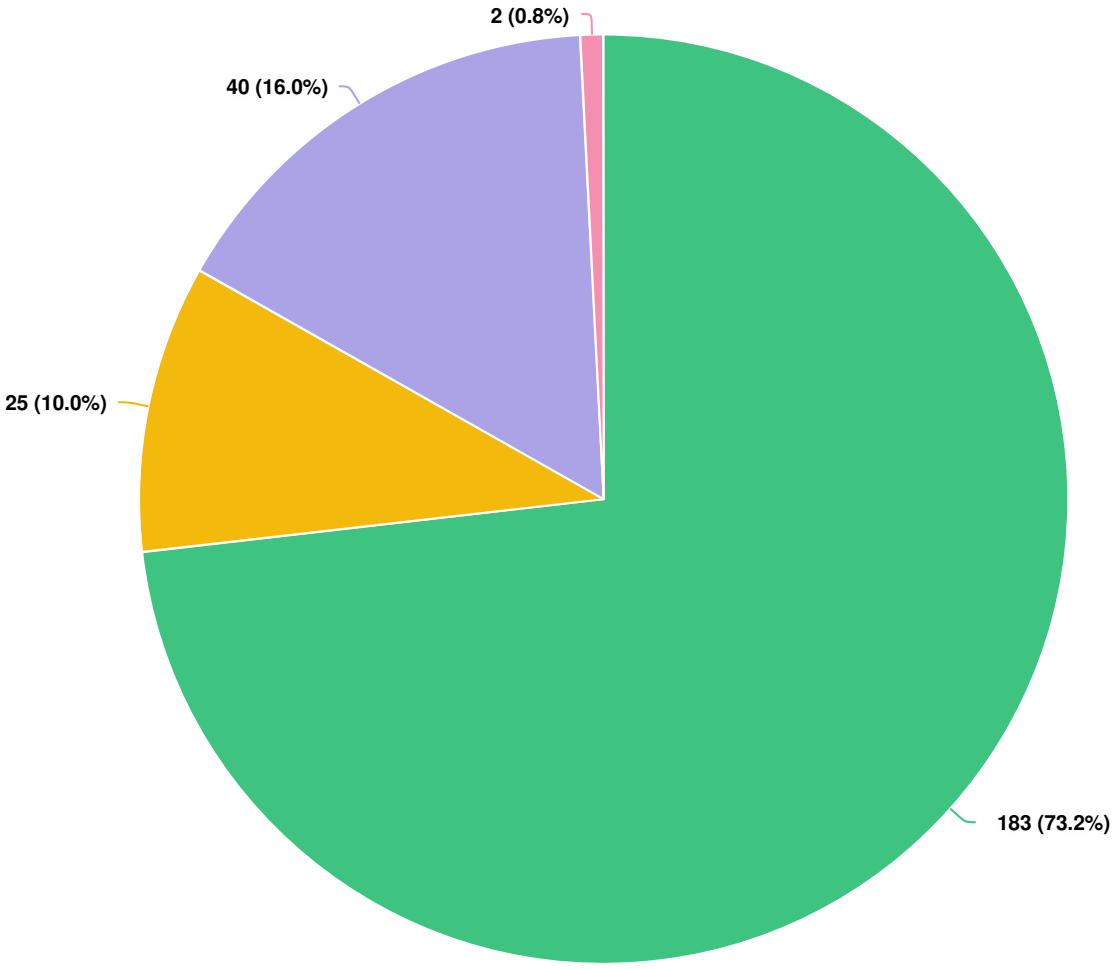
Question options

1 Resident | 2 Residents | 3 Residents | 4 Residents | 5 Residents | 6+ Residents

Optional question (251 response(s), 1 skipped)

Question type: Radio Button Question

What best describes your relationship with your home?



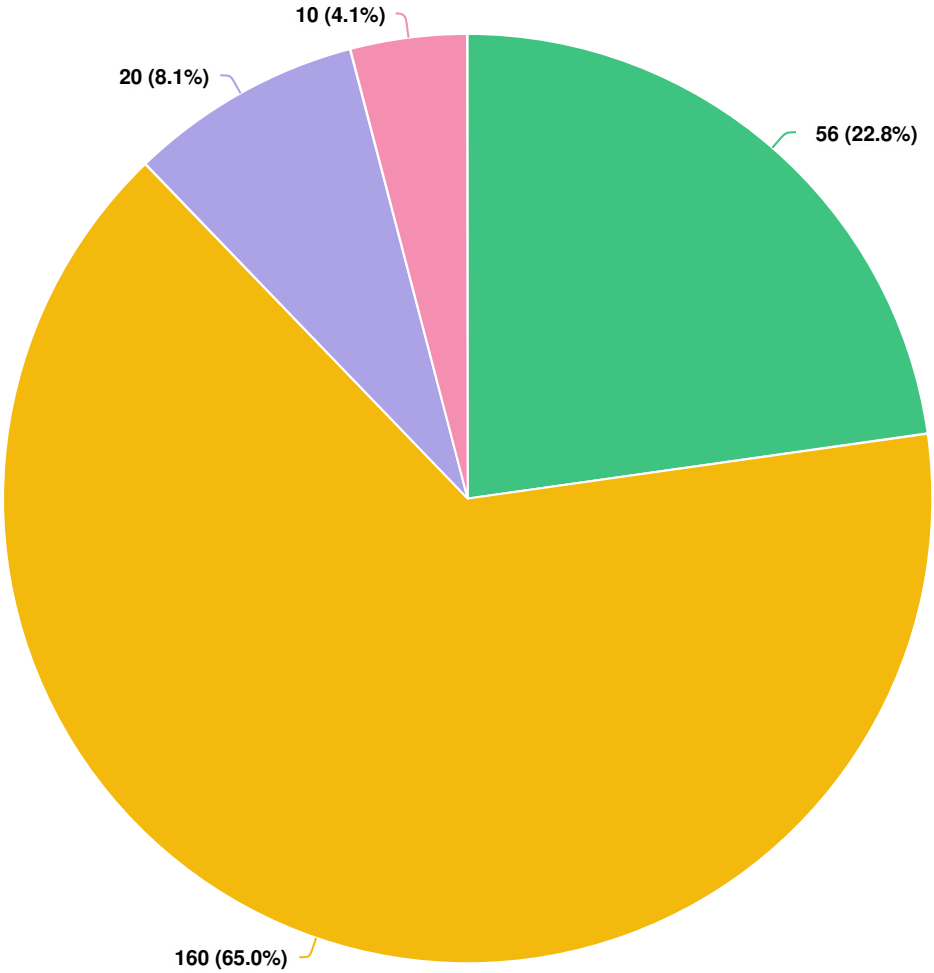
Question options

Owner of home | Renter of home | Residing with family/friends/colleagues | Other (please specify)

Optional question (249 response(s), 3 skipped)

Question type: Radio Button Question

What best describes your employment status?



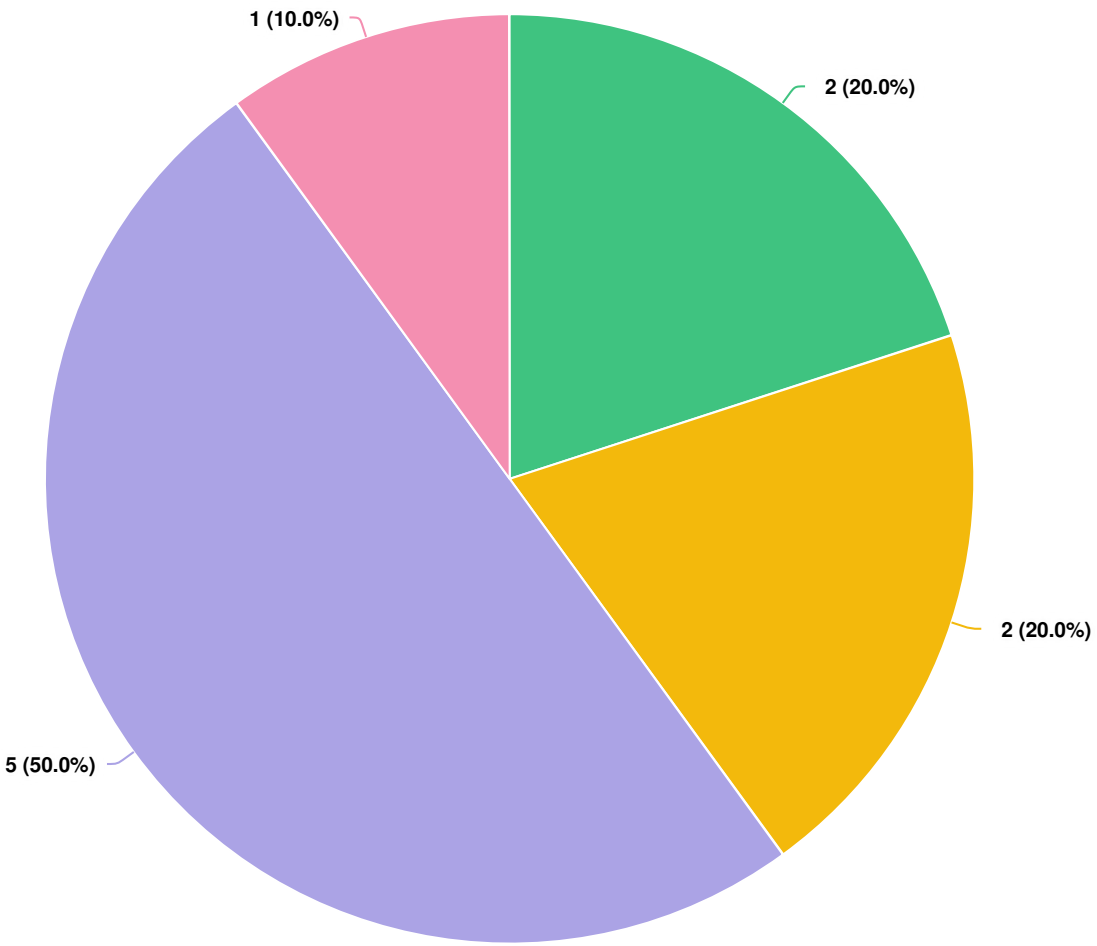
Question options

Unemployed Employed Self Employed Student

Optional question (245 response(s), 7 skipped)

Question type: Radio Button Question

Where do you attend school?



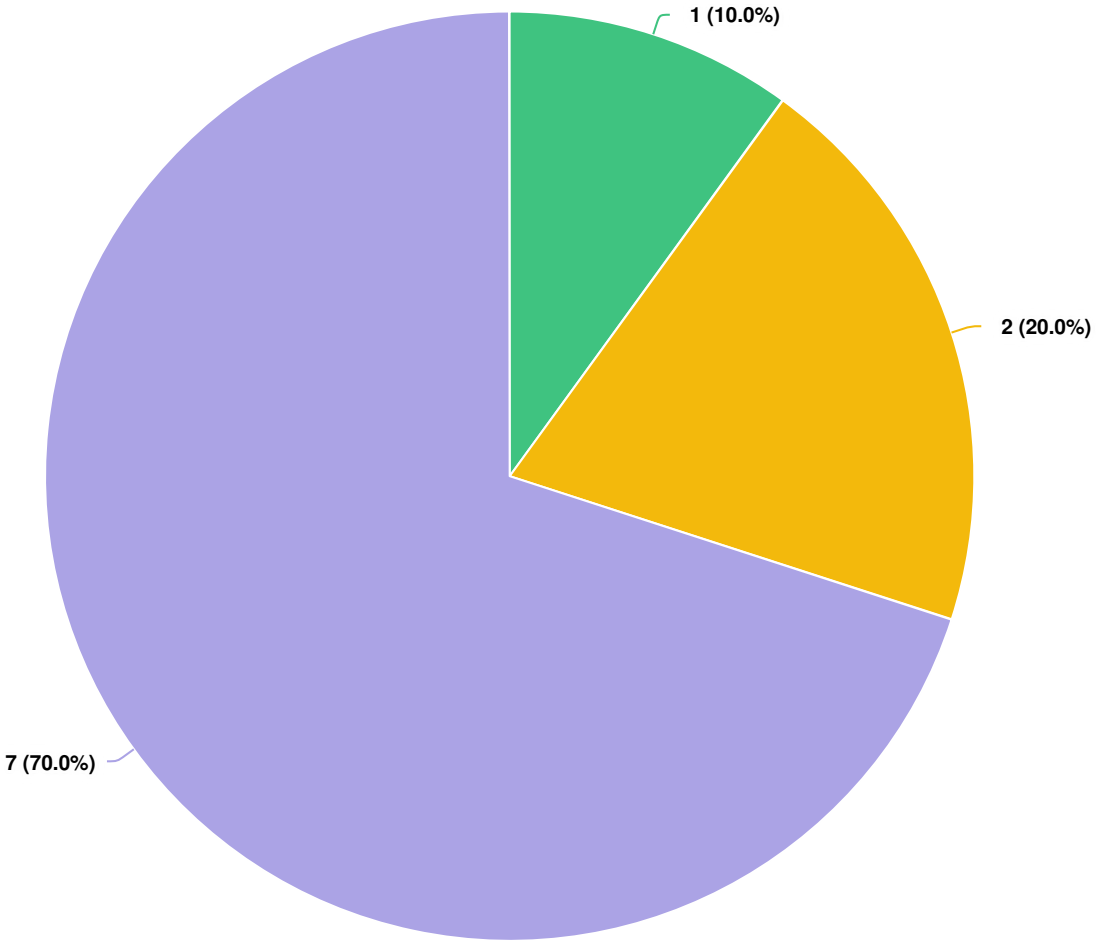
Question options

- Within Ajax
- Outside of Ajax but within Durham Region
- Outside of Durham Region but within the Greater Toronto Area
- Outside of the Greater Toronto Area

Optional question (10 response(s), 242 skipped)

Question type: Radio Button Question

What type of institution are you attending?



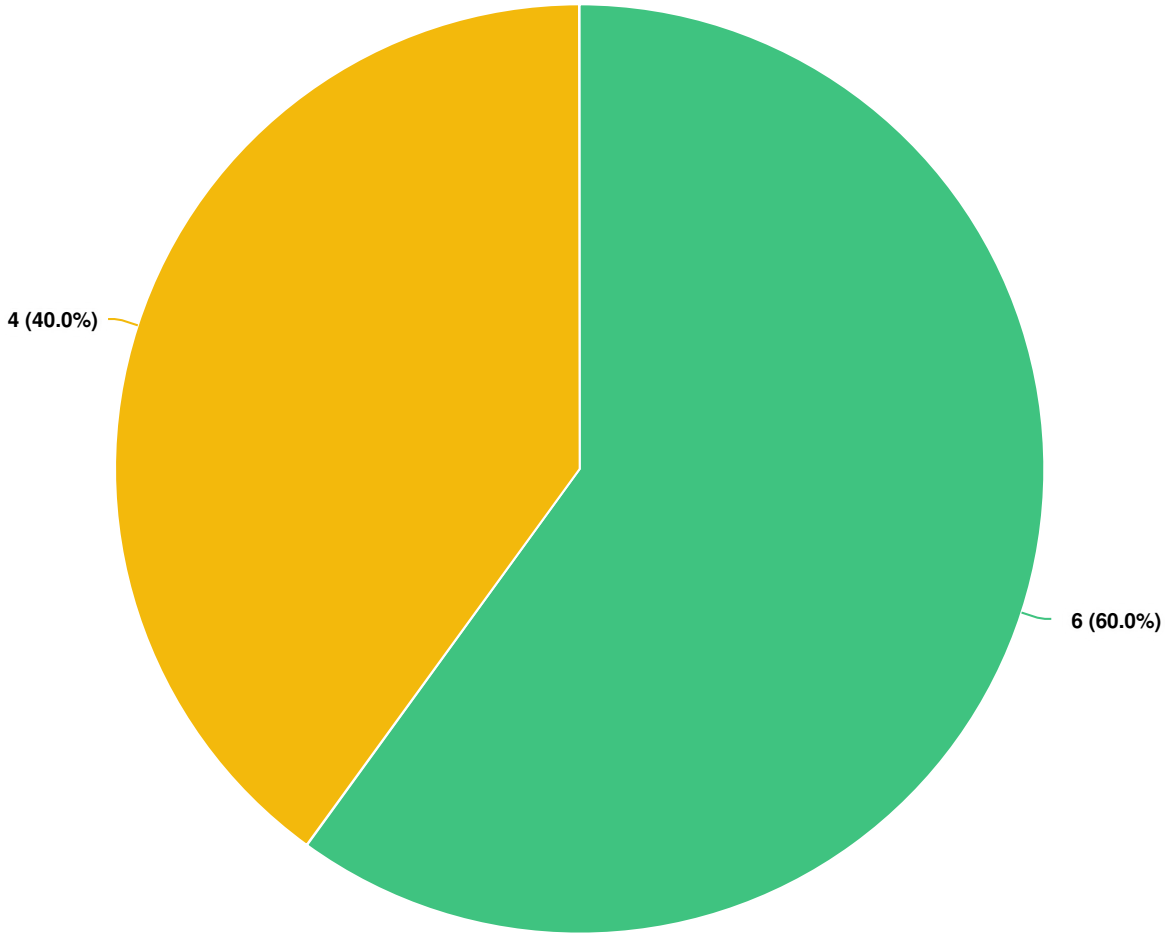
Question options

Elementary School Secondary School Post-Secondary Institution

Optional question (10 response(s), 242 skipped)

Question type: Radio Button Question

Are you employed part-time/full-time while attending school



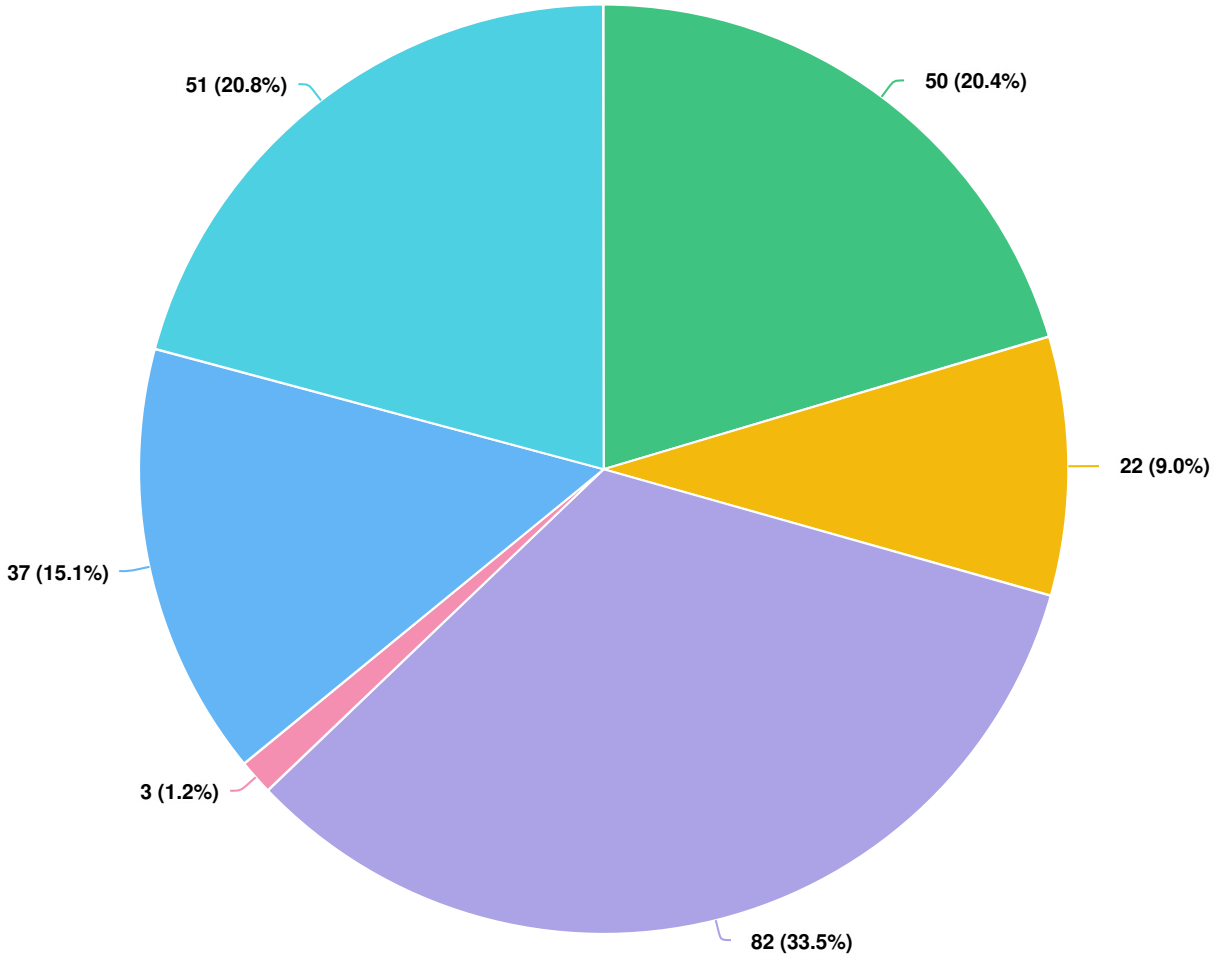
Question options

Yes No

Optional question (10 response(s), 242 skipped)

Question type: Radio Button Question

What best describes the location that you most frequently work?



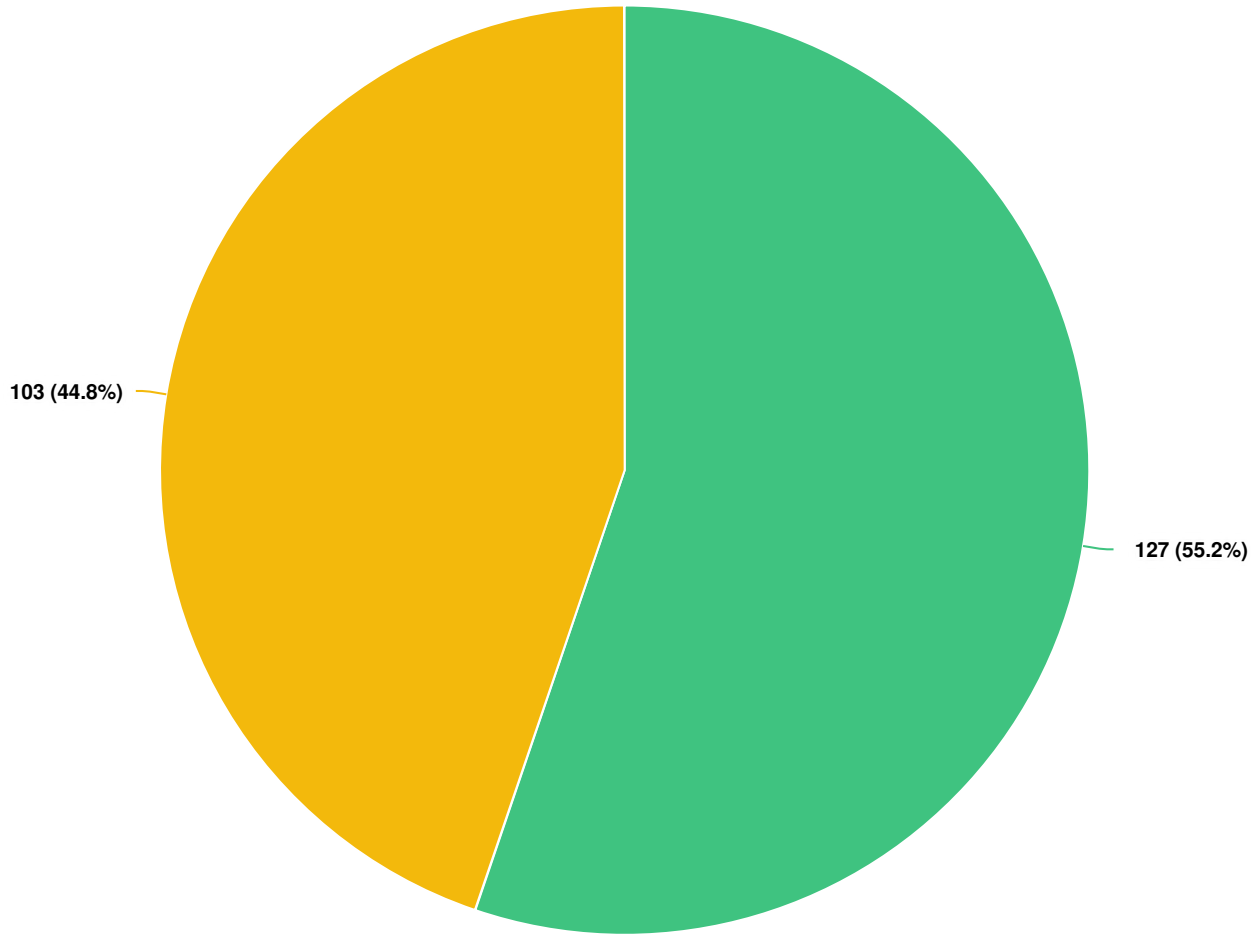
Question options

Within Ajax
Outside of Ajax but within Durham Region
Outside of Durham Region but within the Greater Toronto Area
Outside of the Greater Toronto Area
At home/Online
Not applicable - Do not work

Optional question (244 response(s), 8 skipped)

Question type: Radio Button Question

Is working from home/remotely an option in your line of work/schooling?



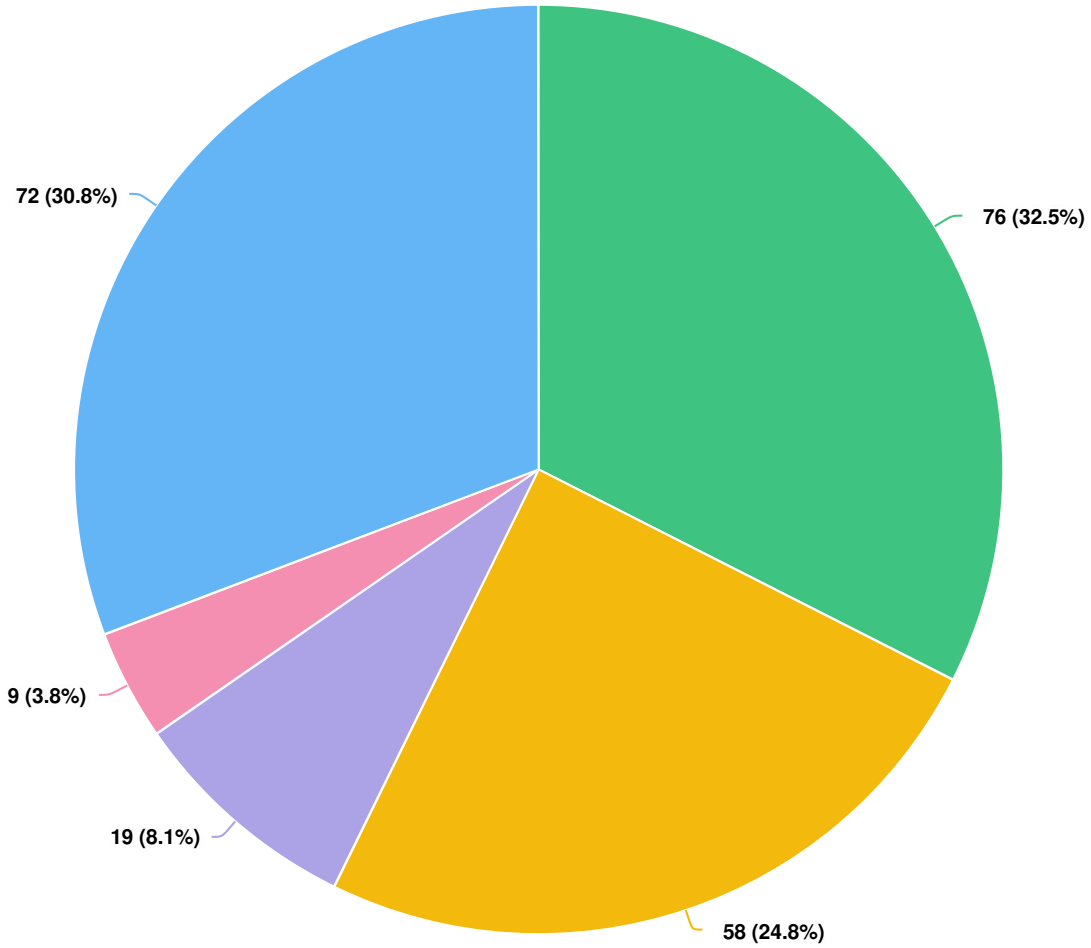
Question options

Yes No

Optional question (230 response(s), 22 skipped)

Question type: Radio Button Question

How frequently do you work/attend school remotely?



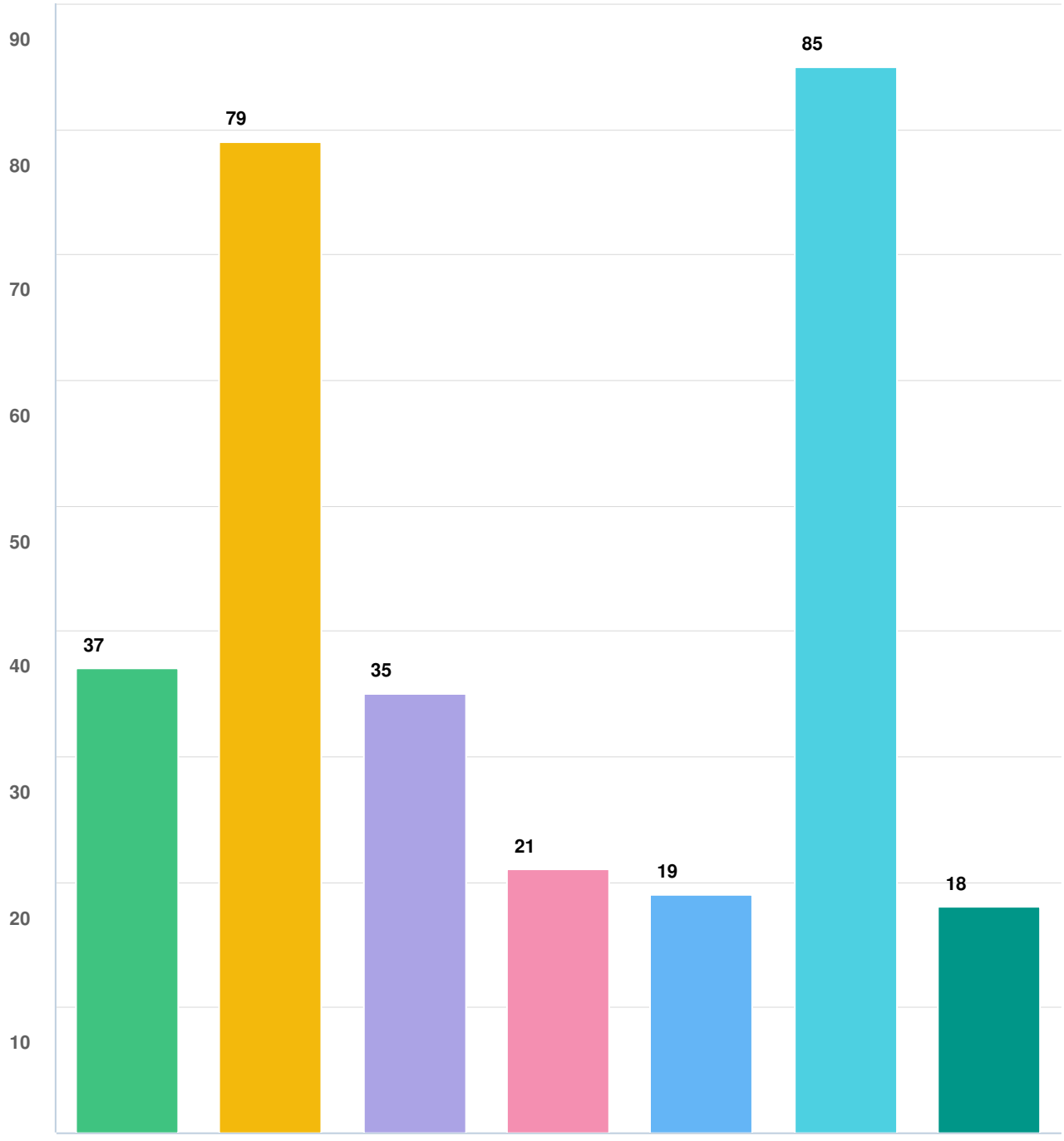
Question options

- Always/Most of the time (4-5 days per week)
- Frequently (1-3 days per week)
- Occasionally (1-4 days permonth)
- Rarely (1 day per month)
- Never

Optional question (234 response(s), 18 skipped)

Question type: Radio Button Question

What barriers are preventing you from working/attending school from home remotely?



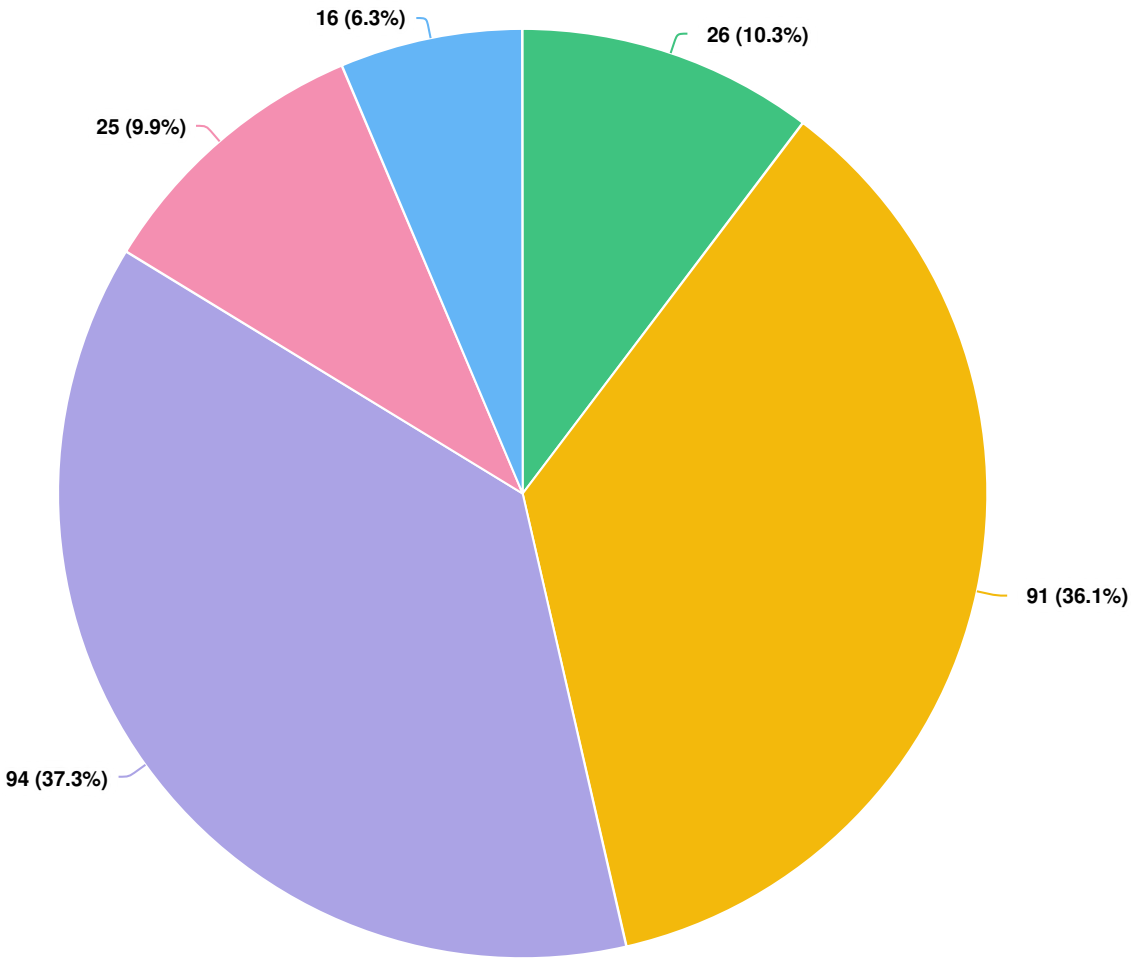
Question options

- Not permitted at my place of employment
- Onsite work is a required component of my job
- I prefer to interact with colleague/people in-person
- I prefer going into my place of employment/school
- I am more productive when attending an institution in-person
- None
- Other (please elaborate)

Optional question (239 response(s), 13 skipped)

Question type: Checkbox Question

How many vehicles are owned by your household?



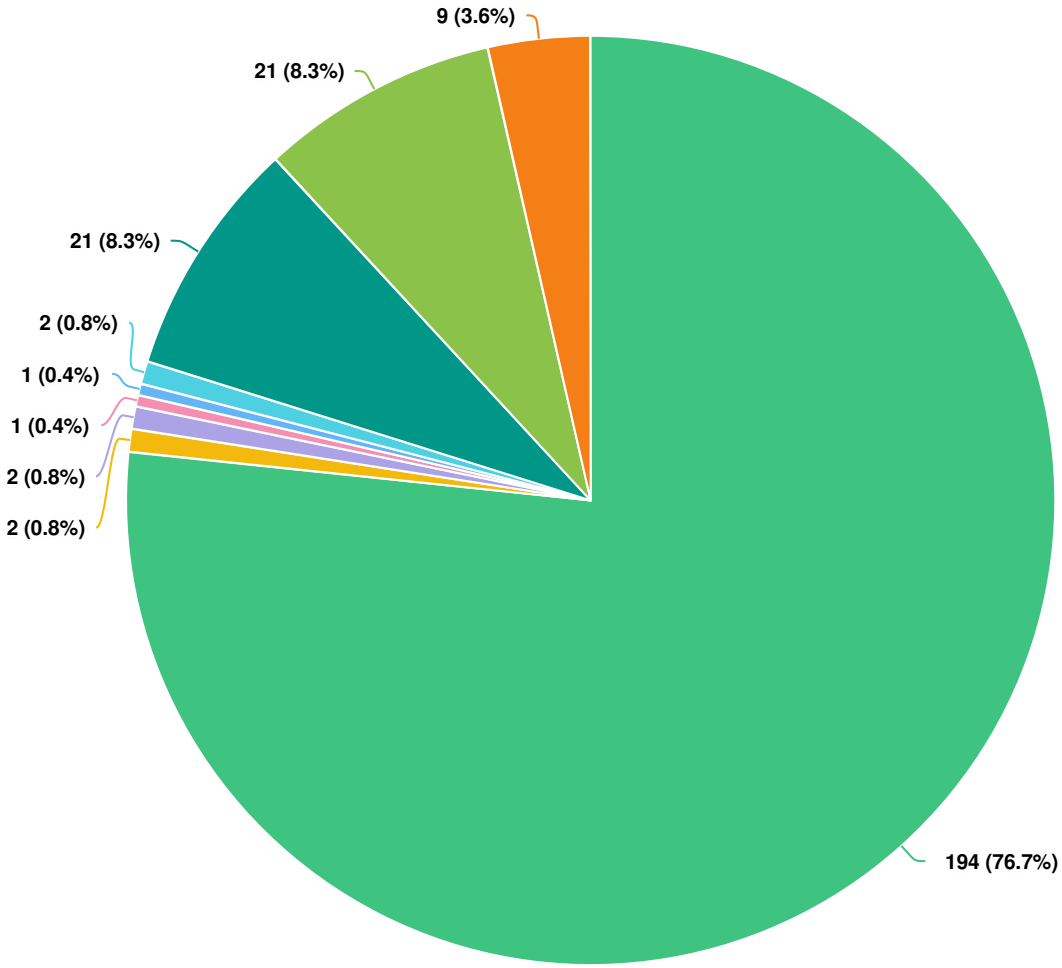
Question options

0 Vehicles 1 Vehicle 2 Vehicles 3 Vehicles 4+ Vehicles

Optional question (251 response(s), 1 skipped)

Question type: Radio Button Question

What type of vehicle/travel option does your household most frequently use for commutes, recreational activities, leisure, and daily trips?



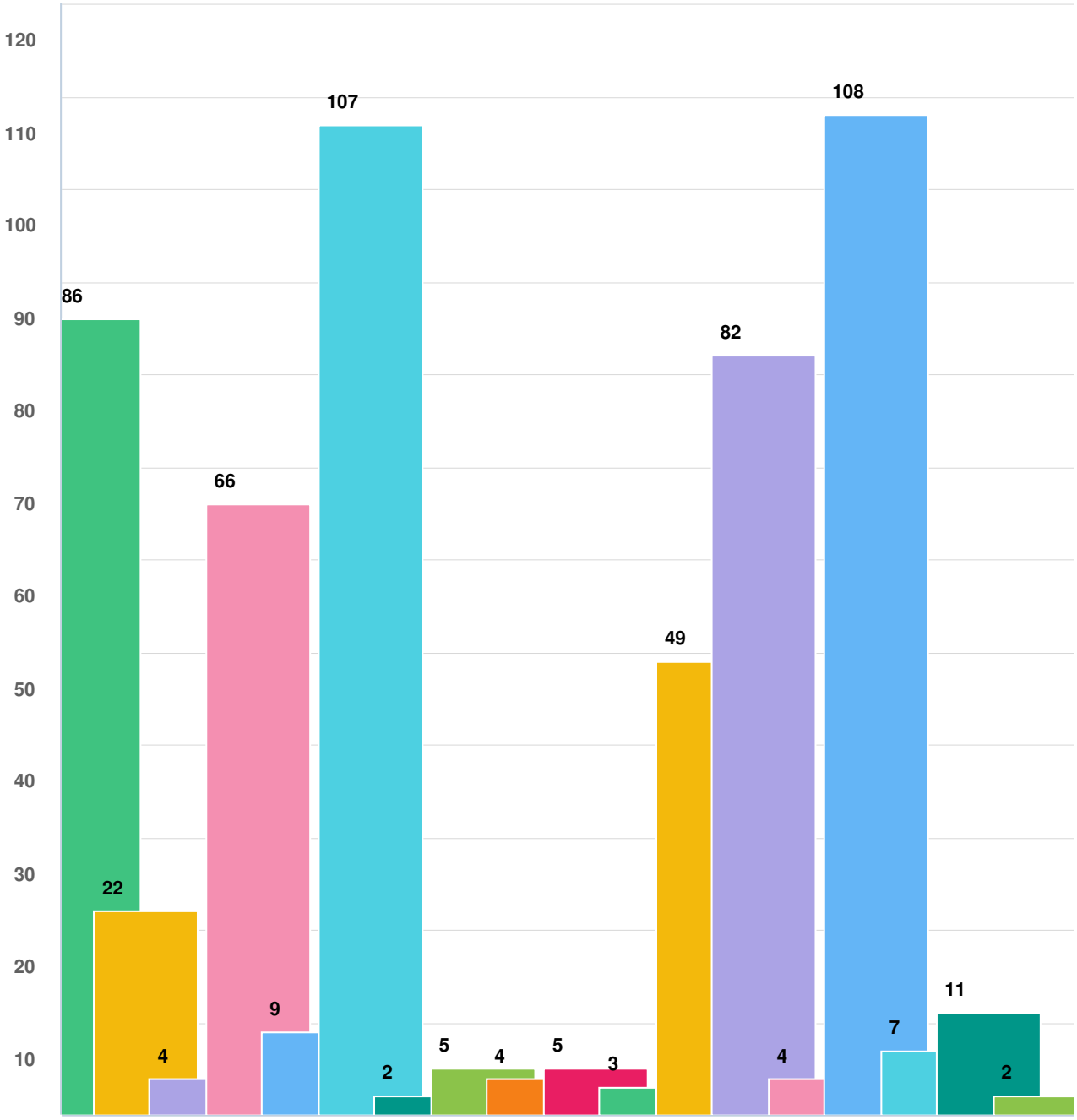
Question options

Personal car | Car pooling | Bicycle | Walking | E-Scooter (Kick style) | Taxi/Uber/Lyft | Durham Region Transit | GO Transit (Bus or Train) | Other (please specify)

Optional question (252 response(s), 0 skipped)

Question type: Radio Button Question

What other transportation modes do you or members of your household frequently use? Select all that apply



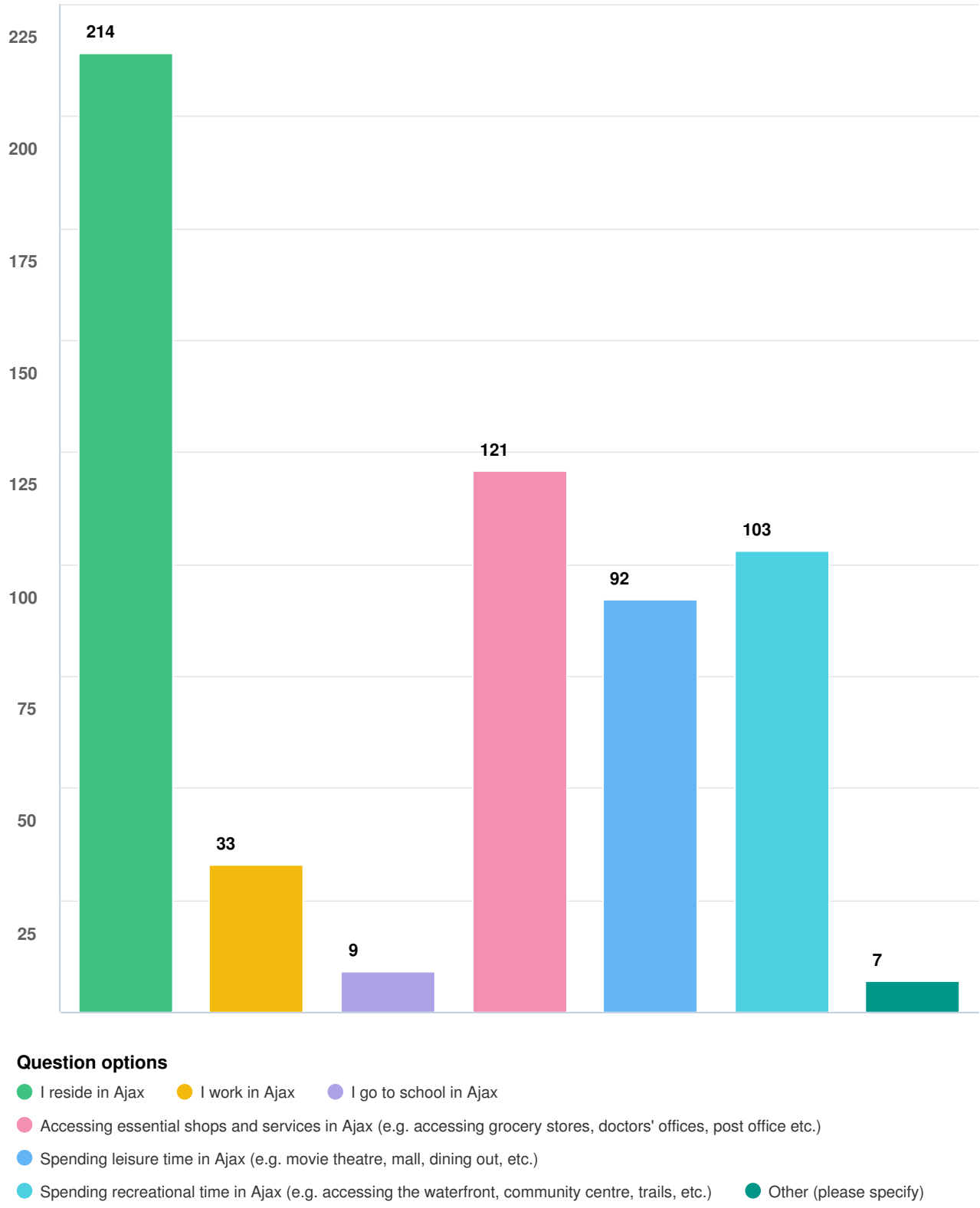
Question options

Personal car | Car pooling | Motorcycle/Moped | Bicycle | E-Bicycle | Walking | Wheelchair/Mobility Device | Scooter (Kick style) | E-Scooter (Kick style) | Skateboard | Rollerblades | Taxi/Uber/Lyft | Durham Region Transit | Durham Region Transit Specialized Services (Paratransit) | GO Transit (Bus or Train) | VIA Rail | None | Other (please specify)

Optional question (246 response(s), 6 skipped)

Question type: Checkbox Question

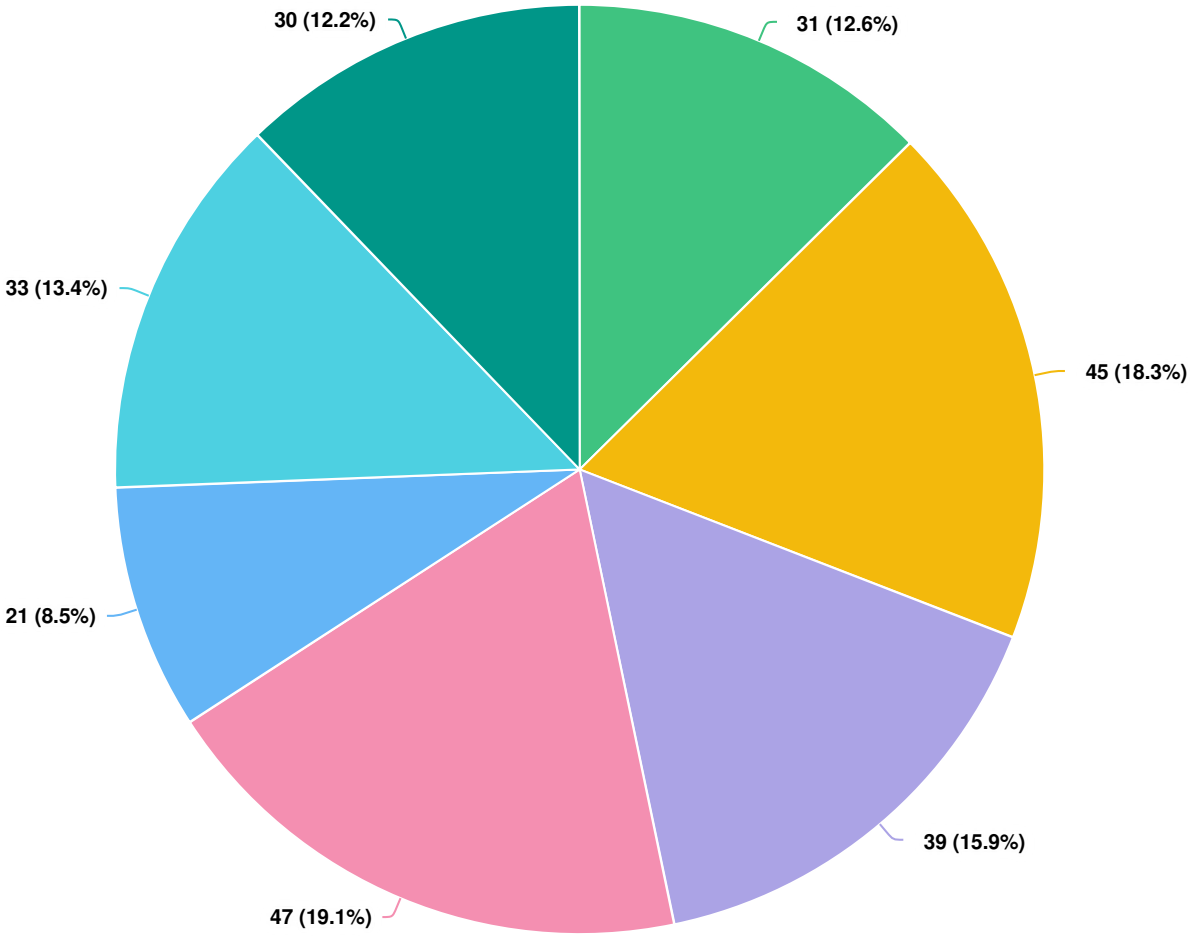
What is your primary purpose for traveling within Ajax?



Optional question (252 response(s), 0 skipped)

Question type: Checkbox Question

On average, what is the cumulative time you spend commuting in an average weekday?



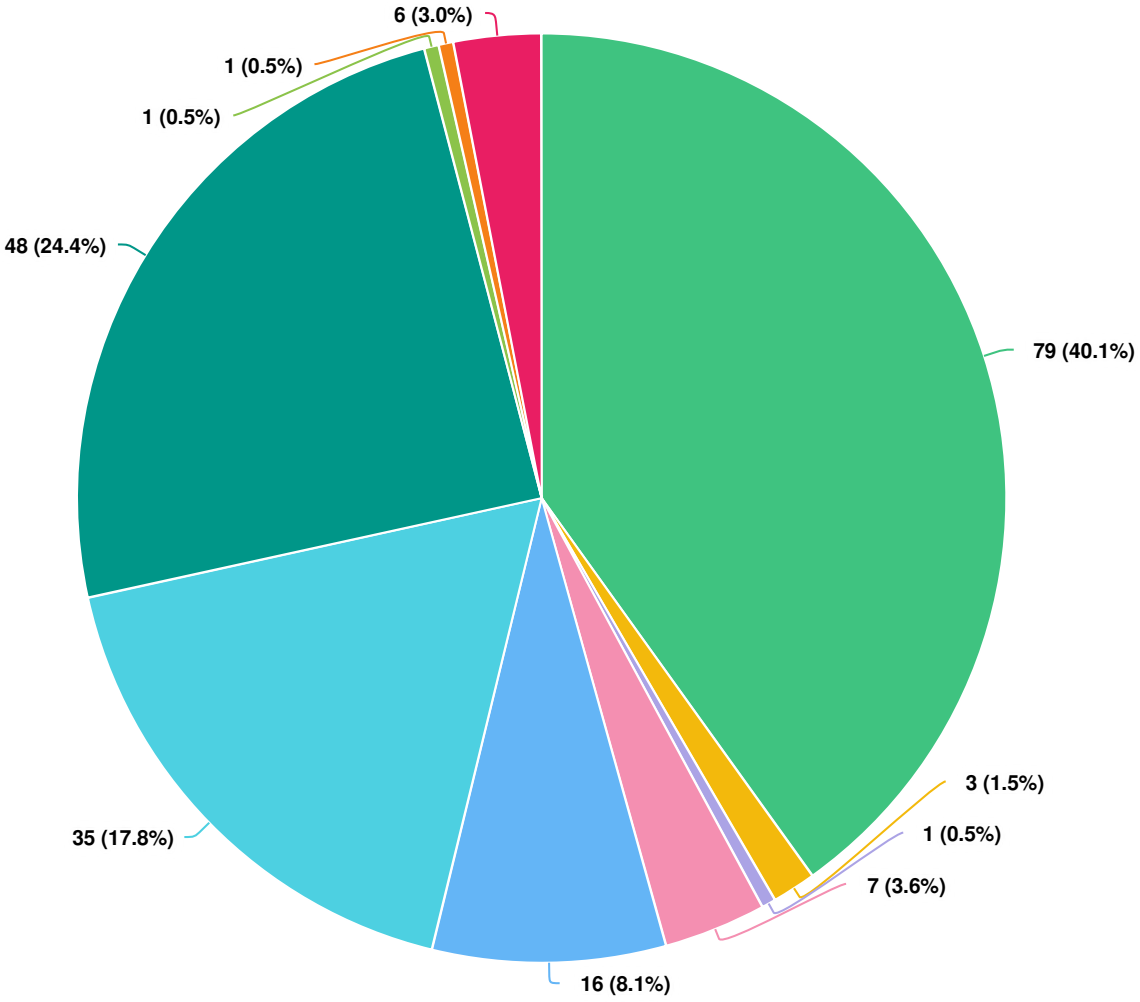
Question options

I work from home and do not have a morning commute | Under 30 minutes | 30-60 minutes | 1-1.5 hours | 2-2.5 hours | More than 2.5 hours | Other (please specify)

Optional question (245 response(s), 7 skipped)

Question type: Radio Button Question

How do you travel to work on a Monday?



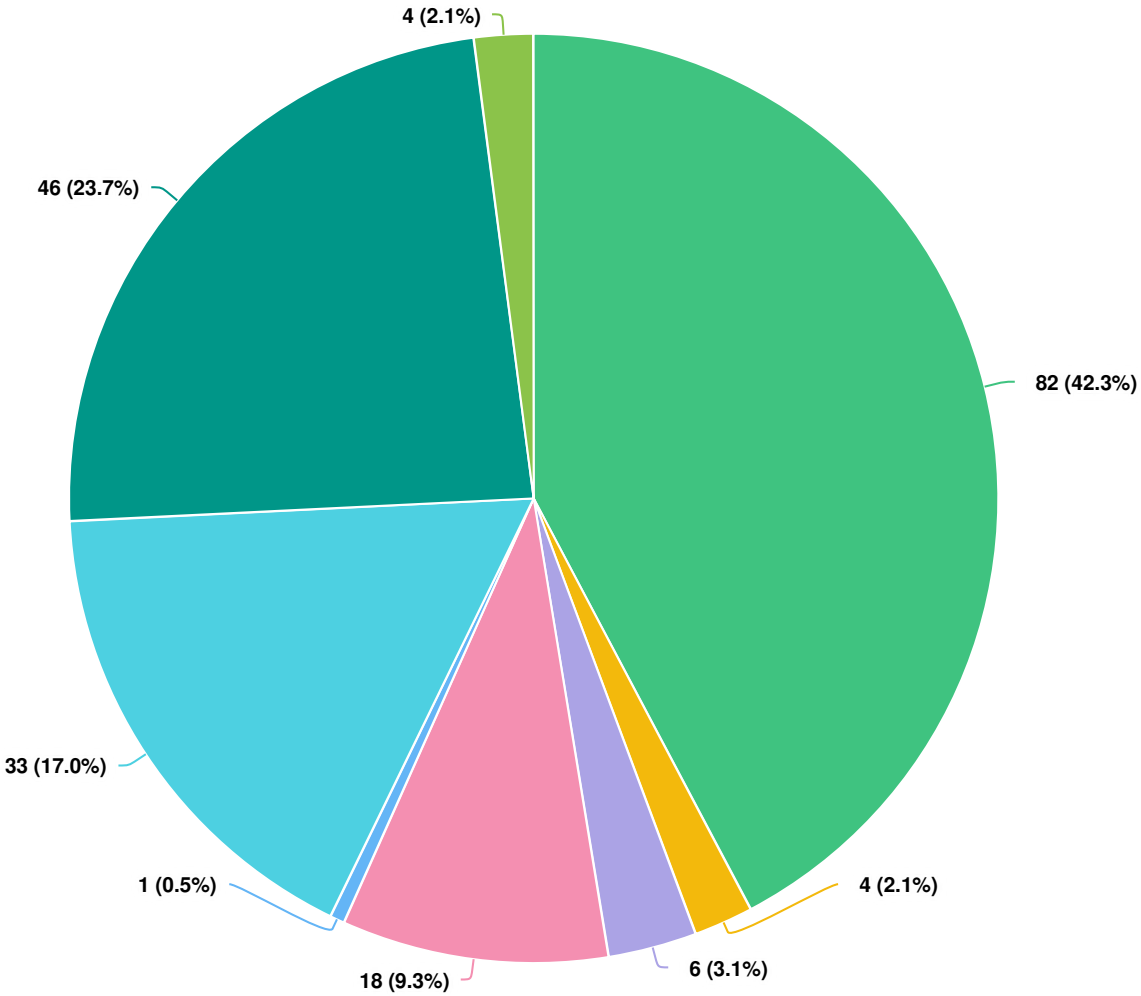
Question options

Car | Bicycle | E-Bicycle | Walking | Durham Region Transit | GO Transit (Bus or Train) | None (work from home or did not go to work) | Ride from a friend, family friend, or neighbor | Ride hailing service (Uber, Lyft) | Other (please specify)

Optional question (196 response(s), 56 skipped)

Question type: Dropdown Question

How do you travel to work on a Tuesday?



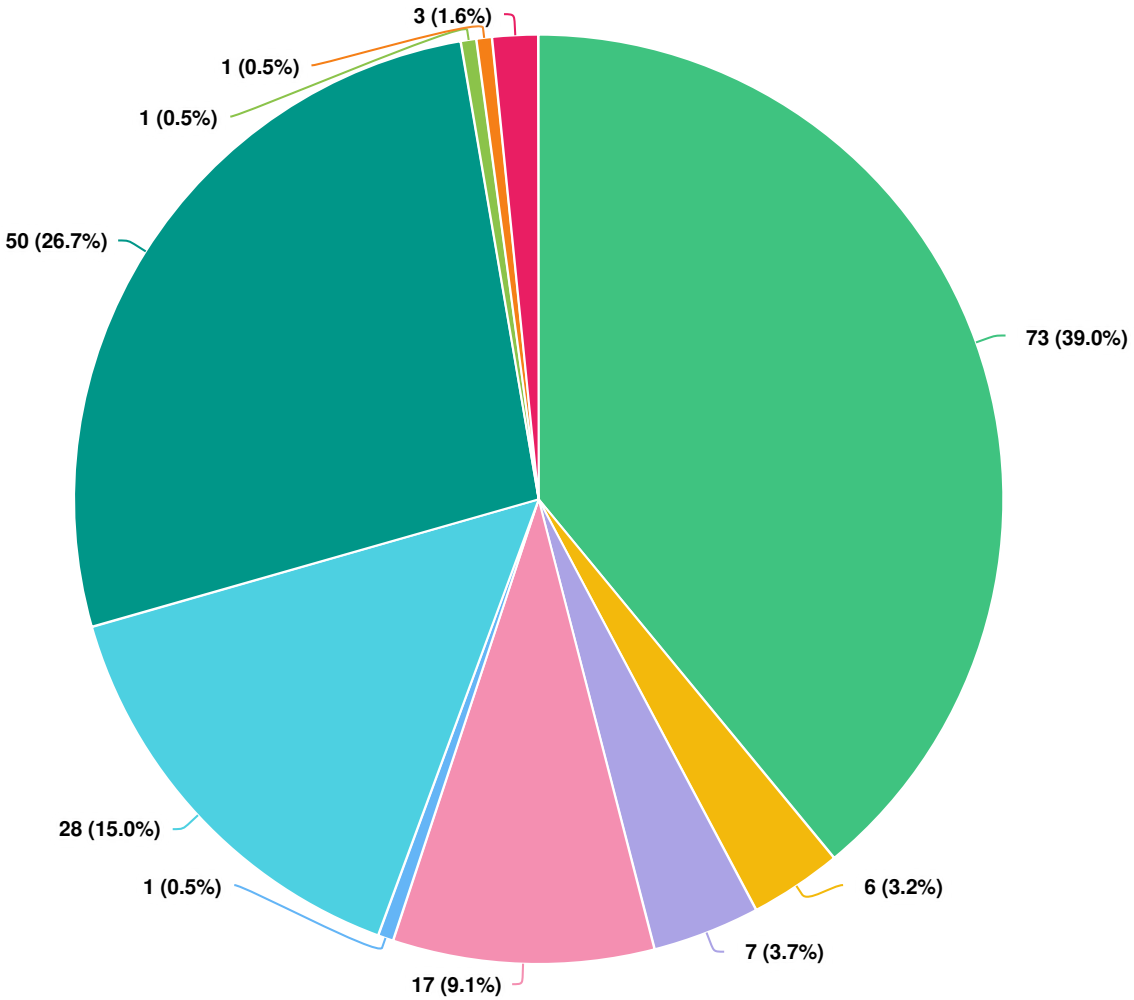
Question options

Car · Bicycle · Walking · Durham Region Transit · Durham Region Transit Specialized Services (Paratransit) · GO Transit (Bus or Train) · None (work from home or did not go to work) · Other (please specify)

Optional question (193 response(s), 59 skipped)

Question type: Dropdown Question

How do you travel to work on a Wednesday?



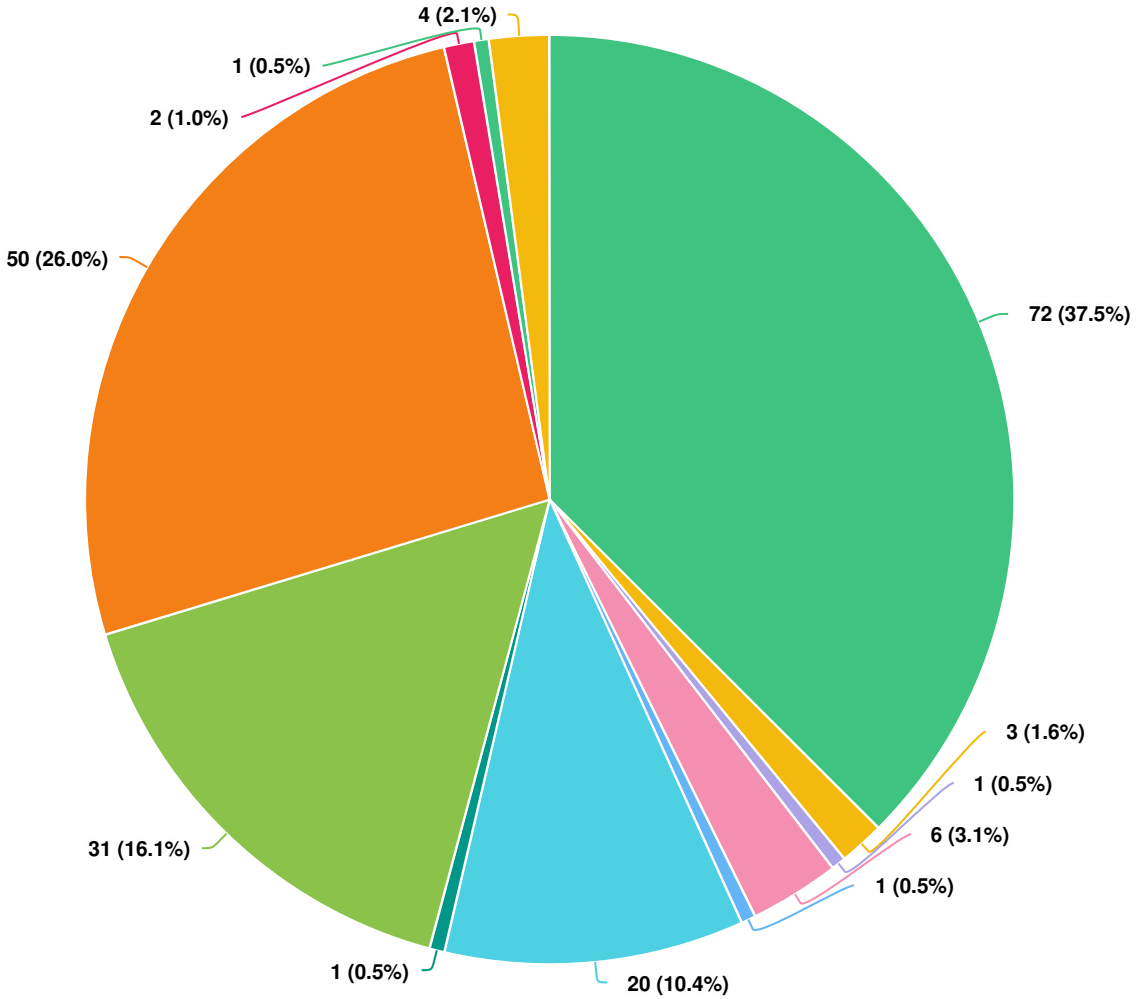
Question options

Car · Bicycle · Walking · Durham Region Transit · Durham Region Transit Specialized Services (Paratransit) · GO Transit (Bus or Train) · None (work from home or did not go to work) · Ride from a friend, family friend, or neighbour · Ride hailing service (such as Uber, Lyft) · Other (please specify)

Optional question (186 response(s), 66 skipped)

Question type: Dropdown Question

How do you travel to work on a Thursday?



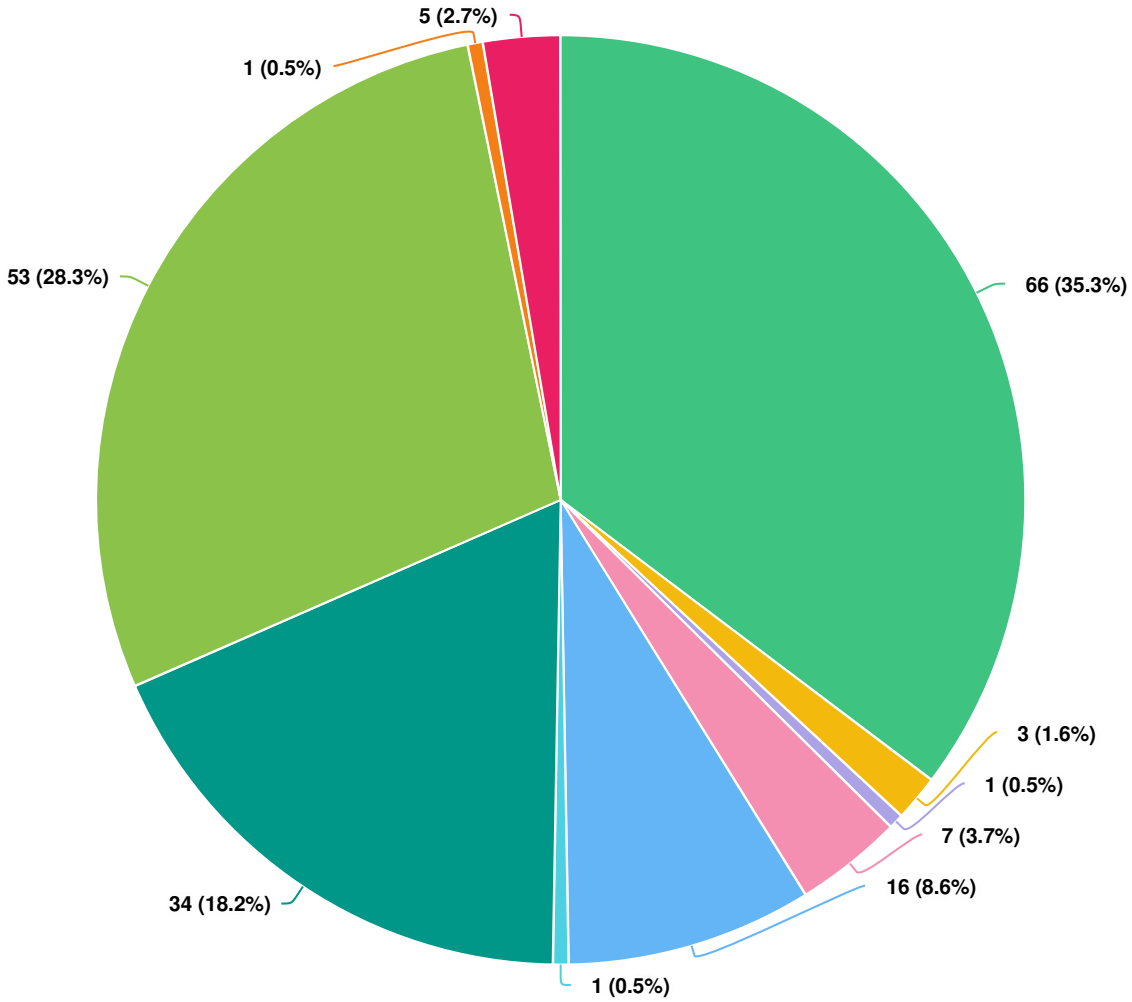
Question options

Car · Bicycle · E-Bicycle · Walking · Skateboard · Durham Region Transit · Durham Region Transit Specialized Services (Paratransit) · GO Transit (Bus or Train) · None (work from home or did not go to work) · Ride from a friend, family friend, or neighbour · Ride hailing service (such as Uber, Lyft) · Other (please specify)

Optional question (191 response(s), 61 skipped)

Question type: Dropdown Question

How do you travel to work on a Friday?



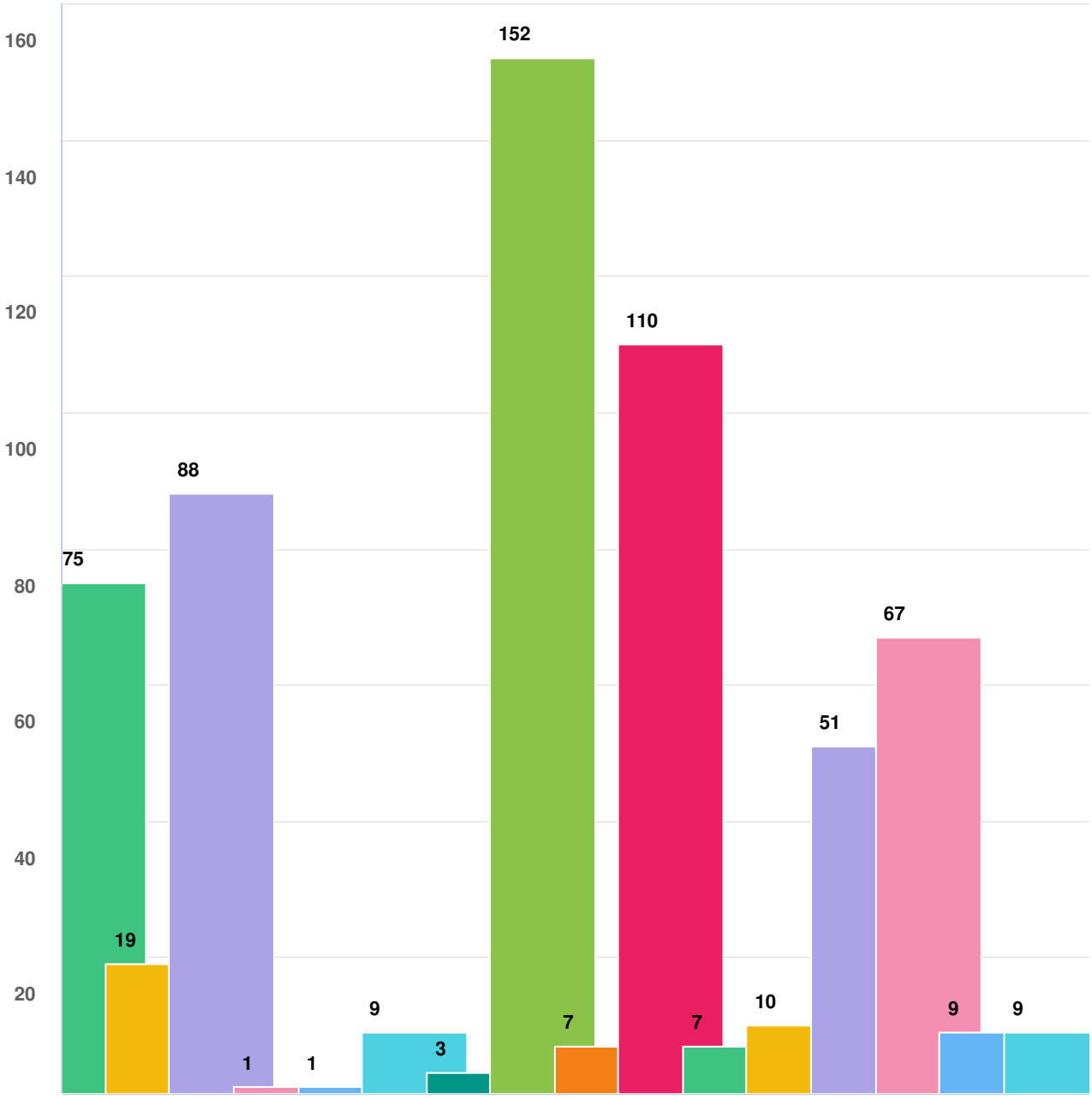
Question options

Car · Bicycle · E-Bicycle · Walking · Durham Region Transit · Durham Region Transit Specialized Services (Paratransit) · GO Transit (Bus or Train) · None (work from home or did not go to work) · Ride from a friend, family friend, or neighbour · Other (please specify)

Optional question (186 response(s), 66 skipped)

Question type: Dropdown Question

If driving a personal car/motorcycle or relying on carpooling services was NOT an option, what transportation option would you use to complete personal trips, commutes, and recreational/leisure trips? Select all that apply



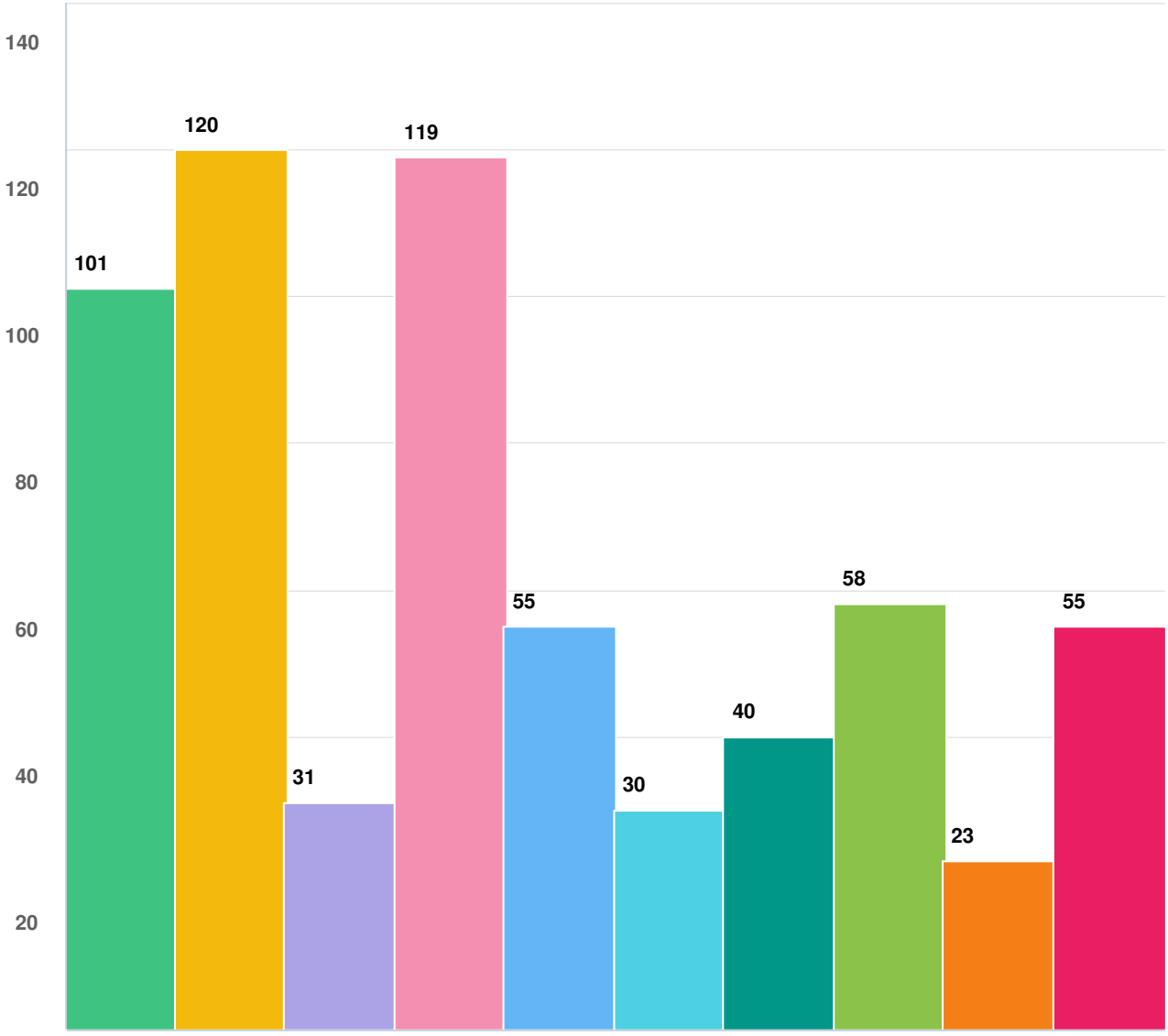
Question options

- Bicycle
- E-Bicycle
- Walking
- Wheelchair/Mobility Device
- Scooter
- E-Scooter
- Rollerblades
- Durham Region Transit
- Durham Region Transit Specialized Services (Paratransit)
- GO Transit (Bus or Train)
- VIA Rail
- None
- Ride from a friend, family friend, or neighbour
- Ride hailing service (such as Uber, Lyft)
- Ride from an organized service (may be paid or unpaid such as Community Care Durham, Seniors on Wheels)
- Other (please specify)

Optional question (247 response(s), 5 skipped)

Question type: Checkbox Question

What can the Town of Ajax do to make cycling a more viable travel method for you? Select top 5 priorities



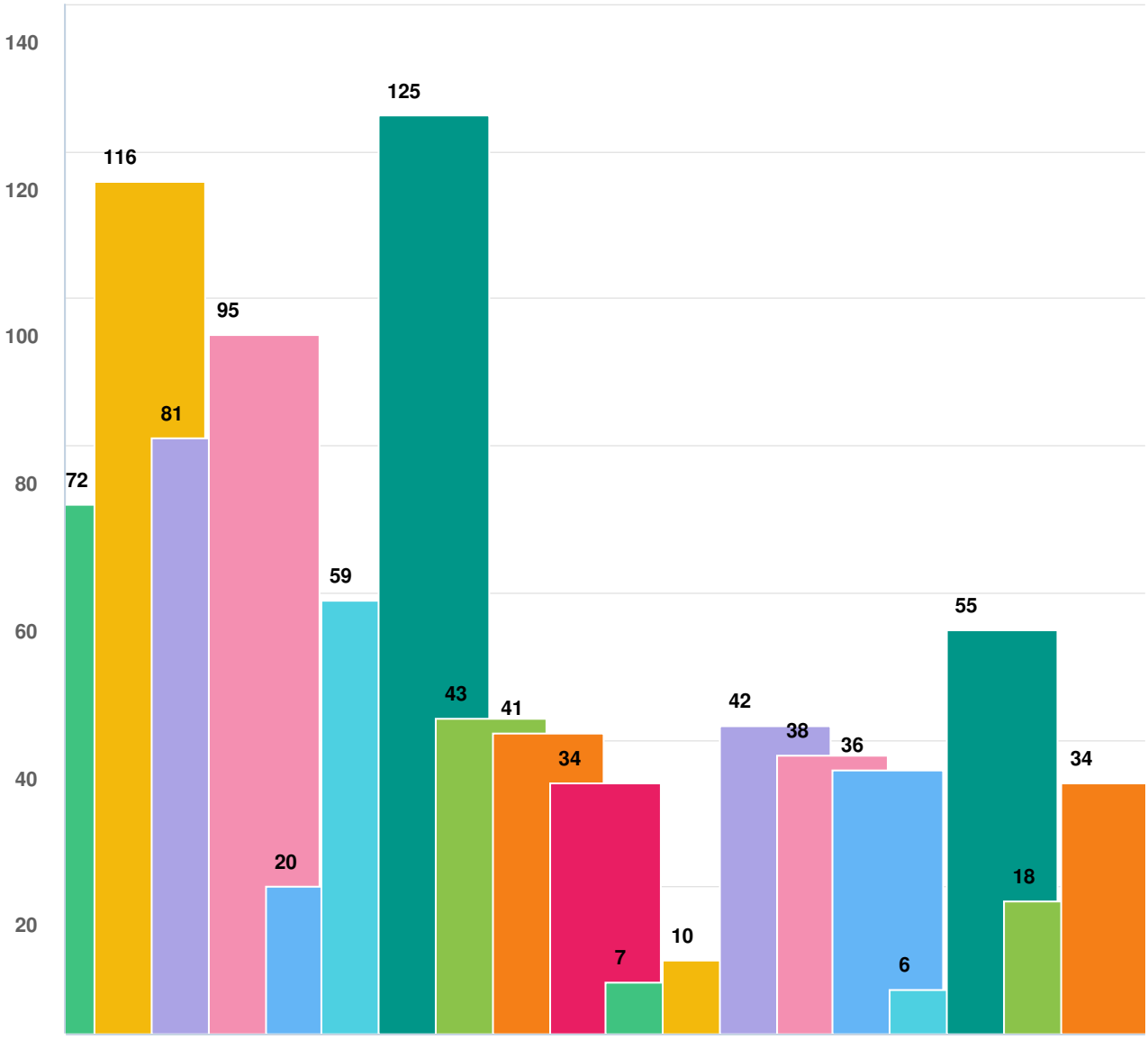
Question options

- Improve secure bicycle storage across Ajax
- Build more bicycle lanes to connect employment, commercial, and recreation areas to residential neighborhoods
- Enhance end of trip facilities (i.e. showers, day-lockers, etc.)
- Improve cycling safety at intersections
- Support bicycle skills and educations programs
- Support cycling donation/recycling programs
- Support community cycling organizations and events (i.e. community bike shops, bike to work week. etc.)
- Make cycling wayfinding signage and maps more accessible
- Access to adaptive bicycles for people with disabilities (i.e. handcycles for wheelchair users, larger trikes for people with down syndrome, etc.)
- Other (please specify)

Optional question (220 response(s), 32 skipped)

Question type: Checkbox Question

What can the Town of Ajax and Durham Region Transit do to make using public transit a more viable travel method for you? Select top 5 priorities



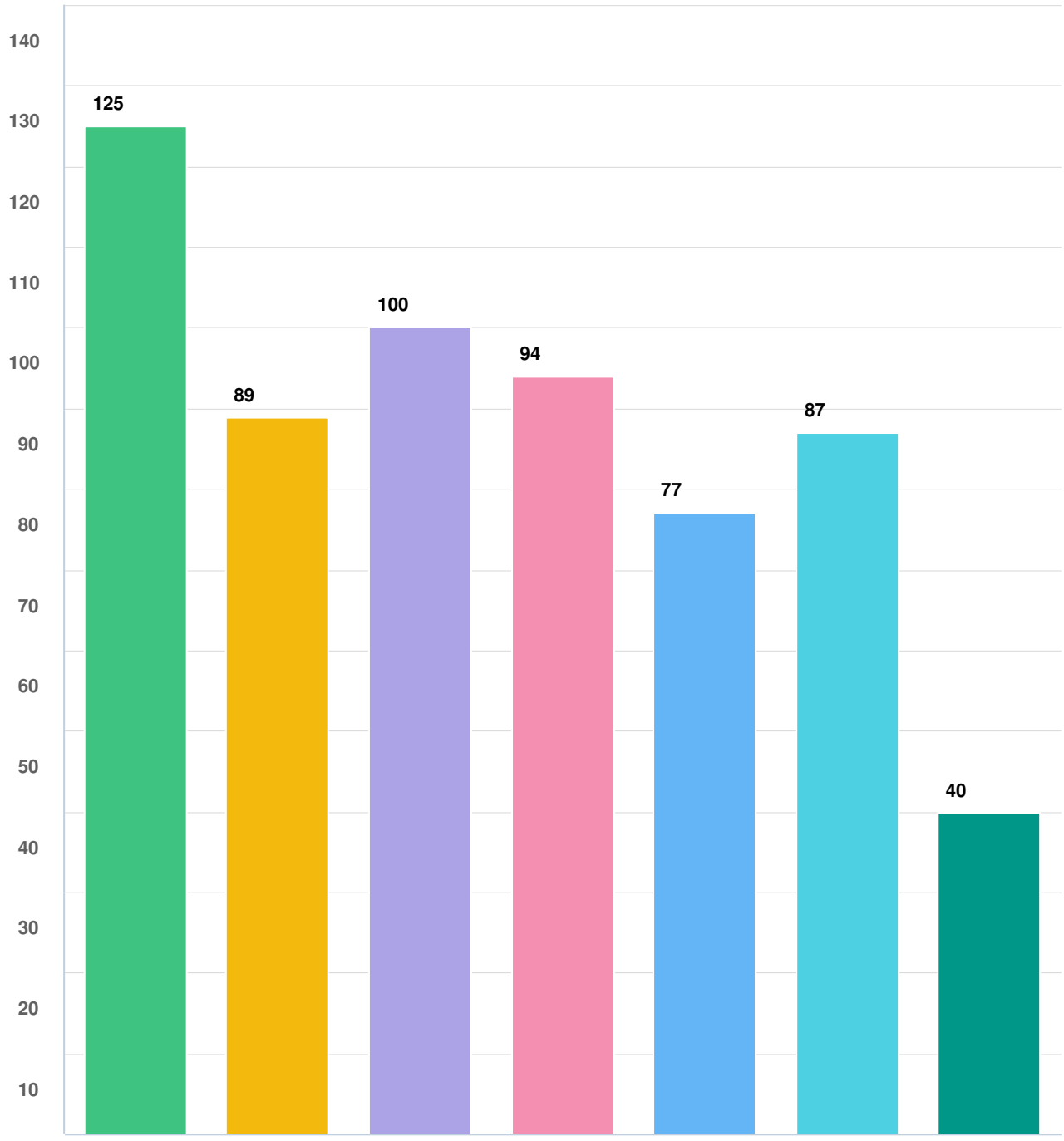
Question options

- Add more bus stops along existing routes
- Improve route directness
- Extend coverage area
- Improve connects to GO Transit service
- Improve connections to York Region Transit service
- Improve connections to Toronto Transit Commission service
- Increase frequency of buses
- Improve the on-time performance of buses
- Enhance "Next Ride" bus tracking at stops
- Add more benches and shelters at bus stops
- Improve bus stop and stop amenity state of repair
- Improve accessibility of buses and stops
- Extend hours of operation
- Improve weekend service
- Adopt an "on-demand" transit service model
- Add more payment options to cover fares
- Lower fares
- Improve maps and wayfinding material
- Other (please specify)

Optional question (234 response(s), 18 skipped)

Question type: Checkbox Question

What can the Town of Ajax do to make walking or rolling a more viable travel method for you? Rolling includes cycling, skateboarding, inline skating, kick-scooter or other form of human-powered transportation. Select top 5 priorities.



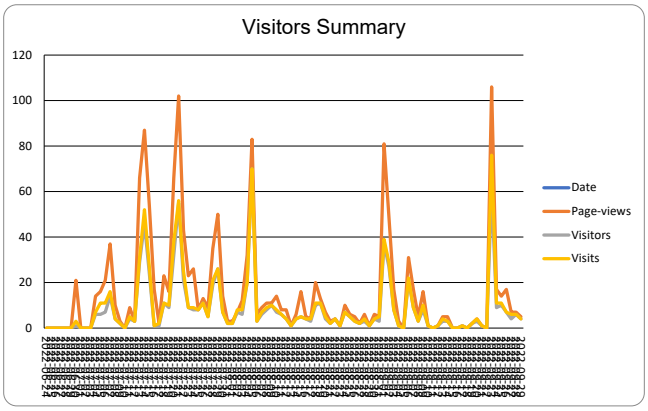
Question options

- Improve sidewalk connections between employment, commercial, and recreation areas to residential neighborhoods
- Improve intersections
- Pave sidewalks
- Add more sidewalks
- Pave trails
- Add more pedestrian crossings (Crosswalks, pedestrian controlled intersections, etc.)
- Other (please specify)

Optional question (219 response(s), 33 skipped)

Question type: Checkbox Question

Project Report:	Transportation Demand Management Plan	01-May-2022	to	29-Sep-2022



Project Highlights

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New Registrations	1
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Photo Views	0
Document Downloads	0

Admin Notes

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Contributed on Forums	0	0	0
Participated in Surveys	2	0	231
Contributed to Newsfeeds	0	0	0
Participated in Quick Polls	0	0	0
Posted on Guestbooks	0	0	0
Contributed to Stories	0	0	0
Asked Questions	0	0	0
Placed Pins on Places	0	0	0
Contributed to Ideas	0	0	0

INFORMED PARTICIPANTS	318
Informed Actions Performed	Participants
Viewed a video	0
Viewed a photo	0
Downloaded a document	0
Visited the Key Dates page	11
Visited an FAQ list Page	0
Visited Instagram Page	0
Visited Multiple Project Pages	80
Contributed to a tool (engaged	233

AWARE PARTICIPANTS	676
Aware Actions Performed	Participants
Visited at least one Page	676

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Qandas	0	Quick Polls	0	Stories	0	Survey Tools	1		

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DOCUMENTS	0	PHOTOS	0	VIDEOS	0	FAQS	0	KEY DATES	1

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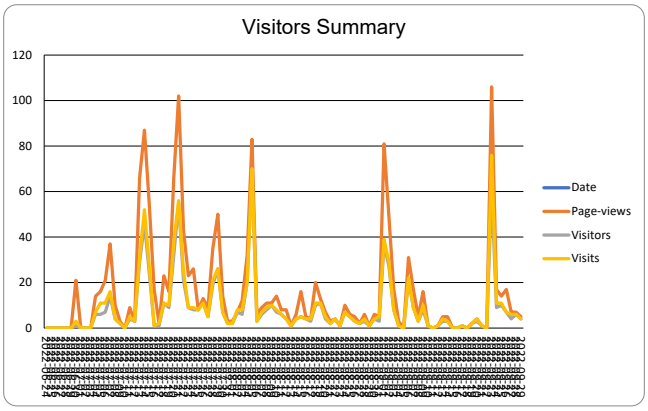

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SurveyTools	Ajax TDM Survey	Draft	338	2	0	231

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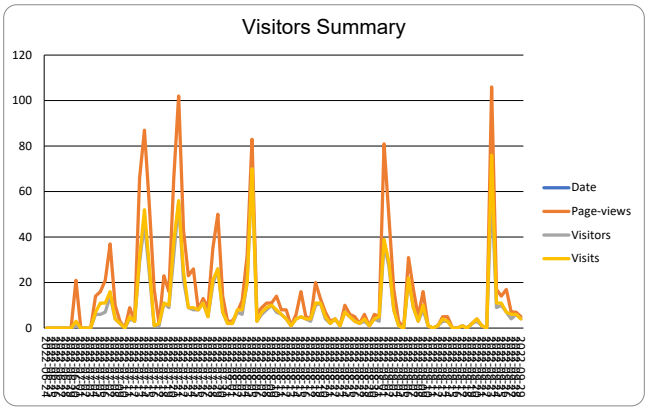

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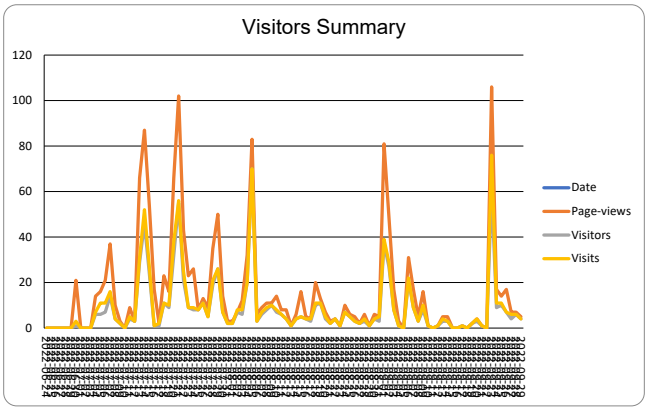

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APPENDIX B

Bicycle Parking Design Guidelines

Bicycle Parking Design Guidelines

February 2023

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1.0 INTRODUCTION

A key component to a successful cycling community is the availability of bicycle parking at key locations. Cyclists need to be sure that when they arrive at their destination, there will be a safe and secure location for them to leave their bicycle while they shop, eat, study, or work. Making high quality bike parking a normal part of our built environment supports the notion that cycling is a normal transportation option, and that the choice to ride is supported and valued. It also helps make everyday cycling visible, providing a nudge that "I could bike here". Providing high quality bike parking is an important component of a cycling-supportive environment where everyday cycling is normal, valued, supported and visible. The intention of these guidelines is to provide planners, developers, and property owners with the appropriate information to support the design, installation, and management of high quality bicycle parking facilities.

Cycling is good for health, for the community, for the environment, and even for business. Cycling can also be combined with other modes of transportation including walking and public transit to get from origin to destination conveniently and efficiently. The Town of Ajax continues to grow its cycling network, and cycling as a means of transportation is becoming more popular for both recreation and commuting. These guidelines will assist in the design and development of high-quality bicycle parking facilities that meets the needs of Ajax residents and visitors to support everyday cycling.

2.0 DESIGN GUIDELINES FOR BICYCLE PARKING

These guidelines primarily deal with the placement and type of bike parking that is required, and consider different types of bikes for users with different types of needs including those with disabilities. **Figure 1** shows some examples of bicycle types and attachments that bicycle parking should accommodate.

Bicycle Parking Definitions

Figure 1 - Common Bike Types and Attachments[1]

Bicycle parking is separated into two categories:

Long-term Bicycle Parking:

Long-term bicycle parking facilities provide a secure parking location where cyclists can leave their bicycle overnight, for a full workday or for several days at a time. Long-term parking is typically used for durations longer than two hours and in a pattern that is more predictable. For example, employees, residents, and transit users during peak hour periods. Typically, long-term bicycle parking facilities are located near washrooms and change facilities, if possible. Long-term bicycle facilities should also provide protection from the elements as well as having a high degree of security such as cameras.

[1] Image courtesy of the City of Boston Bike Parking Guidelines – January 2021

The definition of long-term bicycle parking according to the Town's Zoning By-law is: *a secured and weather-protected bicycle parking space reserved for long-term use by a dedicated user group, such as residents or employees. Long-term bicycle parking spaces shall not be provided within a dwelling unit.*

Examples of appropriate long-term bicycle parking include:

- Bike lockers;
- Enclosed bicycle racks or hooks; and
- Racks in a secured area with controlled access.

Figure 2 - Bike Lockers at a Toronto Transit Station[2]

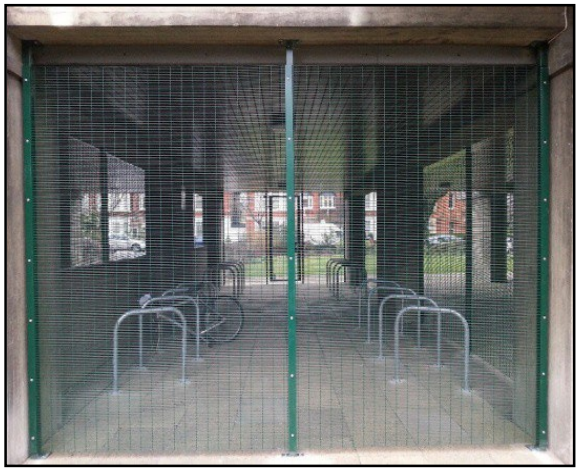

Figure 3 - Bicycle Cage[3]

Short-Term Bicycle Parking

Short-term bicycle parking facilities provide an easily accessible and visible location for a cyclist to secure their bicycle when shopping, eating, attending an appointment, visiting residents, businesses, institutions, or any other activity that requires short term parking, typically under two hours. While it is preferred that these facilities provide some protection from the elements through placement under building canopies, this is not a requirement.

The definition of short-term bicycle parking according to the Town's Zoning By-law is: *a bicycle parking space available for short-term use by any user group. External short-term bicycle parking may or may not be partially protected from the weather. Internal short-term bicycle parking must be provided in a publicly-accessible section of a building with clear and convenient access from the exterior of the building.*

Figure 4 - Covered bicycle parking at Town Hall

Figure 5 - Post and Ring Bicycle Parking

[2] http://www.bikingtoronto.com/blogger/labels/infrastructure.html
[3] http://www.cyclehoop.com/product/cycle-shelters-and-canopies/steel-cycle-shelter/

Criteria for High Quality Bicycle Parking

While the design of bicycle parking is up for some interpretation and creative freedom, there are a number of main criteria that must be satisfied to ensure the bicycle parking meets the needs of cyclists. Good bicycle parking needs to be accessible, convenient, safe, and secure.

In order to be **accessible,** bicycle parking should be:

- Within 35 m of a building entrance, typically;
- Level with the ground or accessible by ramps or elevator;
- Unobstructed by stairs or steep slopes; and
- Identified with way-finding signage.

Figure 6 – Bike Parking Next to Accessible Ramp

Safe and secure bicycle parking should be:

- Made from high quality, durable materials;
- Firmly secured to the ground, floor or wall;
- Visible to staff and patrons inside the building/business;
- Located in a well-lit area;
- Short-term: located in a busy, public area; and
- Long-term: located in an enclosed space with controlled access.

To ensure the bicycle parking is **convenient**, it should be:

- Easy to locate and access;
- Easy to use; and
- Situated close to bicycle friendly routes.

3.0 SHORT-TERM BICYCLE PARKING

Short-term parking is designed for visitors who need to park their bicycles for a few minutes or a few hours. It also prevents people from using trees to lock their bikes. There are a number of requirements that need to be met to ensure safe, secure and accessible short-term parking.

Design

There are a variety of bicycle parking racks available, but several features make some designs more desirable than others.

Materials: Bicycle racks made from galvanized steel or other industrial grade materials are the most durable and less susceptible to crime or vandalism. Avoid the use of materials such as wood, soft or untreated metals to avoid cracking or rust. For rack designs that have welded joints, avoid stainless steel as it does not weld strongly and can be easily broken.

Finishing: Ensure that the bicycle rack has a smooth outer finish that will not scratch or damage the bicycle frame.

Features: The following features are necessary to provide quality bicycle parking facilities:

Two points of contact between the bicycle and the rack: Maximum security is achieved when both the frame and one wheel can be locked to the rack. This also provides good support for the bike while it is parked.

Upright support for the bicycle: Old style “wheel-bender” racks are built to hold only the front or rear tire and do not provide support for the bicycle to remain upright when stored. Bicycles attached to these racks are also more easily stolen because only one wheel is locked and the frame is not.

Figure 7 - “Wheel-Bender” racks provides no upright support for bikes[4]

Sufficient Space for Parking: Rack design should provide adequate space between bikes and allow for parking and locking, while at the same time be able to hold several bikes in one rack.

Examples of good short-term bicycle parking types:

[4] http://www.peml.com/index.php?id=bike-racks-ground-loop

Figure 8 – Examples of good short-term bicycle parking include ring and post and inverted U type designs

Creating Attractive Bicycle Parking

The installation of short-term bicycle parking provides an excellent opportunity to beautify the community and create bicycle parking that doubles as public art pieces. Bicycle parking racks can be designed to coordinate with and compliment site design, existing street furniture, or other amenities on site. Bicycle racks can be custom built to suit specific needs and can add an eye-catching and creative feature to the exterior of a building. Creativity is encouraged in all developments as long as the function of the bicycle parking remains.

Figure 9 - Examples of creative bike parking in Toronto

Installation

Selecting a location for bicycle parking can impact the effectiveness of the parking facility and the extent to which it is used by cyclists.

Sheltered Parking:

Sheltered racks increase the quality of bicycle parking by protecting the bicycles from the elements. Installing racks under existing building canopies, awnings, or providing a free-standing shelter are ways to ensure that bicycles remain dry through rain or snow.

Fixing the Racks to the Ground

It is important that all bicycle racks are firmly secured to the ground or floor by being bolted to a hard surface or fixing them in concrete. For maximum security, concrete is the preferred surface, but other surfaces may also be appropriate.

Spacing

Appropriate spacing of bicycle parking will ensure that cyclists have sufficient room to lock their bicycles without disturbing other bicycles and the surroundings. A typical bicycle footprint is approximately 1.8 m x 0.6 m, whereas cargo bikes and bikes with trailers could extend to 3 m or longer.

A bicycle rack should have a minimum spacing of **0.6 m** between each bicycle parked in a horizontal position.

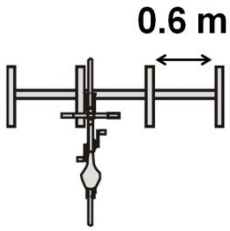

Figure 10

For racks that hold multiple bicycles (>2):

1) Distance between the rack and an obstacle with single sided bike access should be a minimum of **0.6 m.**

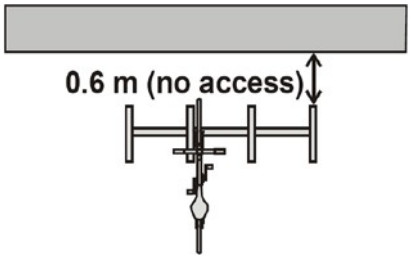

Figure 11

2) The distance between a rack and an obstacle
    a. Parallel to the parked bicycles should be a minimum of **0.45 m**
    b. Perpendicular to the rack with double-sided access should be a minimum of **2.5 m**
3) The distance between bike racks installed parallel to one another should be a minimum of **4.2 m**. This would allow for a minimum walking space on **1.8 m** if one bicycle was parked in each one.
4) The distance between bicycle racks placed in a row should be a minimum of **0.9 m.**

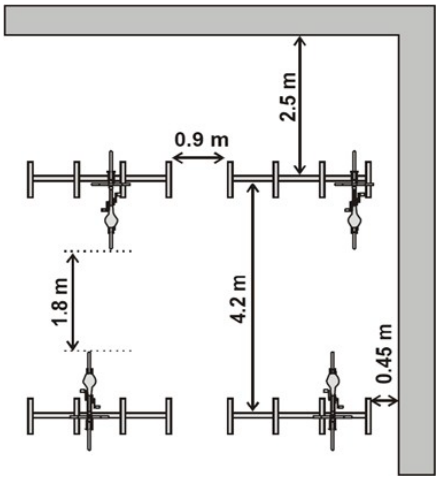

Figure 12

For racks that hold only two bicycles (mostly post-and-ring bike stands):

1) The distance between the rack and a wall, curb or other obstacle should be:
    a. A minimum of **1.5 m** for racks perpendicular to an obstacle
    b. A minimum of **0.7 m** for racks parallel to an obstacle
2) The distance between individual racks placed perpendicular to the wall but parallel to one another is a minimum of **1.0 m**.

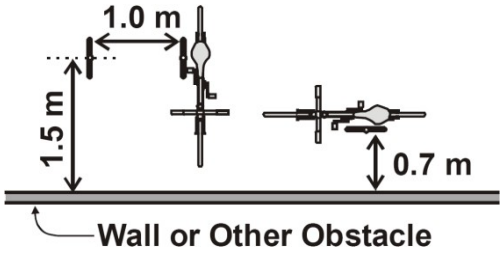

Figure 13

3) The distance between individual racks parallel to an obstacle should be a minimum of **2.5 m** (3.5 m is preferred when high bicycle turnover is anticipated)

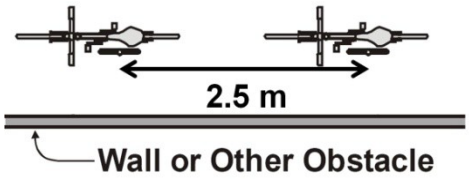

Figure 14[5]

Pedestrian movements should always be considered in the decision-making process for bicycle parking locations. Bicycle racks should never impede the flow of pedestrians or limit access to their destination and should **maintain a minimum 1.8 m pedestrian clearance**.

[5] Images courtesy of the City of Toronto Guidelines for the Design and Management of Bicycle Parking Facilities – May 2008

Accommodating Cargo Bikes and Other Mobility Devices

There should be wider spacing between bicycle racks to accommodate cargo bicycles and adapted cycles. For a space to be more inclusive, it is important to accommodate a range of mobility devices including scooters, handcycles, tricycles, and tandems. **A minimum 1.5 m turning diameter** should be provided for mobility devices to accommodate those that may need to reverse/turn into the space.

Scooter Space Dimensions

The space size should have a minimum length of 1.8 m and a minimum width of 0.9 m. The space offers additional maneuvering spaces to accommodate users that have diminished physical dexterity. It provides sufficient maneuvering clearance for seated transfer.

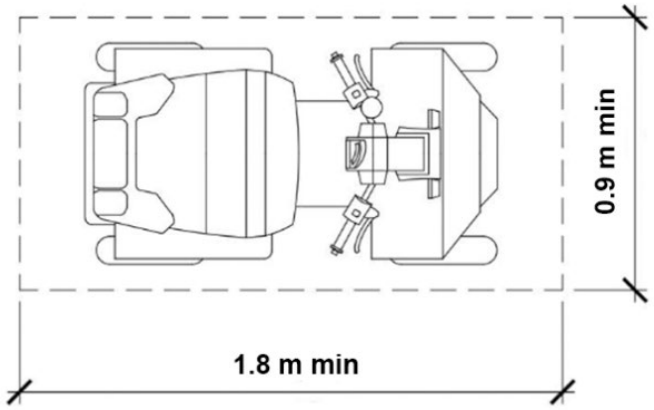

Figure 15 - Scooter Space Dimensions

Handcycle Space Dimensions

The space dimensions for handcycles should have a minimum length of 2.2 m and a minimum width of 1.0 m.

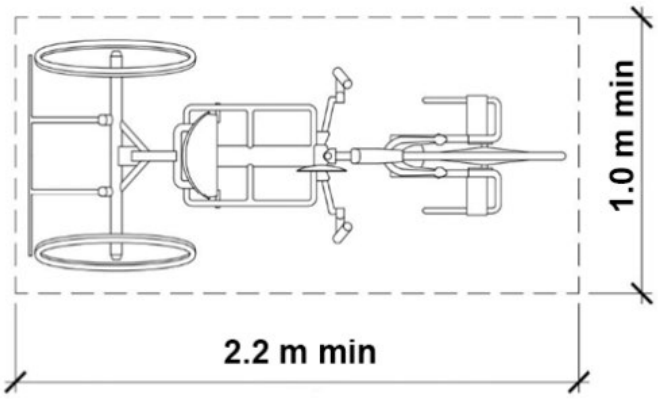

Figure 16 - Handcycle Space Dimensions

Tricycle Space Dimensions

The space dimensions for each tricycle should be a minimum in length of 1.8 m and a minimum width of 1.0 m.

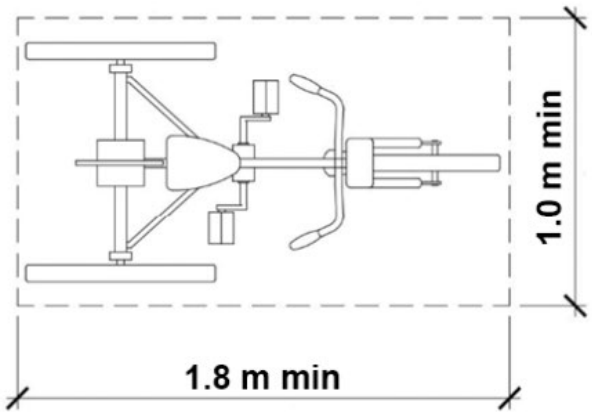

Figure 17 - Tricycle Space Dimensions

Tandem Bicycle Space Dimensions

The space dimensions of each tandem bicycle should be a minimum length of 2.4 m and a minimum width of 0.75 m.

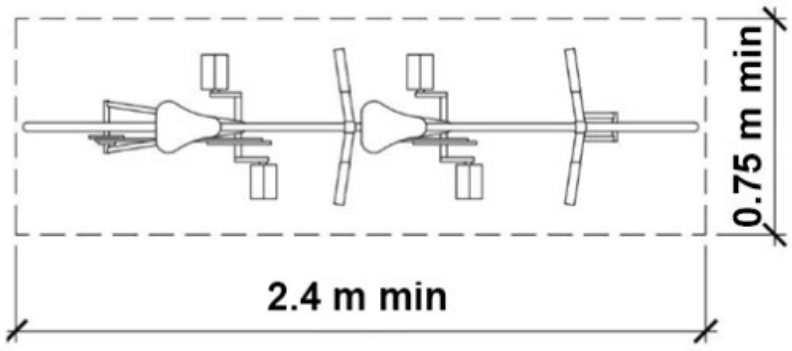

Figure 18 - Tandem Bicycle Space Dimensions[6]

[6] Images courtesy of International Code Council (2017). IDG NYC 2nd Edition.

4.0 LONG-TERM BICYCLE PARKING

Long-term parking is designed for a period of several hours to several days. The following section provides an overview of the different types of long-term bicycle parking and key considerations.

Accommodating Disabled Cyclists

Cycle storage units (e.g. lockers and hangars) are not often designed to take into consideration the accommodation needs of disabled cyclists and dimensions of non-standard cycles. The cost of a non-standard cycle is more expensive in comparison to a typical bicycle, hence it becomes a target for theft. It is vital to have access to step-free, safe, secure storage facilities for disabled cyclists. Existing cycling storage units could be adapted and retrofitted to accommodate larger cycles. Buddy schemes is a community initiative that pairs disabled and non-disabled cyclists to share the same cycle storage that could be implemented within the Town of Ajax.

The long-term bicycle parking facility should also have a greater turning radius to be able to serve adapted cycles. Bicycle parking intended for the use of standard two-wheeled bicycles are generally placed very close to each other, to the point where a three-wheeled cycle cannot fit in between. Dedicated two-way bicycle aisles with a minimum width of 1.75 m shall be provided to access long-term parking areas, and there shall be no obstacles, such as stairs, obstructing users from accessing parking spaces. Parking stands that require use of strength or dexterity to access or operate while standing (e.g. hydraulically-assisted double staking racks) may create barriers for disabled cyclists.

Spacing requirements are the same as short-term parking, but consider including some stalls to accommodate larger bikes and mobility devices, as shown in **Figure *19***.

Figure 19 - Wider parking spaces intended for use by non-standard cycles (bottom right of photo)[8]

[8] https://wheelsforwellbeing.org.uk/wp-content/uploads/2017/11/v2-Nov-2017.pdf

Half-height Stands

The majority of non-standard cycles are either a self-standing (tricycles) or have a stand (cargo bikes). For these types of cycles, it is recommended to have a half-height, longer length stand that is low enough to prevent a standard two-wheeled bicycle leaning against it, as well as not being lower than half-height (since some people might have issues bending down). Half-height stands should be built in groups that are well-marked, well-lit, preferably sheltered to reduce misuses and tripping hazards.

Figure 20 - Half-height stands[9]

Enclosed Bicycle Racks or Hooks

Enclosed bicycle parking facilities should have at least one fully transparent wall to enable visibility from both inside and outside of the facilities. This feature is vital for the security of both bicycles and users. There are different types of bike racks to cater to specific users and bicycle types that have a variety of attachments and needs. For instance, cargo bicycles and adapted cycles such as handcycles and tricycles are used as mobility aids that generally have a wider wheelbase and are not designed to be lifted off the ground.

Bicycle Lockers

Bicycle lockers are individual storage units that are weather-protected, enclosed and operated by a controlled access system. Some systems are set up for multiple users (coin operated or the use of an individual lock) and others are accessed using keys, swipe cards or access codes. Bike lockers located below the ground level should have vertical access via an elevator or ramp (max slope of 5%) where stairs would otherwise be required to reach parking, to be able to accommodate disabled cyclists. There are several bike locker designs available, but costs and quality vary. The highest priority for bicycle lockers should be security and durability because of the low turnover and extended length of time bicycles will be parked. Lockers should be fixed in places using anchor bolts and should be located close to building entrances for convenience and security.

Figure 21 – Bicycle locker

[9] https://wheelsforwellbeing.org.uk/wp-content/uploads/2017/11/v2-Nov-2017.pdf

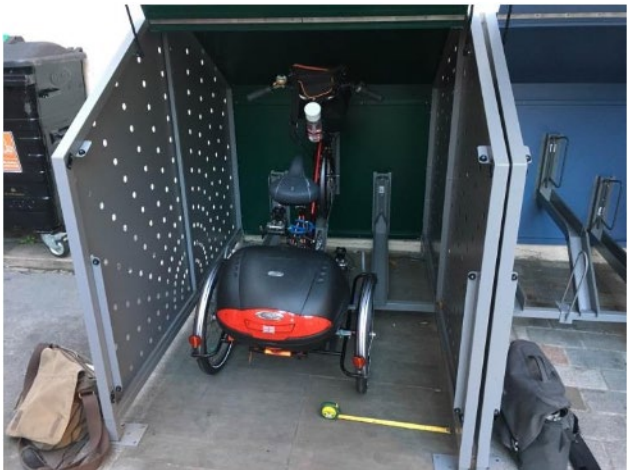

Figure 23 - A Tricycle in a Bicycle Locker[9]

Figure 23 – Bicycle Locker in London, UK9

Bicycle Cages

Bicycle cages are another form of secure bike parking with restricted right of entry. Cages can be accessed using keys, access codes, cards or locks. Bicycle racks are placed inside a fenced off and covered area. These cages can be less expensive than traditional bicycle lockers, have increased capacity while providing an enhanced level of security.

Figure 24 – Bicycle cage

Smaller cages provide better bicycle security as fewer people are entering and exiting the cage throughout the day. For large developments with high demand for bicycle parking, it is preferred to have several smaller cages with separate accesses. The materials used for the walls need to be made from strong metal or mesh to prevent attempts to cut through. These cages can be installed inside or outside a building, or within a parking garage depending on space accommodations. It is important to consider providing sufficient space for cyclists to maneuver their bicycle in and out of the racks and the cage when designing and constructing.

Indoor Bicycle Parking

Indoor bicycle parking should be provided on the first level of underground parking to ensure ease of accessibility and limit the interactions between cyclists and automobiles within the garage.

If underground parking facilities have controlled access, they are an excellent option for protected, long-term bicycle parking. If the entrance is uncontrolled, bicycles need to be secured in some other way, whether by bicycle cage or separated bicycle room within the building.

Showers / Change Facilities

Barrier-free/universal changing facilities and showers that are equipped with shower stalls at non-residential developments can be a strong incentive to encourage bicycle use for commuters. Students or employees that may face long commutes on their bike and have to comply with a professional dress code would be more likely to cycle if they knew these facilities were available.

The number of shower stalls should reflect the amount of long-term bike parking provided.

5.0 PARKING REQUIREMENTS

Ensuring that the appropriate number of parking spaces are provided to meet the needs of a development is a key component of a Transportation Demand Management (TDM) Plan. Demand for bicycle parking varies by development type and use. The following parking requirements outline the minimum number of bicycle parking spaces that are required for development within the Town of Ajax, though the Zoning By-Law is the primary source for bicycle parking rates.

Table 1 - Bicycle Parking Guidelines by Development Type

Land Use	Minimum Number of Bicycle Parking Spaces Required
Residential Uses	
All multi-residential uses that do not include a private garage for each unit	Intensification Areas: 1.0 bicycle parking spaces per dwelling unit, except in a building having fewer than 10 dwelling units, where the minimum number of bicycle parking spaces required shall be zero. Other Areas: 0.75 bicycle parking spaces per dwelling unit, except in a building having fewer than 10 dwelling units, where the minimum number of bicycle parking spaces required shall be zero. Of the total number of bicycle parking spaces required, a minimum number of 10% of the required bicycle parking spaces required shall be designated as short-term bicycle parking spaces and a minimum of 70% as long-term bicycle parking spaces. Provide at least 5% of the long-term bicycle parking spaces as accessible, and no less than one. **Ajax Green Standards for high-density residential buildings:** long-term bicycle parking spaces are required for 30% or more of the building's units.
Commercial Uses	
All Commercial uses excluding Office uses	1.0 space per 1,000 m^2 of GFA up to 6,000 m^2, plus an additional 1.0 space per every 3,000 m^2 of GFA in excess of 6,000 m^2. Consider higher short-term bicycle parking for eating establishments, retails stores and personal service shops. **Ajax Green Standards for non-residential and mixed-use buildings**: long-term bicycle parking spaces for 10% of employees and short-term bicycle parking for 7.5% of peak visitors.
Office Uses	
Business / Medical office	1.5 spaces per 1,000 m^2 of GFA up to 4,000 m^2, plus an additional 1.0 space per every 2,000 m^2 of GFA in excess of 4,000 m^2 **Ajax Green Standards for non-residential and mixed-use buildings**: long-term bicycle parking spaces for 10% of employees and short-term bicycle parking for 7.5% of peak visitors.
Industrial Uses	
All industrial uses	1.0 space per 2,000 m^2 of GFA up to 12,000 m^2, plus an additional 1.0 space per every 6,000 m^2 of GFA in excess of 12,000 m^2 **Ajax Green Standards for non-residential and mixed-use buildings**: long-term bicycle parking spaces for 10% of employees and short-term bicycle parking for 7.5% of peak visitors.
Institutional[1]	
School, Elementary	3.0 per class
School, Secondary	6.0 per class
School, Post-Secondary	1.0 per 10 students / employees
Community Centre	2 spaces plus 1.0 for every 250m^2 of GFA

1. Under Section 3.4.1 of the Official Plan, Daycare can be provided in all schools as well as residential, mixed use, and employment designations. For daycare facilities, it recommended to consider providing at least one space suitable for cargo bicycles, trailers, or trailer bikes.

Commercial, Industrial, or Institutional Uses

Commercial, industrial, and institutional developments require a higher number of short-term parking spaces for both visitors and employees compared to residential land uses. Covered, outdoor parking close to building entrances offer customers, clients, and employees an important service and raises the profile of active transportation. Bicycle parking can also reduce the number of surface parking spaces that need to be provided.

Uses with High Bicycle Parking Demand

Some land uses have the potential to generate a higher than average demand for bicycle parking spaces. Examples of those uses include:

- **Schools** – all levels of educational facilities (elementary, high school, colleges, and universities) generate a large number of active transportation trips. Also, the addition of sufficient bicycle parking can encourage students to choose active modes.

- **Transit Hubs, Hospitals and Large Factories** - these are destinations for a large number of visitors, customers and employees who may require access to parking facilities at all hours due to shift work or late-night travel schedules.

- **Places of Assembly** – community centres, sports stadiums and facilities, and concert halls receive a large number of visitors during special events. The demand for parking is often high at designated times but can be lower at other periods.

- **Places of Worship** – similar to places of assembly, they receive a large number of visitors at one time and increase the demand for bicycle parking. Section 3.4.2a of the Official Plan mentions that places of worship can be permitted under residential or mixed use.

6.0 SITE DESIGN STRATEGIES

There are additional factors that need to be considered to ensure that the site design of a development is bicycle friendly and also remains accessible to all.

Accessibility

While it is important for bicycle parking to be accessible to cyclists, consideration also needs to be given to the impact bicycle parking will have on accessibility for pedestrians with impairments. Pedestrian access must not be hindered by bicycle racks including those who may be visually impaired or use mobility aids such as walkers, scooters, or wheelchairs. Bicycle racks should also be located in areas that will help to minimize interactions between automobiles and bicycles on the site. More permanent bicycle parking, including bicycle lockers and cages can create blind spots and block sightlines. These impacts should be considered and mitigated as best as possible. The design of a parking structure should also be cane-detectable and adhere to the CNIB's "Clearing Our Path" Guidelines for Protruding Objects and other Obstacles[10].

Improvements to accessibility can be made by providing parking facilities for non-standard cycles on ground level or offering step-free access, for example a shallow ramp or large accessible lift. Another method would be to install cycle parking bays for people with non-standard cycles to ride into and out of, which requires no need for reversing, turning, or lifting of a cycle.

Stairways

Stairways can be an impediment to cyclists as they are often required to carry their bicycles up or down. A ramp or small channel that fits a bicycle wheel can be installed on the edge of a stairway to eliminate this issue.

Signage

Clear signage can help cyclist easily locate parking on site and indicates to pedestrians and motorists that they should expect bicycle traffic on the site. These visual indications of parking should be strategically placed and provide directional arrows to be most effective.

A few ways that could improve signage are in the following:

- Place signs that clearly denote cycle parking for non-standard cycles, for example “reserved for cargo and non-standard cycles. Priority to disabled cyclists”;
- Blue and white paint is recommended to delineate the zone that non-standard cycle bay occupies, and could potentially include a logo that depicts a disabled cyclist and cargo cycle;
- Blue and white paint could also be used on stands/poles to help differentiate the areas; and
- Signage should be in a large-sized (at least 36 pt.) sans-serif font, not use block capitalization of all letters wherever possible, and use plain language and symbols for instructions.

Safety and Security

Bicycle parking facilities should maximize safety and security to both cyclists and their property. There are a number of strategies that can be used to accomplish this however some may be more suitable than others depending on the situation and site.

[10] https://www.clearingourpath.ca/3.2.0-protruding-objects_e.php#:~:text=To%20be%20cane%20detectable%20by,on%20the%20child's%20size%2Fage

Safety and security measures include:

- Installing security cameras in bicycle parking areas;
- Locating bicycle parking close to building entrances, while also placing non-standard cycle parking bays nearest to the entrance and providing shelter to protect from the elements;
- Installing bright lights (at least 100 w) and/or convex mirrors to minimize blind spots and dark corners that will also help people with poor vision to read signage;
- Locating bicycle parking within view of parking lot attendants, building security, or in a busy area close to other public amenities;
- Dedicated (cyclist-only) entrances with limited access to indoor parking facilities and outdoor bike cages (security card access or keys);
- Installing a “panic button” in bicycle parking areas that would provide a direct line to security in the event of an emergency. The installation should include help for deaf people using sign language, text, and a face for people to lip-read; and
- Monitor reserved cycle parking for disabled cyclists using a Blue Badge scheme to differentiate bicycles and cyclists that are entitled to park at the facility.

7.0 POLICY CONTEXT

The Town of Ajax policy documents support the development of a connected and convenient cycling network. This section provides a brief overview of the Town's policy documents as they pertain to cycling.

Town of Ajax Official Plan

The Town's 2022 Official Plan (OP) supports the evolution towards a more active and sustainable community through a number of policies. Bicycle parking facilities are one of the many sustainable urban design elements supported by the OP. The OP policies support development proposals that contribute to healthier, and safer living environments. The OP also has policies that supports Green Building and Environmental Design through development applications, the Town will support green industries green building technologies and construction practices that includes bicycle parking as well as secure bicycle storage facilities. For transportation policies, the OP encourages the incorporation of bicycle parking and/or storage within all new developments and offering change and shower facilities where appropriate. Active Transportation policies of the OP highlight the need to provide bicycle parking and/or storage facilities at primary destinations as well as ensuring there is adequate and secure bicycle parking available on park property where appropriate.

Town of Ajax Integrated Transportation Master Plan Update

The Town of Ajax Transportation Master Plan was updated in 2019. This document envisions the enhancement of the sustainable and multi-modal transportation system for the Town of Ajax. One of the ways this goal can be achieved is through the planning, designing, constructing, and maintaining the transportation network to support pedestrians, cyclists, transit, and vehicular traffic. The goal is to provide accessibility to all users and all modes of transportation. This document speaks to the effort of reducing car-dependency and vehicle road congestion by providing a convenient network for residents which includes accessible and safe bicycle parking locations. The improvement of quality and increased quantity of bicycle parking in the Town will help the Town in meeting the needs of Ajax cyclists while supporting the TMP vision.

Ajax Green Standard - Green Development and Environmental Design Guidelines (GDEDG)[11]

The GDEDG is a tool designed to aid in evaluating development applications through the lens of sustainability and climate change mitigation and adaptation by providing a suite of mandatory and optional metrics. It highlights the need for pedestrian and bicycle friendliness to promote sustainable mobility. The Municipal Green Development Standards requires bicycle parking in conformance with the Town Zoning Bylaw, safe and direction active transportation routes, and standards for trip-end facilities like showers that encourage active transportation. Requirements include:

- **High-density residential buildings**: long-term bicycle parking spaces for 30% or more of the building's units
- **Non-residential and mixed-use buildings:** long-term bicycle parking spaces for 10% of employees and short-term bicycle parking for 7.5% of peak visitors
- **Non-residential and mixed-use buildings**: one trip-end facility for each gender for every 30 bicycle parking spaces provided

[11] https://www.ajax.ca/en/business-and-growth/ajax-green-standard.aspx#:~:text=The%20Ajax%20Green%20Standard%20%28AGS%29%20is%20a%20tiered,development%20and%20redevelopment%20in%20the%20Town%20of%20Ajax

8.0 CONCLUSION

The Town of Ajax is committed to building an inclusive community that is conducive to cycling with a sufficient amount of bicycle parking to help increase the number of trips, whether for recreation or commuting. Many cyclists are deterred from riding because of the concern that there will not be a safe place to store their bike when they arrive at their destination. The intention of these guidelines is to provide planners, developers, and property owners with the appropriate information to support the design, installation, and management of high quality bicycle parking facilities.

	TDM Initiative	Requirements Description / Examples	Land Use Category																								Implementation Cost	Effectiveness of Initiative	Point Value
			Residential			Commercial Office				Commercial Retail				Mixed-Use			Institutional				Educational		Industrial						
			Single Unit	Low-Medium Density	High Density	Single Tenant - Low Density	Single Tenant - High Density	Multiple Tenant - Low Density	Multiple Tenant - High Density	Single Tenant - Small / Med	Single Tenant - Large	Multiple Tenant - Small Plaza	Multiple Tenant - Large Plaza	Low Density Res - Comm. Frontage	High Density Res - Comm. Frontage	High Density Res - Large Comm.	Municipal Building	Hospital / Health Services	Place of Worship	College / University	Elementary School	High School	Light Industrial - Small	Lignt Industrial - Large	Heavy Industrial - Small	Heavy Industrial - Large			
Category 1: Walking & Cycling																													/30
1.1	Pedestrian-Friendly Site Design Strategies	Buildings are oriented toward the street; pedestrian linkages between site access points and building entrances are accessible, direct and continuous	■	■	■	■	■	■	■	□	■	□	■	■	■	■	■	■	■	■	■	■	■	■	■	■	$$$	●●●	TBD
1.2	Pedestrian Route Enhancements	Major pedestrian routes are enhanced with landscaping and lighting	■	■	■	□	■	□	■	□	■	■	■	■	■	■	■	■	■	■	■	■	□	■	□	■	$$$	●●●	TBD
1.3	Multi-Modal Network Connections	The development provides sidewalks and pathways that connect pedestrians and cyclists to adjacent transit, ped, cycling and rec trails	■	■	■	■	■	■	■	□	■	□	■	■	■	■	■	■	■	■	■	■	■	■	■	■	$$$	●●●	TBD
1.4	Short-Term Bicycle Parking	Quantity meets or exceeds the minimum set out on the applicable zoning by-law; If more than 10 spaces are provided, 50% are weather protected	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	□	□	□	$$	●●●	TBD
1.5	Long-Term Bicycle Parking	Quantity meets or exceeds the minimum set out on the applicable zoning by-law; access to long-term bicycle parking is restricted		■	■		■	□	■		■	□	■	□	■	■	■	■	■		■			□	■	□	■	$$	●●●	TBD
1.6	Showers and Change Rooms	Quantity meets or exceeds the minimum set out on the applicable zoning by-law					■	□	■		■	□	■	□	□	■	■	■	■		■	■			■		■	$$	●●●	TBD
1.7	Other End-of-Trip Facilities	Other end-of-trip facilities, such as e-bike charging stations, bicycle repair stations, etc., will be provided					■	□	■		■	□	■	□	□	■	■	■	■		■	■			■		■	$$	●●	TBD
Category 2: Transit																													/40
2.1	Transit-Supportive Site Design Strategies	Buildings are oriented toward streets; primary building entrances are located close to transit stops; bicycle parking is located close to transit stops		■	■	■	■	■	■	□	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	$$$	●●●	TBD
2.2	On-Site Transit Stop Enhancements	Shelters, lighting and benches are provided for any on-site transit stops	□	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	$$	●●	TBD
2.3	Off-Site Transit Stop Enhancements	Where the development abuts an off-site transit stop and insufficient space exists for a transit shelter in the public right-of-way, protect land for a shelter and/or install a shelter	□	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	$$	●●	TBD
2.4	Enhanced Pedestrian Connections to Transit	Primary pedestrian connections between building entrances and transit stops are enhanced with landscaping and lighting		■	■	■	■	■	■	□	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	$$$	●●●	TBD
2.5	Transit Information	Displays, screens or kiosks displaying transit information are integrated into common areas, near entrances or at transit stops	■	■	■	□	■	□	■	□	■	■	■	■	■	■	■	■	□	■	□	■	□	■	□	■	$	●	TBD
2.6	Transit Priority Measures in Close Proximity to Site*	Prioritize the movement of transit vehicles over other vehicles through signage/signal modifications or dedicated bus lanes / queue jump lanes	■				■	□	□		□	□	□								□	■					$$$	●●	TBD
2.7	Shuttle Services**	Connect residents / employees to major transit stations or stops; enable access to transit where walking distances to transit stop are significant and travel patterns not aligned with routes		■	■		■		■		■		■		■	■	□	■	□	■	□	■		■		■	$$$	●●●	TBD
Category 3: Parking																													/20
3.1	Reduced Parking Rate	The amount of vehicle parking provided does not exceed the quantity required in the applicable zoning by-law	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	$	●●●	TBD
3.2	Site Design Strategies for Parking Areas	Parking is located at the back of buildings or in an underground parking structure		■	■	□	■	■	■	□	■	■	■	□	■	■	■	□	■	□	■	■	□	■	□	■	$$$	●●●	TBD
3.3	Shared Parking	A shared parking strategy will be used to reduce the number of parking spaces required before the minimum quantity required in the applicable zoning by-law		□	■	□	■	□	■	□	■	■	■	□	■	■	■	□	■	□	□	■	■	□	■	■	$	●●●	TBD
3.4	Unbundled Parking	Parking has been unbundled from residential units and will be sold / leased separately		□	■									■	■	■											$	●●●	TBD
3.5	Preferential Carpool Parking	Carpool parking will be provided in visible, convenient locations close to building entrances; ideally, carpool parking should be located next to accessible parking				■	■	■	■	■	■	■	■	□	□	■	■	■	■	■	■	■	■	■	■	■	$	●●●	TBD
3.6	Carshare Parking	Carshare parking will be provided in visible, convenient locations		■	■		■	□	■		■	■	■		■	■	■	■	□	■	□	□					$	●●●	TBD

TDM Initiative		Requirements Description / Examples	Land Use Category																								Implementation Cost	Effectiveness of Initiative	Point Value
			Residential			Commercial Office				Commercial Retail				Mixed-Use			Institutional				Educational		Industrial						
			Single Unit	Low-Medium Density	High Density	Single Tenant - Low Density	Single Tenant - High Density	Multiple Tenant - Low Density	Multiple Tenant - High Density	Single Tenant - Small / Med	Single Tenant - Large	Multiple Tenant - Small Plaza	Multiple Tenant - Large Plaza	Low Density Res - Comm. Frontage	High Density Res - Comm. Frontage	High Density Res - Large Comm.	Municipal Building	Hospital / Health Services	Place of Worship	College / University	Elementary School	High School	Light Industrial - Small	Lignt Industrial - Large	Heavy Industrial - Small	Heavy Industrial - Large			
3.7	Reduced Minimum Parking Requirements Based on Proximity to Transit*	Reduction of parking requirements to account for anticipated transit usage		■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	$	●●●	TBD
3.8	Cash-in-Lieu of Parking to Fund Public Parking or Fund Sustainable Transportation*	Employees have a choice to keep a parking space, or accept payment in exchange for giving up a designated space and committing to a sustainable mode of travel		□	■		■		■	■	■	■	■		■	■	□	■	□	■	■	■	□	■	□	■	$$$	●●●	TBD
3.9	Paid Parking*	Charging for parking (an amount at least as much as the cost of transit) can encourage and influence people to use sustainable modes of transportation				□	■	□	■	□	■	■	■	■	■	■	□	□	□	■	■	■	□	■	□	■	$	●●●	TBD
Category 4: Education, Programming & Incentives																													/10
4.1	Marketing Materials Highlighting TDM	Marketing materials highlight elements such as proximity to trails, transit, cycling network routes, neighbourhood walk score, or TDM-supportive infrastructure such as bike storage	□	■	■		■	■	■		■	■	■		■	■	■	■		■	■	■		■		■	$	●	TBD
4.2	Smart Commute Durham Membership	Large commercial office, industrial or institutional tenants (100+ employees) will be encouraged to join Smart Commute				■	■	■	■	■	■	■	■	□	□	■	■	■		■			■	■	■	■	$$	●●	TBD
4.3	Information Packages	Packages containing information about transportation alternatives will be distributed to new tenants / employees / residents	■	■	■	□	■	□	■	□	■	■	■		■	■	■	■		■	□	■	□	■	□	■	$	●●●	TBD
4.4	Transit Passes	1 month of transit passes will be provided with new home purchases, or with office or residential rental agreements	■	■	■	□	■	□	■	□	■	□	■	■	■	■	■	■	□	■	□	■					$$	●●●	TBD
4.5	Subsidized Carshare and/or Bikeshare Memberships*	Carshare / bikeshare memberships will be provided with new home purchases, or with office or residential rental agreements	■	■	■	□	■	□	■	□	■	□	■	■	■	■	■	■	□	■	□	■					$$	●	TBD
4.6	On-Site Carshare Vehicles*	Provision of dedicated carshare vehicles	□	■	■		■	□	■		■	■	■		■	■	■	■		■		□					$$	●●	TBD
4.7	On-Site Bikeshare Vehicles*	Provision of dedicated bikeshare docking station and bikes	□	■	■	□	■	□	■	□	■	■	■		■	■	■	■		■		■					$$	●●	TBD
4.8	Wayfinding Signage*	Assists people that are walking or cycling to navigate and locate transportation services, infrastructure, and key destinations in the area	□	■	■		■	□	■		■	□	■		■	■	■	■	□	■	□	■	□	□	□	□	$$	●●	TBD
4.9	School Travel Plan*	STP is a cost-effective planning approach that aims to get more kids walking and cycling to school by addressing ongoing transportation and traffic safety problems																			■	■					$$	●●●	TBD
4.10	Awards / Recognition Program**	Recognition of organizations, developers, individuals and businesses for support and implementation of TDM initiatives; can be used as a selling feature for proposed developments		■	■		■	□	■		■	□	■		■	■	■	■	□	■	□	■	□	■	□	■	$	●	TBD
4.11	Cycling Skills Course**	Offer cycling skills courses for new residents, students, and employees interested in trying cycling focusing on cycling fundamentals, safety, and guidelines for cycling individually or in a group		■	■		■	□	■		■	□	■		■	■	■	■	□	■	■	■					$$	●●	TBD
4.12	Emergency Ride Home Program**	Provides a subsidized trip or reimburses commuters that use a sustainable transportation option to get to work or school and experiences an unplanned emergency and requires a way home quickly					■	□	■		■	□	■		■	■	■	■	□	■	□	■	□	■	□	■	$$	●●	TBD
4.13	Micromobility Services**[1]	Provide micromobility options such as bike share or kick-style e-scooters to residents, tenants / employees, students and community members	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	□	■	■	■					$$$	●●●	TBD
4.14	Support for Remote Work**	Applicants should take increasing rates of remote working into consideration by offering amenities such as coworking spaces in a residential development		□	■	■	■	■	■								■	■	□	■	□	■	■	■	■	■	$	●●	TBD

Initiatives listed based on Durham Region TDM Guidelines for New Development (Draft), December 14th, 2021, unless otherwise noted

* Initiative Added From Town of Ajax Transportation Demand Management Plan, Appendix A: TDM Development Guideline (June, 2015)

** New Initiative Added (Absent from Draft Durham Region/Ajax Guidelines)

Notes

1 Micromobility Services is dependent on the Town's incoming micromobility project on whether to permit e-bikes and e-scooters.

Legend

■ Higher Priority Initiative
□ Lower Priority Initiative

Implementation Cost

$ Low
$$ Moderate
$$$ High

Effectiveness of Initiative

● Low
●● Moderate
●●● High

THE TOWN OF AJAX

TRANSPORTATION DEMAND MANAGEMENT PLAN UPDATE

SUMMARY OF BEST PRACTICES

March 2022

Revision History

First issue	March 3, 2022			
Prepared by	Reviewed by	Approved By		
Pegah Tootoonchian, Senior Transportation Planner	Shawn Smith, Senior Project Manager	Shawn Smith, Senior Project Manager		
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April 7, 2022	**Address Town comments**			
Prepared by	Reviewed by	Approved By		
Pegah Tootoonchian, Senior Transportation Planner	**Shawn Smith, Senior Project Manager**	**Shawn Smith, Senior Project Manager**		
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<Date>	<Remarks>			
Prepared by	Reviewed by	Approved By		
<preparer, title>	<reviewer, title>	<approver, title>		

Contributors

WSP Transportation Planning

Pegah Tootoochian

Enya Huang

Shawn Smith

Brett Sears

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1. Introduction

Efficient transportation networks are the result of a balance of supply and demand. Governments typically provide the "supply" in the form of road infrastructure and transit service in order to meet residents' and visitors' "demand" for travel. The traditional approach to accommodating demand has been to increase the supply of transportation infrastructure, by building or widening roadways. This is a costly endeavour, and one that may only exacerbate the problem. A recent paper from the Victoria Transport Policy Institute[1] reports that when a roadway is widened, additional traffic is generated by trips that would have not have otherwise been made. The result of this is increased external costs and reduced social, environmental, and economic benefits from the widening project.

The Victoria Transport Policy Institute further describes examples of individual travel decisions that are more likely to generate traffic. For the purposes of this example, generated traffic refers to the additional peak-period trips that occur when capacity is increased (widenings), whereas congested roads have the potential to defer trips that are not urgent, influence alternative modes, and forego avoidable trips. The examples from the paper include:

1. **Consumers choose closer destinations when roads are congested and further destinations when traffic flows more freely**. *"I want to try the new downtown restaurant but traffic is a mess now. Let's just pick up something at the local deli."* This also affects long-term decisions. *"We're looking for a house within 40-minute commute time of downtown. With the new highway open, we'll consider anything as far as Midvalley."*
2. **Travelers shift modes to avoid driving in congestion**. *"The post office is only five blocks away and with congestion so bad this time of day, I may as well walk there."*
3. **Longer trips may seem cost effective when congestion is light but not when congestion is heavy.** *"We'd save $5 on that purchase at the Wal-Mart across town, but it's not worth fighting traffic so let's shop nearby."*

What the above reference describes is that transportation demand is flexible and can be influenced by policy, guidelines, and improved infrastructure focused on changing the way of travel.

Transportation Demand Management is a strategy that focuses on increasing the efficiency of transportation systems with policies, programs, and initiatives that are targeted towards changing travel behaviour. The goal is to redistribute demand for travel, either by shifting trips to different times of day or to more efficient modes or, reducing demand by allowing them to not be made at all (such as by teleworking).

The Town of Ajax has expressed a clear goal of increasing the share of peak period trips made by sustainable modes to support alleviating traffic congestion, reducing reliance on the single-occupant automobile, and

[1] Victoria Transport Policy Institute. Generated Traffic and Induced Travel (2021). https://www.vtpi.org/gentraf.pdf

providing a more balanced and equitable transportations system. While increased people movement may have relied upon roadway widenings in the past, this is no longer desirable in the Town of Ajax. At the same time, automobiles are the least efficient method of moving people from a space and environmental perspective and reallocating or improving space for other modes can have major benefits.

In light of the Town's rapid growth in population to 137,000 persons and 41,000 jobs by the year 2031, as well as the considerable growth forecasted for Durham Region, corresponding shifts and transportation measures are required to meet the increased demand on the local transportation system. The following Summary of Best Practices is intended to inform the Town of Ajax's Transportation Demand Management Plan Update.

This Summary of Best Practices aims to explore and answer the following questions:

- What is Transportation Demand Management (TDM)?
- How are jurisdictions similar to the Town of Ajax incorporating TDM principles and actions into their planning, construction, and programming activities?
- How can the Town of Ajax take its efforts in TDM to the next level to achieve a substantial sustainable mode shift in its transportation network?

Figure 1 - Passenger boarding a bus in Ajax serviced by Durham Region Transit

What is Transportation Demand Management?

Transportation Demand Management (TDM) is the application of programs, policies and services to influence shifts in travel behaviour and improving the efficiency of the transportation network. The combination of these elements is intended to support long-term use of sustainable travel options such as walking, cycling, transit (**Figure 1**), micromobility (such as e-scooters) and/or carpooling. TDM can support municipalities with decision-making tools that can guide residents towards equitable, accessible, and sustainable travel options. The array of policies, strategies, and programs can, for example, influence development patterns and how communities are planned. For example, a key success factor in walkable neighbourhoods is the ability for people to reach essential services on foot or by bicycle.

Transportation Demand Management addresses transportation challenges through soft measures, such as programs, and hard measures, such as supportive infrastructure. On the one hand, programs and measures can influence behaviour change amongst residents to rely on alternative modes of travel other than the automobile. Similarly, the infrastructure approach, such as interconnected on-road and off-road

active transportation facilities, bicycle parking and amenities, and safety features can encourage an uptake of active travel options for daily travel.

TDM investments can be viewed as place-based or systematic. Place-based strategies are either tangible and located in a specific place (e.g., a new bike facility or transit service), or less tangible but designed to benefit a specific place (e.g., promotion of carpool groups in a specific corridor). Systemic strategies are more regional and planned with goals and objectives to influence and desired outcome (e.g., promotion of public transit in an entire media market).

Some examples of TDM measures include:

- **Measures to reduce peak-hour demand or to shift it to off-peak periods,** such as congestion pricing, variable parking rates, and peak-period transit fares.
- **Measures to eliminate certain trips altogether,** such as mixed-use development and work-from-home promotional programs, including Smart Commute.
- **Measures to shift single-occupancy vehicle (SOV) trips into more efficient modes,** such as employer-funded transit passes, active school travel programming, limits placed on the maximum number of parking spaces permitted in new developments, increased parking fees, improvements to walking and cycling infrastructure, improvements to transit service, high-occupancy vehicle (HOV) lanes, TDM measures checklists for new developments, and carpool matching programs.

What is the Town of Ajax Currently Doing?

Current TDM Action Plan

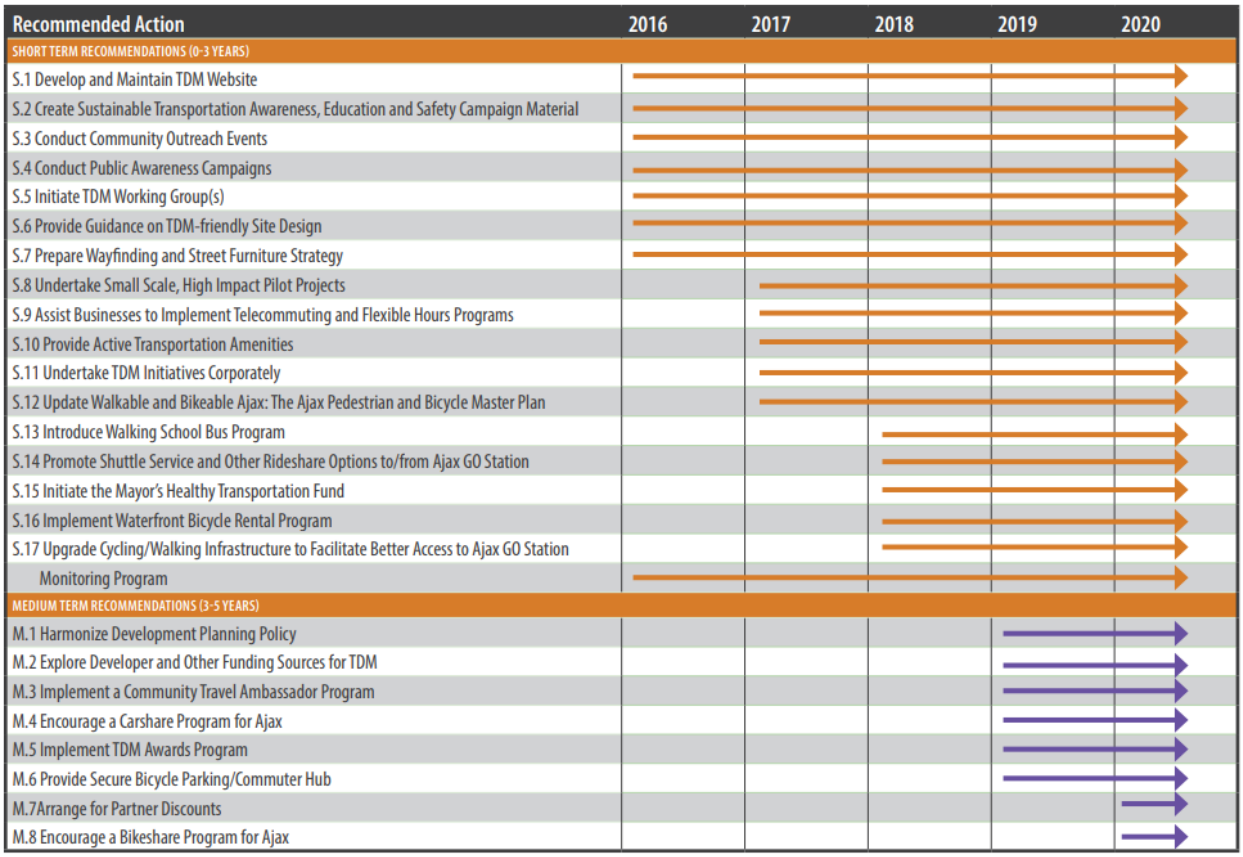
The Town of Ajax adopted a Transportation Demand Management Plan in June 2015, inclusive of recommended measures categorized in short term (0 to 3 years), medium term (3 to 5 years), and long term (beyond 5 years) horizons based on implementation considerations, constraints and resources required.

Short term recommendations are targeted towards Town specific corporations and within the community. Medium term measures are focused on building upon short term measures, as well as policy and programs that will promote sustainable transportation modes, and more options for travel. Long term measures are dependent on the implementation of earlier initiatives, and require further consideration around costing and funding sources.

Proposed Timelines

Timelines for each specific recommendation are shown in the **Table 1**, displaying from the year 2016 to 2020 and beyond.

Table 1 - Timeline of Recommended Actions - short term and medium term (Source: Town of Ajax TDM Action Plan)



In the present, the recommendations vary in different action statuses ranging from "not started", "ongoing" and "complete". **Table 2** shows the details for each short-term recommendation and **Table 3** shows the details for medium-term recommendations. **Table 4** shows the TDM Plan aim targets.

The following recommendations have been completed:

- S.2 Create Sustainable Transportation Awareness, Education and Safety Campaign Material;
- S.3 Conduct Community Outreach Events (#GetAjaxMoving);
- S.4 Conduct Public Awareness Campaigns;
- S.6 Provide Guidance on TDM-friendly Site Design;
- S.7 Prepare Wayfinding and Street Furniture Strategy
- S.10 Provide Active Transportation Amenities;
- S.12 Update Walkable and Bikeable Ajax: The Ajax Pedestrian and Bicycle Master Plan; and
- S.13 Introduce Walking School Bus Program.

Ongoing work on recommendations include:

- S.1 Develop and Maintain TDM Website;
- S.8 Undertake Small Scale, High Impact Pilot Projects;
- S.11 Undertake TDM Initiatives Corporately;
- S.17 Upgrade Cycling/Walking Infrastructure to Facilitate Better Access to Ajax GO Station, and;
- Monitoring Program (Chapter 9).

The recommendations that have not begun include:

- S.5 Initiate TDM Working Group(s);
- S.9 Assist Businesses to Implement to Implement Telecommuting and Flexible Hours Programs;
- S.14 Promote Shuttle Service and Other Rideshare Options to/from Ajax GO Station;
- S.15 Initiate the Mayor's Healthy Transportation Fund; and
- S.16 Implement Waterfront Bicycle Rental Program.

Table 2 - Short Term Ajax TDM Action Status and Details

Recommendation	End of 2016	End of 2017	End of 2018 and beyond	Status	Details
S.1 Develop and Maintain TDM Website	Website	Maintenance	Maintenance	In progress: #GetAjaxMoving webpage to be reviewed and updated	Active and Safe Routes to School (ASRTS) is updated, but other content needs to be revamped. Communications role is filled.
S.2 Create Sustainable Transportation Awareness, Education and Safety Campaign Material	2 Videos and 2 Leaflets	1 Video and 2 Leaflets	1 Video and 2 Leaflets	Complete. Multiple bike safety videos, toolkits available	Needs to be promoted annually.
S.3 Conduct Community Outreach Events	Program Launch plus 5 Events	5 events	5 events	#GetAjaxMoving launch complete. Average of 3 in-person community events each year, 1 in-person event each year from 2020-2021, 1-2 virtual events with attendees from 2020-2021.	Next steps in 2022: Trailfest (recreation) will be held, one in-person event and one week long self- guide campaign. Around 120 people attended the event in 2019. There will be a pop-up event on the Waterfront Trail. It is an effort from Transportation & Recreation staff and volunteers to provide bike bells and trail etiquette.
S.4 Conduct Public Awareness Campaigns	3 Campaigns	3 Campaigns	3 Campaigns	Complete. Over 3 campaigns per year since 2016. Ongoing campaigns.	Awareness Campaigns include School Campaigns such as Winter Walk Day, Walktober and Bike Month.
S.5 Initiate TDM Working Group(s)	Working Groups and 1 Meeting	2 Meetings	2 Meetings	Not started	Consider having Durham Region lead this initiative.
S.6 Provide Guidance on TDM-friendly Site Design	Awareness	Regularly Commenting	Regularly Commenting	Complete. Regularly Commenting	Zoning by-law is being updated.
S.7 Prepare Wayfinding and Street Furniture Strategy	RFP and Consultant Selection	Strategy	Implementing Strategy	Complete. Currently Implementing Strategy	There is signage to the GO Station and to the Waterfront Trail. This is included in the Wayfinding Strategy.
S.8 Undertake Small Scale, High Impact Pilot Projects	-	3 projects	3 projects	Ongoing: pop-up temporary bike parking at 1-2 events	Currently not available at the Town. In 2020-2021, the Town's economic development department worked to encourage businesses to become bicycle friendly businesses.

Recommendation	End of 2016	End of 2017	End of 2018 and beyond	Status	Details
S.9 Assist Businesses to Implement Telecommuting and Flexible Hours Programs	-	Program Material	3 Businesses	Region coordinates Smart Commute program for businesses	Town promotes through offering telecommuting toolkits aimed at residents.
S.10 Provide Active Transportation Amenities	-	2 Stations	Maintenance	Complete. 2 bike repair stations installed 2021. More to be purchased and installed in 2022.	Town keeps a bike parking inventory.
S.11 Undertake TDM Initiatives Corporately	-	Transit Pass and Fuel Cards	Transit Passes, Fuel Cards and Town Bike Fleet	Investigated bike fleet. Challenging due to space. Staff investigating the feasibility of creating a bike library.	-
S.12 Update Walkable and Bikeable Ajax: The Ajax Pedestrian and Bicycle Master Plan	-	RFP and Consultant Selection	Approved	Complete. Currently Implementing ITMP	-
S.13 Introduce Walking School Bus Program	-		1 Walking School Bus with Leader	Completed at 1 school, no longer available	Lessons learned include providing incentives or rewards to the people that lead a walking group. May need to look into a substitute at one of the schools since walking bus leaders (students) are no longer available when sports season arrived.
S.14 Promote Shuttle Service and Other Rideshare Options to/from Ajax GO Station	Monitor Town of Milton as a case study	Program Outline	Shuttle Bus	Not started	Recreation department organizes shuttle buses for large events from GO station to the site. The Town is not aware of any Durham Transit initiatives.
S.15 Initiate the Mayor's Healthy Transportation Fund	-	Program	Inaugural Grants	Not started	Town offers funding to organizations through the IMPACT Ajax fund.
S.16 Implement Waterfront Bicycle Rental Program	-	Program	Program Operating	Not started	Bike shops such as Northern Cycle could be considered as partners for this recommendation.
S.17 Upgrade Cycling/Walking Infrastructure to Facilitate Better Access to Ajax GO Station	-	Design	Implementation	In progress. Challenging due to property impacts.	Town is working with Metrolinx, the multi-use path (MUP) is proposed on Westney in the capital forecast.
Monitoring Program (Chapter 9)	Baseline Survey	Annual Monitoring	Conferences, Training and Delegations	Challenging due to staff capacity.	Data of eco-counter is available, the ones on Harwood were recently replaced. Equipment needs to be repaired – within budget. No eco-counter data available from 2019 onwards. Automatic Traffic Recordings and Turning Movement Counts at school crossing are still with the same schools and needs to be evaluated every few years. Transit ridership could be obtained from GO transit and Durham Region Transit.

Table 3 - Medium Term Ajax TDM Action Status and Details

Recommendation	End of 2016	End of 2017	End of 2018 and beyond	Status	Details
M.1 Harmonize Development Planning Policy	Policies Revised and Adopted	Policies Being Implemented	-	In Progress (Zoning by-law update)	-
M.2 Explore Developer and Other Funding Sources for TDM	Review Completed	New Funding Sources	-	Use of development charges and various grants for implementing the ITMP.	Development charges funding for multi-use paths and active transportation infrastructure is available.
M.3 Implement a Community Travel Ambassador Program	Program Established and 5 Ambassadors Hired	5 Ambassadors Operating	-	Not started	Goal is for all communities within the Town.
M.4 Encourage a Carshare Program for Ajax	Existing Program Extended and Operating	Program Operating	-	Not started	-
M.5 Implement TDM Awards Program	Program and Inaugural Ceremony	Ceremony	-	Not started	Can link to existing community award program, consider a volunteer appreciation award from the Town.
M.6 Provide Secure Bicycle Parking/Commuter Hub	Equipment Purchased and Set-up	Program Operating	-	Bicycle parking installed at all schools and multiple businesses. Program to be revised.	Bike parking is available at all Town facilities.
M.7 Arrange for Partner Discounts	-	Partners Approached and Trial Program Operating	-	Not started	The Town's economic development department can be coordinated to investigate incentives for TDM program participation such as transit passes.
M.8 Encourage a Bikeshare Program for Ajax	-	Existing Program Extended and Operating	-	Not started - currently investigating the feasibility of creating a bicycle library	Investigate the feasibility of bicycle library and in the process of creating a proposal. Researching case studies that have community bike hubs including Markham cycles, and Scarborough cycles.

Table 4 - TDM Plan Aim Targets

Term	Recommendation	By End of Year	Status	Details
Short Term				
1S	Engage with 10% of the population of Ajax, with high levels of positive awareness	5	Need metric for this.	Town is tracking social media, in-person events, website check-ins, and program participation.
Medium Term				
1M	50% of all employers and 80% of the biggest 10 employers in Ajax actively participating in the Town's TDM Program and having implemented at least one TDM measure	3	2 Smart Commute businesses with over 50 employees	Coordinating with Smart Commute Durham.
2M	All of the 10 largest employers in Ajax actively participating in the TDM Program and having implemented at least one TDM measure	3	2 Smart Commute businesses with over 50 employees	-
3M	Achieve a reduction in single-occupant vehicle use by 5% compared with Year 1 baseline	5	1% increase in single-occupant vehicle use but also 1-2% increase in walking/cycling (TTS data, 2011-2016, 6-9AM). Waiting for 2021 Census data results	-
Long Term				
1L	All organizations with over 50 employees/100 daily visitors or users in Ajax to be active participants in the TDM Program	10	2 Smart Commute businesses with over 50 employees	-
2L	Achieve a reduction in single-occupant vehicle use by 10% compared with Year 1 baseline	10	1% increase in single-occupant vehicle use but also 1-2% increase in walking/cycling (TTS data, 2011-2016, 6-9AM). Waiting for 2021 Census data results	-
3L	Achieve an increase in modal share of transit, cycling and walking trips in the AM peak hour to 30% of all trips (from about 20% of all trips in 2011). Ajax Transportation Master Plan target is 30% by the year 2031.	10	2011-2016: 2% increase for trips made by residents, 1% increase for trips made to Ajax by residents of TTS area (TTS)	Coordinating with Durham Transit, and review of master plans.

Funding Sources and Opportunities

The following are some opportunities that could potentially support the funding of TDM programs and initiatives for the Town of Ajax. The Federal Gas Tax Fund allocation and the general levy, also known as property tax, typically are funding sources for programs. Reviews of the Development Charges By-law explores the opportunity for developers to fund TDM measures. Additional TDM measures that could potentially fund TDM strategies include parking enterprise, Federation of Canadian Municipalities Green Municipal Fund, private grants, as well as partnerships.

Parking Enterprise consists of a self-funded, sustainable parking model that includes the operation, structural, maintenance and expansion of future municipal parking procedures. Typically, municipalities that adapt the enterprise model possess a substantial amount of parking infrastructure, specifically garages, since those types of infrastructure requires operating and maintenance costs in the long term. A few municipalities that have a Parking Enterprise Fund use parking revenues to fund active transportation programs. For instance, Toronto Parking Authority assists with the financing of Bike Share Toronto using the revenues received from parking.

Federation of Canadian Municipalities Green Municipal Fund is the result of the Government of Canada providing an asset of $550 million to the Federation of Canadian Municipalities (FCM) to create the Green Municipal Fund (GMF). Providing below-market rate loans and grants to municipal initiatives that enhance air, water, soil quality, while protecting the environment. The fund could be utilized to finance costs related to implementing the TDM plan.

Private Grants could help fund TDM initiatives such as health related organizations, for instance the Heart and Stroke Foundation focuses on funding research related to active travel modes and sustainability. Ontario Trillium Foundation (OTF) is another source of possible funding to operations, projects, and capital.

Partnerships create opportunities to raise funds from other levels of government, public health partners, businesses, as well as schools. In the past, businesses, public health organizations and schools have helped support initiatives including Bike to Work Day and Go Green programs. Businesses will more likely participate if they gain recognition, in which they would assist with advertising, and cash payments that cover the costs pertaining to TDM initiatives. Metrolinx is another potential partner that could help fund projects with staff and finances available that could support active transportation. Municipalities within Durham Region and the regional government will most likely be interested in collaborating to ensure that TDM programs run smoothly. For instance, the TDM programs implemented in the Town would influence travel demands in nearby municipalities such as City of Pickering and Town of Whitby.

Municipal Policy Structure and TDM Measures

The Transportation Demand Management Plan must align with the Town's Integrated Transportation Master Plan as well as the Official Plan. The following section is an overview of TDM context-specific considerations

and how they have been incorporated into these documents. Updating the TDM Plan at this time presents an opportunity to incorporate recommendations from the 2019 ITMP and to inform the Official Plan Review, scheduled to begin in 2023.

Integrated Transportation Master Plan (ITMP)

The ITMP supports the TDM Plan's vision to develop a balanced transportation system and builds upon the objectives of the Plan through a three-pronged approach focused on infrastructure, promotion, and normalization.

i. **Infrastructure:** Includes roads, trails, and transit facilities required for moving people by modes other than the single-occupant vehicle. To meet the overarching transportation goals set out by the Town, the infrastructure will require regular and appropriate maintenance, and must be designed to be safe and accessible by all users. Similar to the outcomes of the jurisdictional scans discussed below, a key component of successfully implementing TDM measures is integration of land use planning in transportation infrastructure projects, promoting development with short-distance trips accessible by sustainable modes (walking, cycling, transit).
ii. **Promotion**: Includes information, incentives, educational programs, and other events that encourage users to consider the ways they can integrate sustainable modes into their travel options. Catering messaging to all users is important for ensuring the mobility needs of all persons are met. Examples can include championing sustainable travel through initiatives such as group rides or mentorship programs focused on normalizing modes such as walking, cycling, and public transit.
iii. **Normalization**: Includes social, legal, and/or legislative updates to support shift in travel behaviour. The primary focus of normalization will be the inclusion of TDM actions in key municipal documents, including the Traffic Impact Study Guidelines and the Official Plan and Land Use By-law.

Official Plan

The goals and recommendations from the 2019 ITMP and the forthcoming TDM update can serve as policy inputs into the Official Plan update in 2023. At the present time, the Town's Official Plan defines TDM as "the application of policies and strategies that serve to reduce the travel demand on roadways with respect to single-occupant vehicles". TDM goals and policies are primarily found in Section 4.0 of the Official Plan Consolidation, specifically in Vibrant Streets where the intent is to design and implement a transportation network that is safe and efficient for moving both people and goods. One of the overarching goals of Vibrant Street is TDM initiatives that promote alternative modes of transportation, with emphasis on increasing transit ridership, walking, and cycling. The policies are then designed to serve the needs of local residents and visitors with a focus on parking strategies, transit-supportive, and active transportation supportive land use planning and design principles.

The Official Plan also articulates a series of policies geared towards Parking (Section 4.2.7), recognizing the needs for a multi-modal transportation system. The Parking policies are focused on support supply that meets the needs of both businesses and residents, and balancing with good urban design, compact development, and an overall approach that encourages use of alternative modes.

Through the Official Plan update, consideration should be given to the following policy updates:

- Reviewing and updating policy framework to reduce parking requirements, in favour of infrastructure and facilities that encourage use of alternative modes of travel (i.e. bicycle parking as part of new developments)
- Replacing parking minimums with parking maximums
- Encouraging intensification and mixed-use development to help reduce travel times
- Encouraging neighbourhood retailing
- Integrating first and last mile solutions

Further building off the ITMP, the Official Plan should include Complete Streets policies for its role in strengthening and supporting implementation of TDM measures. Although the Town of Ajax was an early adopter of Complete Streets Policy in Canada, the existing Official Plan does not speak to how, as a legislated document, it will implement or support this policy framework. During the next Official Plan update, it is recommended Complete Streets policies be added to apply its concepts to all existing, retrofit, and new transportation projects, recognizing its importance for multi-modal, accessible, and safe travel opportunities.

Town of Ajax Bicycle Parking Design Guidelines

The Town of Ajax has an extensive cycling network that it plans to continue expanding as the Town experiences change in population and shifts in travel demand. For the use of cycling facilities to be viable, the availability of bicycle parking is critical at key locations. The Town of Ajax has introduced the Bicycling Parking Design Guidelines as part of its current Transportation Demand Management Plan to improve the quality of bicycle parking at existing and future development sites, as well as to provide planners, developers, and property owners with guidance on the installation and management of appropriate facilities. Examples of key locations that include bicycle parking are the Ajax Pickering Hospital, Town parks, Town Hall, and Ajax GO Station.

The advantage of providing high quality bicycle parking facilities is the ability to connect the Town's cycling network with other modes of transportation such as public transit, and encourage more cycling, especially for commuting purposes.

The Bicycle Parking Design Guideline are categorized into long-term and short-term parking facilities with the intent of assisting with the design and development of high quality parking facilities. Examples of long-term parking include bike lockers, bicycle racks in designated access-controlled spaces. Short-Term facilities

include covered bicycle parking in public spaces or parking and ring bicycle parking, which are intended to be easily accessible and visible to date.

Part of ensuring sufficient and accessible parking facilities is provided, the Town of Ajax has developed parking requirement minimums, broken down by development type (**Table 5**)[2]. Although these minimum requirements have been established, new developments are not including sufficient bicycle parking. Zoning by-law does not reflect these requirements yet.

Table 5 - Bicycle Parking Requirement by Development Type, Town of Ajax

Use	Requirement
Residential Units	
Long Term	0.6
Short Term	0.07
Total	0.67
Office*	
Long Term	0.1
Short Term	0.1
Total	0.2
Commercial*	
Long Term	0.05
Short Term	0.2
Total	0.25
School*	
Long Term	0.06
Short Term	0.06
Total	0.12

*per 100 square metres of space

With the continued improvement to designing bicycle parking facilities, examples of areas where further consideration can be given to its implementation and requirements informing the TDM Updates and ultimately municipal policy documents, include:

i. Specifying gross floor area (GFA) as part of the minimum bicycle parking requirements and design standards for new developments
ii. Expanding zoning requirements to include detached and semi-detached homes as part of "residential"

[2] Town of Ajax. Transportation Demand Management Plan – Appendix F: Bicycle Parking Design Guidelines. (2014)

Parking Management Strategy

The Town's Parking Management Strategy details a series of strategies as part of the broader Transportation Demand Management Plan. Part of the Parking Management Strategy is to encourage a shift in travel mode and reduction in parking requirements compared to conventional standards. The strategy was developed using a planning policy framework focused on reducing the amount of land required for parking, promoting alternative modes of travel other than the single-occupant vehicle, and intensification.

Benefits that can be yielded with a parking management strategy include:

- Creating more accessible and efficient land use patterns, while simultaneously supporting other strategic land use planning objectives;
- Designing and implementing more attractive urban environments that promote livable communities;
- Generating revenue to fund multi-modal parking facilities and transportation infrastructure improvements;

Within the Strategy, the Town has identified a series of principles that guide decision-making processes in planning that support parking management. At a high-level, the principles are focused on providing people with viable parking and travel options; serving multiple users and destinations to maximize facility and minimize the need for land consumption; prioritizing highest and best use of space; parking quality should be prioritized to include aesthetics, security, and accessibility; and importantly, a comprehensive cost benefit analysis for the purposes of examining strengths of parking planning.

Several potential strategies are included as part of the Parking Management Strategy (Table B.1 in the Appendix, and refence **Table 6** below), which also take note of the possible reduction values they can represent as a whole and their degree of applicability to the Town of Ajax. The strategies detailed in the table below can be cross-referenced against the Jurisdictional Scans what new approaches can be introduced be introduced into the Town's TDM Update as well as any modifications required to the parking management strategy.

Table 6 - Potential Parking Management Strategies (Table B.1), Town of Ajax

Management Strategy	Description	Typical Reductions	Applicability in Ajax
Shared Parking	Parking spaces serve multiple users or destinations	10–30%	●
Parking Regulations	Regulations result in more efficient use of parking facilities	10–30%	●
More Accurate and Flexible Standards	Adjust parking standards to more accurately reflect demand in a particular situation	10–30%	●
Parking Maximums	Establish maximum parking supply standards	10–30%	●
Remote Parking	Provide off-site parking facilities and encourage their use	10–30%	○
Smart Growth	Encourage more compact, mixed, multimodal development	10–30%	●
Walking and Cycling Improvements	Improved walking and cycling expands the range of destinations served by a parking facility and reduces vehicle trips	5–15%	●
Increase Capacity of Existing Facilities	Increase parking supply by using otherwise wasted space, smaller stalls, car stackers and valet parking	5–15%	○
Mobility Management	Encourage more efficient travel patterns	10–30%	○
Parking Pricing	Charge motorists directly for using parking facilities	10–30%	○
Improve Pricing Methods	Use better charging techniques to make pricing more convenient and cost effective.	N/A	○
Financial Incentives	Provide financial incentives to shift mode	10–30%	○
Unbundle Parking	Rent or sell parking facilities separately from building space	10–30%	○
Parking Tax Reform	Various tax policy changes that support parking management	5–15%	○
Bicycle Facilities	Provide bicycle storage and changing facilities	5–15%	●
Improve User Information	Provide convenient and accurate information on parking availability and price	5–15%	○
Improve Enforcement and Control	Ensure that parking regulation enforcement is efficient, considerate and fair	N/A	●
Transportation Management Associations	Establish member-controlled organizations that provide transport and parking management services in a particular area	N/A	○
Overflow Plans	Establish plans to deal with occasional excessive demand	N/A	○
Address Spillover Problems	Use management, enforcement and pricing to address spillover problems	N/A	○
Facility Design and Operation	Improved parking facility design and operations to help solve problems and achieve parking management objectives	N/A	●

Legend: ● Very Applicable ○ Somewhat Applicable

Town of Ajax TDM Development Guidelines

The Town has developed TDM Development Guidelines as part of the existing Transportation Demand Management Plan, which outlines different methods to incorporate TDM into new development, redevelopment and into existing buildings within Ajax. The targeted outcomes are:

- Enhance the aesthetics of streetscapes while catering to different users of the road such as motorists, transit, pedestrians, and cyclists.
- Develop communities that fosters various uses, in order for residents to live and travel to work near each other.
- Allow adequate space on streets and public spaces to integrate transportation for different modes of travel, prioritizing active transportation infrastructure such as bike lanes, sidewalks, and safer crosswalks, as well as incorporating transit.
- Encouraging an active lifestyle to residents.

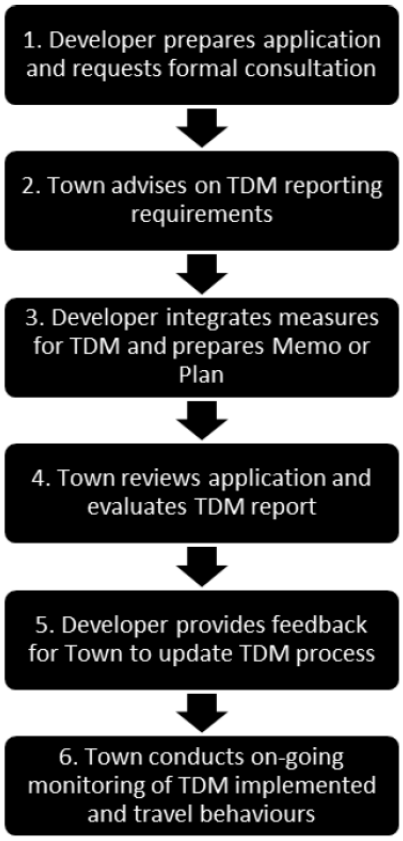


Figure 2 - Ajax TDM Development Guidelines process (Source: Town of Ajax Development Guidelines)

Benefits for Developers

If developers incorporate TDM initiative in development applications, there are benefits that could be awarded including the potential to:

- Develop higher densities;
- Decrease development expenses as a result of lowering parking requirements;
- Use land that would have been parking to be able to provide supplementary units/ amenities;
- Enhance the potential of intensification development plans on smaller pieces of land;
- Make transportation alternatives more accessible to residents that would like options; and,
- Endorse sustainable development objectives, such as LEED status.

Guideline

An overview of the development guideline process and the relationship between the Town and developer(s) during the various stages is shown in the **Figure 2**.

The existing TDM Development Guidelines will be cross-referenced against the Jurisdictional Scans to inform the Plan updates, including considerations around diverse land uses, adaptive bicycles and other active modes of travel for persons of all abilities, and micromobility opportunities that are supportive of the Green Development and Environment Design Guidelines.

The TDM Plan

The applicant must prepare a TDM plan that lists out TDM measures to apply to the development such as criteria involving planning and design, walking and cycling, transit, parking, carshare/bikeshare, wayfinding and travel planning, education/promotion and incentives. The applicant needs to demonstrate trip reductions from the TDM measures, site plan, and proposed monitoring and evaluation of TDM measures.

Town staff assess development applications as well as how TDM measures are integrated into the buildings using a TDM Development Checklist.

2. Jurisdictional Scan

To help inform the Town's TDM Update, a summary of measures implemented across multiple jurisdictions has been completed. The purpose of the jurisdictional scan is to learn about and discuss measures that can be adapted to thrive within the context of Ajax's communities. The jurisdictions were selected for their similar geographical and municipal make-up (single-tier versus two-tier) to the Town, and some for their innovative approaches that could serve to be valuable consideration upon which the Town's TDM Plan can be updated. The jurisdictions range throughout Canada and internationally.

Case Study #1: Metrolinx Transportation Demand Management Background Paper

Recognizing the growth taking place across the Greater Toronto and Hamilton Area (GTHA), Metrolinx launched the 2015 Regional Transportation Plan review as a future ready plan to support an evolving transportation network[3]. The Background Paper focuses on TDM and the application of behavioural tools that optimize the transportation network by changing demand for travel to align with supply.

In the 2008 "The Big Move", TDM was a key focus for achieving regional transportation goals. It included a series of priority actions and supporting policies that lay the foundations for TDM across several municipalities. Although "The Big Move" is now superseded by the 2041 Regional Transportation Plan (RTP), the priority actions and supporting policies are relevant to highlight. They include:

- Developing a TDM policy and strategy for provincial ministries and agencies (such as school boards)
- Establishing guidelines and model policies that support municipalities with developing and implementing TDM in municipalities' Official Plans and Transportation Master Plans
- Encouraging TDM within the private sector
- Subsidized transportation, such as transit, by employers to encourage alternative modes of travel
- Incorporating TDM into financial tools as an investment strategy
- Including TDM strategies as part of planning applications

Through the Regional Transportation Plan review, six overall strategies were identified to support municipalities and agencies with the development of new TDM strategies. They included:

1. Engage communities in travel behaviour change through community-based marketing
2. Inform decisions with usable travel information
3. Support the use of alternative modes through incentives
4. Enable behaviour change by developing and facilitating organizational TDM programming
5. Support the delivery of transport network improvements

[3] Metrolinx. Transportation Demand Management Background Paper (2015). https://www.metrolinx.com/en/regionalplanning/rtp/technical/11_TDM_Report_EN.pdf

6. Using development and financial policy to change travel behaviour

The Background Paper also explored international case studies, based on relevance to the GTHA, to support future development of TDM programming in the region. Through the 2041 RTP, these same case studies are referenced for their significant results with TDM tools.

London, United Kingdom

- Employs well-funded infrastructure and programs for widespread awareness and use of sustainable travel options
- High uptake of alternative travel modes attributed to diverse and convenient public transportation options
- Key initiatives that have influenced travel behaviour include investments in bicycle infrastructure and shared mobility; TDM in development approvals processes; and congestion charges in Central London
- The combination of the congestion charge and the improvements in transit (streetcar shown in **Figure 3**) and bus supply contributed to a 18% reduction in traffic and 30% drop in delays, while generating $2.5 billion CAD over 10 years on further transportation improvements

Figure 3 - Streetcar in London, UK

- **Best practices applicable to the GTHA context:** diversity and convenience of transit alternatives; significant investment in bike infrastructure and shared mobility; congestion charging; and TDM included in the development approval process.

Portland, Oregon, USA

- TDM initiatives are funded under a Regional Travel Options program aimed at improving awareness of sustainable travel options
- Strategic investments in programs and services that promote “non-drive alone” options
- **Best practices applicable to the GTHA context**: grant programs; individualized marketing; and, Transportation Management Associations (TMAs).

San Diego, California, USA

- Established an iCommute program that focuses primarily on the implementation of TDM activities with a comprehensive strategy to engage with employers and an outreach program to increase participation

- Introduction of construction mitigation plans and programs as well as the Sustainable Community Strategy to guide development
- Strengthening of inter-departmental relationships to support efforts in promoting travel behaviour change
- 40-year Regional Transportation Plan that focuses on improving and expanding transit provisions; mandating reduced greenhouse gas (GHG) emissions; and development of TDM programs.
- **Best practices applicable to the GTHA context:** employer outreach; construction mitigation plans; and working across departments to promote and embed TDM.

Tokyo, Japan

- Tokyo Metropolitan Government has benefited from TDM programming in areas of environment, health, equity, and transportation network performance (**Figure 4**)
- Implemented a shortage of parking provisions to minimize car ownership, with low minimum parking requirements for developments. Median parking rate approximately $740/month
- Using major rail stations as anchors for development with connectivity across the region
- **Best practices applicable to the GTHA context:** parking management and regulation; low emissions zones; and, employer transit subsidies.

Figure 4 - Walkable streetscape in Tokyo

Conclusions

Although the Metrolinx TDM Background papers draws upon the 2008 Big Move Plan, much of its principles have been re-introduced and re-invented in the 2041 RTP to make TDM programs more attractive and effective for GTHA municipalities. The Metrolinx case study provides an excellent overview of the **broad number of stakeholders involved in establishing a successful TDM program**. Stakeholders include all levels of government, schools and school boards, transportation and planning departments, employers, transit agencies, and developers. **The review highlights some "soft" approaches including providing better travel information and individualized marketing, as well as more "hard" approaches such as financial incentives, network improvements, and development policy.**

Case Study #2: Region of Waterloo Transportation Demand Management Program Plan

The Region of Waterloo published its 2017-2021 TDM Program[4] to support the recommendations of the Grand River Transit (GRT) Business Plan and help achieve the ridership and sustainability targets of the Regional Transportation Master Plan (RTMP). The program's vision is to "work with employers, residents, and other stakeholders in the community to reduce single occupancy vehicle use and meet 2031 trip targets for transit and active transportation modes." Since the early 2000s, Waterloo's TDM strategy has progressed from one that focused on TDM services for regional staff, to one thinking about and serving local neighbourhoods, and informing residential development and office TDM plans.

A core component of the Waterloo TDM Program plan is participation in the TravelWise Transportation Management Association (TMA), a public-private partnership that provides various services to participating organizations, such as development of workplace cultures and incentives and planning tools, to decrease single occupancy vehicle use. Examples include carpool matching, trip logging software, and discounted corporate transit passes. **Table 7** summarizes the goals and select initiatives from the program.

Table 7 - Goals of Waterloo TDM Program and Selected Ongoing or Proposed Initiatives

Enhance Program Integration and Coordination	- Continue to partner with other TDM stakeholders to coordinate, plan, and promote local TDM initiatives - Participate in Area Municipalities' station area and active transport planning processes
Continue Investing in the TravelWise TMA	- Secure annual dedicated funding and program service agreements for TMA - Targeted TravelWise promotion - Establish and monitor TMA-Wide Mode Split Target - Partner with local community organizations to expand/deliver services to TMA members
Enhance TDM's Involvement in Development Application and Review Process	- Review and comment on development applications as received - Update the Transportation Impact Study Guidelines to include a reference guide and monitoring plan for applicants - Encourage TravelWise Developer category membership for new developments
Develop and Support Residential and School-based TDM Program(s)	- Collaborate with School Travel Planner to support community and school-based sustainable transport initiatives (ex. school-based cycling training) - Promote and support TravelWise for schools, colleges, and high-density residential developments
Develop Comprehensive Performance Monitoring for TDM Program	- Work with external commuter/TDM service providers to share and monitor performance - Use evaluation data collected from TDM and active transportation initiatives, and external data to inform TDM planning and program development

[4] Region of Waterloo. Transportation Demand Management Program Plan (2017). https://www.regionofwaterloo.ca/en/exploring-the-region/resources/TDM---Transportation-Demand-Management-Program-Plan-2017-2021-2122780-v9C.pdf

Enhance ROW's Internal TDM Support for Regional Employees	- Continue to Manage Internal TravelWise Program and Initiatives - Gain political support for, and monitor Regional commuter drive alone reduction target and plan - Seek opportunities to increase/improve commuter options and services for Regional staff
Invest in Strategic Community Outreach	- Continue to develop and implement targeted, mode-specific TDM communications, promotions, and resources in coordination with Area Municipalities - Promote residential access to transit along ION and iXpress corridors
Support the Delivery of Transportation Network Improvements	- Work with the local bikeshare and Municipalities to support integration of bikeshare and transit - Seek opportunities to enhance access to transit, transit alternatives, and active transportation for challenging to serve areas, low income groups, and/or disadvantaged communities

Conclusions

The Region of Waterloo case study provides an example of the **ways that a TMA can support TDM initiatives** in a given jurisdiction. The strategy is exclusively focused on "soft" measures including providing **strategic input on other projects, collaborating with other relevant groups, tracking modal split over time, and running public-facing marketing initiatives and campaigns**.

Case Study #3: City of Kitchener Integrated Transportation Master Plan

The final report was published in June 2013. The recommendations in the TMP are to implement strategic-level management and operational improvements to the City's transportation system by 2033. The largest municipality in the Region of Waterloo is Kitchener. It is forecasted that Kitchener will have approximately 315,000 residents by the year 2031, which is a 90,000 difference from the population forecast of 2011. The projected aging population will significantly affect new urban travel demands.

The main goal of the TMP is similar to other case studies in reducing private automobile dependence by the horizon year of 2031. Many alternatives to private auto use should be provided to achieve this goal, at the same time should be suitable to the local context and needs. Alternative modes of transportation are necessary to have competitive levels of service compared to automobile use, considering factors such as travel cost, time, and comfort.

Transportation Demand Management recommendations in the report include the following:

- **Expand employer TDM programs in the City of Kitchener**

- Use existing TDM tools and services, building upon the TravelWise program to support carpool matching, subsidized transit passes, guaranteed-rode home and outreach programs to promote sustainable modes of transportation to City staff, when commuting to and from work.
- In addition to the downtown core, TDM initiatives should be considered in other areas of the City at major employment areas.

- **TDM Coordinator**
  - The City should work closely with the Region and employers, specifically in downtown Kitchener, to implement TravelWise programs, assist the application of TDM programs including telework. Carbon tracking, as well as guiding TDM appropriate site design of developments.
- **Support Car-share in the City**
  - Increase awareness by coordinating outreach and promotional events that creates more awareness, provide preferred parking to carshare vehicles in order to promote the service, to grow and help the long-term possibility for car-share to stay in the City and Region.
- **TDM strategies to be integrated into site planning and development approval processes**
  - The City could consider developing a TDM checklist for transportation related projects to review and evaluate development applications within the City of Kitchener, the Region and the Province.
  - For new developments and extensive transportation projects, a TDM checklist could include a precondition that TDM plans must be a component of transportation impact studies for new developments.
- **Engage residents by partnering with local groups and the Region**
  - Conduct individualized marketing to encourage sustainable modes of travel for all kinds of trips.
  - Individualized marketing is known to target populations, groups and adjusts the TDM strategies/ programs to meet the needs to offer other modes of travel.

Conclusions

The City of Kitchener's case study gives a list of TDM recommendations that should be implemented, including **employer TDM programs, delegate a TDM coordinator, support car-share, TDM measures that should be incorporated into development approval processes**, and to engage with residents through local groups.[5]

[5] City of Kitchener. Kitchener Integrated Transportation Master Plan (2013). https://www.kitchener.ca/en/resourcesGeneral/Documents/DSD_TRANSPORT_Transportation_Master_Plan_2013.pdf

Case Study #4: Peel Region Sustainable Transportation Strategy

The Peel Regional Council approved its Sustainable Transportation Strategy (STS) in February 2018, setting a goal of a 50% sustainable mode share by 2041. With projections predicting a population increase of 600,000 by 2031, and 45% more vehicles on the road during the morning peak periods by 2041 if current transportation choices persist, TDM was recognized as a key strategic component to achieving the 50% sustainable mode share target, defining "sustainable modes" as all trips except for single occupancy vehicle (SOV) trips.[6]

Exhibit 2-9: Comparison of External Costs for 2011 Baseline, 2041 Trend and 2041 Vision Scenarios

	Cost of...	2011 Baseline	2041 Trend	2041 Vision
		at 9.4 billion vehicle-km	at 11.9 billion vehicle-km	at 9.9 billion vehicle-km
AUTO (Unit = per vehicle-km)	Driving ($0.63/unit)	$5.9 B	$7.5 B	$6.2 B
	Cogestion ($0.22/unit)	$2.1 B	$2.6 B	$2.2 B
	Collision Externalities ($0.08/unit)	$750 M	$950 M	$790 M
	Criteria Air Contaminants ($0.002/unit)	$20 M	$25 M	$20 M
	Carbon Emissions ($0.04/unit)	$380 M	$480 M	$400 M
		at 1.2 billion passenger-km	at 3.5 billion passenger-km	at 4.2 billion passenger-km
TRANSIT (Unit = per passenger-km)	Service ($0.20/unit)	$240 M	$700 M	$840 M
		at 50 million person-km	at 96 million person-km	at 146 million person-km
WALKING (Unit = per person-km)	Health Benefits ($2.96/unit)	$150 M	$280 M	$430 M
		at 11 million person-km	at 12 million person-km	at 90 million person-km
CYCLING (Unit = per person-km)	Health Benefits ($1.48/unit)	$16 M	$18 M	$130 M
	NET COST	$9.2 Billion	$12 Billion	$9.9 Billion

Figure 5 - External costs comparison for two different scenarios

[6] Region of Peel. Sustainable Transportation Strategy (2018). https://www.peelregion.ca/pw/transportation/residents/sustainable-transportation-strategy.asp

In setting its vision for 2041, Peel also conducted an analysis of external costs comparing the implementation of the envisioned 50% sustainable mode share to the "business as usual" scenario, finding that implementing the vision could save over $2 billion in external costs by 2041, as shown in **Figure 5**.

The Region of Peel developed a Transportation Demand Management Implementation Plan, proposing a variety of actions to increase sustainable mode share by promoting by multimodal travel, walking, cycling, transit, carpooling, and telework. **Table 8** includes the high-level themes included in the plan. The full document includes a series of concrete actions for the Region associated with each theme.

Table 8 - Key themes in the STS related to TDM for travel modes promoted[7]9

Multimodal	- Influence the shape of development - Strengthen multimodal function of Regional roads - Make roads safer for vulnerable users - Influence personal travel decisions - Strengthen Region's leadership role
Walking	- Provide comfortable, continuous walking routes - Improve winter maintenance of walking facilities - Promote walking across the Region
Cycling	- Provide comfortable, continuous cycling facilities - Expand bicycle facilities and end-of-trip facilities - Promote cycling across the Region
Transit	- Make regional roads more transit-supportive - Improve connections to transit - Explore new technologies and business models to support transit - Promote transit use across the region
Carpooling	- Expand carpool lots - Explore new technologies and business models to support carpooling - Promote carpooling in key markets
Telework	- Promote flexible work arrangements as a win-win-win solution - Help workplaces support flexible work arrangements

[7] Adapted from: Region of Peel. Transportation Demand Management Implementation Plan (2018). https://www.walkandrollpeel.ca/projects/sts/pdf/peel-TDM-plan.pdf

The TDM implementation plan identified short-term actions to be achieved between 2018-2022 to shape the sustainable development and transportation habits of the Region. Each short-term action outlined is accompanied by a description, five-year timeline, identification of roles and responsibilities of the Region and its partners, and a year-by-year projection of some of the associated costs to the Region. Examples include:

- Feasibility studies and pilots for satellite workplaces
- Alternative transit services
- New HOV lanes and public vanpooling
- Stakeholder engagement and co-learning
- Updates to the Region's workplace TDM measures
- Policy updates & advocacy for changes outside of the Region's jurisdiction

In order to achieve the outlined actions, the Sustainable Transportation team also proposes to seek partnership opportunities with other functions of the Region, local municipalities, Metrolinx, the Ontario Ministry of Transportation, business sector partners, and private transportation network companies (e.g. Uber, Lyft).

The TDM implementation plan also included guidelines on performance management, outputs (resources applied, activities) and outcomes (effects on behaviour & travel, downstream effects on environment, economy, society) to be tracked. The plan notes that the collection of performance measurement data should be built into initiatives at the outset as the cost of obtaining high quality and sufficient data post-implementation can be difficult.

In December 2021, Peel Region released a report to Council with an implementation update on the Sustainable Transportation Strategy (STS). Using the key themes from the STS related to promoting TDM, the Region has reported 21.7 kilometers of active transportation infrastructure implemented up to the end of 2021; leading or supporting active transportation program such as Bike Month, Community Cycling Programs, Ontario Active School Travel Fund Project, and Peel Safe and Active Routes to School; as well as transit improvements in collaboration with the local municipalities to support complete communities and first- and last-mile connections.

Conclusions

The Peel case study is a holistic example of a **suite of solutions, strategies, and detailed actions that can be implemented by an upper tier municipality to support a major sustainable mode shift**. The plan incorporates a range of "hard" and "soft" measures including **changing processes and policies to better support sustainable modes, taking ownership of responsibilities such as maintenance of in-boulevard active facilities, and updating engineering standards**.

Case Study #5: City of Brampton TDM

City of Brampton launched the Transportation Master Plan Update on March 2015. The TDM recommendations for the TMP Update include:

1. Recruit staff that would be able to lead the TDM program for the City of Brampton as well as collaborate with Smart Commute Brampton/ Caledon, plus other stakeholders.
2. Brampton staff should partner up with the Region of Peel TDM staff to assist with the development of engagement strategies, create partnerships, and manage TDM policies and programs.
3. The City should resume the collaboration with Smart Commute Brampton/Caledon in order to encourage TDM and join forces in the advancement of more TDM programs including the marketing for ride hailing services as well as the emergency ride home initiative.
4. Coordinate with Brampton Transit to raise awareness of new, extended transit services that are defined in the Transit Service Plans. With the implementation of new communities, it should simultaneously be planned with new services, stated in the Future Transit Service Plans.
5. The Brampton Transit Five Year Business Plan (2013-2017) and Servicing Plan Working Paper should be further implemented to ensure that transit service is integrated in the development process early on, such as brand new or expanded paths in the new neighbourhoods. Once the new service has been established, awareness should be brought to the developers and new residents to learn about the new service. When the residential outreach programs are finalized, it should encompass these new regions.
6. Build a detailed action plan that aligns with the timeline of the TMP Update after the short-term actions have been introduced by staff, budget, and partnerships. It should outline the application of the development checklist specifically for new developments.
7. Have suitable partners and stakeholders participate in the residential travel planning initiatives and aid in funding.
8. Partner up with post-secondary institutions, for instance Sheridan College campus to create TDM programs that are campus focused.
9. Continue implementing and monitoring all of the TDM programs. As time progresses, the City can consider growing the programs. The programs will need to be flexible enough to apply to new programs, populations, and the increase of employment.

TDM Toolkit Recommendations

1. Update the Brampton Work Place Commuter Program.
2. The Sustainable Mobility Coordinator should coordinate with Brampton Transit staff to build marketing and promotional resources to promote residents, workers to utilize the new service when a new transit service is introduced.
3. Transit services should continue to be delivered when building new developments.

4. Create and market a emergency ride home system, specifically for residents that use travel modes that are sustainable.

TDM in Planning and Development Applications Recommendations

1. Build tools that are customizable and can integrate TDM measures into development applications.
2. Improve the TDM checklist that will be used in the development application review phase.
3. Ensure that there are conditions when including TDM programs in development approvals.
4. Encourage new communities to be built.
5. Minimize parking regulations to the maximum and take into consideration minimal parking requirements in places that are in close proximity to transit, which could result in fewer cars in the region. [8]

Conclusions

The City of Brampton case study has TDM measures within its TMP with recommendations that mainly involve **coordination between different partners, building a detailed action plan, improving the TDM toolkit, as well as integrating TDM in planning and development applications**.

Case Study #6: York Region TDM Plan

York Region is currently focused on maximizing the efficiency of the Regional Road network by targeted programs and strategies to enhance sustainable travel options such as walking, cycling, public transit, carpooling and other alternative forms of transportation.

York's TDM programming related TDM initiatives are summarized below:

- The **Smart Commute Program,** which offers individualized marketing and encouragement campaigns to support sustainable commuting in the community and to workplaces (**Figure 6**). This is being implemented in partnership with the City of Markham, City of Richmond Hill, City of Vaughan, and Town of Newmarket,

Figure 6 - Example of Smart Commute Program Supporting Sustainable Commutes

[8] City of Brampton. Transportation Master Plan Update (2015). https://www.brampton.ca/EN/Business/planning-development/transportation/Documents/Final_TMPU_Sept2015.pdf

supporting residents, employers, and employees by exploring and encouraging walking, cycling, transit, carpooling, and teleworking.

- The **Smart Commute Tool,** smart phone application and web-based tool for finding "travel partners" for commuting and other trips across the GTHA.
- **MyTrip,** York Region's TDM program for new and growing communities, which started as a pilot project in 2017. It is delivered as a TDM program available to new and growing communities with the intent of encouraging sustainable travel options. It has completed outreach at 20 building across the Region and over 1,000 household have completed the initial travel survey.
- **Active School Travel Programming,** collaborating with local and Regional stakeholders to encourage active school travel. Includes infrastructure and materials that support school commutes to reduce single-occupant trips.

TDM is also a component of other York documents and strategies, including:

- **Transportation Mobility Plan Guidelines (2016),** a document for guiding how proposed developments are reviewed from a transportation perspective across the Region.
- **New Communities Guideline (2013),** a document to assist local municipalities and the development industry to implement sustainable community design.
- **Designing Great Streets (2019),** a guidelines document intended to promote better integration of road design with land use context and support livable and healthy communities.
- **Transportation Master Plan (2016),** which included TDM as an objective, with focuses on collaboration, workplaces, schools, new and growing communities, and development approvals.

Conclusion

York Region is focusing its Transportation Demand Management efforts on enhancing personal travel options through walking, cycling, public transit, carpooling. The intent for the Region is to redistribute the travel demand across the transportation network by shifting the demand to a mix of travel modes and away from the busiest travel times. **The Region has introduced a series of programs and incentives focused on workplaces, schools and new communities, as well as better planning and design guidance toward more complete streets.**

Case Study #7: City of Ottawa TDM Strategy

The City of Ottawa published a Transportation Demand Management Strategy in May 2012, yet the transportation system has transformed since then. Many factors have shifted including the construction and extension of the O-Train LRT system, increase of micro-mobility (e-scooters and bikes), expansion of transportation network companies (Uber and Lyft), smartphones are widely used, and a pandemic altered travel patterns.

The Transportation Master Plan – Draft for Public Review was released in December 2021 to provide programs that shape the economic, social, and physical factors behind individuals travel choices. The TMP

offers guidance for raising the effectiveness of the City's TDM attempts, for instance using technology to its full extent, concentrating on trips to school, and supporting multi-modal trips. School Travel Planning and "Let's Bike to Transit" promotional campaigns, university transit passes, and the TDM Checklist for new land development are all examples of TDM programs. Several TDM programs are produced in coordination with private and public sector associates to maximize the effectiveness and to match sustainable transportation goals. The City has a program in place called TravelWise that collaborates with organizations throughout Ottawa to promote workers to walk, cycle, take transit, and carpool to and from work.

The update to the TDM Strategy produces a structure for permitting pilots at the same time assessing the usefulness of TDM tools. The collaboration between community organizations, private sector, and different levels of government help facilitate innovation within TDM.

TDM values will continue to be included into a variety of City-led initiatives. For instance, new transit and active transportation infrastructure could be endorsed by TDM campaigns and specific events. Walking, cycling and transit use could be supported by Public health campaigns, libraries and recreation centre programs. "I Dig Elgin" is an example of a construction mitigation initiative that the City could consider, as well as supplying bike valet facilities at important occasions, and utilize strategic urbanism approaches to allow the public to re-examine how people interact with streets.

Prioritize Active School Trips

A way to help students decide on an environmentally friendly mode of travel to school is the Active School Travel Planning program, which was designed for this purpose. The program makes the school areas safer by decreasing car traffic in the vicinity of schools during the beginning and end of school hours. Students at participating schools have seen a growth in walking and cycling through the program. With the combination of new pedestrian infrastructure and "Walk-a-Block" initiatives, which are programs that give a designated vehicular drop-off zone away from the front of schools, there have been major results.

Improvements made to the transportation and operations are needed to make walking and cycling safe in neighbourhoods. The City considers "School Streets" (car-free zones in school frontages during peak school traffic hours) where needed. It is highly recommended for school staff to also walk, bike, or use transit to get to school, avoiding pressure when looking for school parking and to foster safer environments for walking/cycling.

Encourage Sustainable Travel Options for City of Ottawa Employees

The City supports employees to choose sustainable modes of travel. Bike parking and changerooms are ways that could encourage active transportation. To shift travel mode behaviour, paid parking is a tool used to allow employees to consider other travel options and can aid the costs of maintaining City parking amenities.

Increase Support for Multimodal Trip Planning Tools

There are resources that help people plan trips having a clear understanding of their journey options, known as multimodal planning tools. The tools display different modes of travel, as well as using many transit systems, including cycling and walking, micro-mobility choices such as e-scooters, carshare, taxis, and ride hailing services shown in **Figure 7**. Private sector companies provide multimodal planning tools that also shows weather, accidents, and places that are under construction. City of Ottawa will continue to give the public real-time data of O-train service, to ensure that multimodal planning tools that rely on this data are accurate.

With the advancement of the digital platforms of the various modes of travel, it could potentially lead to the integration of all the platforms, as a single service called Mobility-as-a-Service (MaaS). MaaS is a single platform that allows for the payment and planning of all the services combined. All the separate platforms need to work closely together to implement this effective planning and payment system. The City plans to look into improving payment and service integration. [9]

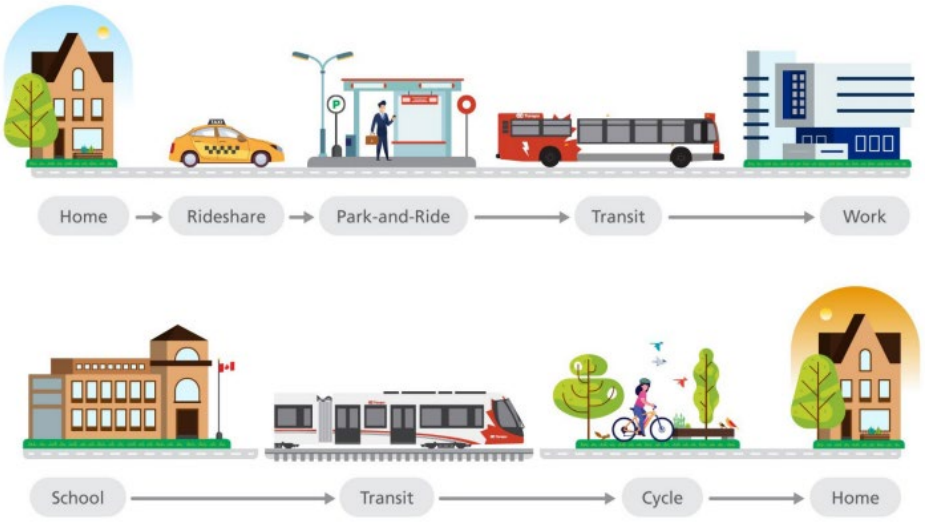


Figure 7 - Examples of Multimodal trips (Source: City of Ottawa TDM Strategy)

Conclusions

The City of Ottawa case study concentrates **on TDM programs that address trips to and from school, encouraging employees and employers to use sustainable travel options, and multimodal trips**. The City hopes to influence travel behaviour through these programs and to reduce demand for single-occupant

[9] City of Ottawa. Transportation Master Plan – Draft for Public Review (2021) https://engage.ottawa.ca/transportation-master-plan

vehicle trips with the help of **leveraging technology**.

Case Study #8: City of Whitehorse TDM Plan

Like the case studies that have come before, the City of Whitehorse identified growth and changing travel needs as drivers for introducing mobility management strategies. The intent behind Whitehorse's strategies has been to minimize the need for road widenings and instead implementing strategies that use their existing transportation resources more efficiently. This combined with their goal of reducing driving trips from 75% to 50% by 2036, focusing on carpooling, walking, cycling, and public transit[10].

The City of Whitehorse identified six strategies for guiding implementation of the Plan. Within each strategy, there are 5 to 7 action items that are listed as short-, medium-, and long-term. They include:

i. **Coordinate TDM Implementation and Promote the Options.** Consider the careful application of policies, programs, and knowledge to encourage shift in travel behaviour. The action items detailed within this strategy are focused on coordination, cooperation, and awareness. Examples include: **hiring a TDM coordinator** to oversee the implementation, coordinate data collection and monitoring; creating **TDM partnerships** with employers, institutions, and community organizations; creating a **webpage** that is a one-stop shop for accessing transportation tools to remain actively involved in travel options available to residents and visitors.

ii. **Improve Active Transportation Options.** Facilitate active transportation to achieve modal share targets as well as improve safety, reliability, convenience, and accessibility. Examples include: **developing and regularly updating and Active Transportation plan** that informs future planning and design decisions while prioritizing investments in active transportation infrastructure; identifying and improving **priority routes** between neighbourhoods and downtown areas; working collaboratively between City departments and NGO's to maintain network of trails, paths, roads,

Figure 8 - Citizen Trail Maintenance

[10] City of Whitehorse. Whitehorse Transportation Demand Management Plan (2014). https://www.whitehorse.ca/home/showpublisheddocument/4250/635736090644570000#:~:text=The%20TDM%20Plan%20was%20an,and%20Trails%2C%20and%20Operations%20staff.

and sidewalks with regular **review of maintenance practices** and establish a **citizen trail maintenance** program (i.e. adopt a trail - **Figure 8**).

iii. **Improve Public Transit**. A realistic alternative to the private automobile that supports both long and short commutes, is comfortable, and reduces wear and tear. Examples of actions include: creating a **long-term transit plan**; expanding **transit pass programs** that incentivize increased ridership; **improve downtown transit exchange** to enable convenience and access to multiple routes at one centralized location.

iv. **Support Employees to Use Alternative Travel Modes**. Recognizing that commuters are a captive audience with consistency travel behaviour that can be targeted to influence change. Focusing on how employees commute to and from work as well as work-related travel by exploring **alternative work arrangement** such as teleworking, compressed work weeks, and flexible work schedules that are outside peak hours; making alternative modes more attractive than driving by providing **incentives** to employees for participating in alternative transportation (i.e. green business recognition program between the City and the Chamber of Commerce. Recognizing businesses who have implemented strategies to reduce single-occupant vehicle use by creating a trip-end facility); **greening City's vehicle fleet** in collaboration with Federation of Canadian Municipalities. The plan includes a fleet and fuel management system that tracks and evaluates fuel usage, asset tracking, and life cycle optimizations as well as policies focused on using hybrid vehicles and/or alternative fuels.

v. **Maximize Existing Vehicle Infrastructure and Discourage Single-Occupant Vehicles**. Accommodate vehicles but focus on prioritizing opportunities for shared vehicle travel and more fuel efficient vehicles. To achieve this action, the City of Whitehorse has identified **parking management** (on-street, off-street, and policies) as an integral piece to their TDM Plan for its ability to manage supply, pricing, and convenience. Examples of other actions include, allocating a portion of **gas tax funds** to TDM; **increasing congestion thresholds** for assessing intersection capacity in coordination with capital investments prioritization between road infrastructure and active transportation infrastructure.

vi. **Encourage Supporting Land Use**. Thoughtfully planning land use enhances the attractiveness, convenience, and safety of alternative travel modes while not compromising vehicle travel efficiency. Examples of actions include, **regulating TDM in new developments** with specifications around bicycle facilities, electric vehicles, and reduced parking supply; planning for and creating **complete streets** to give comfortable access to pedestrians, cyclists, transit-users, and mobility-impaired individuals with prescriptive policies and guidelines; and **developing a TDM + LEED Credit Program**. TDM measures were identified to provide opportunity for gaining points towards LEED certification in new construction and neighbourhood developments.

Conclusions

The City of Whitehorse's Transportation Demand Management Plan speaks to a range of TDM principles and actions that are targeted towards facilitating increase travel using alternative modes that also support social, economic, and environmental benefits. This case study provides a **series of action items from infrastructure prioritization, funding sources for allocating towards infrastructure and programming, as well as supporting policy frameworks and guidelines that further reinstate implementation of action plans and shift in travel behaviour.**

Case Study #9: City of Coquitlam

The City of Coquitlam has identified a City-wide target of 30% of all trips to be made by sustainable modes, with Transit-Oriented Developments achieving a higher mode shift (from 30% to over 50%) to overcome the imbalance across neighbourhoods whose access to frequent and reliable transit is low[11]. One of the overarching goals of the TDM Plan is to support future transit-oriented development neighbourhoods with integrated policies and strategies that enhance sustainable transportation and livability.

The City has identified TDM as an essential component of their Official Community Plan as well as Strategic Transportation Plan, embedding programs and policies that encourage alternative modes that incentivize, educate, and balance future needs. The TDM plan emphasises integrating land use and transportation planning, taking a leadership role in promoting shift in travel choices by promoting alternative modes of travel amongst own staff, developing and promoting education and awareness programs, as well as encouraging private sector agencies and stakeholders to promote TDM initiatives.

In January 2022, the City updated its Transportation Demand Management Guidelines for New Developments to be applied to all new residential, multi-family, commercial, and office projects. Within the guidelines, the City includes a TDM structure breakdown, as shown in **Table 9**[12].

[11] City of Coquitlam. City of Coquitlam Transportation Demand Management Guidelines for New Developments (2022). https://www.coquitlam.ca/DocumentCenter/View/5281/Transportation-Demand-Management-Guidelines-for-New-Development-PDF

[12] City of Coquitlam. City of Coquitlam Transportation Demand Management Guidelines for New Developments (2022). https://www.coquitlam.ca/DocumentCenter/View/5281/Transportation-Demand-Management-Guidelines-for-New-Development-PDF

Table 9 - City of Coquitlam TDM Structure Overview

TDM Measure Rate	• $1,100 per apartment or townhouse dwelling unit. • $1,700 per 100 m² GFA for new commercial and/or office space.
TDM Monitoring Fund	• $20 per apartment or townhouse dwelling unit. • $25 per 100 m² GFA for new commercial and/or office space.
Additional TDM Measures	• Provision of TDM travel information packages for all new residential, commercial and office units. • Provision of directional signage to on-site parking spaces for residential, commercial, and office visitors.

Current Development TDM Initiatives include: annual carshare memberships and driving credits to users, subsidized public transit passes for residents and employees, and electric bicycle options to residents. The overarching goal of these initiatives is to reduce traffic congestion as well as demand for parking.

Carshare memberships is intended to be complimentary to sustainable travel modes. As part of new developments, the City is focusing on encouraging partnership with carshare providers to provide memberships to future residential, commercial, and/or office units.

Subsidized public transit increases accessibility and mobility to those without an access to a car. Future development should focus on arranging with subsidized transit passes with service providers.

Electric Bicycle option to encourage cycling as one form of sustainable travel, promoting health, environmental, and economic benefits. Similar to the carshare memberships, future developments should consult with electric bicycle providers to equip residents with access to this service.

Other complementary measures include TDM Monitoring, which is essential to tracking the effectiveness of the proposed measures. The City of Coquitlam uses the funds collected through new developments to complete on-going evaluations with the intent of updating or introducing new measures.

Conclusions

The City of Coquitlam is a comprehensive example of TDM measures for promoting shift in travel behaviour and an uptake of sustainable modes of travel, particularly **in transit oriented development neighbourhoods**. The case study demonstrates the value of a **well-defined guideline for new developments**, focusing on initiatives that have potential to reduce traffic congestion, reduce demand for parking, and lastly, reduce reliance on the single-occupant automobile.

Case Study #10: Arlington Country Commuter Services TDM Plan

Arlington County of Virginia is a U.S. leader in evidence-based Transportation Demand Strategy. Since the launch of its TDM programming in the 1980s, the County's initiatives have included "Commuter Stores" providing residents with information and customer service, sophisticated marketing, the Arlington Transportation Partners program for employers and residents, and an online transit portal. In the 2000s, the TDM strategy expanded to include walking and cycling programming, the TDM for Site Plan Development Program and the Capital Bikeshare. Recognizing the importance and potential of TDM for transforming transportation, the County also developed its research program into the public-facing Mobility Lab™, described as "a platform to promote and disseminate understanding and best practices associated with TDM and demonstrate its value through scientific research."[13]

Key to the County's success has been its emphasis on evidence-based policy and strategy, and commitment to monitoring the effects of their initiatives. Twenty years of data collection, research and thought leadership have positioned the County and the Mobility Lab as go-to advisors for state and federal policy makers on TDM.

As a reflection of the elevated status of Arlington's TDM program and the funding and authority it commands, Arlington County Commuter Services (ACCS) now exists as its own Bureau under the Department of Environmental Services, and works closely with the County's Bicycle, Transit and Pedestrian Advisory Committees and the Transit Commission.

The range of TDM activities the ACCS offers includes ridematching, vanpool assistance, residential outreach, employer outreach, telework, marketing and promotion, guaranteed ride home, bikesharing, carsharing, commuter stores and school pool.

[13] Mobility Lab. The Arlington Story (2021). https://mobilitylab.org/2021/05/10/the-arlington-story-how-transportation-demand-management-provides-more-choices-and-encourages-behavior-change/

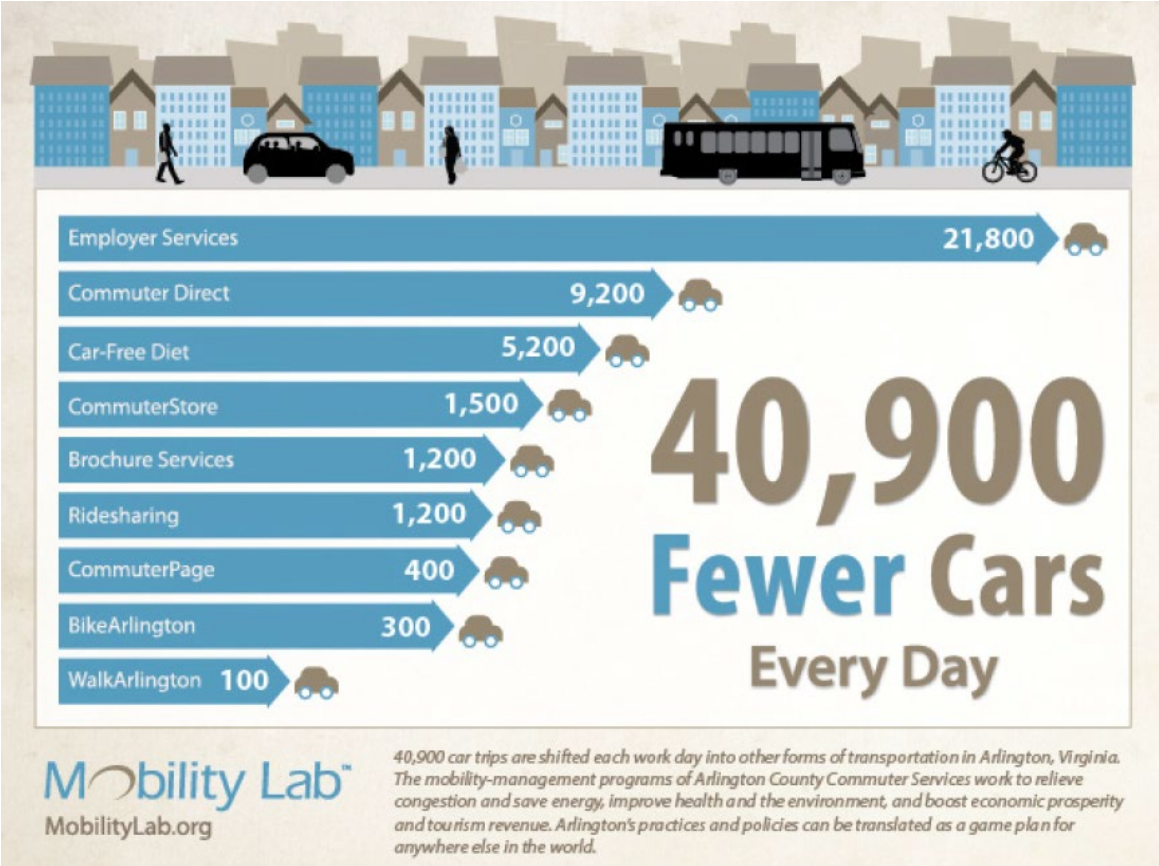


Figure 8 - An Arlington County Mobility Lab report on the impact of its TDM program

Through its TDM efforts, Arlington County claims to prevent over 40,000 solo car trips each workday, as seen in **Figure 8**. The residential and employer outreach programs reduce car trips by 40% for residential buildings[14] and result in office workers taking 39-79% fewer weekly trips than would be expected without TDM. Recipients of transit subsidies were also twice as likely to commute by transit. [15] Through the TDM for Site Plan Development Program, Arlington decreased driving from 71% to 28% where parking charges were added and made it possible to unbundle parking from density requirements, resulting in 13% lower car ownership by adult residents in the affected developments **(Table 10**).

[14] Mobility Lab. Annual Report (2018). https://1105am3mju9f3st1xn20q6ek-wpengine.netdna-ssl.com/wp-content/uploads/2018/05/Residential-Aggregate-Analysis_Final-Report.pdf
[15] Mobility Lab. Office Building Transportation Performance Monitoring Study (2016). https://mobilitylab.org/research-document/2016-office-building-study/

Table 10 - Average Drive - Alone Commute Trips at the Building Level

	Drive alone share of commute trips		Percent of commuters who always drive alone	
TDM Measure◊	**Offered**	**Not Offered**	**Offered**	**Not Offered**
Travel Information (64%)	49%	47%	39%	47%
Transit Support (16%)	50%	46%	41%	37%
Bicycle Amenities (69%)	49%	NA**	40%	NA**
Shuttle to Metro* (31%)	61%	64%	49%	54%
Unbundled Parking	48%	54%	38%	45%

*comparison is only to buildings outside the Metro walking catchment

** Measure offered in all buildings

◊Number in parentheses indicates the percent of respondents who believe the TDM measure is available in their building.

Arlington County Commuter Services' Priority Strategies and Expansion Programs for 2018-2023 include:

- Strengthen Arlington's Regional Competitiveness
- Reduce Single Occupant Vehicle Usage
- Advance Equity in the Use of Transportation Options
- Expand ACCS's Organizational Capacity
- Extend the Reach of TDM
- Integrate First Mile/Last Mile Technologies and Services
- Utilize New Electronic Information to Foster Transportation Behavior Change
- Update TDM County Policies
- Strengthen Capital Bikeshare as Transportation Service
- Utilize Return on Investment Measures for TDM
- Conduct Research Activities to Advance the Effectiveness and Use of TDM

Conclusions

Arlington is a comprehensive example of a **successful operational TDM program**. The case study demonstrates the value of **rigorous tracking and evaluation, producing measurable results that the organization can use to justify requests for more funding and expansion of its mandate**.

Case Study #11: San Francisco TDM Plan

The San Francisco TDM Plan was published in 2017. The City experiences obstacles to build infrastructure for its expected growth in specific areas since the land is mostly developed already. Building key infrastructure is associated with major costs, hence the City has decided to build upon existing structures and to concentrate on advancing significant projects such as the Central Subway and the Van Ness Bus Rapid Transit.

Figure 9 - Road Sharing Infrastructure Initiatives (Source: San Francisco TDM Plan)

Figure 10 - Road Sharing Infrastructure Initiatives (Source: San Francisco TDM Plan)

Equity

To be able to support lower transportation costs, it is vital to integrate walking, cycling, public transit, and car-share where necessary and to implement it from the TDM Plan. The strategy in the TDM Plan is to allow residents that are residing in low-income communities, whether they have limited access to internet, inadequate technology, language barriers to still be able to make informed travel choices.

San Francisco TDM Plan has an equity focused TDM measure, which is Strategy 10: Facilitate transportation equity through targeted provision of education and encouragement for minority, low-income, disabled, and senior populations to take transit, walk, bike, use rideshare or carshare.

Implementation strategies with key actions include:

- Encourage pass-based programs which provide a discount to Muni fares for low/ moderate income residents, seniors, residents with disability, and youth under 18 years of age.
- Build a strategy on understanding the unique transportation wants and different methods of serving multiple socio-economic and demographic groups within the city, and to allocate for those that travel to and from the city.
- Look into the feasibility of distributing free bikeshare memberships to those that qualify within the San Francisco social service agencies.
- Have ongoing outreach, education, and encouragement to take all forms of transit, target various populations to reduce the costs, as well as getting residents to use transit and bikeshare.
- Create transit etiquette materials that aim to give safe transit experiences for all passengers, but especially to seniors and people with disabilities.

- Encourage data collection and the analysis of paratransit trips to understand if the services provided are ideal for serving seniors and people with disabilities.
- Site management practices should include and take into consideration transportation options to accommodate low to moderate income populations.
- Explore solutions for vehicle sharing programs to be responsibly priced for people in all income backets to access and without the worry of primary language spoken.
- Help the implementation of the pilot shuttle program for Bayview residents to use this transportation service.

Audience

The City is expected to grow 5% in the next few years, with major shift in demographics in the age brackets of 14 years old and under, as well as 60 years and older. These individuals are looking for more choices in transportation and housing rather than car-oriented lifestyle. Demographics show that there should be a focus on TDM initiatives targeted towards children, elderly, and families to provide sustainable travel modes. The TDM Plan should also focus on alleviating economic inequality and have the City's diverse residents get involved by using culturally sensitive ways of communication.

Employment Centre

There was a projection of a 7.9% increase of employment from the year 2015-2020 In addition, the City will add about 190,000 jobs and 100,000 households by the year 2040. This would put pressure on the transportation system, mainly being used by San Francisco residents, but also being used by workers from neighbouring cities that travel into San Francisco for work. It is recommended that the City use TDM to enable employees to fully utilize all the transportation options, for example, transit from local and regional governments to be included in TDM programs, policies, and services for employees/employers.

Monitoring & Evaluation

It is important to continue monitoring and enforcing TDM measures in development project approval processes to ensure that developments have TDM measures in place to serve residents, tenants, and employees. There has been a lack of monitoring in the past due to limited staff capacities, however moving forward, there should be partnerships between agencies to monitor and ensure compliance.

The TDM Plan has a TDM measure that focuses on monitoring, evaluating, and enforcing the practices into the development project approval process. Strategy 4: Monitor, evaluate and enforce conditions of development project approval.

Implementation strategies with key actions include:

- Maintain a standardized monitoring system, reporting practices, and tools that can be applied to existing and future development projects, incorporating data tracking.

- Enforce and monitor TDM-related developer conditions.
- Create and implement evaluation programs that can be used to identify the effectiveness of TDM initiatives related to land use development.
- Resume the enhancement of TDM team's role in informing developers building new development to be aware of and to consider TDM measures.

TDM program evaluation should report transportation behaviour changes as it affects transportation-related metrics, for instance vehicle miles traveled (VMT) and greenhouse gas emissions.

The success of TDM initiatives in reducing single occupancy vehicles (SOV) is shown in the following:

1. An increase in walking, cycling, transit users, and shared rides are all factors that may be a result of TDM efforts. Since there are many reasons that could justify why these modes of travel have grown, for example an addition of new bike lane, and walking challenges. Which makes it difficult to verify which TDM programs are comparable to the impact of change.
2. TDM program achievement is all of the efforts (programs and policies) added together and should be assessed at its entirety. However, individual programs should still be evaluated.[16]

Conclusions

The San Francisco case study provides TDM measures, plus implementation strategies that **focus on equity**, changing demographics, employment growth and the impacts it has on the existing transportation system. Also outlining ways to monitor and evaluate TDM measures in existing and future developments. The study lists out implementation strategies that could help reduce single occupancy vehicle use while outlining that TDM measures should be evaluated individually and as a whole.

[16] San Francisco TDM Plan (2017). 11-7-17_item_11_transportation_demand_management_plan_0.pdf (sfmta.com)

3. Summary of Findings

The case studies reviewed provided a broad sample of best practices in Transportation Demand Management in the GTHA and beyond. The findings will be used to inform the Town of Ajax TDM Plan update. The following are key themes:

Understanding the stakeholders

A holistic change to modal shift requires the involvement and coordination of many stakeholders. The Metrolinx case study provides a useful overview of some of those organizations including all levels of government, municipal transportation and planning departments, schools and school boards, developers, employers, and community organizations. The City of Kitchener example demonstrates the importance of engagement between both local groups the Region as a means of meeting TDM targets and goals in the long-term. Any TDM strategy should consider how to involve these stakeholders and work with them to achieve mutually beneficial outcomes (such as encouraging less parking in a new development, in exchange for providing discounted transit passes to residents/workers where service is provided, as seen with the City of Coquitlam). The York Region case study touches on a specific stakeholder collaboration, targeting active school travel amongst students. Part of the coordination involved here is providing infrastructure and materials that support safe and accessible school commute, that in turn reduces single-occupant trips during school pick-up and drop-off periods.

Assigning responsibility where authority exists

While a third-party can be relied upon for providing effective TDM programming ("soft" measures), these organizations have limited ability to implement "hard" measures such as policy changes or infrastructure-related initiatives. The Peel case study assigns responsibility and actions to the groups with the greatest ability to make change; for example, by calling upon transportation staff to implement a multimodal level of service (MMLOS) evaluation framework for new roadway projects. TDM must be core to an organization's operation to be successful.

Funding Sources and Measuring and Monitoring Reports

Regular evaluation and reporting are needed to ensure that the value of various TDM measures are understood and communicated to the public, as noted through the City of Coquitlam, San Francisco, and Arlington case studies. When tangible results are produced, as in the case of Arlington, they can be used to justify funding increases or taking on a broader mandate. Without results, TDM risks being perceived as a "nice-to-have" that can be cut during a tight budget cycle. Measuring and monitoring tools are also critical to updating and/or introducing new measures as opportunities become available for communities. San Francisco evaluates some TDM initiatives by analyzing vehicle miles travelled, green house gas emissions, and reduction of single occupancy vehicles (SOV).

It is also important that TDM be recognized as a long-term, value-adding initiative for achieving sustainable mode share goals, rather than a series of marketing initiatives. The Arlington case study shows the value of robust funding requirements, such as government-funded programs, and evaluation tools that assess the impacts of TDM services to see a positive return on investment, in contrast to Waterloo's public-private partnership which relies more on short-term funding. Similarly, the City of Whitehorse identifies specific funding sources, such as gas tax funds, as a means to allocate financial means towards maximizing existing infrastructure with opportunities for transit and active transportation.

Alternative Work Arrangements

Most impactful to travel demand strategies is reducing or eliminating the need for travel altogether. Implementing alternative work arrangements can yield outcome-oriented productivity amongst employees, increasing both the social and economic benefits for both the employer and employee. Both the York Region and City of Whitehorse case studies speak to measures that qualify as alternative travel and work arrangements, including teleworking or even flexible work schedules outside peak working hours to spread out travel demand over time.

Integrating TDM in New Developments

Establishing a method for quantifying the TDM benefits of a project and incorporating an evaluation process into planning documentation. Regulating evaluation ensures that projects are meeting the minimum TDM threshold recognized by the municipality. The City of San Francisco uses monitoring and evaluation techniques related to development projects that range from reporting practices, incorporating data tracking, identifying the effectiveness of TDM measures, and informing developers. The City of Kitchener outlines strategies to integrate TDM into site planning and development approval processes, by using a TDM checklist. For instance, that applications for new developments must include a TDM Plan as a component of transportation impact studies. The City of Brampton and the City of Coquitlam are also examples where minimum thresholds and development application tools, such as a TDM checklist, are used to evaluate elements such as parking requirements to minimize dependency on single-occupant vehicles and instead encourage use of sustainable modes with alternative incentives such as bicycle parking thresholds and subsidized transit passes.

Next Steps

This Summary of Best Practices report has reviewed and summarized existing Town plans, policies, and guidelines related to TDM as well as national and international examples of TDM measures. Moving forward, additional inputs will be collected and analyzed through the proposed Communication and Engagement Plan for the TDM Update. The Communication and Engagement Plan will focus on agency consultations, a public open house, and a social climate analysis. The intent is to ensure the updated Plan is implementable by the Town, as well as equitable and accessible to all users.

Upcoming Deliverables beyond the Summary of Best Practices includes:

- Updates to the TDM Development Guidelines, Parking Management Strategy, and Bicycle Parking Design Guidelines, where appropriate
- Development of a Measuring and Monitoring Program
- An updated TDM Plan for the Town of Ajax, inclusive of program targets, timelines, and a proposed comprehensive list of measures and guidelines on overall implementation

Town of Ajax Report

Report To:	General Government Committee
Prepared By:	Jason McWilliam, Manager of Legislative Services/Deputy Clerk
Report #:	LIS-2023-07
Subject:	**Privacy Risk Management Policy**
Ward(s):	All
Date of Meeting:	April 11, 2023
Reference:	N/A

Recommendation:

1. That the Privacy Risk Management Policy, found as Attachment #1, be approved; and,
2. That the Town Clerk, or designate, be delegated authority to amend the policy as required to incorporate industry best practices and support continuous improvement to the Town's privacy risk management program.

Background:

Since its introduction in 1990, the *Municipal Freedom of Information and Protection of Privacy Act* (MFIPPA and/or the Act) has established basic requirements guiding the collection, use, retention, and destruction (processing) of personal information by municipalities and other public sector entities. While minor amendments have been enacted from time to time, its requirements have failed to keep pace with changes occurring in society, including evolving public, legal and regulatory expectations, industry best practices and the increased reliance upon technology and third-party solutions in municipal operations. Due to these broad changes, organizations in both the public and private sector have adopted policies and supporting practices to meet expectations that exceed the basic requirements outlined in legislation.

Historically, much of the Town's focus has been on meeting the basic legislative requirements; however, following significant investments made in the Town's information technology capabilities and the heightened awareness and concern for the security of personal information, it has become apparent that there are opportunities to improve the management of personal information processed during the Town's business activities. To support these improvements, a Privacy Risk Management Policy has been developed to provide a foundation for changes that will improve the security and management of personal information processed by the Town's business activities and support the continued development of a culture of privacy.

At a high level, the policy defines privacy principles guiding the processing of personal information by the Town and articulates the organization's risk appetite as it pertains to the use of personal information, imposes a requirement to complete of Privacy Impact Assessments (PIAs) prior to the processing of personal information, and clarify roles and responsibilities and reporting thresholds when privacy incidents and breaches occur within the organization. These

changes will not fully eliminate privacy incidents and breaches; however, they should result in improved controls and technical safeguards applied to personal information and assist staff in better understanding their responsibilities when working with personal information.

Discussion:

The proposed Privacy Risk Management Policy contains several sections, with the key elements identified and discussed in greater detail below:

- Privacy Principles
- Privacy Risk Appetite Statement
- Privacy Impact Assessments
- Privacy Incident and Breach Reporting
- Third Parties
- Roles and Responsibilities
- Complaints and Inquiries

Privacy Principles:

The new policy identifies nine privacy principles that inform and underpin the policy and the Town's overall privacy program. These principles are drawn from work completed by the Organization for Economic Co-operation and Development (OECD) to define minimum standards for privacy and data protection and are reflected in most privacy legislation, including MFIPPA, the Personal Information Protection and Electronic Documents Act (PIPEDA), Canada's private-sector privacy legislation and Europe's General Data Protection Regulation (GDPR), recognized as the most stringent privacy legislation currently in effect anywhere in the world.

These principles define the Town's commitment to those whose personal information is collected and used in our activities. It should be noted that in certain circumstances, current technology or business processes may limit our ability to fully meet the expectations established by the principles. Where this occurs, future work undertaken by staff will seek to close these gaps and harmonize our activities with these principles.

Privacy Risk Appetite:

The information and data collected and used by our activities are critical data assets and Town staff are increasingly expected to engage in data-driven decision-making and derive insights, where appropriate, to drive innovation and improve the efficiency and effectiveness of Town services. However, in order to do so, staff must be aware of the risks the organization is willing, or unwilling, to assume in pursuing these objectives. Accordingly, a privacy risk appetite statement has been prepared to guide staff in their work, while communicating to Town residents and stakeholders the types of risks that the organization is willing to assume.

The statement that has been developed is conservative in nature, recognizing our obligations under MFIPPA, our role as a trusted steward of personal information and an inability for individuals to seek out competing alternatives for the services provided by the Town. It identifies that there is no appetite for the misuse, loss or theft of personal information and commits the organization to implementing appropriate administrative controls and technical safeguards to prevent such events. However, recognizing the imperative to make more effective use of our informational assets, it permits the use of data, including de-identified and aggregated personal information, in data analytics activities to improve Town operations, to enhance the efficiency or capacity of Town programs, or to derive insights that improve the customer or client experience.

Privacy Impact Assessments:

The most significant and important change resulting from the adoption of this policy will be a new requirement to complete Privacy Impact Assessments (PIAs) on any new or revised business activity that utilizes personal information, or data containing de-identified or aggregated personal information. PIAs are a risk management tool use to assess, document, and accept risks associated with the processing of personal information and will provide for more effective management awareness and oversight of business activities. The completion of PIAs will ensure that activities align with regulatory requirements, the Town's privacy risk appetite and privacy principles. Finally, they will provide evidence of the Town's compliance with MFIPPA in the event of a privacy compliant and/or investigation by the Information and Privacy Commissioner of Ontario (IPC). By adopting a standard methodology and process to assess, mitigate and risk-rate business activities, it will be possible to establish Key Performance Indicators (KPIs) and Key Risk Indicators (KRIs) and measure efforts to minimize risk over time and provide additional datapoints that can be used to prioritize investments in people, processes and technology.

Privacy Incident and Breach Reporting:

As noted earlier in this report, MFIPPA has failed to keep pace with developments occurring in the privacy space and incorporate best practices that have been identified in other jurisdictions. Included amongst these are mandatory breach reporting requirements and standards assisting organizations in understanding the circumstances where notification to affected individuals and privacy regulators is, and is not, required.

To provide clarity to Town staff and community stakeholders, the policy will require notice to affected individuals where a breach poses a real risk of significant harm. Taking cues from PIPEDA and the GDPR, staff will assess the incident and take into consideration the sensitivity of the information involved and likelihood that the information is or may be misused before determining whether notification is required. Where the analysis determines that there is a risk of significant harm, including bodily harm, humiliation, damage to reputation or relationships, loss of employment, business or professional opportunities, financial loss, identity theft, negative effects on the individual's credit score or damage to or loss of property, staff will provide notice to the affected individual(s) and the IPC as soon as the circumstances permit.

It is important to highlight that this approach may result in some privacy incidents not being reported to affected individuals or the IPC, particularly where the information involved is not particularly sensitive (for instance where personal information is limited to name and telephone number or email address) or where there is evidence that there is a low probability that personal information will be misused (for example, when information is disclosed internally, or written confirmation has been received that the information has been destroyed/deleted by the recipient). This approach is consistent with industry best practices and will avoid potential stress and anxiety where the likelihood of harm to an individual is low.

Third Parties:

Finally, the policy will result in improvements to the way third parties are assessed and onboarded before their services are leveraged by the Town. Residents and other stakeholders have come to expect to engage with the Town when and how they see fit, often requiring solutions that leverage cloud-based technologies. While these solutions improve the user experience, in many cases they result in less direct control over the configuration of these solutions and reduced oversight over how personal information is processed.

To address these challenges, the policy establishes requirements for contracts executed between the Town and third parties, particularly where the third party will have access to, or process personal information on behalf of the Town. These requirements will also include clauses defining breach notification timelines and prohibitions on the secondary use of personal information for a vendors own business purposes.

Financial Implications:

There are no direct financial implications associated with the recommendations contained in this report. Any costs associated with the implementation of the policy and associated requirements will be accommodated within existing operating budgets.

Communication Issues:

To support the rollout of the Privacy Risk Management Policy, internal communications and training will be delivered to Town staff to ensure they are fully aware of the new requirements imposed by the Policy. Additionally, information will be included on the Town's website identifying the privacy principles underpinning the Town's privacy management program.

Relationship to the Strategic Plan:

This report supports the pillar of Modernizing our Community and Priority 4.2 – Readying our Organization for the Future.

Conclusion:

As a result of numerous changes in the privacy and regulatory landscape, gaps in the Town's current program, as well as the heightened importance of privacy in the lives of Canadians, including Ajax residents and stakeholders, it is now time for the Town to implement a more structured privacy program. The adoption of a comprehensive Privacy Risk Management Policy is a first step in improving the Town's risk management practices within the privacy domain and a clear action demonstrating the Town's commitment to privacy.

Attachments:

ATT-1: Privacy Risk Management Policy

Prepared by:

Jason McWilliam – Manager, Legislative Services/Deputy Clerk

Submitted by:

Nicole Cooper – Director, Legislative and Information Services

Approved by:

Shane Baker – Chief Administrative Officer

TOWN OF AJAX POLICIES AND PROCEDURES

SUBJECT: **Privacy Risk Management**

COUNCIL ISSUED:	COUNCIL REVISED: Click here to enter a date.	AUTHORED BY:	REVISION #:	Page 1 of 8	POLICY #:

1. POLICY OBJECTIVE

1.1. The objective of this policy is to define a framework to minimize legal, operational, privacy, regulatory and reputational risks associated with the improper creation, collection, use, sharing, retention or destruction ("processing") of personal information in the course of the Town's business activities.

2. SCOPE

2.1. This policy applies to all Town of Ajax employees, contractors, volunteers and Members of Council and activities involving the processing of personal information.

3. PROCEDURE

3.1. Definitions

Data Subject means the individual who personal information is about.

Employee means in the context of this policy, all Full-Time, Part-Time, Contractual and Seasonal employees, as well as Members of Council and Volunteers.

IPC means the Information and Privacy Commissioner of Ontario.

PIA means a Privacy Impact Assessment.

Personal Information means information about, or that could be used to infer the identity of, an identifiable individual.

Privacy Incident means an event where an individual believes personal information may have been subject to unauthorized processing.

Privacy Breach means an event where an investigation has confirmed that personal information has been subject to unauthorized processing.

De-identified Information means personal information which has been aggregated or processed to remove identifiers which can be used to directly identify an individual, or which can be used to infer the identity of an individual.

Risk of Re-identification means the risk that de-identified or aggregated data can be linked back to an identifiable individual.

MFIPPA means the Municipal Freedom of Information and Protection of Privacy Act.

Privacy Risk means the risk of improper collection, use, sharing, retention or destruction of Personal Information.

Processing means the creation, collection, use, sharing, retention or destruction of personal information.

Re-identification means the use of alternative datasets, including third party and publicly available data, to reveal or discover the identity of individuals contained in de-identified information.

Significant Harm includes bodily harm, humiliation, damage to reputation or relationships, loss of employment, business or professional opportunities, financial loss, identity theft, negative effects on an individual's credit record and damage to or loss of property.

Third Party means any non-Town of Ajax entity that delivers, or supports the Town in delivering, products or services, whether a fee for service is involved or not, or whether they have been engaged through the Town's procurement processes or not.

Town means the Corporation of the Town of Ajax, including its constituent departments, boards and committees.

3.2. Privacy Principles

The Town of Ajax has adopted the following principles to serve as a foundation for its privacy program and to guide its activities when processing personal information. Where they cannot be fully achieved immediately, they will serve as objectives to be achieved as the Town's works to enhance the management of personal information. These principles are supplemented by the implementation of administrative controls and technical safeguards to ensure compliance with applicable statutory requirements, regulatory expectations, and privacy best practices.

3.2.1. Accountability

The Town has delegated to the Director, Legislative and Information Services/Town Clerk the role as Head for the purposes of the Municipal Freedom of Information and Protection of Privacy Act ("MFIPPA"), including the duty to oversee the Town's compliance with MFIPPA and other applicable privacy regulations. At all times the Town will remain accountable for the ways in which personal information is processed by the Town, its employees and any third parties we may choose to support it in conducting its business activities.

3.2.2. Transparency and Purpose Limitation

When collecting personal information, the Town will identify the purposes for which it will be processed. Once collected, information shall be processed only for the reasons identified at the time of collection, or that are consistent with the original purpose and are reasonably anticipated by the data subject or are otherwise permitted by law.

3.2.3. Data Minimization

The Town limits collection of personal information to that which is needed to fulfill our business requirements. Where achievable, the Town will employ de-identified information instead of identifiable information in our business activities. At no time will the Town attempt to re-identify aggregated or de-identified data supplied by a third party or that is publicly available. Before making aggregated or de-identified datasets available to the public, the Town will assess the risk of re-identification and refrain from releasing data where the risk is deemed excessive.

3.2.4. Collection Limitation

At all times, the Town will only collect personal information where permitted by law, using fair and lawful means. Personal information will only be collected directly from the data subject unless indirect collection is necessary in the circumstances and permitted by law.

3.2.5. Individual Rights

The Town will support data subjects as they seek to exercise their rights, including the right to access their personal information, under applicable legislation.

3.2.6. Data Quality

The Town will strive to ensure that activities processing personal information use information that is up to date and accurate and will seek to implement solutions that make it easy for data subjects to update or correct inaccurate information.

3.2.7. Retention Limitation

The Town will only retain personal information as long as necessary to fulfill the purpose for which it was collected, including any related statutory or regulatory requirements, including our internal records retention requirements.

3.2.8. Integrity and Safeguarding

The Town will ensure the processing of personal information is conducted securely and appropriate controls, including administrative policies, and physical and technical controls to restrict unauthorized processing, are implemented. The robustness of our controls will also take into consideration the sensitivity of the personal information that is processed and the harms that could result from the inappropriate processing of such information.

3.2.9. Third Parties

Where required, the Town may engage Third Parties to process personal information on its behalf. Where this occurs, the Town will only provide access to personal information that is required to perform the services requested and contractually bind the Third Party to protect personal information. The Town will also prohibit the secondary use of personal information by third parties for their own purposes wherever possible.

3.3. Privacy Risk Appetite

The Town of Ajax recognizes that all activities involve some degree of risk. To assist employees and members of the public to understand the general nature of the risks that the Corporation is willing to accept in the privacy domain, the following risk appetite statement has been prepared.

Privacy risk spans Legal and Regulatory, and Data Security risk domains. Failure to adhere to our legal, regulatory and data protection obligations may lead to legal challenges, financial penalties, regulatory scrutiny and reputational harm. Accordingly, the Town of Ajax has a low tolerance for activities using personal information that do not align with the expectations of our stakeholders or the letter or spirt of the laws and regulations that apply to the Town and no tolerance for the loss, or unauthorized or accidental processing, of personal information.

The Town of Ajax has a high tolerance for the use and re-use of de-identified and aggregated personal information in data analytics activities, including by machine learning and artificial intelligence algorithms, that improves the operational efficiency or capacity of Town programs, or results in actionable insights that improve the customer/client experience. To support these objectives, the Town will develop and implement operational frameworks that empower staff to derive insights and drive innovation while ensuring the responsible and ethical use of data.

3.4. Corporate Privacy Program Requirements

3.4.1. Privacy Impact Assessments

Privacy Impact Assessments are a risk assessment completed to identify, evaluate, and mitigate risks associated with the processing of personal information.

Privacy Impact Assessment's (PIA) are required for all new activities processing personal information, or de-identified data, for existing activities where there is a material change in the activity, or where a third party is engaged to support the delivery of services by, or on behalf of, the Town. A material change includes, but is not limited to, the processing of new personal information, a change in the way personal information is processed, or a change to the systems or third parties used to process personal information by or on behalf of the Town of Ajax. The PIA will assess the activity to ensure that the processing of personal information:

- is permitted by applicable law;
- is consistent with the Privacy Principles and Privacy Risk Appetite Statement;
- is necessary to achieve the stated objectives for which personal information has been collected;
- aligns with the expectations of the data subject and the public more broadly;
- implements appropriate administrative and technical safeguards to protect personal information; and,
- does not result in risks that outweigh the benefits afforded to the individual.

PIAs must be completed, reviewed, and approved before personal information is processed. PIAs must be completed by the Department processing personal information. PIAs must be reviewed by privacy specialists within Legislative and Information Services to assess the inherent and residual risks associated with the activity. Privacy specialists may make recommendations to minimize any identified risks which must be reviewed by the relevant Department Head. All recommendations and outstanding risks must be accepted by the relevant Department Head before personal information may be processed by the activity. Where activities do not adhere to the Town's privacy principles or pose significant privacy risk to the individual's whose information is being processed, or reputational risk to the organization, and where consensus cannot be reached between LIS and the relevant department on mechanisms to appropriately mitigate such risks, the proposed activities may be escalated for review and approval or rejection by the Chief Administrative Officer.

3.4.2. Privacy Incident and Breach Reporting

A privacy incident is any instance where personal information may have been subject to unauthorized collection, use, sharing, retention, or destruction and includes instances where personal information has been used in a way that is inconsistent with the purposes communicated to an individual at the time of collection. A privacy incident may become a privacy breach after an investigation has confirmed a failure to safeguard or otherwise protect personal information. Most incidents occur due to human error but may arise from the malicious activities of a third party, system failures, or weaknesses in controls applied to business processes and systems.

Privacy incidents must be reported as soon as possible to minimize potential harms to those impacted by the incident, protect the interests of the Town and its stakeholders and fulfill the Town's obligations under the *MFIPPA* and other applicable law. All incidents must be investigated as soon as possible to contain the incident, determine the root cause and implement remedial actions to mitigate the occurrence of similar events. All incidents must be managed to ensure compliance with any breach reporting requirements outlined in applicable privacy law or Town procedures.

Legislative and Information Services will coordinate the Town's activities associated with all reported privacy incidents. This includes conducting investigations with the support of internal and external partners, determining whether an incident constitutes a privacy breach and the circumstances where notifications to regulators and individuals impacted by a privacy incident are required, making recommendations to improve the handling or safeguarding of personal information, and managing all interactions with the Office of the Information and Privacy Commission of Ontario. Town Departments are responsible for supporting investigations, completing notification activities, where required, to individuals impacted by privacy breaches originating within their department, and implementing remedial actions required to minimize the recurrence of similar events. Where appropriate, other internal or external stakeholders may be engaged to support investigatory activities, coordinate notification efforts, support implementation of corrective actions or undertake other activities necessary to respond to a privacy incident or privacy breach.

Privacy incidents will be reported to applicable regulators and individuals where the breach poses a real risk of significant harm to those impacted. When assessing whether the incident poses a real risk of significant harm, the Town will take into consideration:

- The sensitivity of the information impacted by the breach; and,
- The probability that information has been, is being, or will be misused.

Where an incident is determined to pose a real risk of significant harm to any individual, the Town will undertake notification activities as soon as the circumstances permit.

3.4.3. Third Parties

At all times, the Town of Ajax remains accountable for the personal information processed during its business activities. From time to time, third parties may be engaged to deliver, or support delivery of, services to the Ajax community and stakeholders. Prior to sharing personal information with a third party, a contract containing data protection clauses requiring, at minimum, the following must be in place:

- Requirements mandating, at maximum, notification within 48 hours of the third party becoming aware of a privacy incident or privacy breach;
- Implementation of controls and safeguards to protect personal information;
- Monitoring requirements to identify authorized and unauthorized processing of personal information;
- Audit rights, including the right to access business and system records prepared by the third party to monitor access to personal information supplied by, or collected on behalf of the Town of Ajax.

Additional controls or operational conditions may be required over and above the contractual requirements identified above. As required by Section 3.4.1 of this Policy, all business activities engaging a third party must complete a PIA before personal information may be shared with, collected or processed by, the third party on behalf of the Town.

3.4.4. Complaints and Inquiries

Departments will be the first point of contact and responsible for addressing department-specific complaints and inquiries respecting the processing of personal information from members of the public. Legislative and Information Services will respond to general inquiries respecting the management of personal information by the Town, serve as an escalation point for complaints and inquiries from members of the public that cannot be resolved by Departments and manage all interactions with the IPC or other regulatory bodies on behalf of the Town respecting the processing of personal information.

3.4.5. Roles and Responsibilities

Employees are responsible and accountable for:

- maintaining the privacy and confidentiality of personal information at all times;
- processing personal information in accordance with established procedures;

- completing mandatory privacy training on a routine basis to ensure they understand the requirements of the Town's privacy management program; and,
- reporting all privacy incidents as soon as possible.

Departments are responsible and accountable for:

- understanding where, why and how personal information is processed by or on behalf of the department;
- developing and implementing business processes and associated controls that conform with this policy and any other applicable policies or standards;
- ensuring that access controls are applied to systems containing personal information and that access to personal information is subject to regular and ongoing monitoring;
- completing PIAs on new or revised business activities involving the processing of personal information;
- completing appropriate due diligence prior to sharing personal information with any internal or external party;
- monitoring the ongoing processing of personal information by employees and third parties to ensure it remains consistent with the purpose(s) stated at the time of collection, or is otherwise permitted by law;
- ensuring employees complete mandatory privacy training required by this policy;
- supporting investigations conducted into all reported privacy incidents;
- providing notice to individuals impacted by privacy breaches; and
- responding to inquiries from the public respecting the ways in which personal information is processed in the department's business activities.

Department Heads are responsible and accountable for:

- championing the importance of adhering to the Town's Privacy Management Policy to departmental staff;
- ensuring that new and revised business activities are assessed by a Privacy Impact Assessment prior to implementation;
- accepting residual risks identified by PIAs that are associated with the processing of personal information undertaken by their Department; and,
- reviewing internal reporting on key risk indicators and the state of privacy risk management within their Department.

Legislative and Information Services are responsible and accountable for:

- reviewing and assessing PIAs, including the adequacy of controls applied to personal information, risk rating inherent and residual privacy risk and making recommendations to mitigate potential risks and harms associated with the processing of personal information;
- developing and delivering mandatory privacy training;
- coordinating the Town's response to all reported privacy incidents or privacy breaches, in partnership with relevant internal and external stakeholders where appropriate;

- managing all interactions with the IPC, and other regulatory bodies, respecting privacy complaints and other inquiries associated with the processing of personal information by the Town;
- establishing and monitoring key risk indicators and reporting on the state of privacy risk management across the Town.